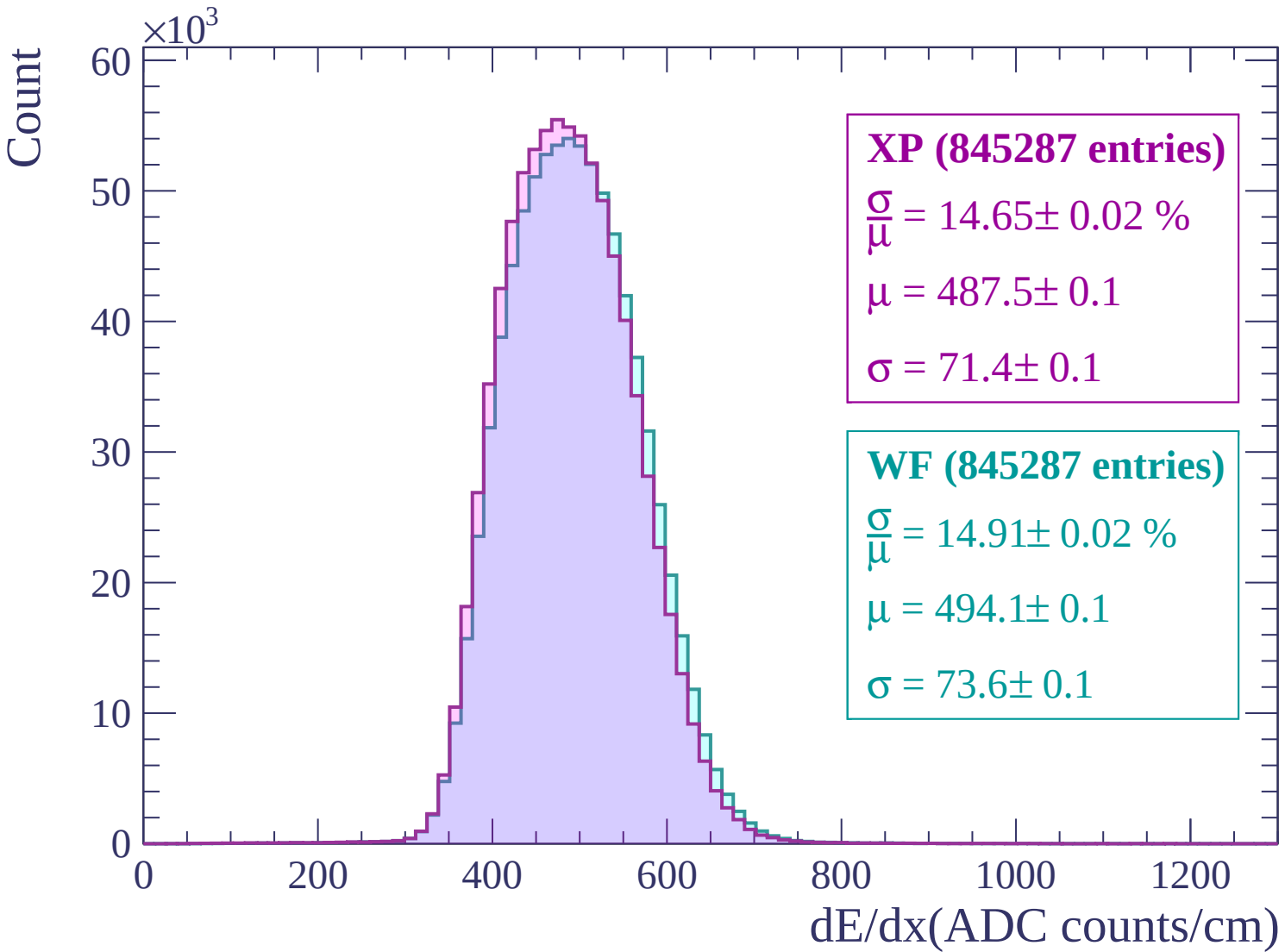
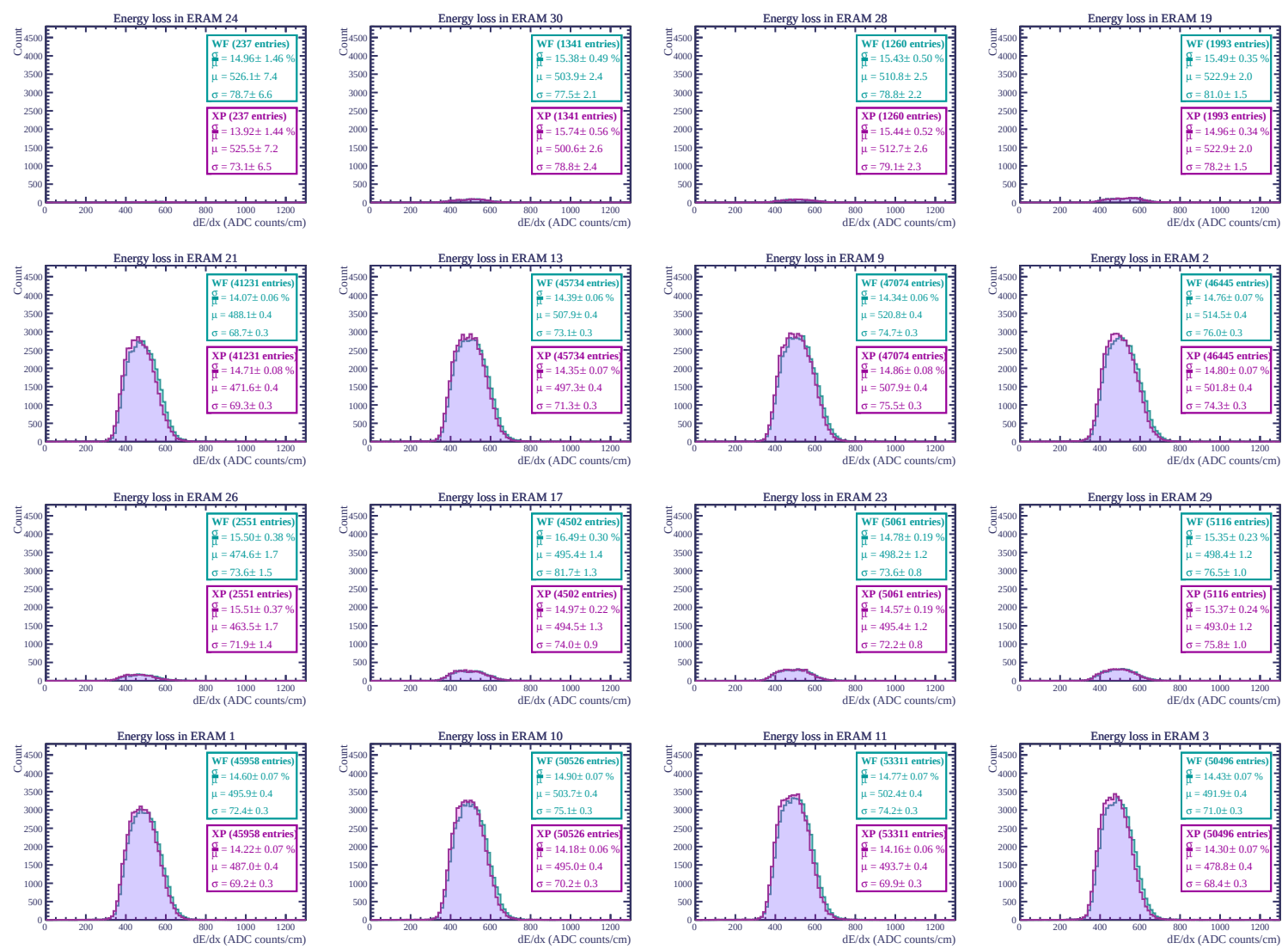
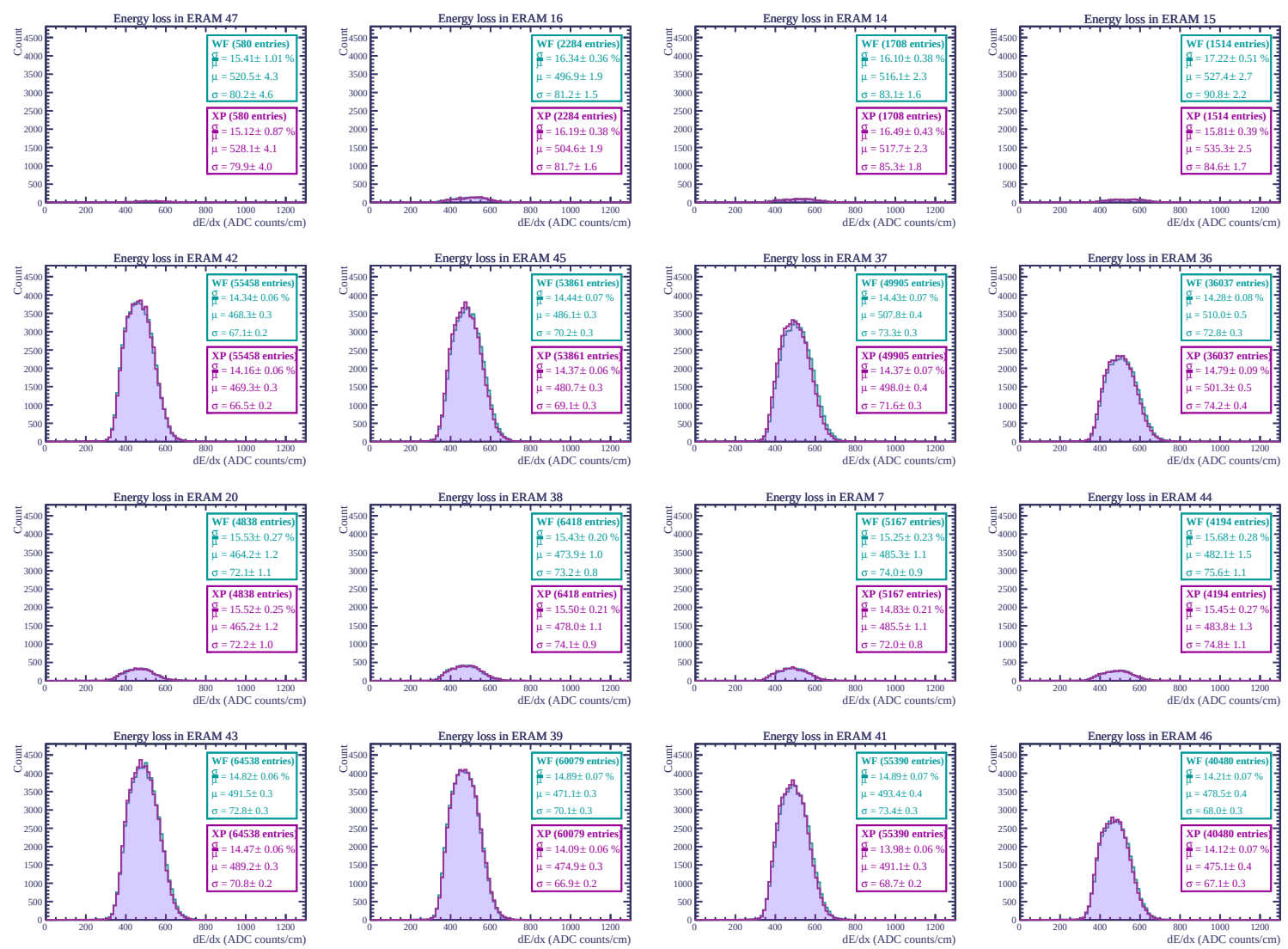
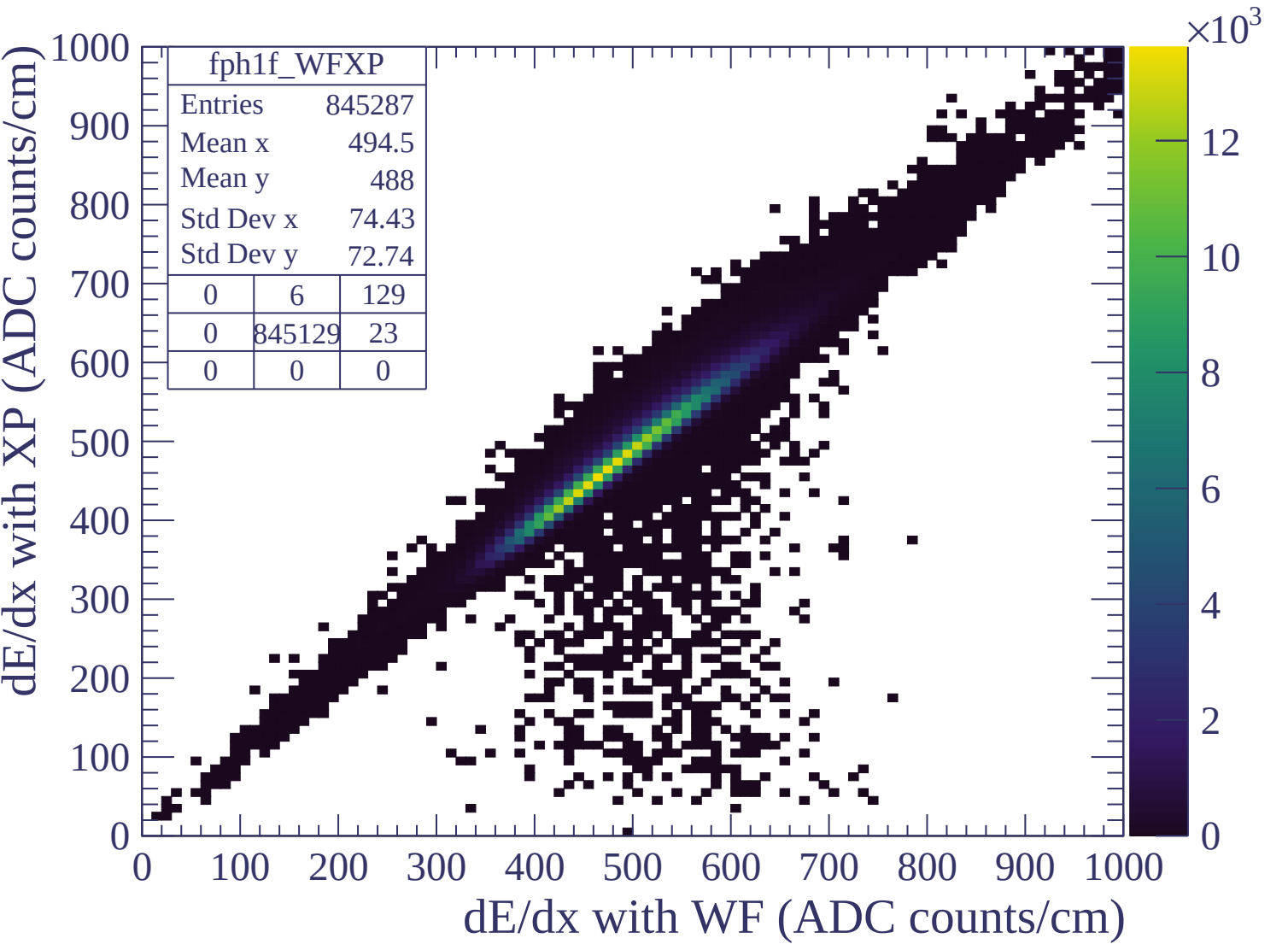
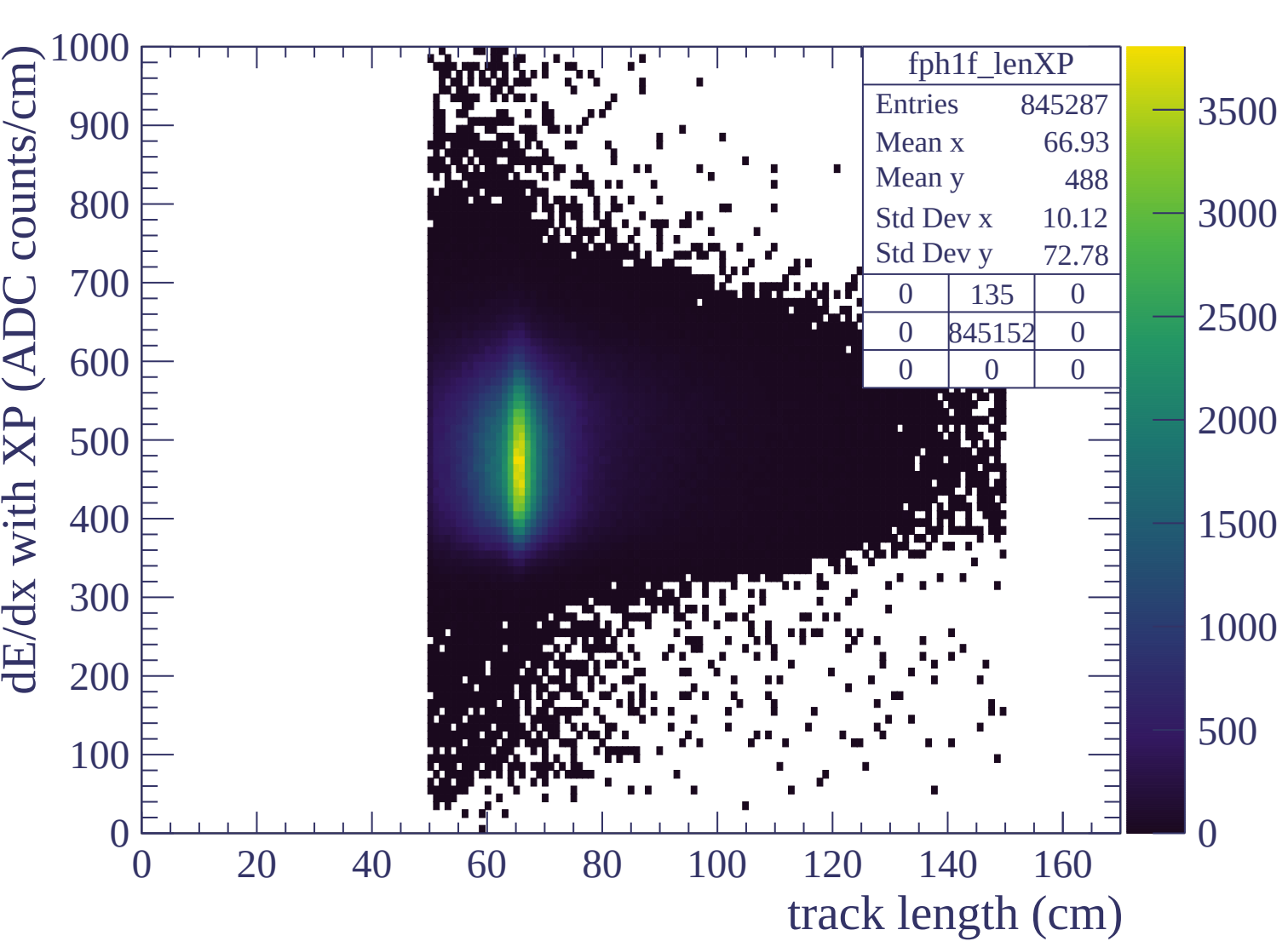


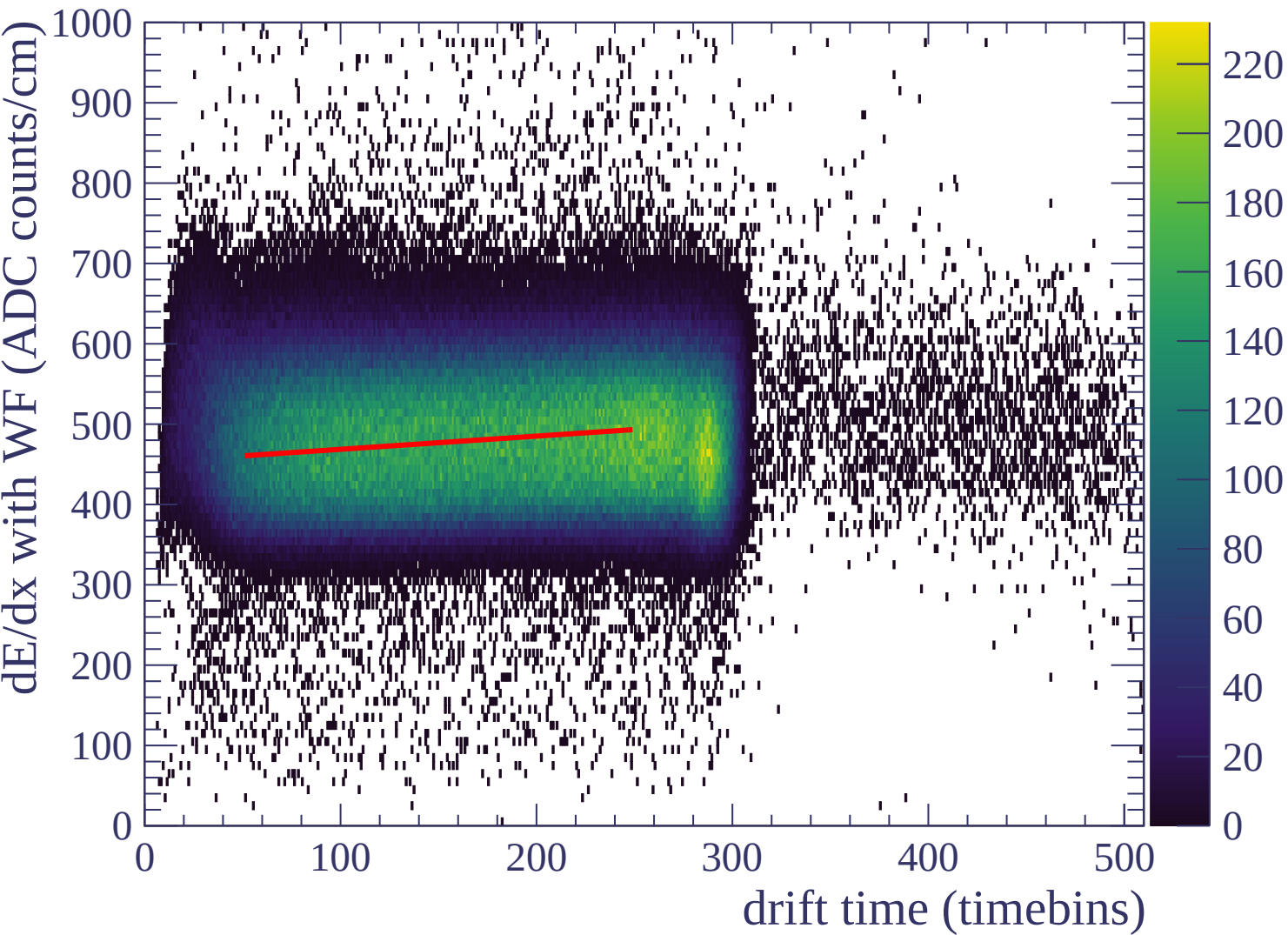


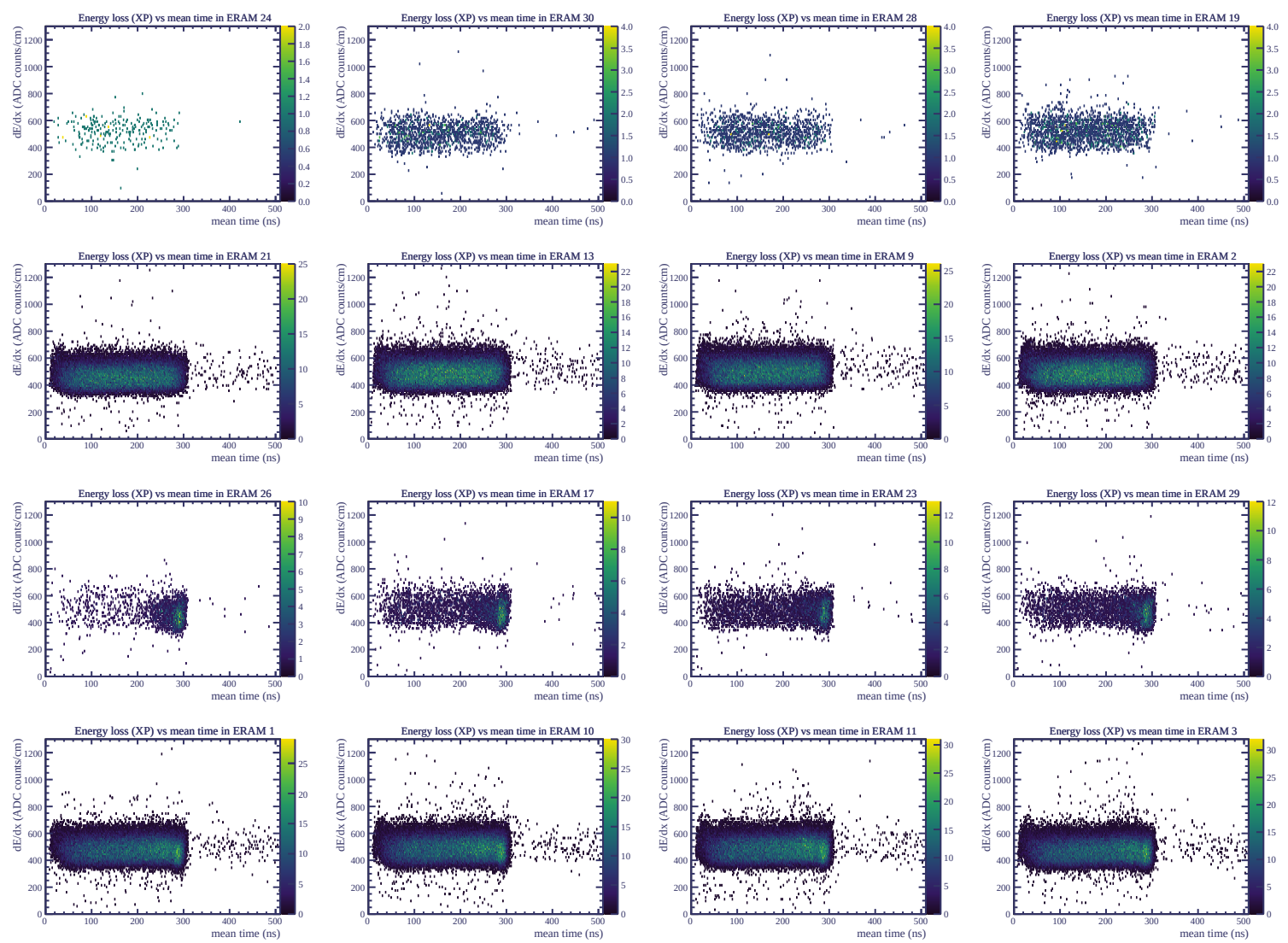


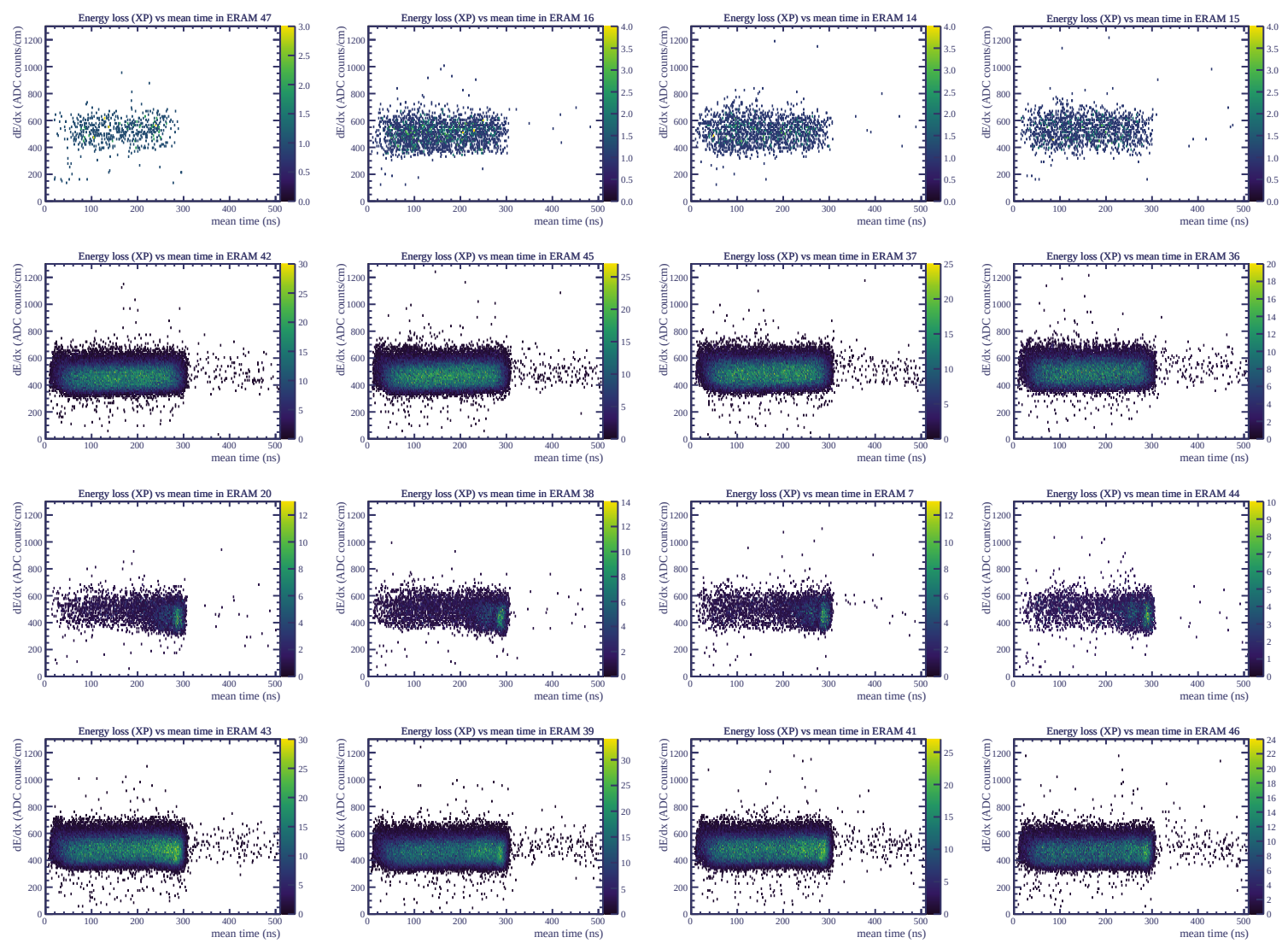






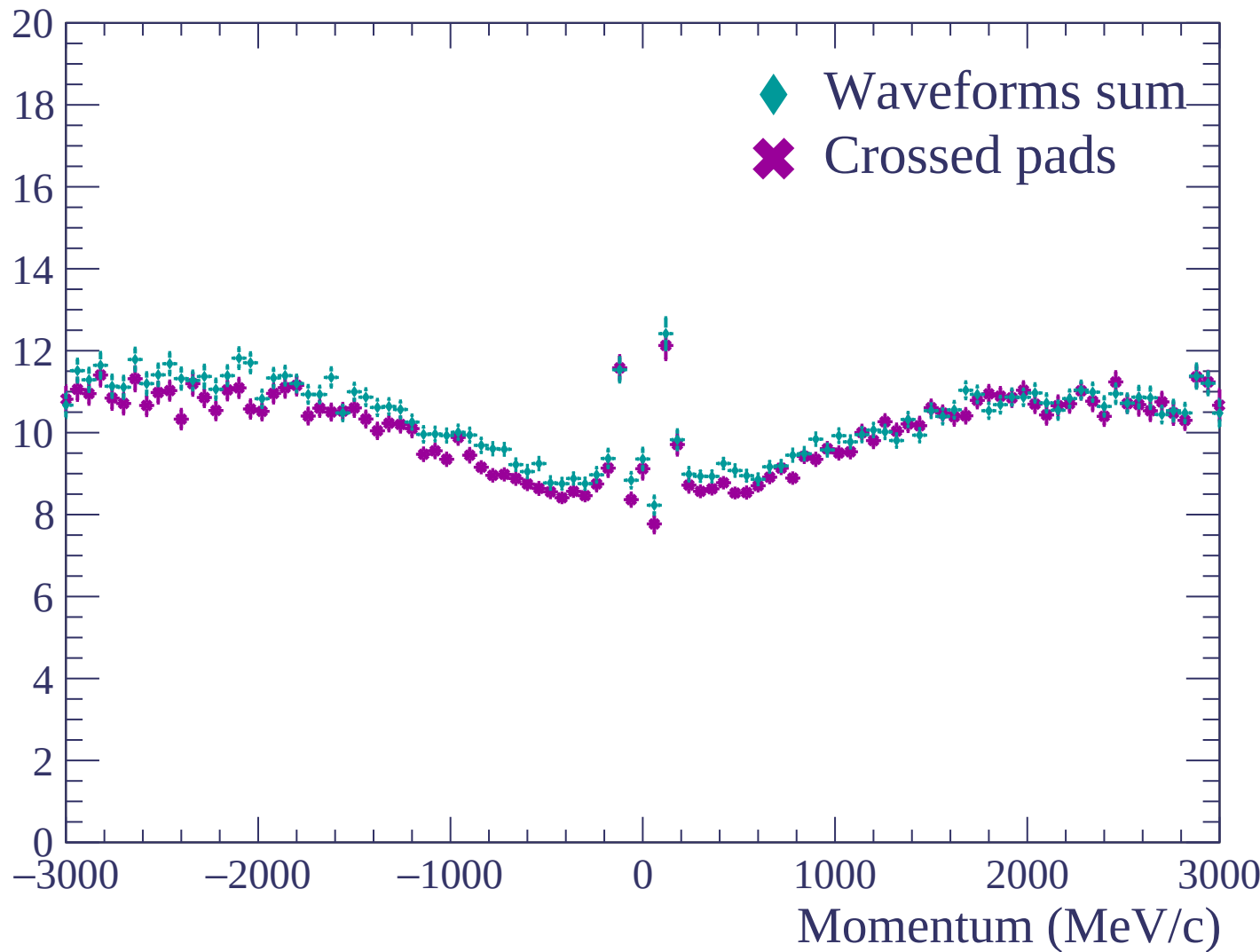


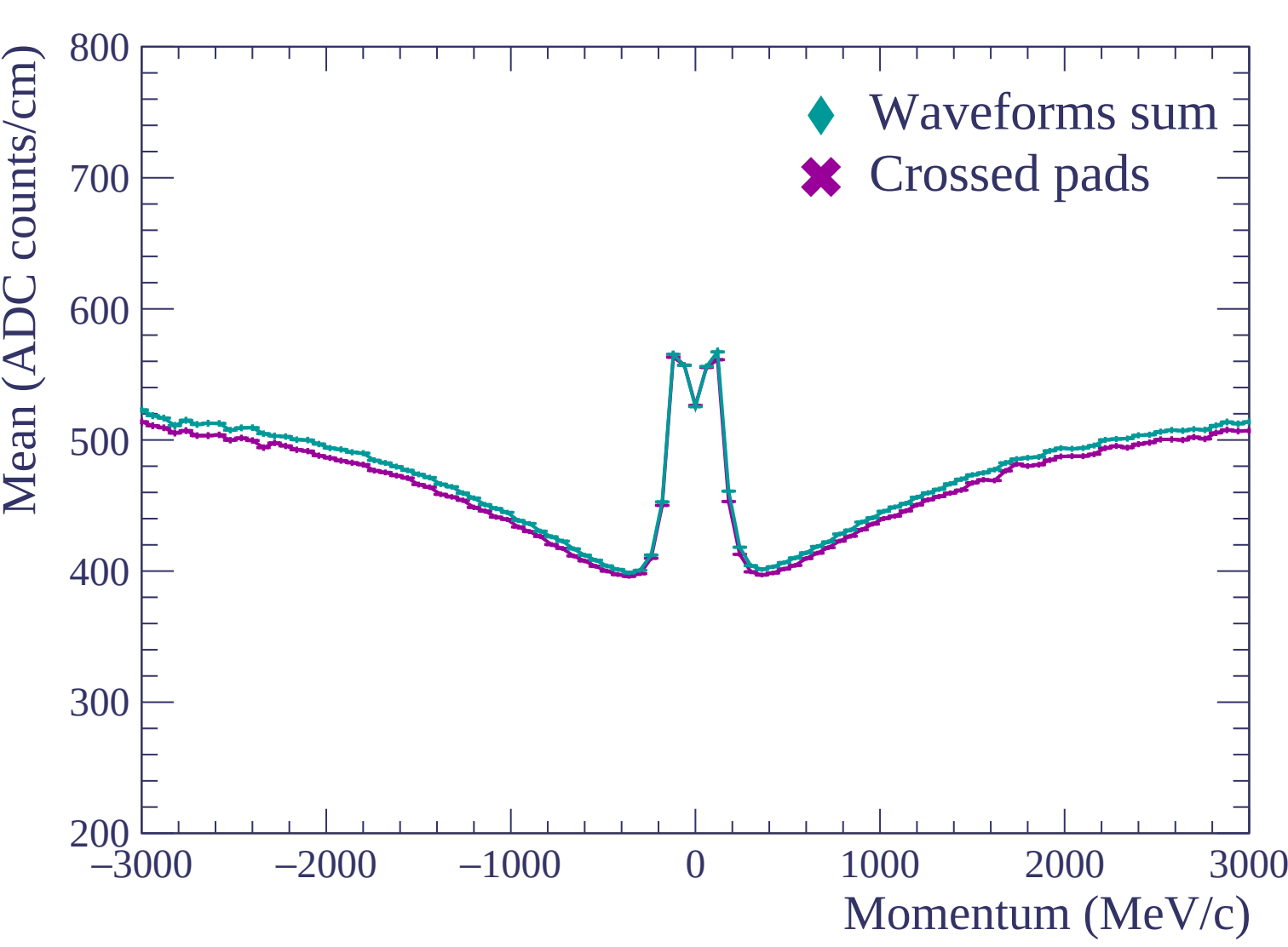


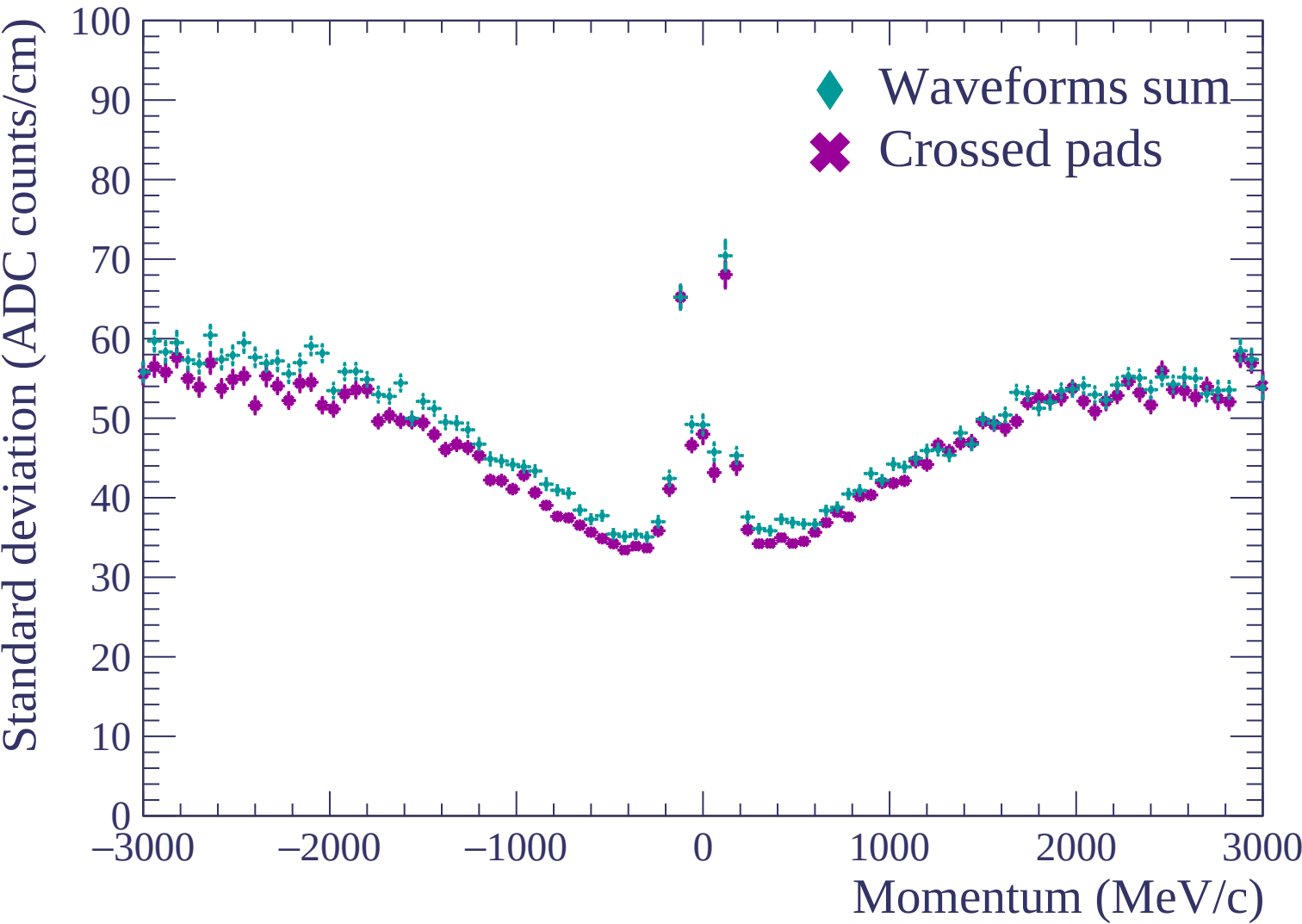


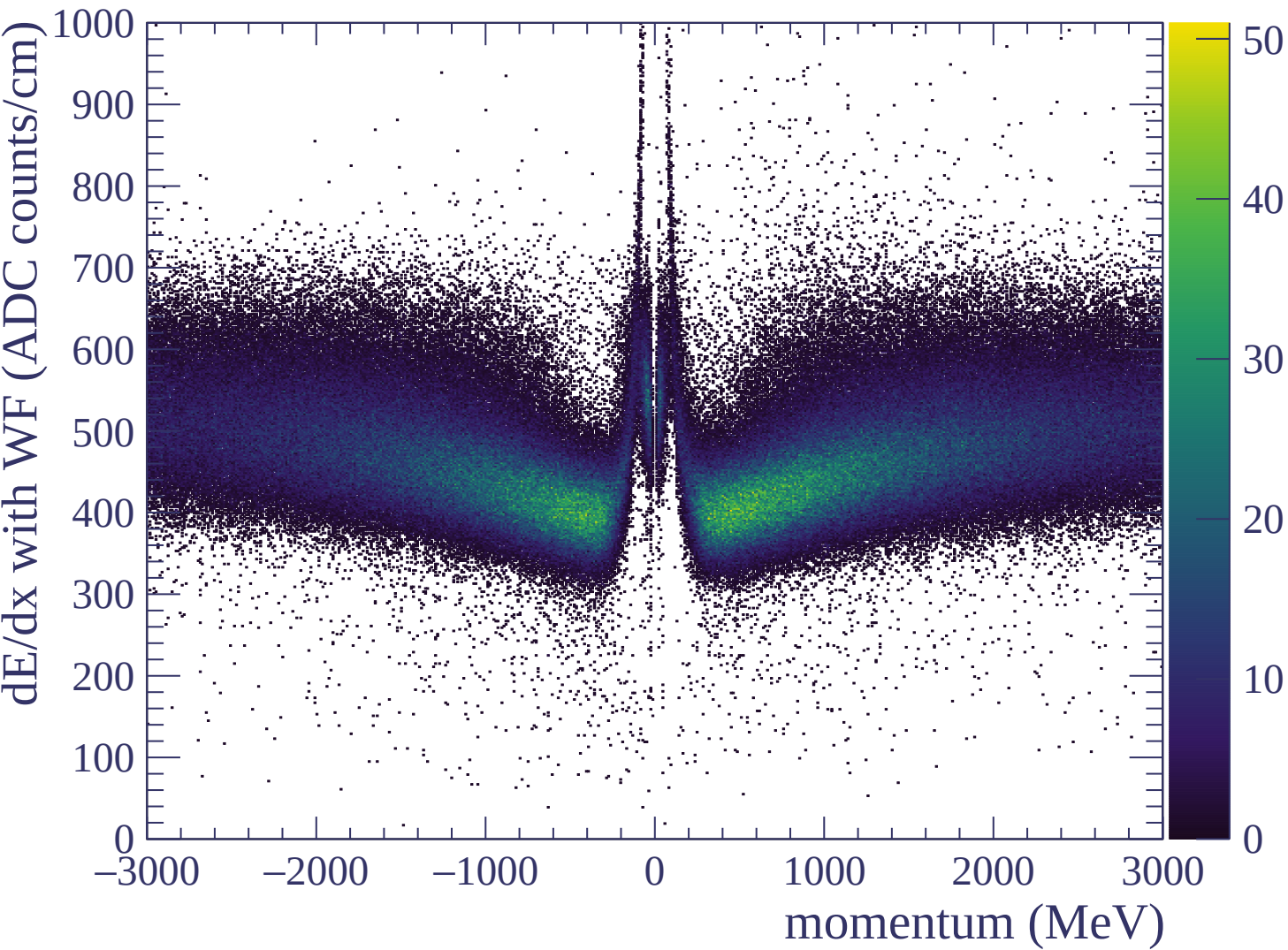
Resolution (%)

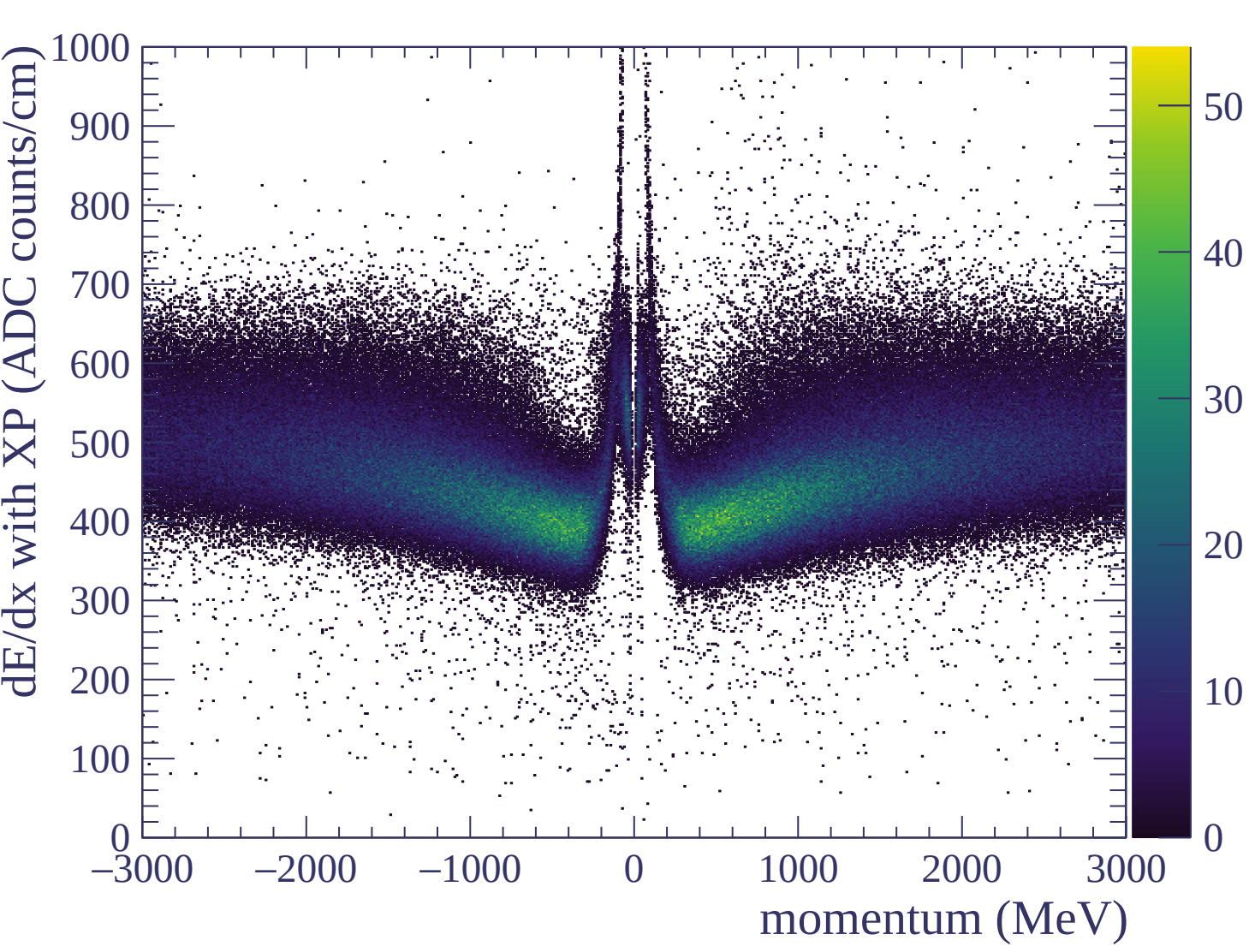
◆ Waveforms sum
✕ Crossed pads

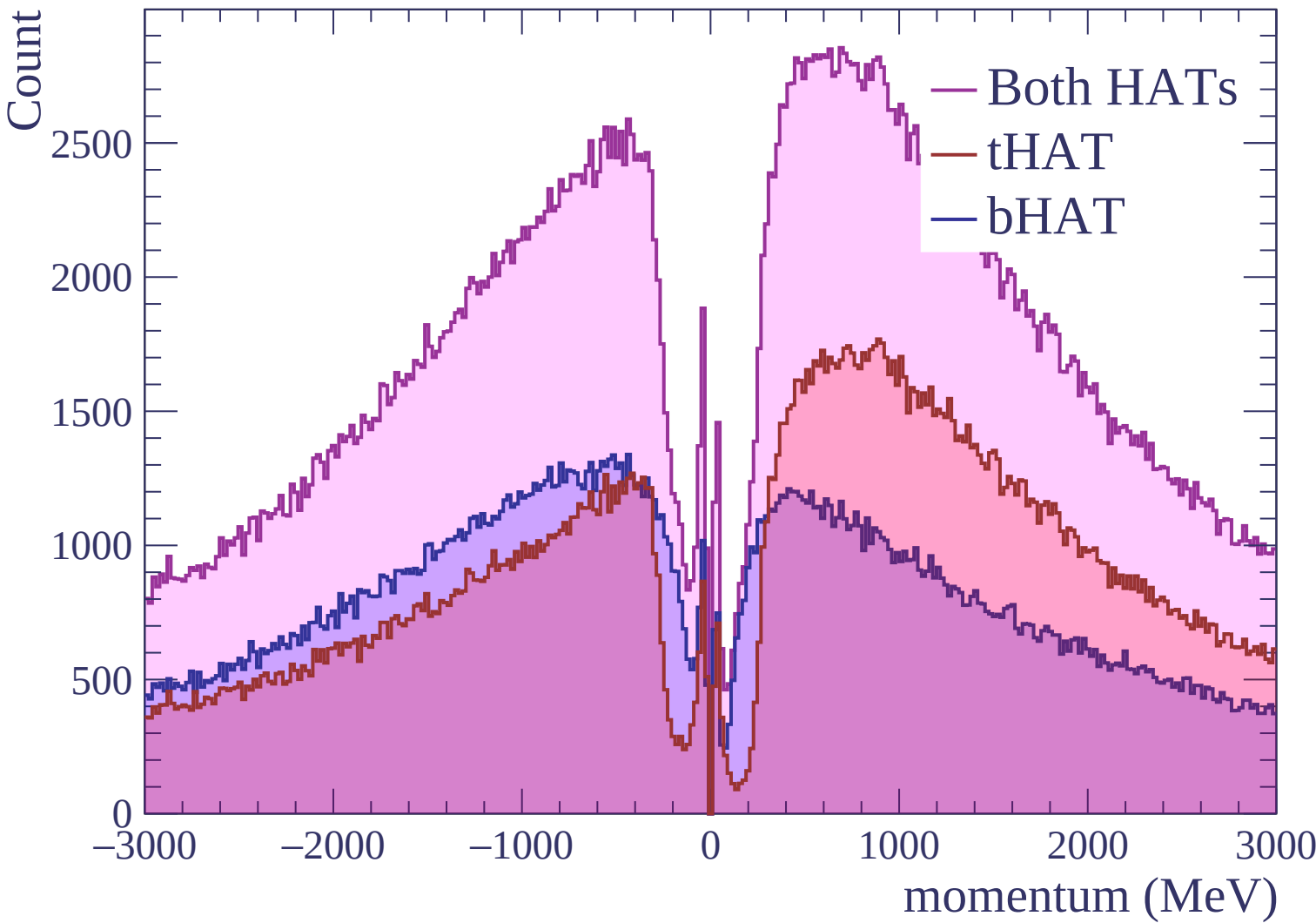


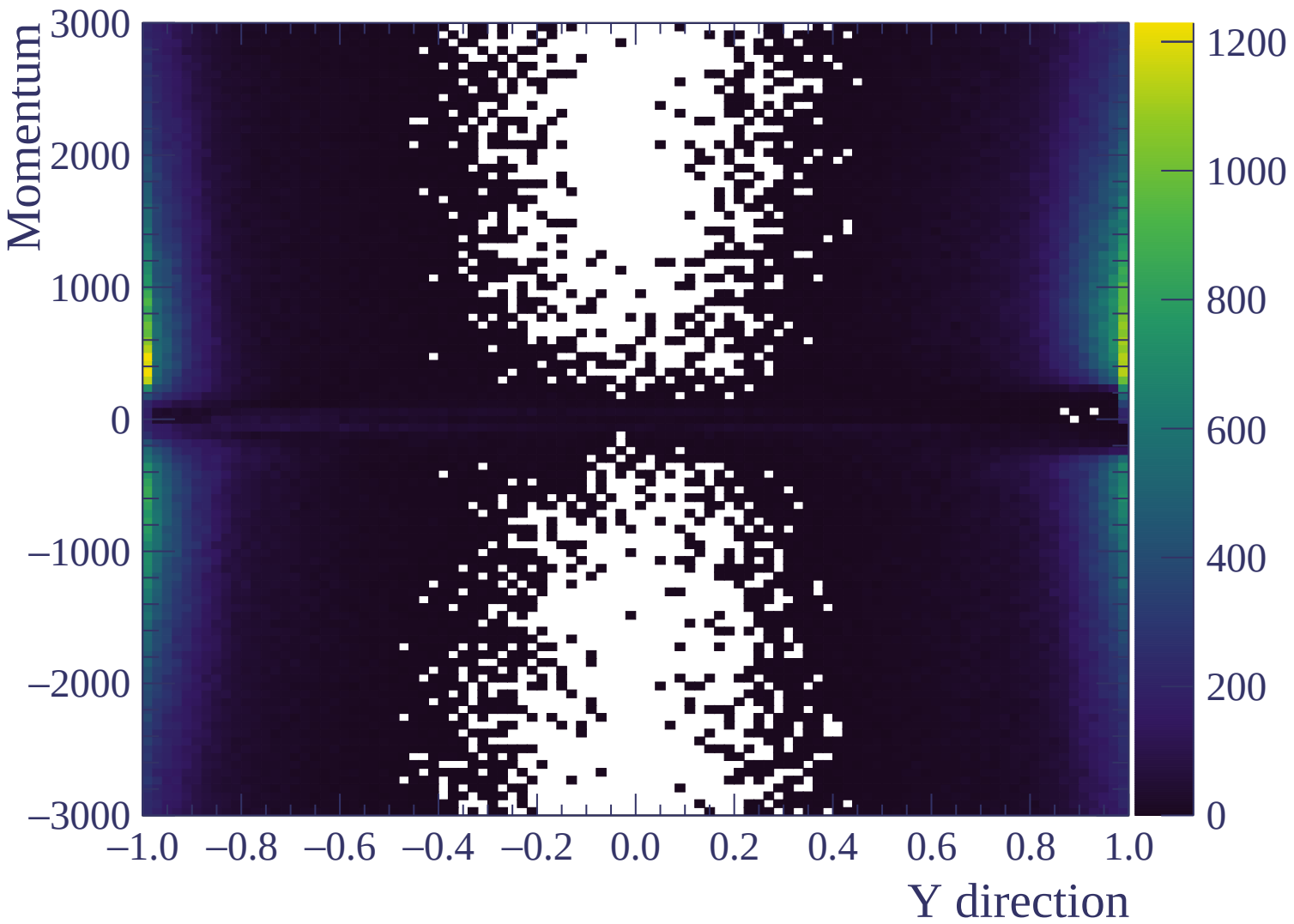


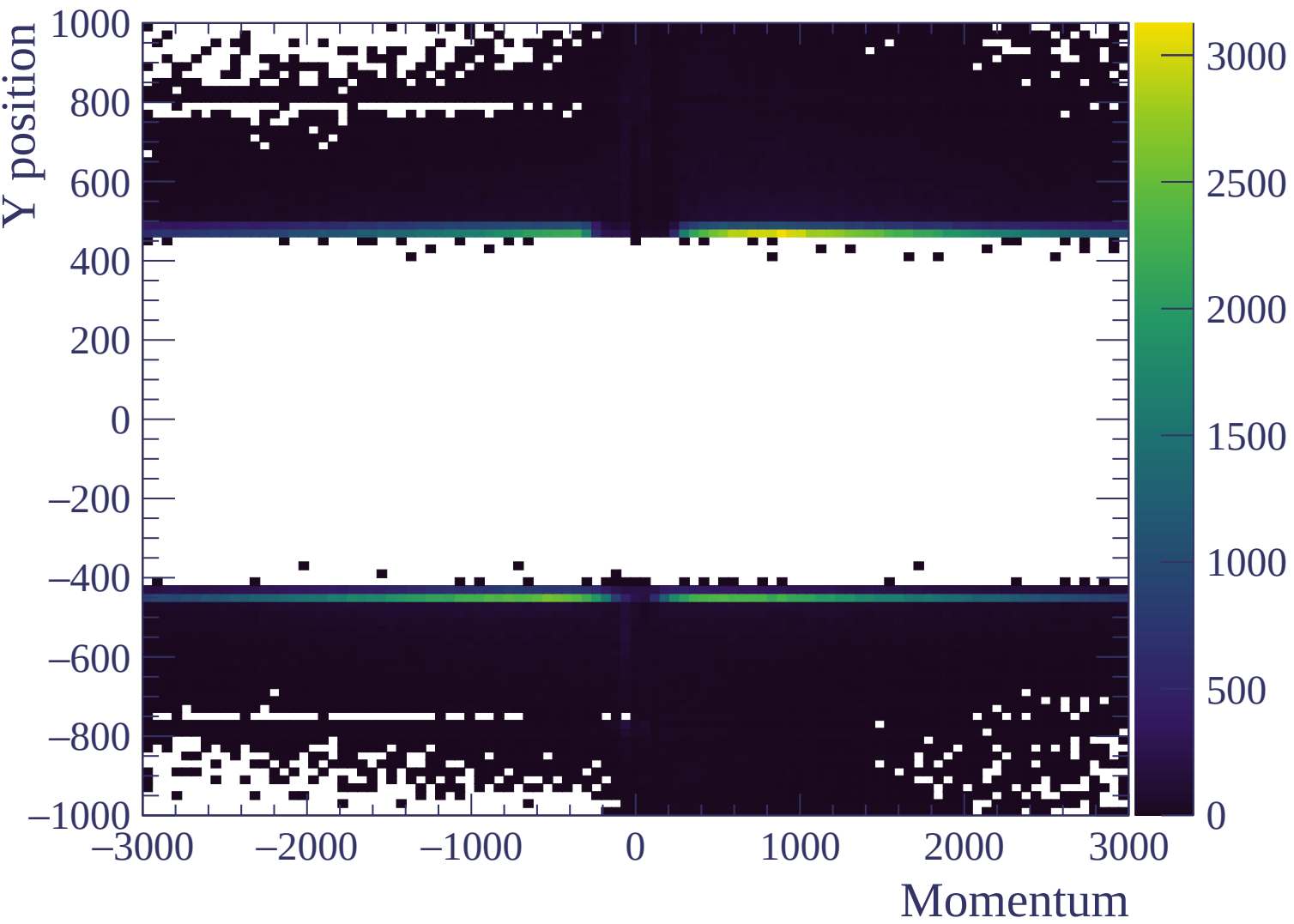


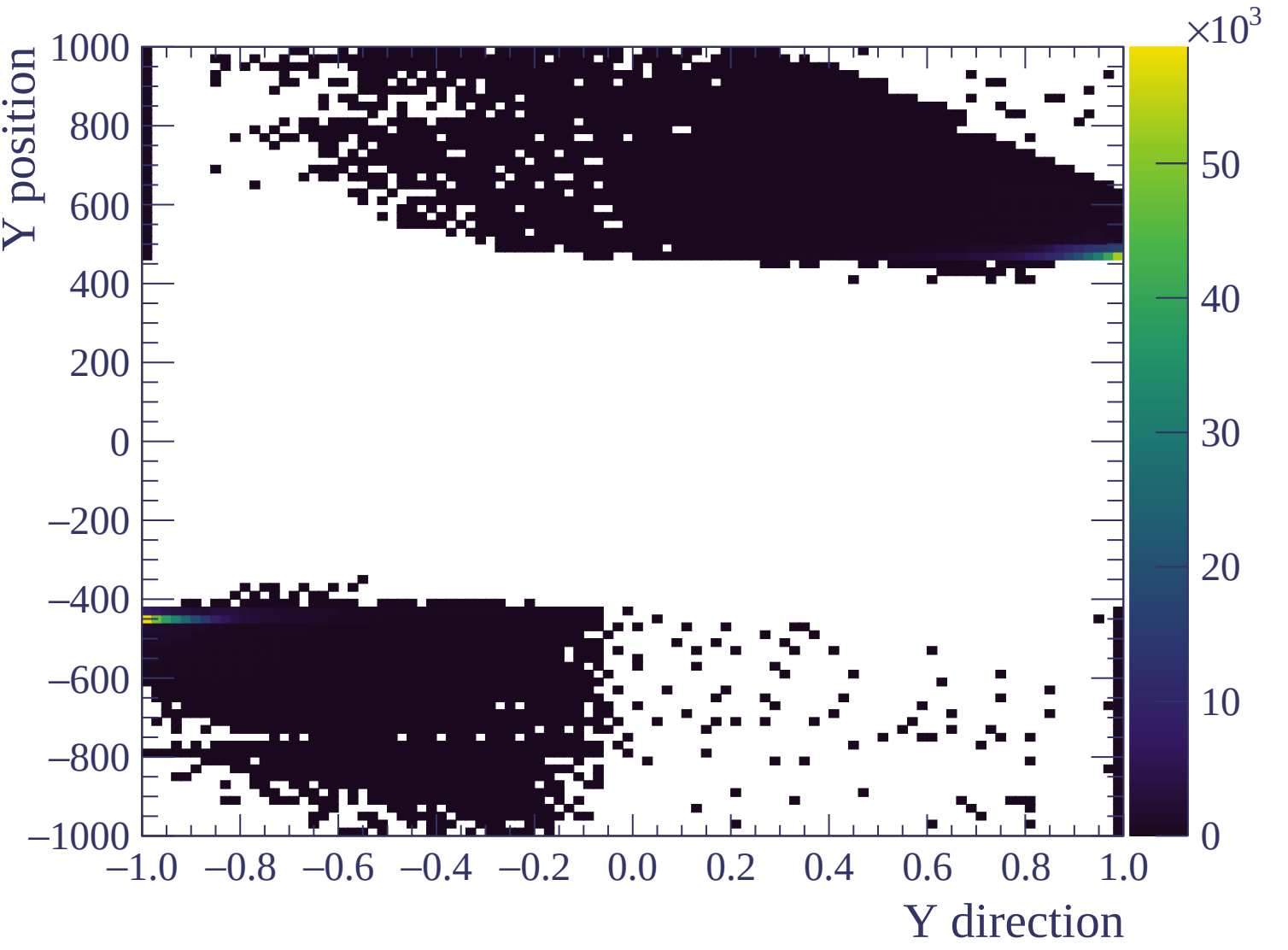




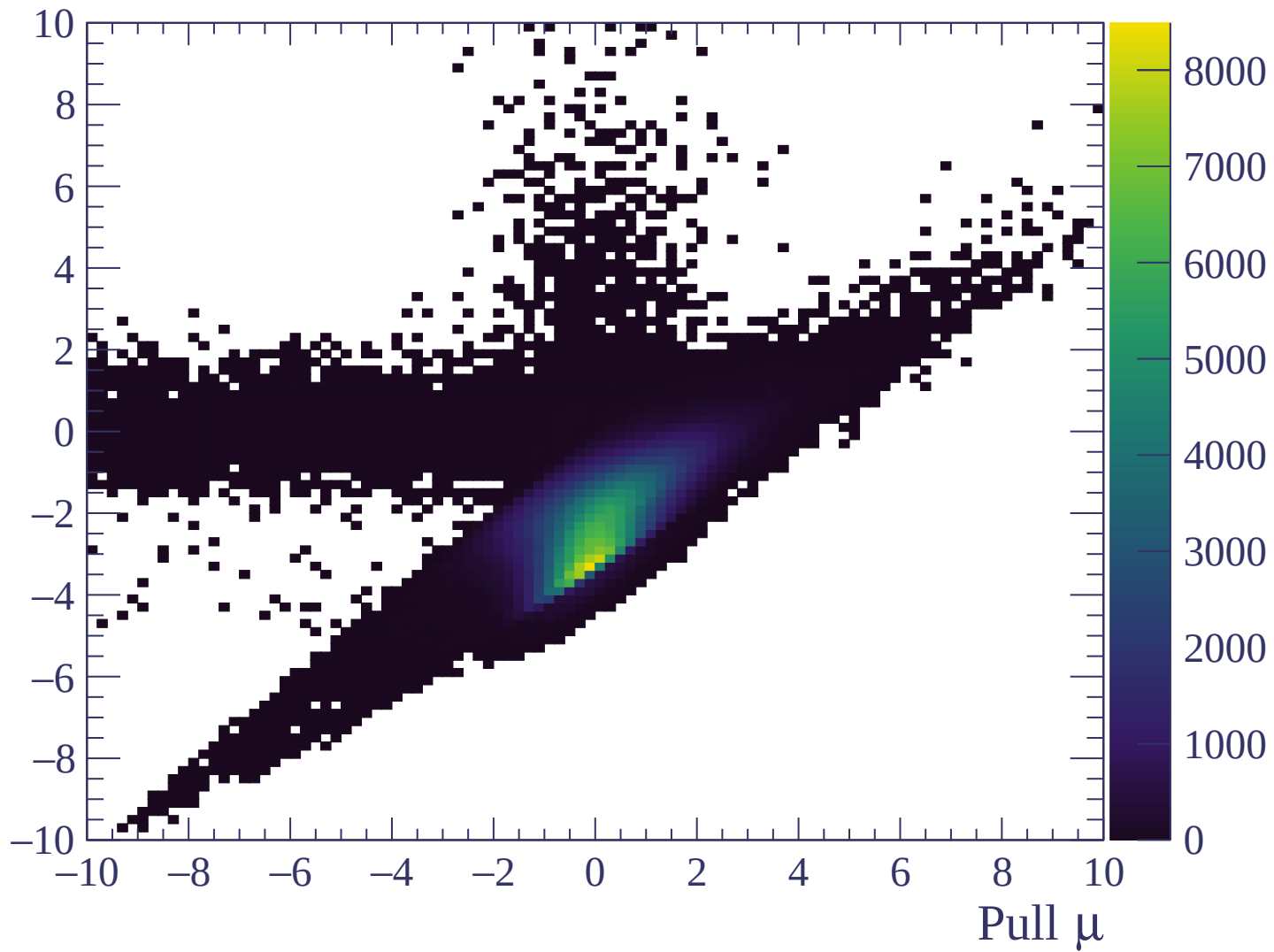






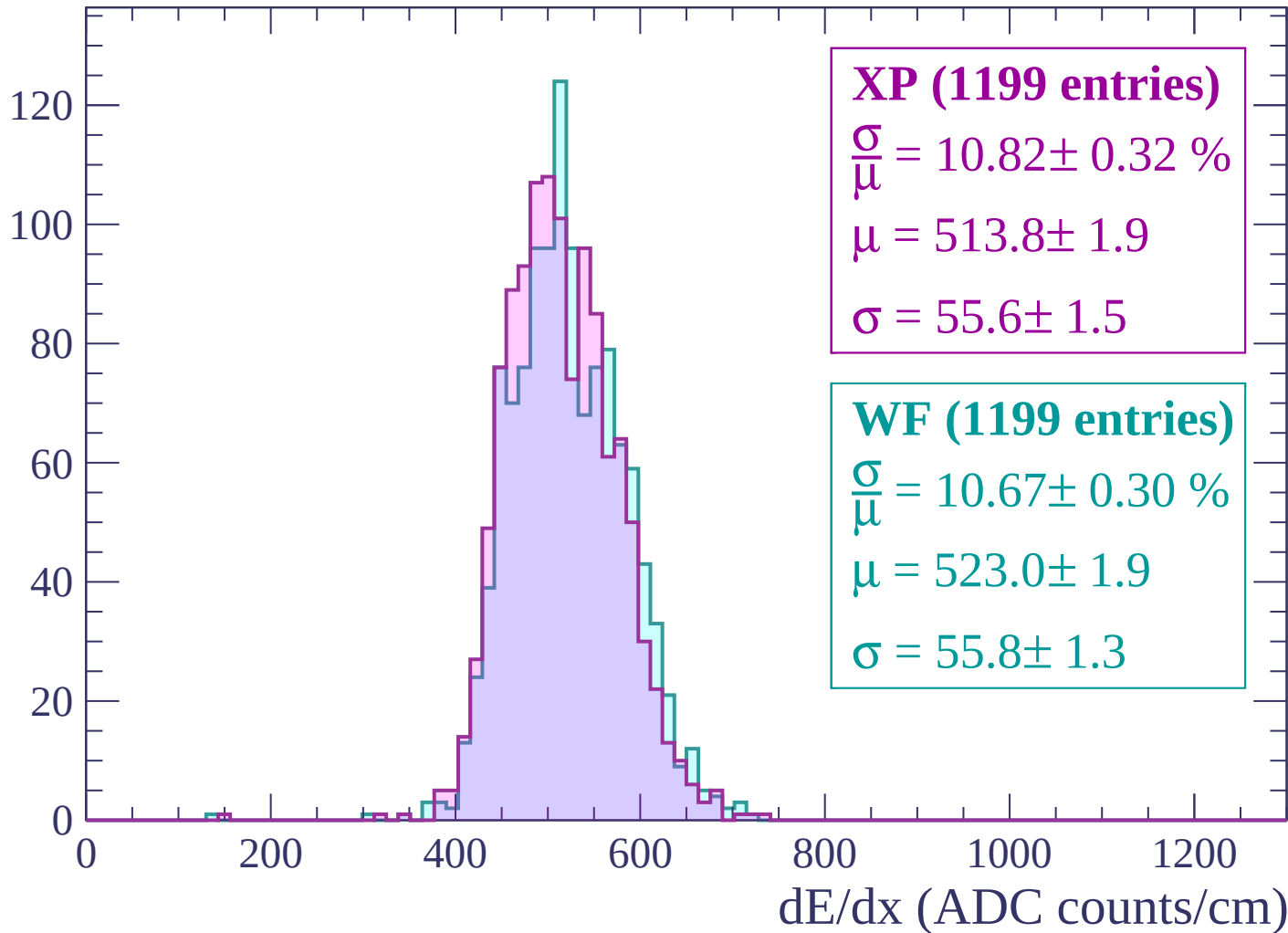


Pull e



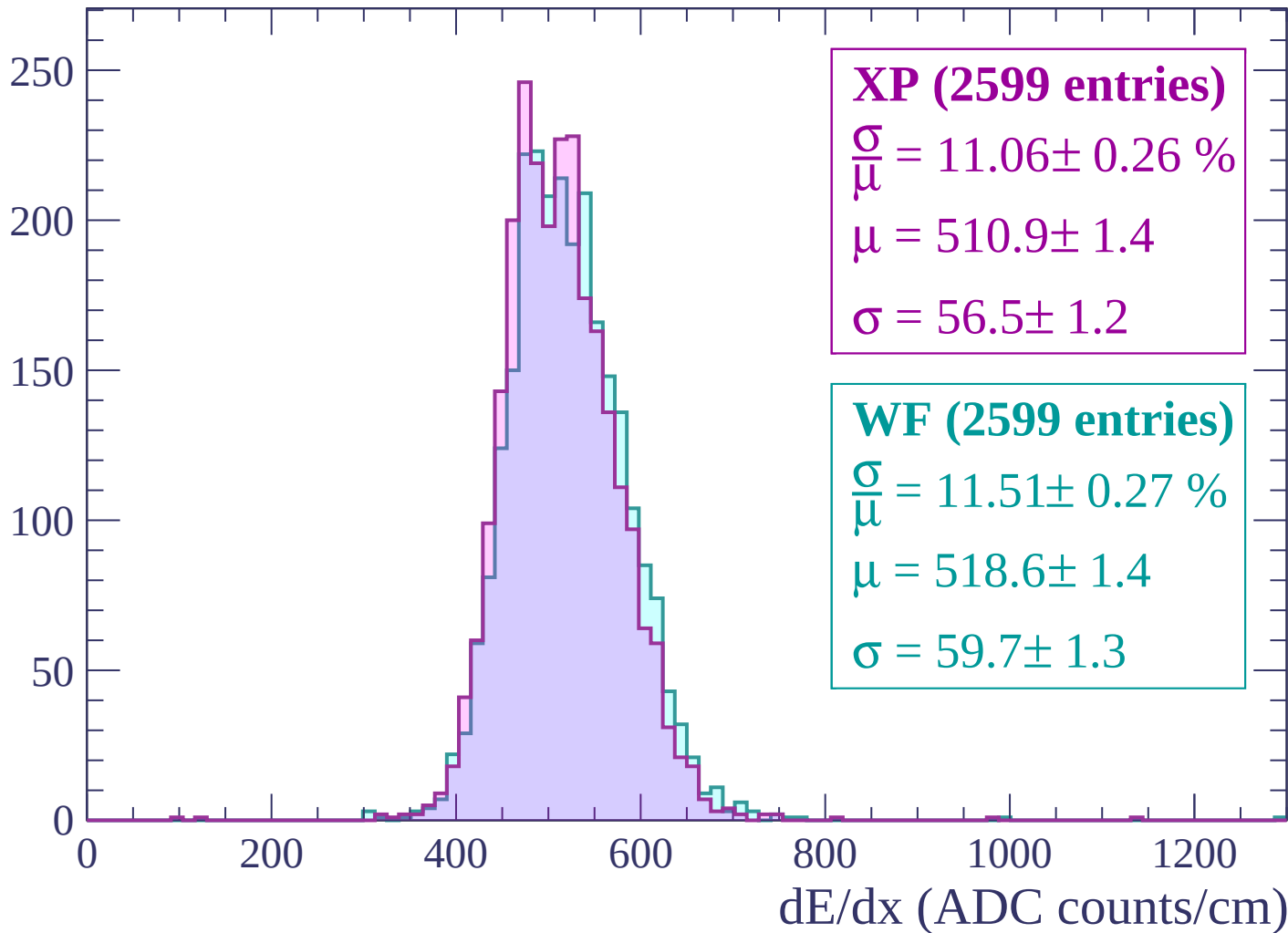
Energy loss | $-3000 < p < -2940$

Count



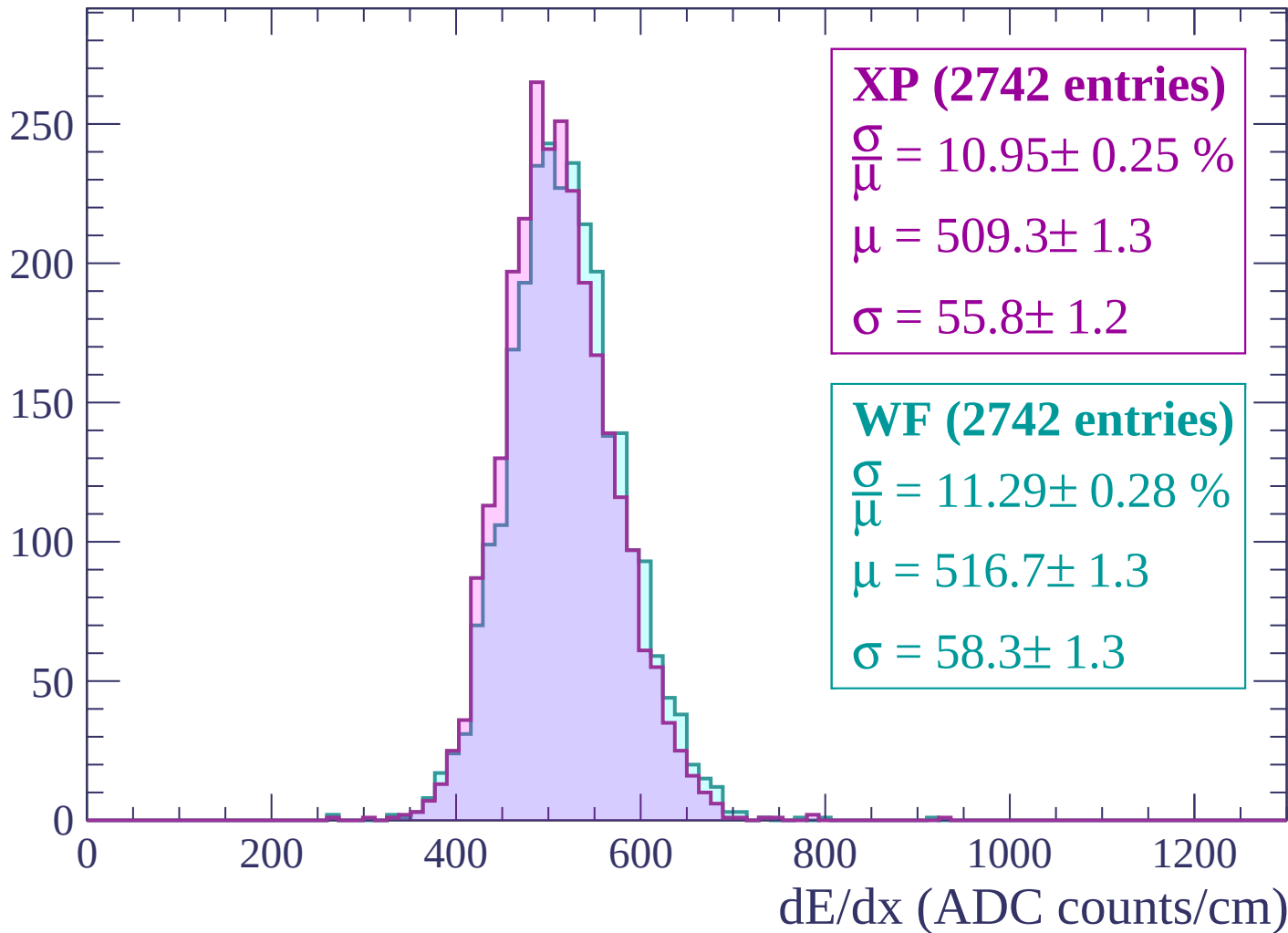
Energy loss | $-2940 < p < -2880$

Count



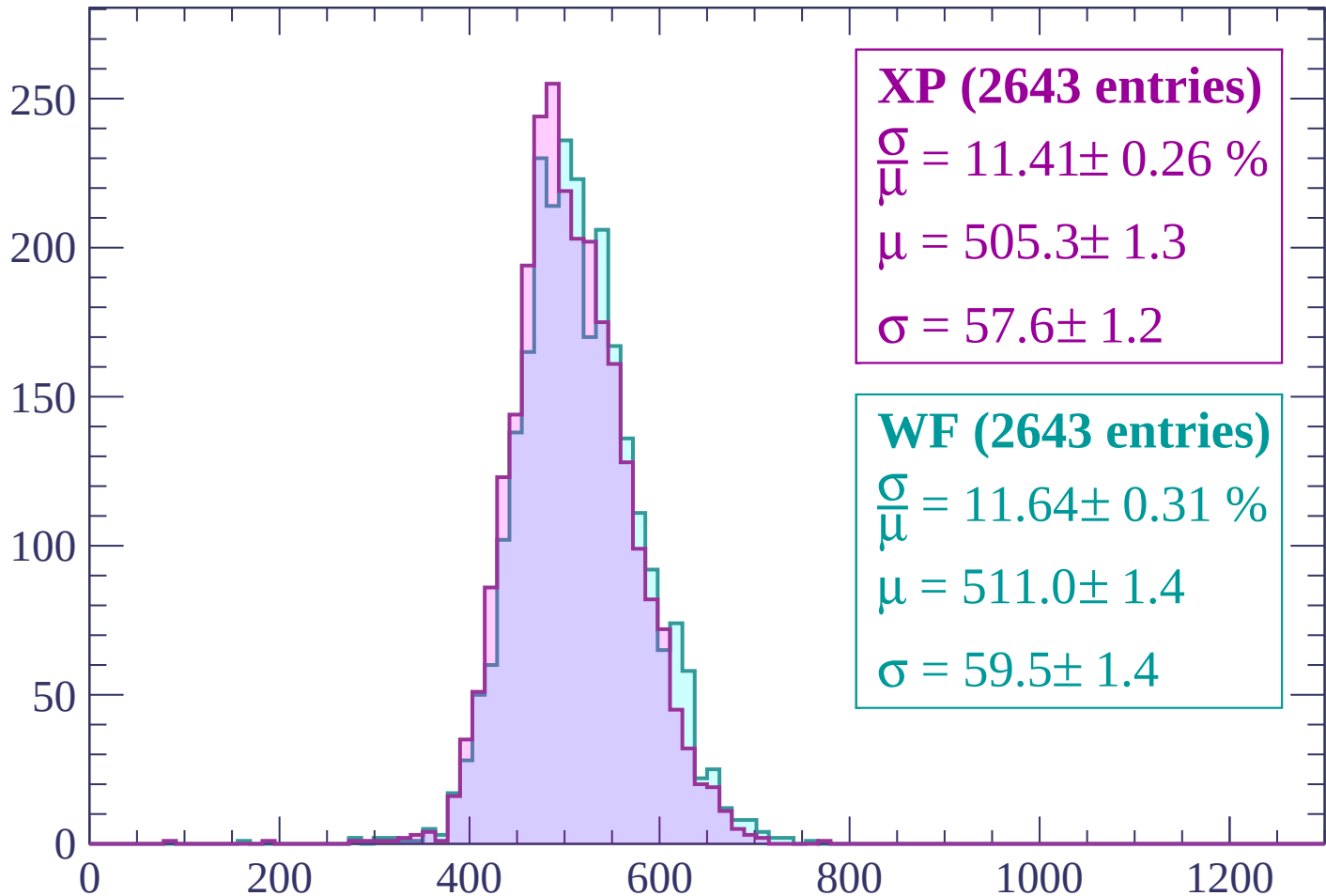
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP (2643 entries)

$\frac{\sigma}{\mu} = 11.41 \pm 0.26 \%$

$\mu = 505.3 \pm 1.3$

$\sigma = 57.6 \pm 1.2$

WF (2643 entries)

$\frac{\sigma}{\mu} = 11.64 \pm 0.31 \%$

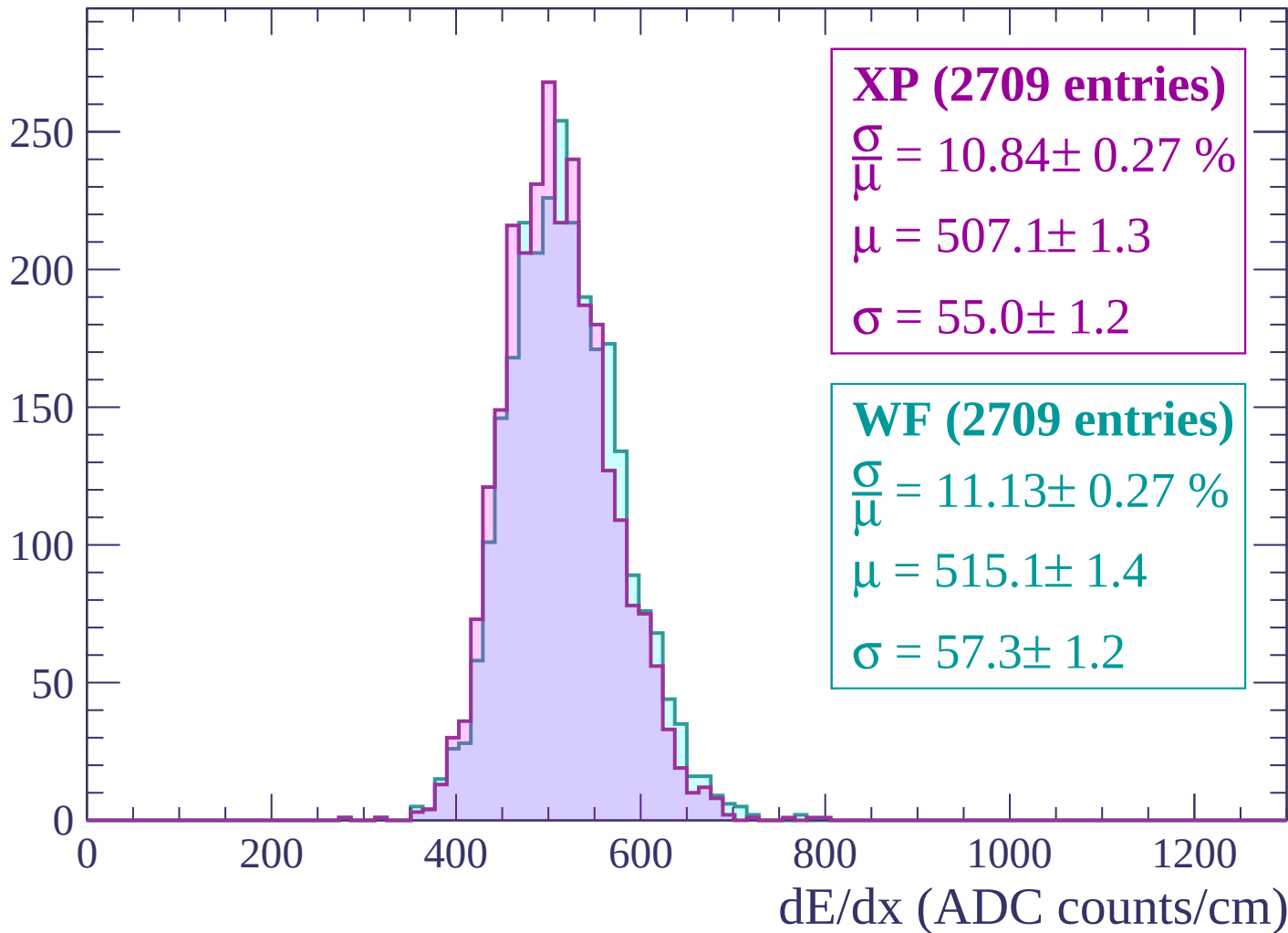
$\mu = 511.0 \pm 1.4$

$\sigma = 59.5 \pm 1.4$

dE/dx (ADC counts/cm)

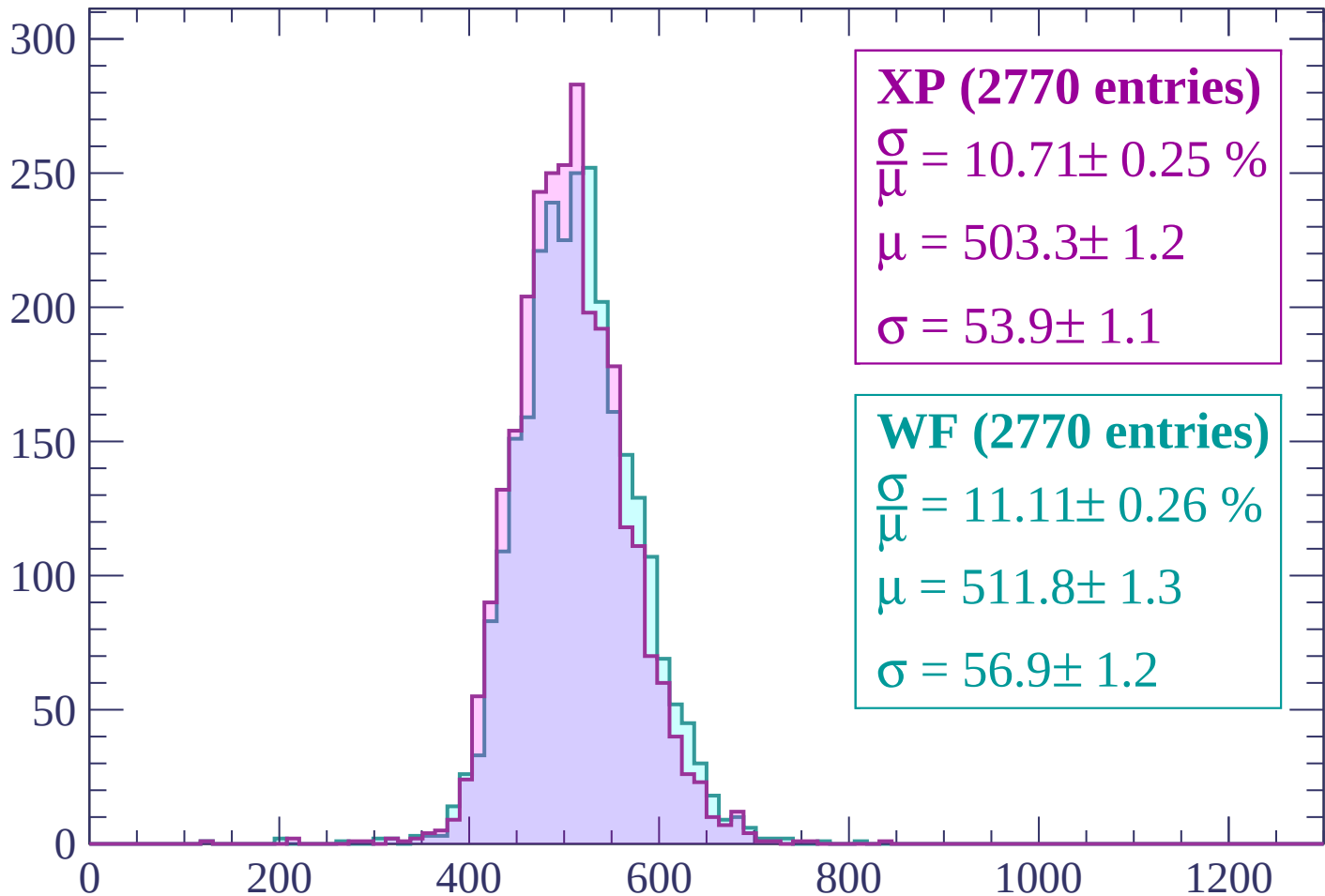
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (2770 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.25 \%$

$\mu = 503.3 \pm 1.2$

$\sigma = 53.9 \pm 1.1$

WF (2770 entries)

$\frac{\sigma}{\mu} = 11.11 \pm 0.26 \%$

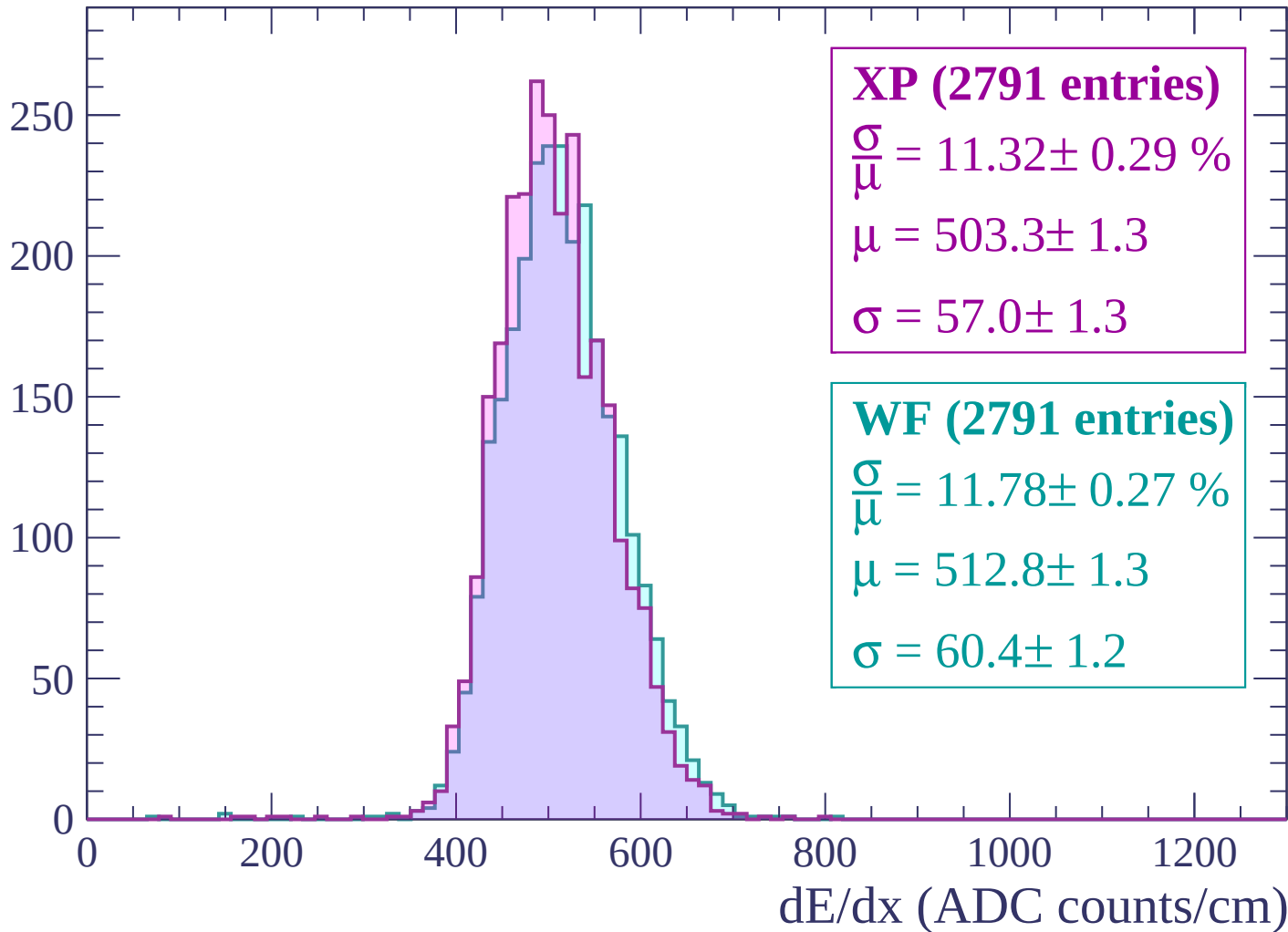
$\mu = 511.8 \pm 1.3$

$\sigma = 56.9 \pm 1.2$

dE/dx (ADC counts/cm)

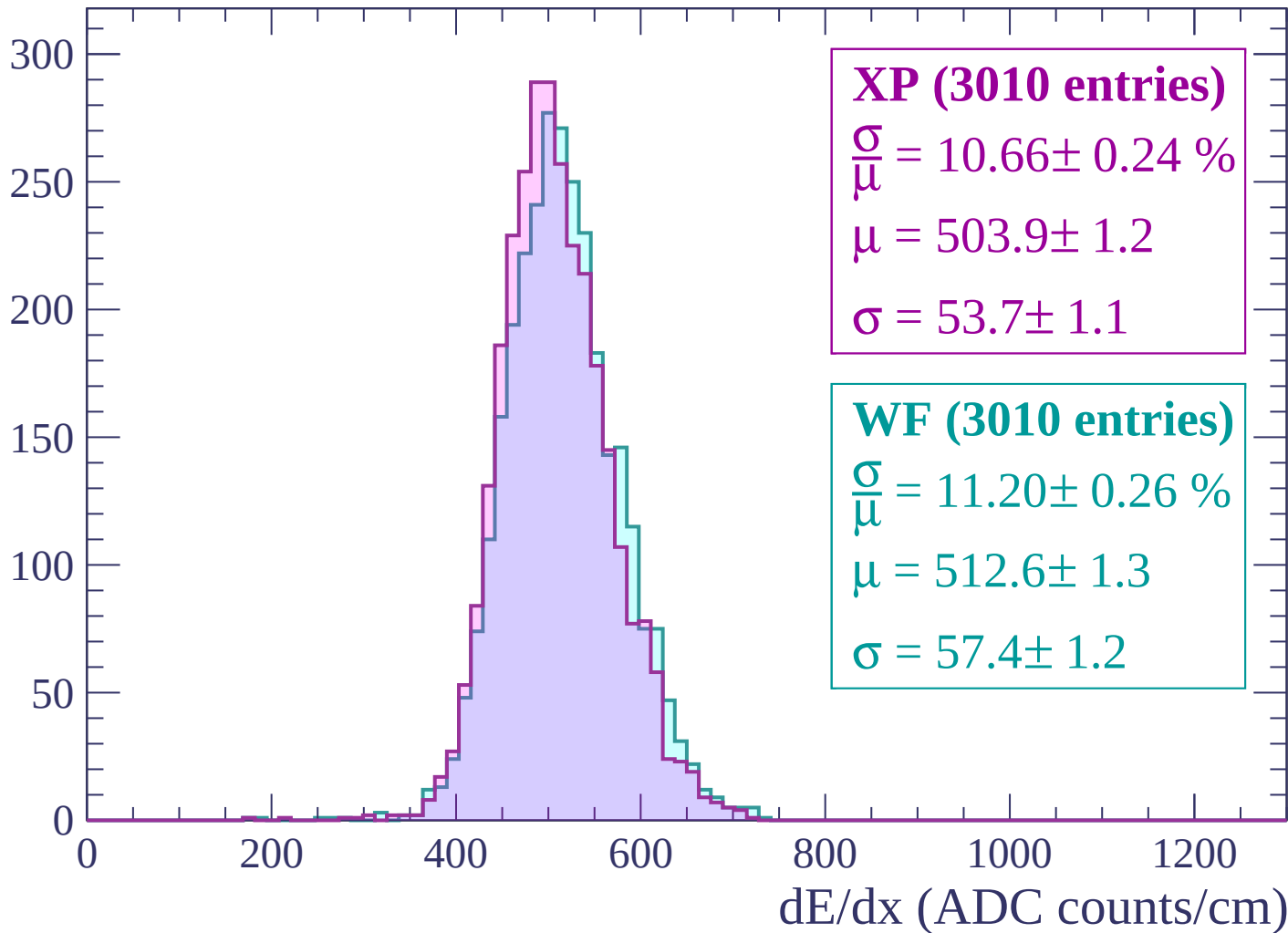
Energy loss | $-2640 < p < -2580$

Count



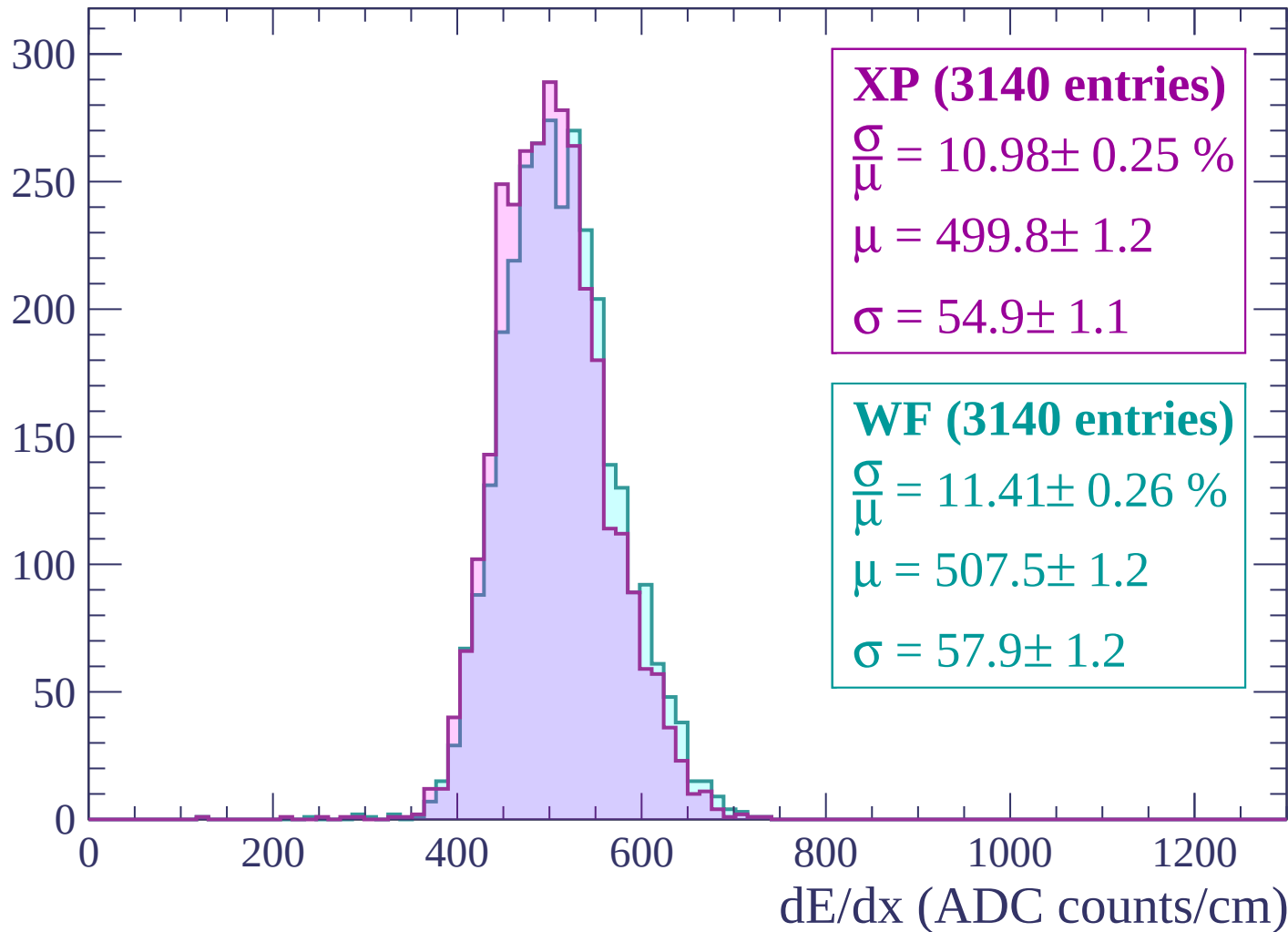
Energy loss | -2580 < p < -2520

Count



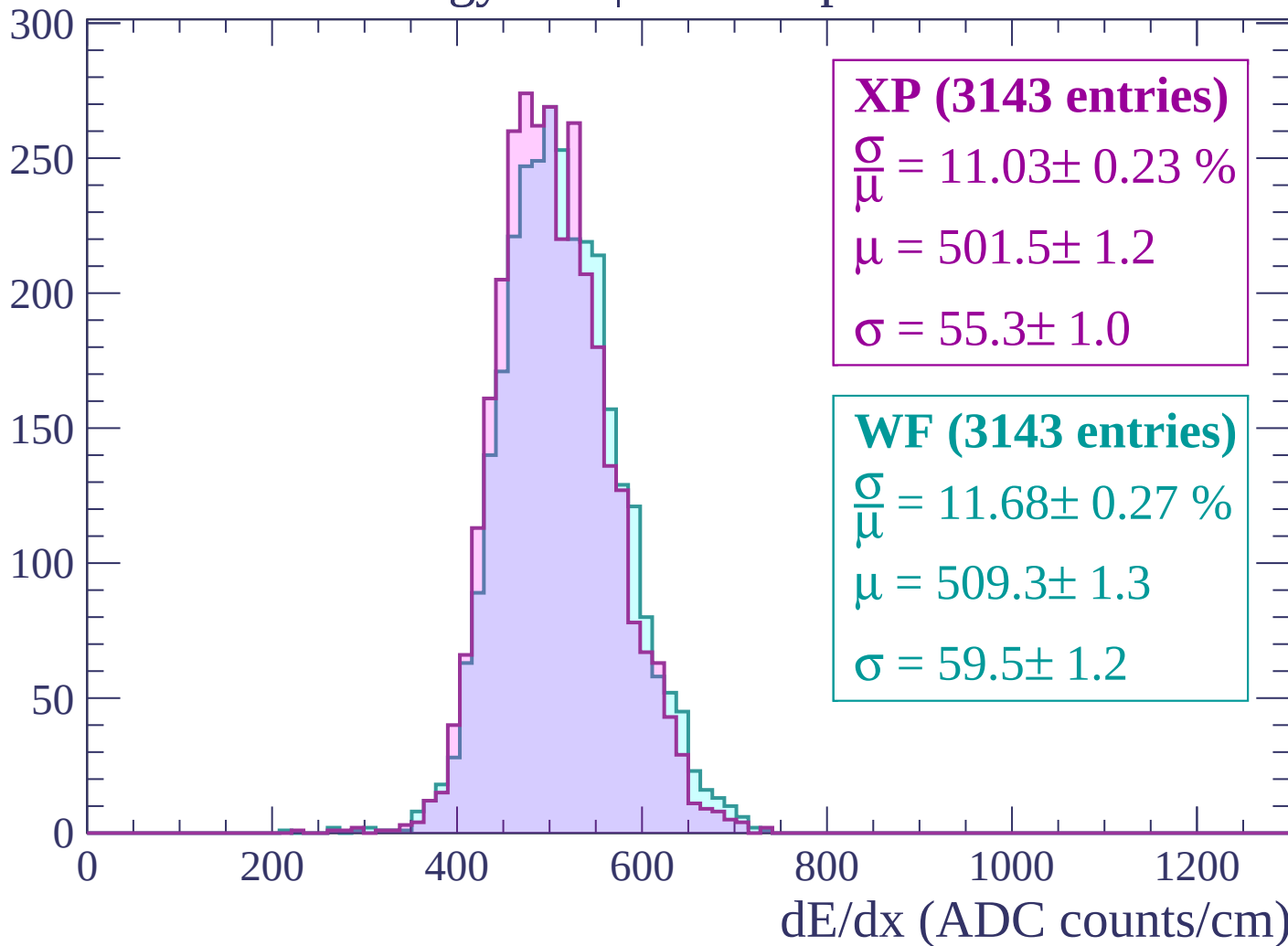
Energy loss | -2520 < p < -2460

Count



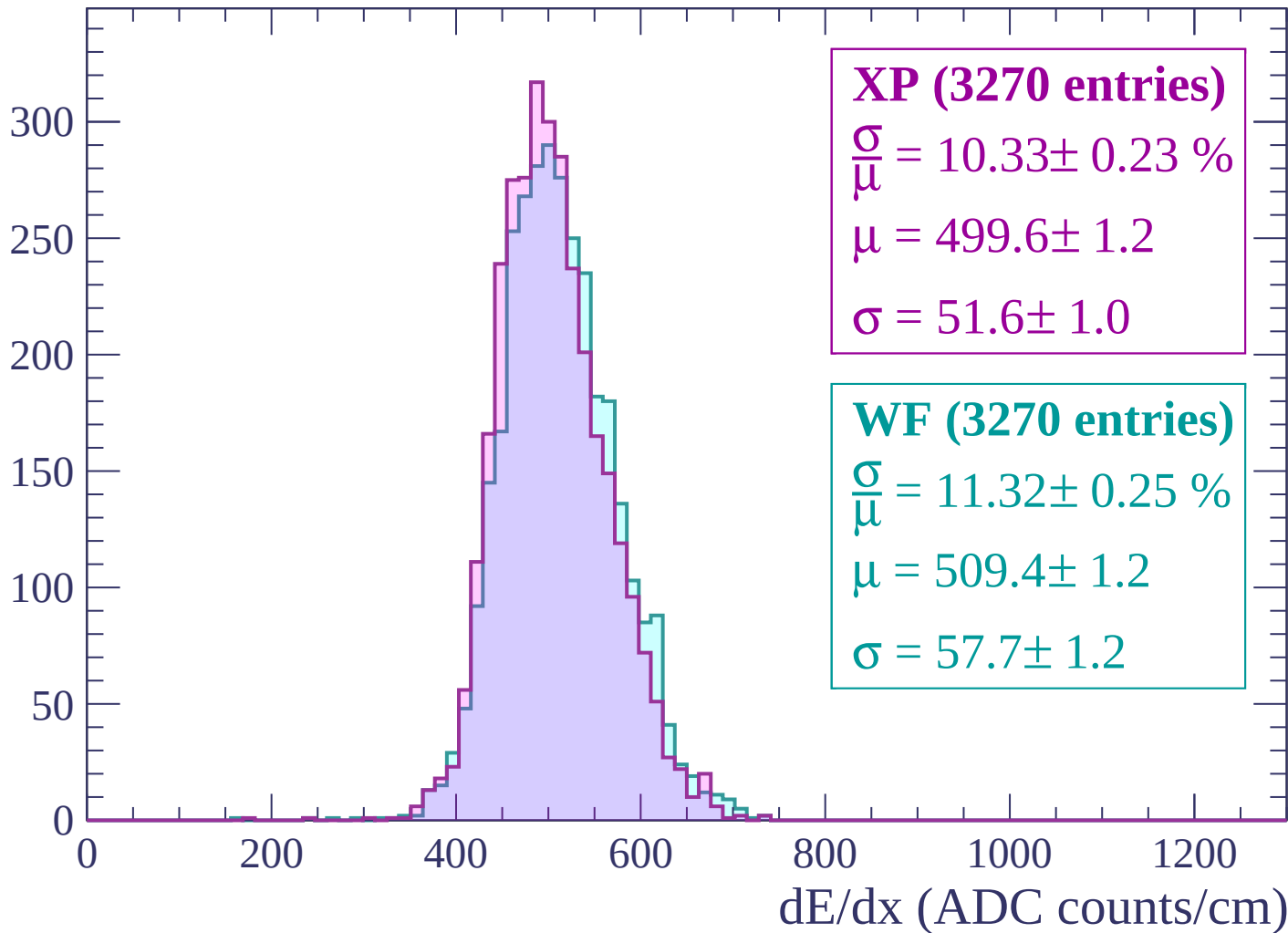
Energy loss | $-2460 < p < -2400$

Count



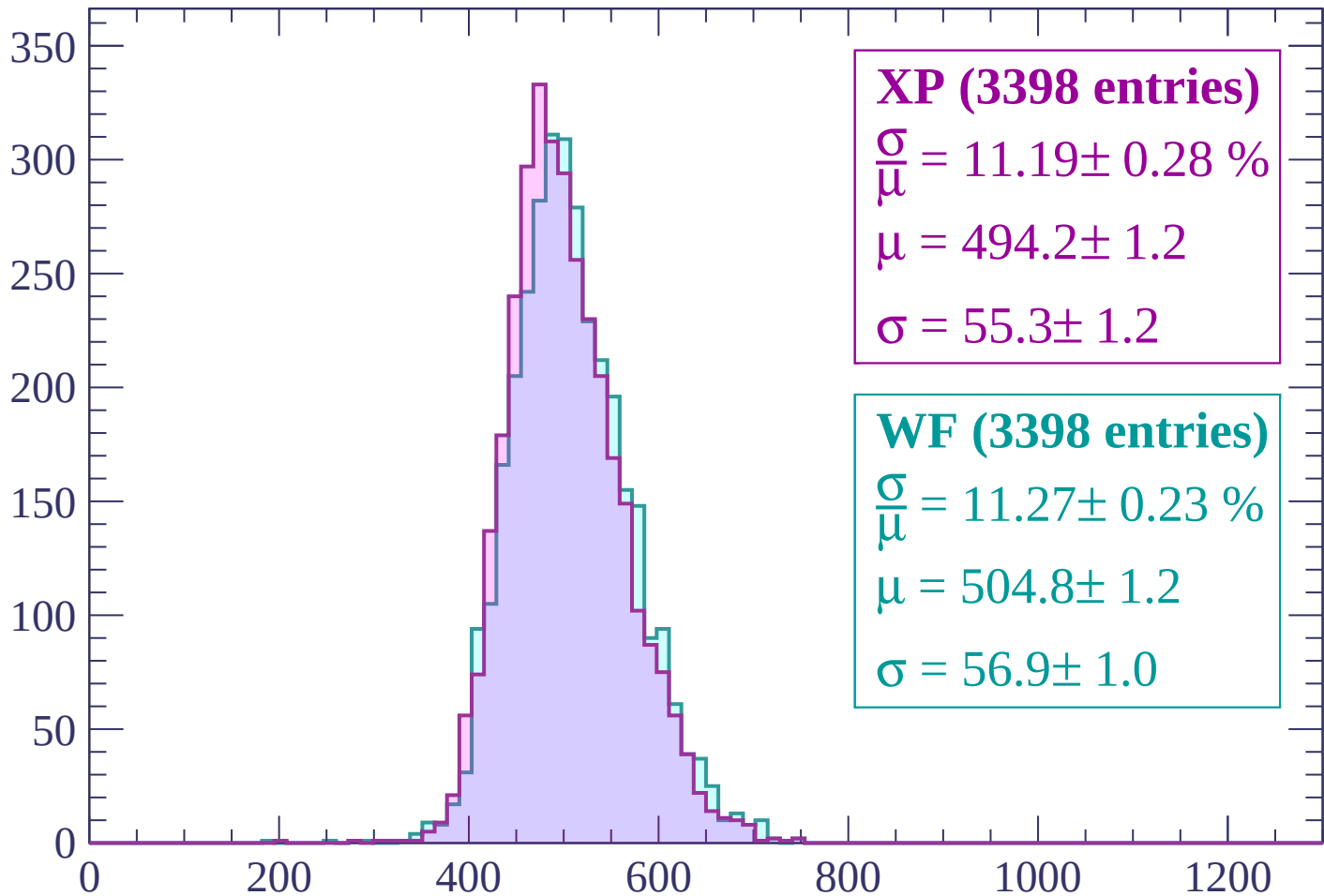
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (3398 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.28 \%$$

$$\mu = 494.2 \pm 1.2$$

$$\sigma = 55.3 \pm 1.2$$

WF (3398 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.23 \%$$

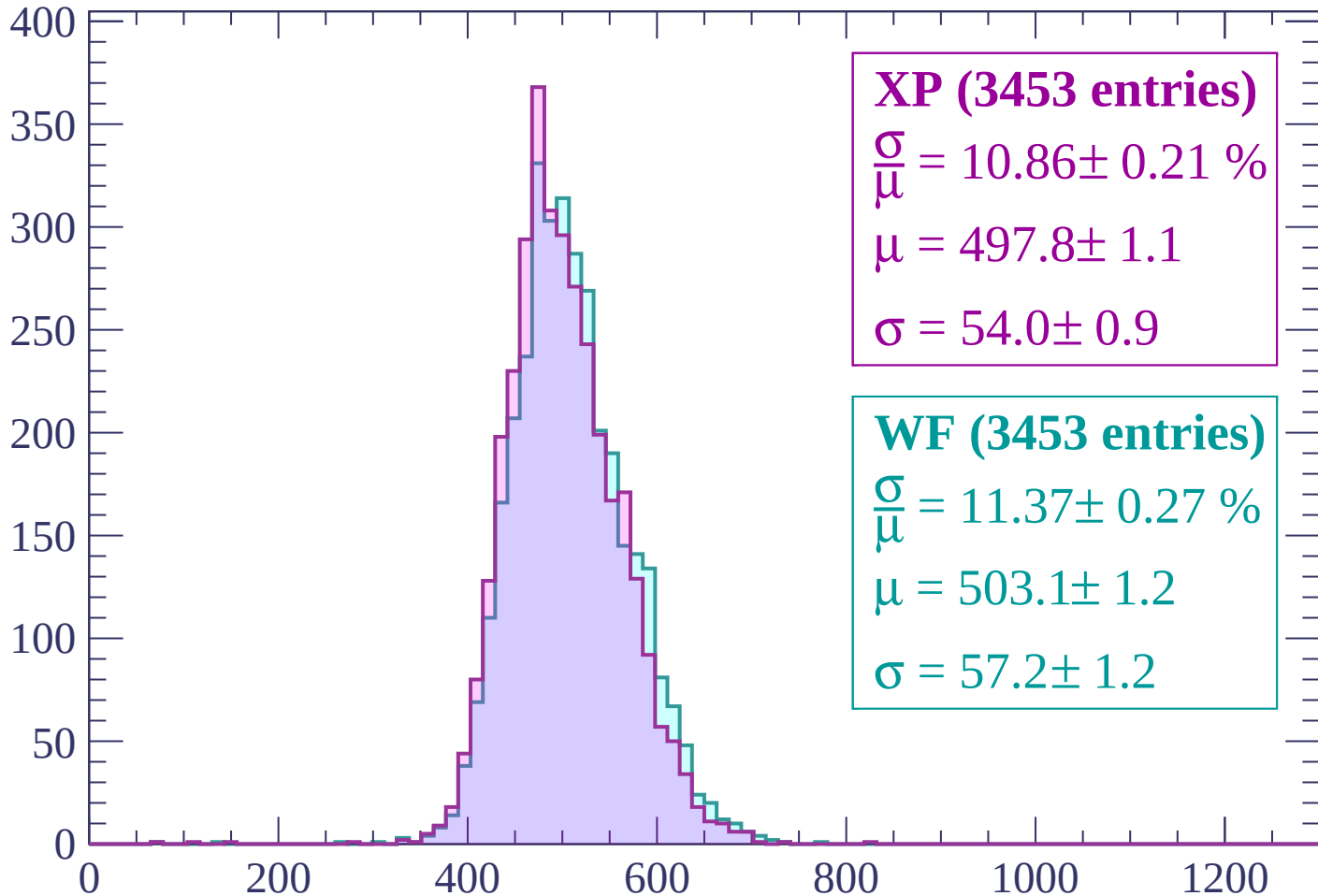
$$\mu = 504.8 \pm 1.2$$

$$\sigma = 56.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP (3453 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.21 \%$$

$$\mu = 497.8 \pm 1.1$$

$$\sigma = 54.0 \pm 0.9$$

WF (3453 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 0.27 \%$$

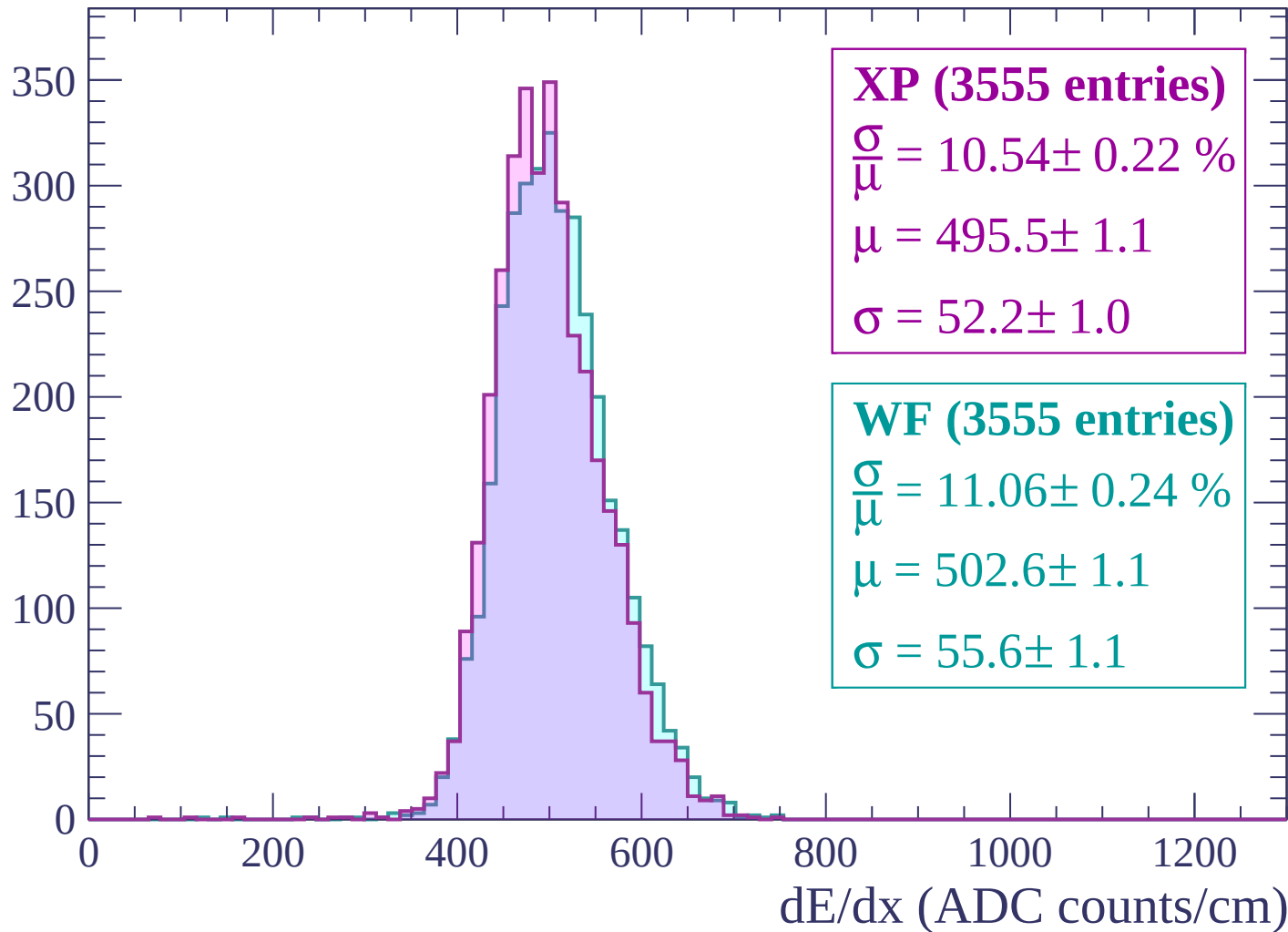
$$\mu = 503.1 \pm 1.2$$

$$\sigma = 57.2 \pm 1.2$$

dE/dx (ADC counts/cm)

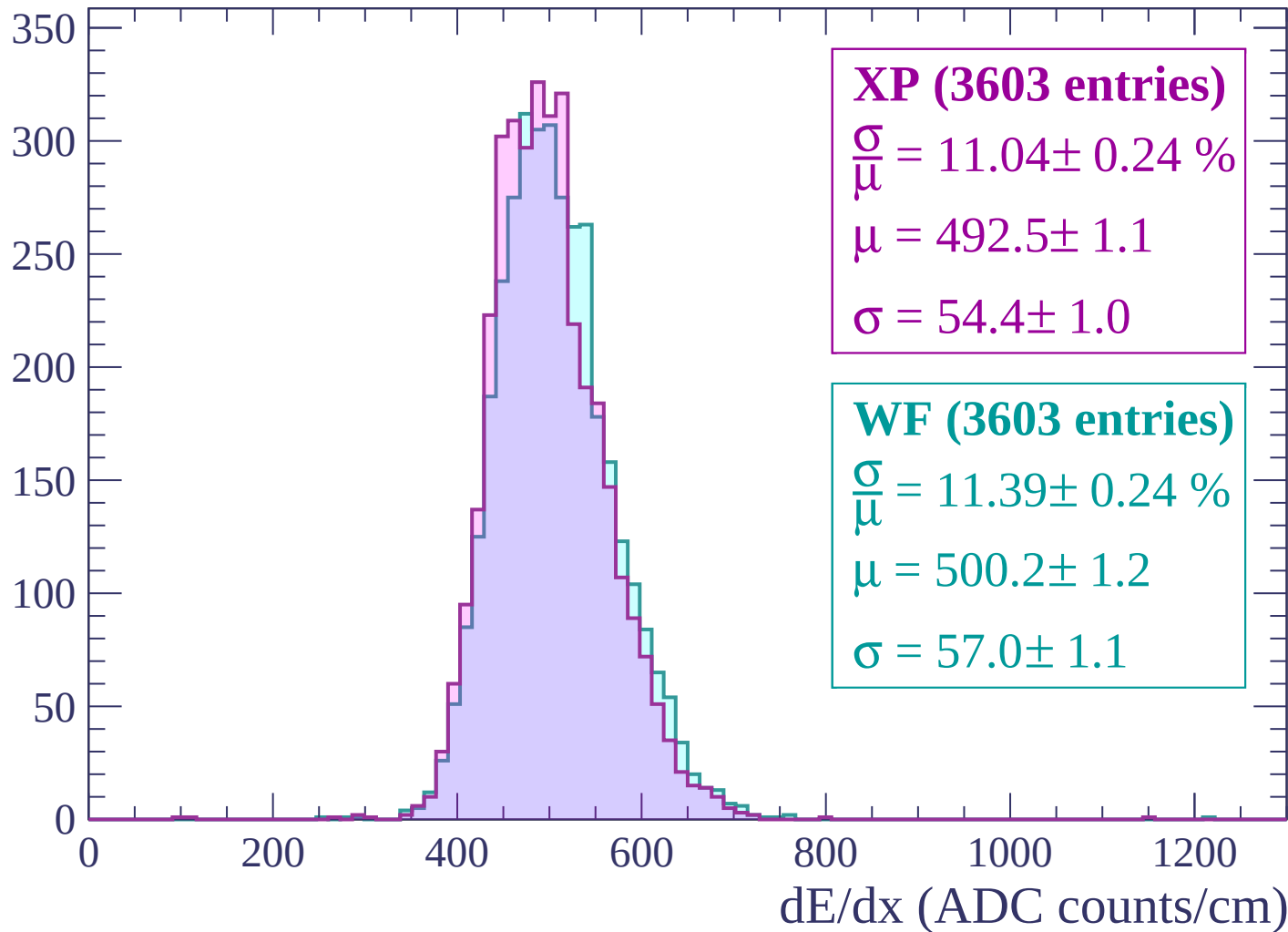
Energy loss | -2220 < p < -2160

Count

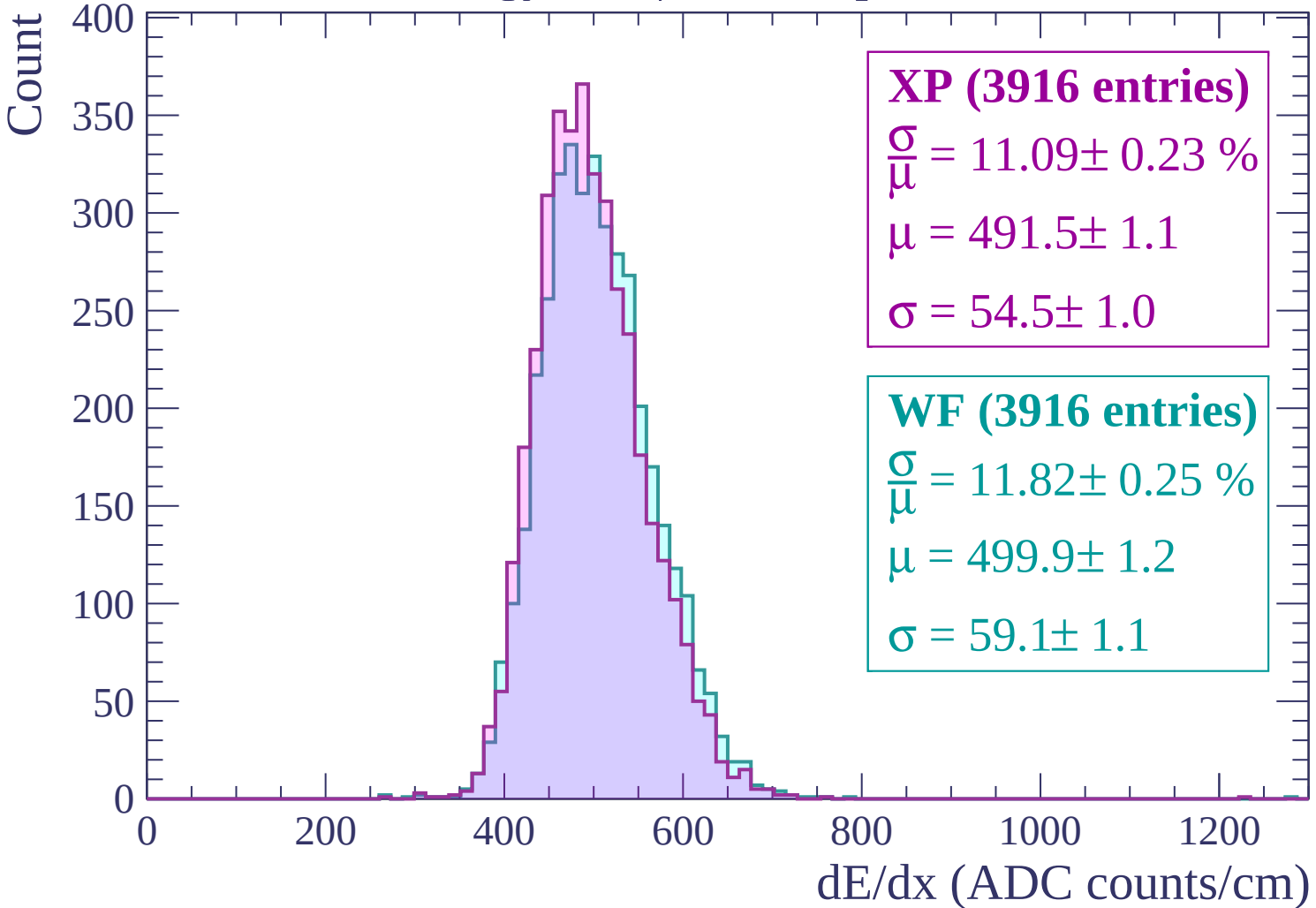


Energy loss | -2160 < p < -2100

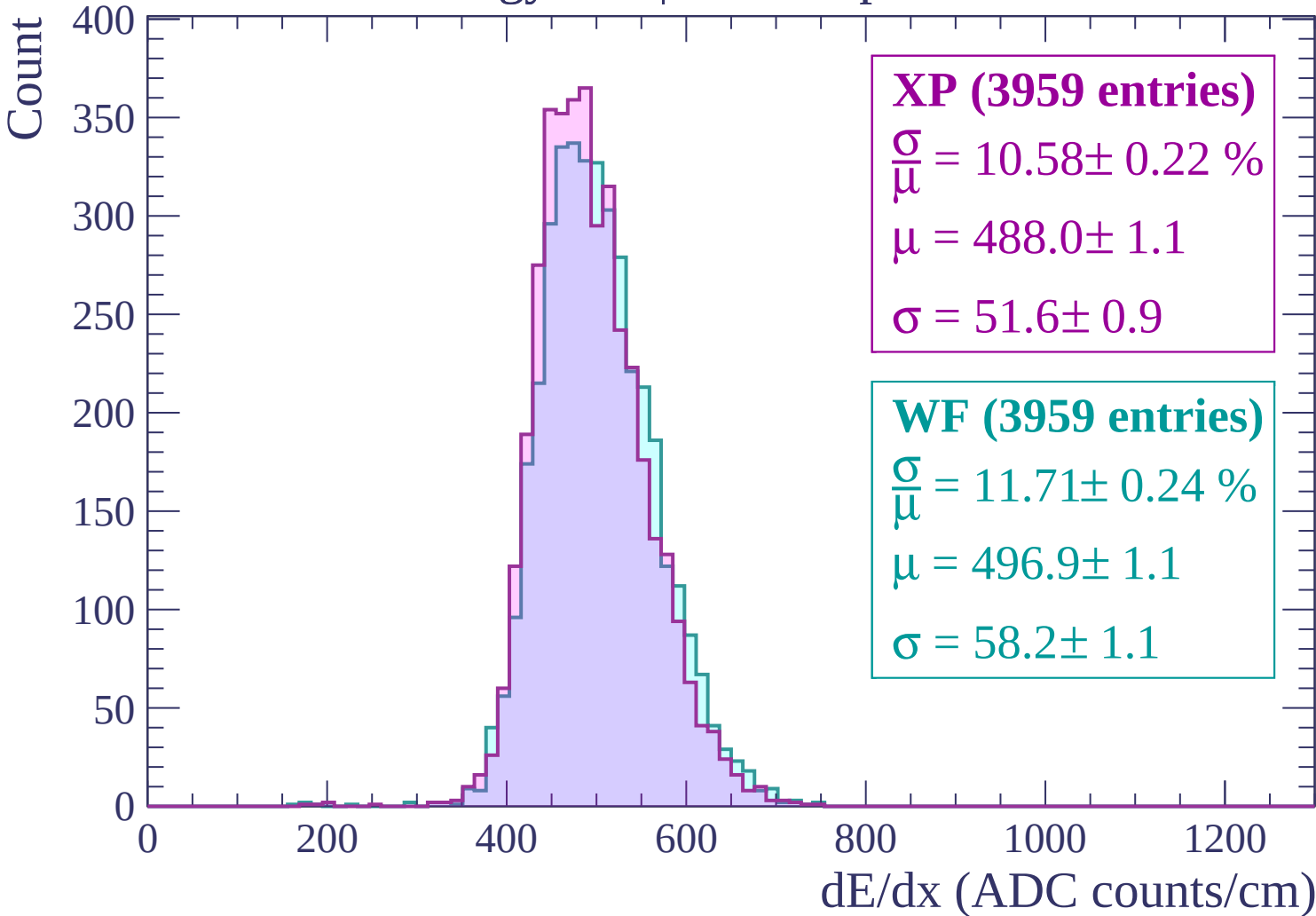
Count



Energy loss | -2100 < p < -2040

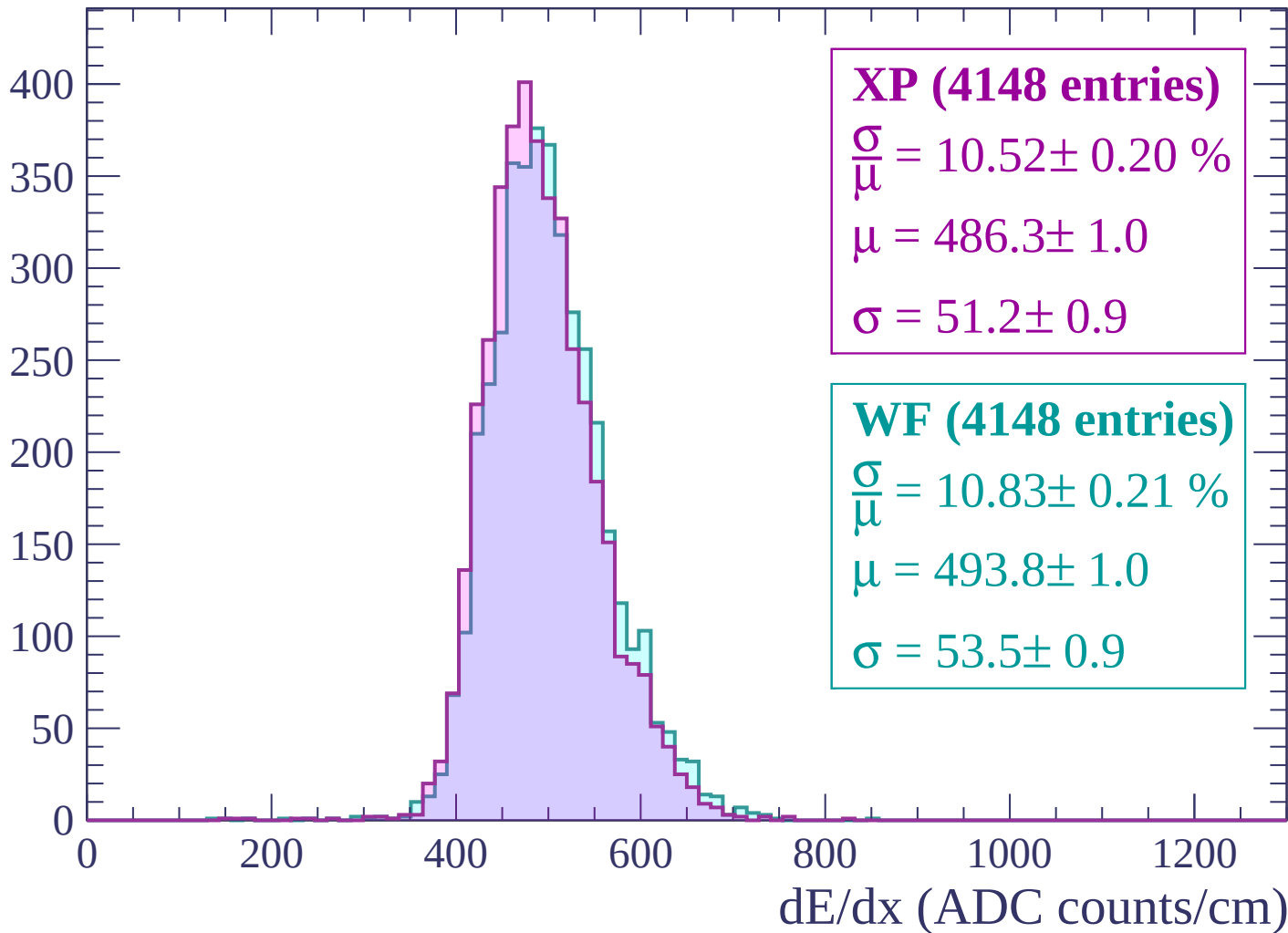


Energy loss | -2040 < p < -1980



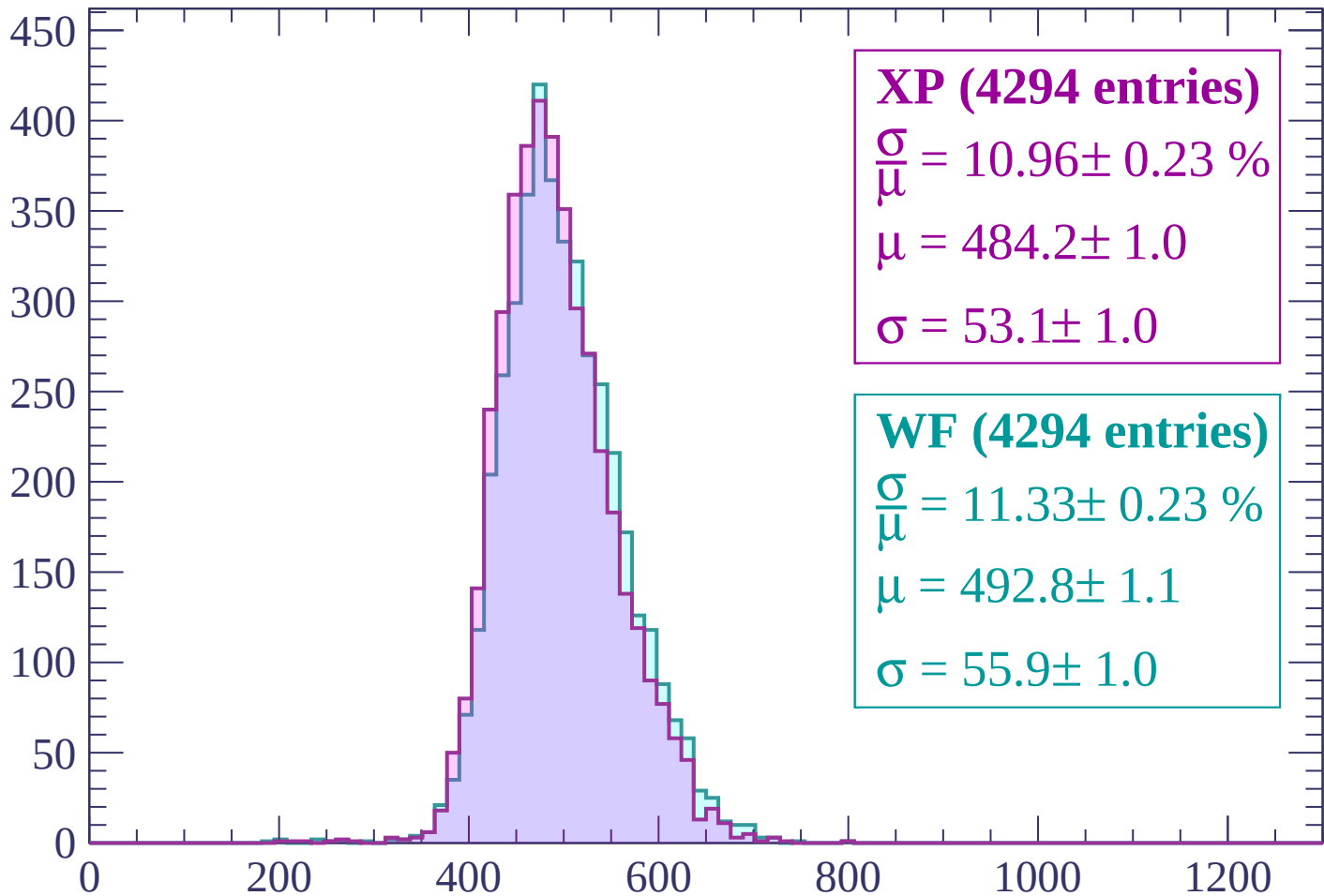
Energy loss | -1980 < p < -1920

Count



Energy loss | -1920 < p < -1860

Count



XP (4294 entries)

$$\frac{\sigma}{\mu} = 10.96 \pm 0.23 \%$$

$$\mu = 484.2 \pm 1.0$$

$$\sigma = 53.1 \pm 1.0$$

WF (4294 entries)

$$\frac{\sigma}{\mu} = 11.33 \pm 0.23 \%$$

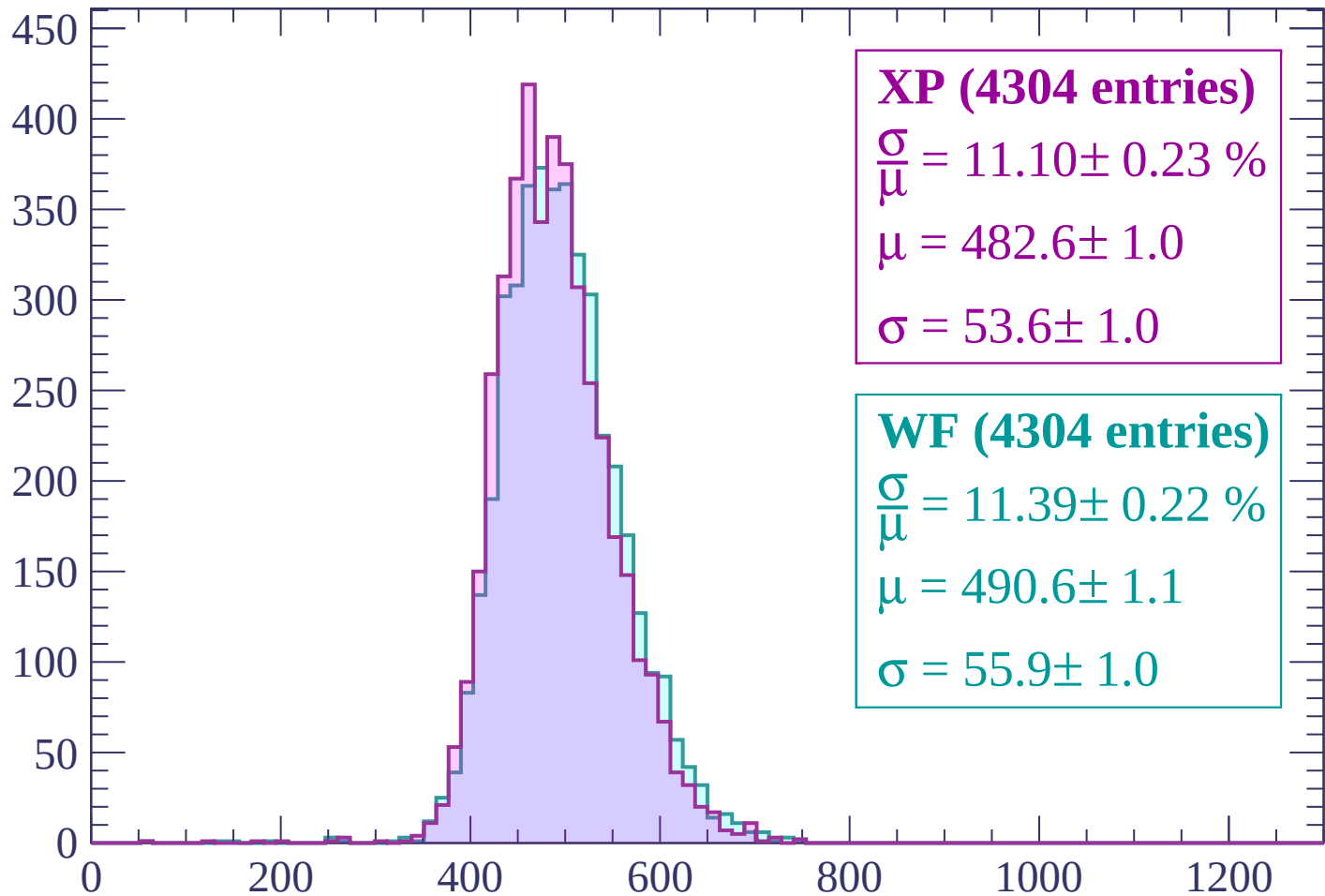
$$\mu = 492.8 \pm 1.1$$

$$\sigma = 55.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1860 < p < -1800$

Count



XP (4304 entries)

$\frac{\sigma}{\mu} = 11.10 \pm 0.23 \%$

$\mu = 482.6 \pm 1.0$

$\sigma = 53.6 \pm 1.0$

WF (4304 entries)

$\frac{\sigma}{\mu} = 11.39 \pm 0.22 \%$

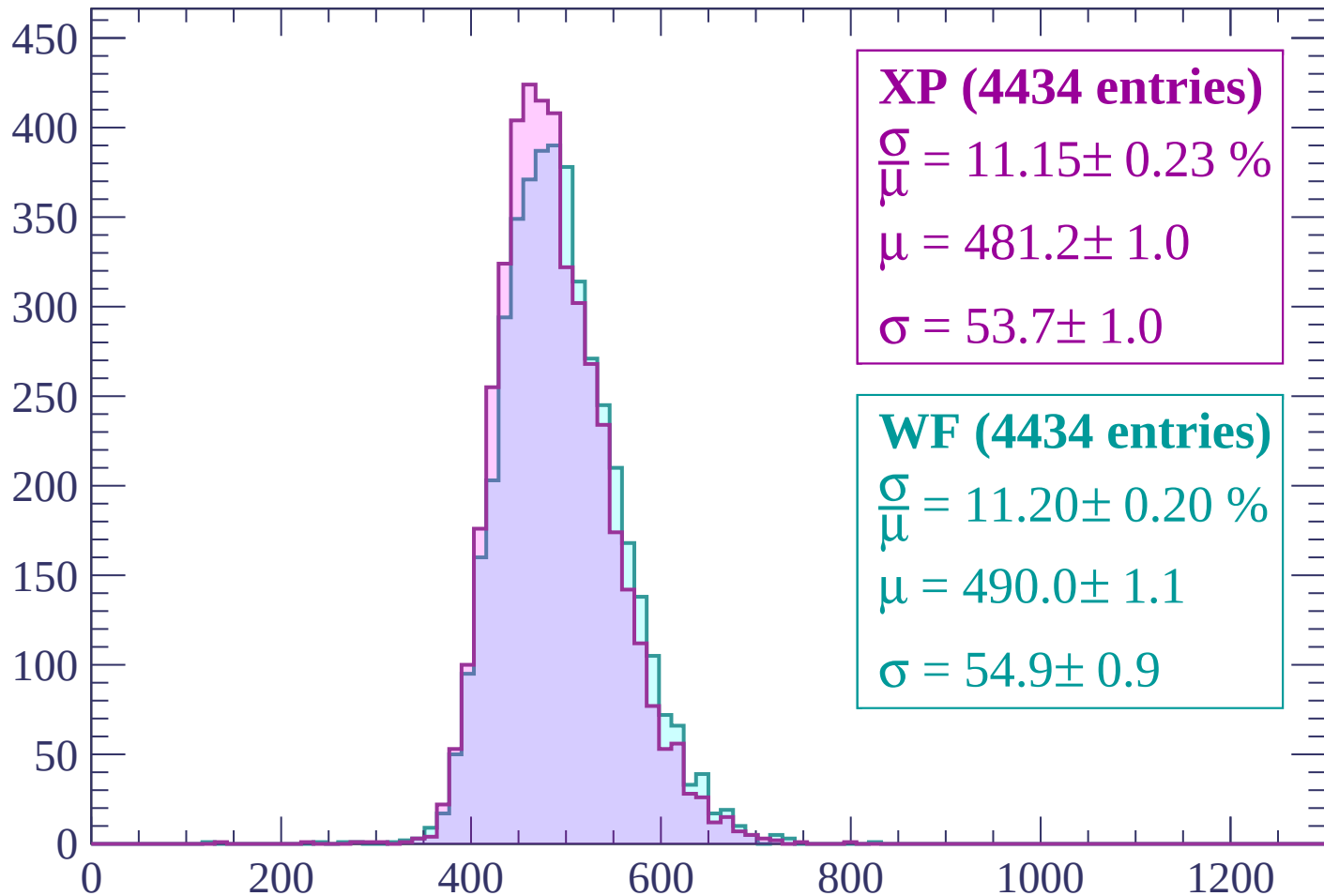
$\mu = 490.6 \pm 1.1$

$\sigma = 55.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count



XP (4434 entries)

$\frac{\sigma}{\mu} = 11.15 \pm 0.23 \%$

$\mu = 481.2 \pm 1.0$

$\sigma = 53.7 \pm 1.0$

WF (4434 entries)

$\frac{\sigma}{\mu} = 11.20 \pm 0.20 \%$

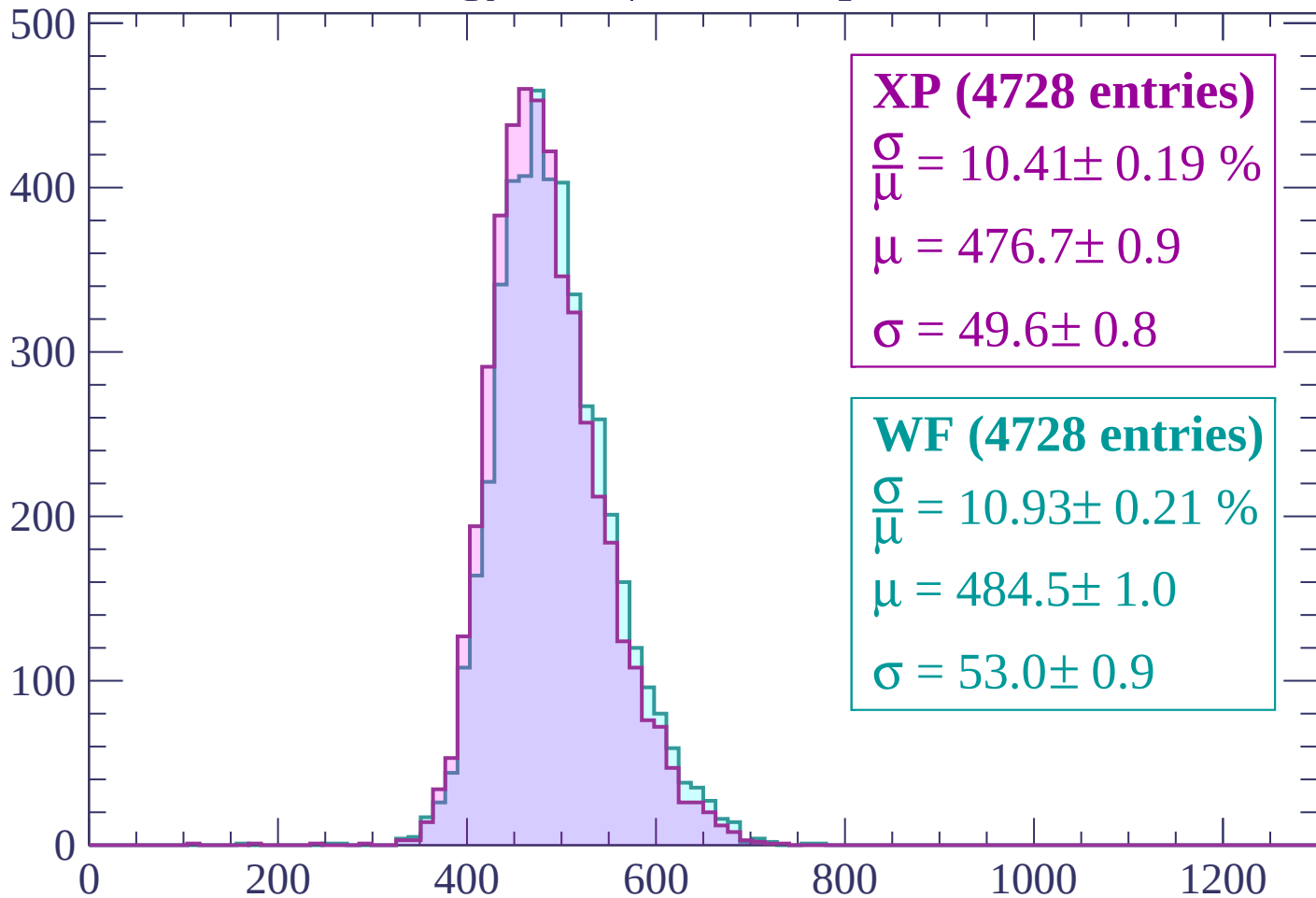
$\mu = 490.0 \pm 1.1$

$\sigma = 54.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (4728 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.19 \%$

$\mu = 476.7 \pm 0.9$

$\sigma = 49.6 \pm 0.8$

WF (4728 entries)

$\frac{\sigma}{\mu} = 10.93 \pm 0.21 \%$

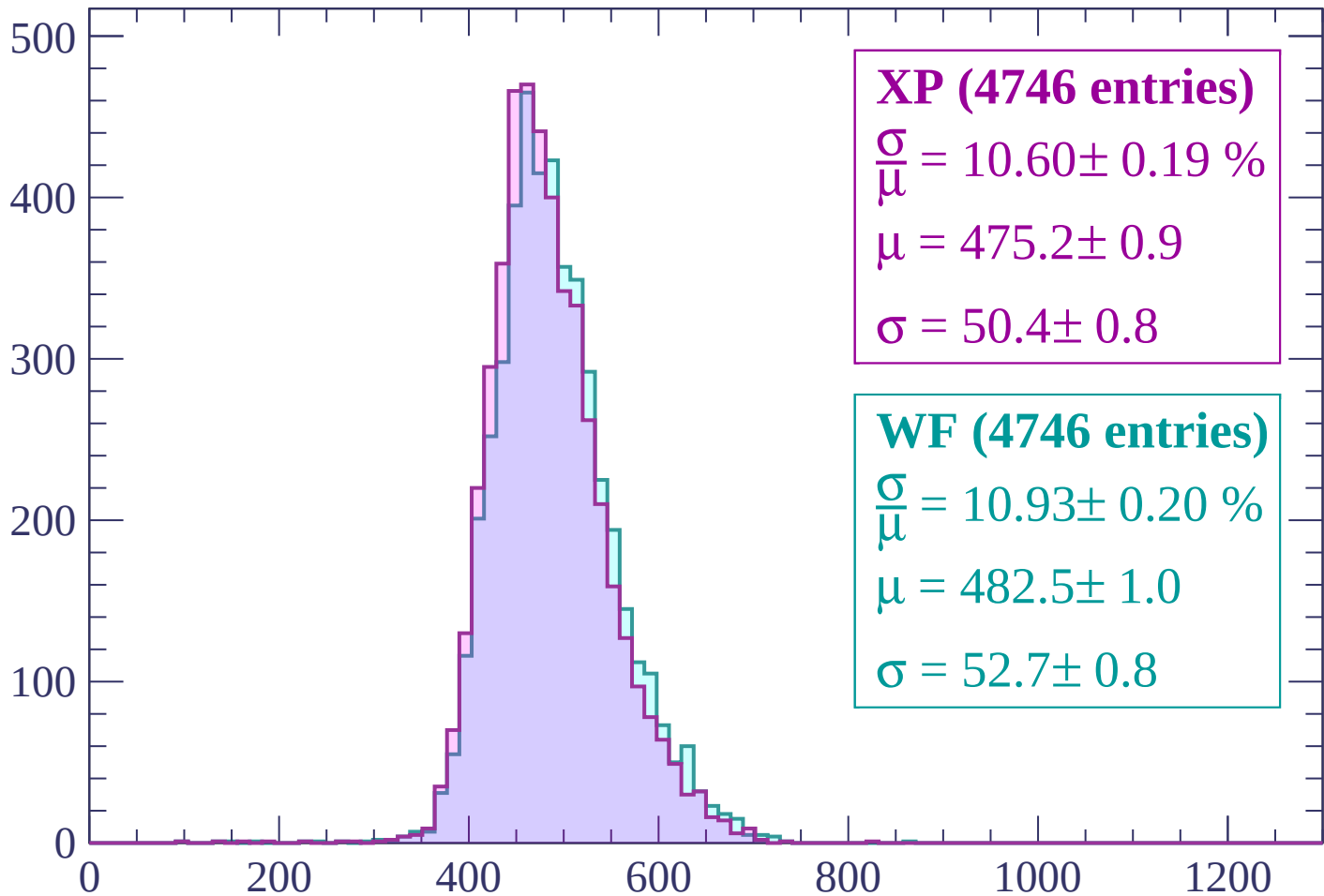
$\mu = 484.5 \pm 1.0$

$\sigma = 53.0 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4746 entries)

$$\frac{\sigma}{\mu} = 10.60 \pm 0.19 \%$$

$$\mu = 475.2 \pm 0.9$$

$$\sigma = 50.4 \pm 0.8$$

WF (4746 entries)

$$\frac{\sigma}{\mu} = 10.93 \pm 0.20 \%$$

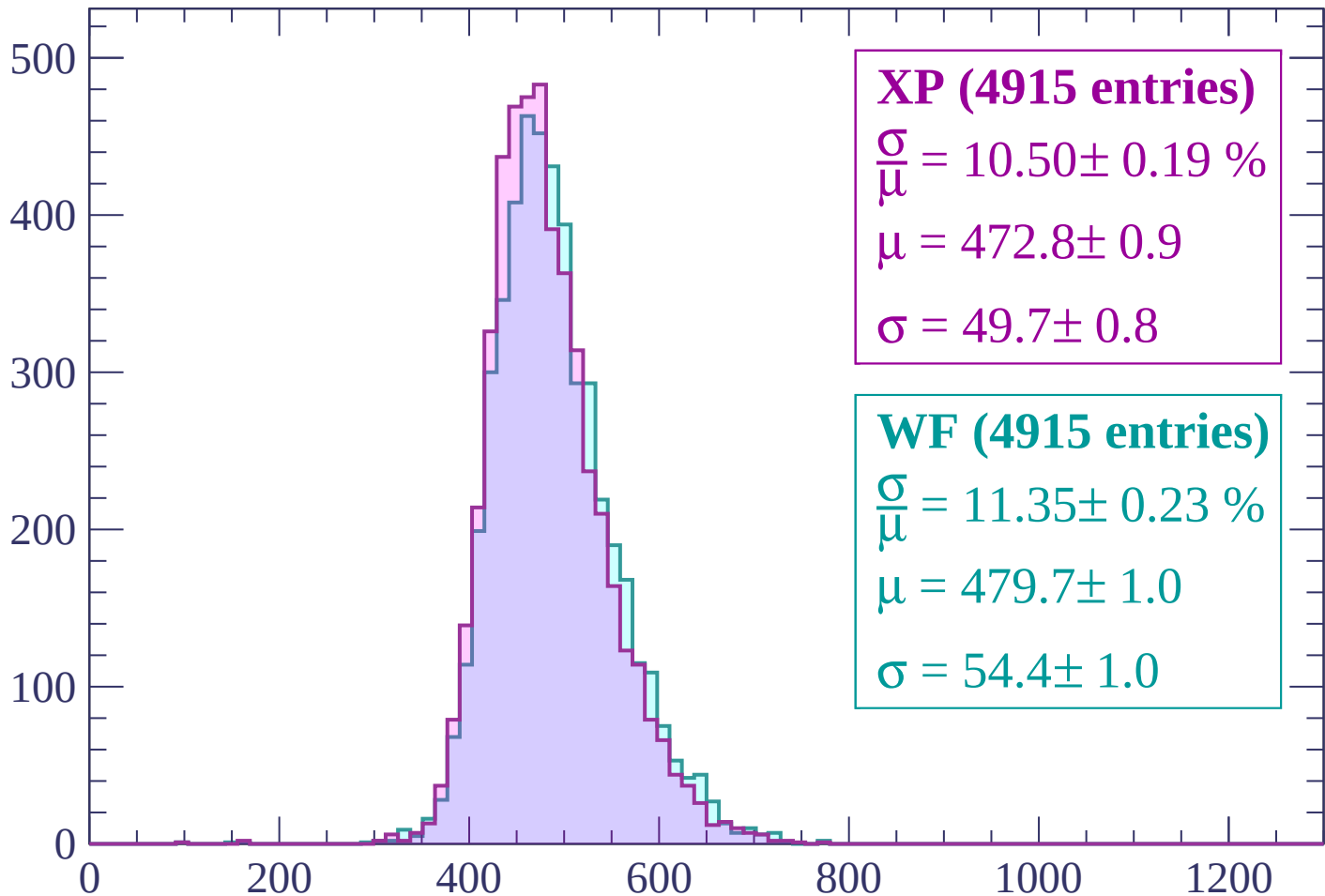
$$\mu = 482.5 \pm 1.0$$

$$\sigma = 52.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count



XP (4915 entries)

$$\frac{\sigma}{\mu} = 10.50 \pm 0.19 \%$$

$$\mu = 472.8 \pm 0.9$$

$$\sigma = 49.7 \pm 0.8$$

WF (4915 entries)

$$\frac{\sigma}{\mu} = 11.35 \pm 0.23 \%$$

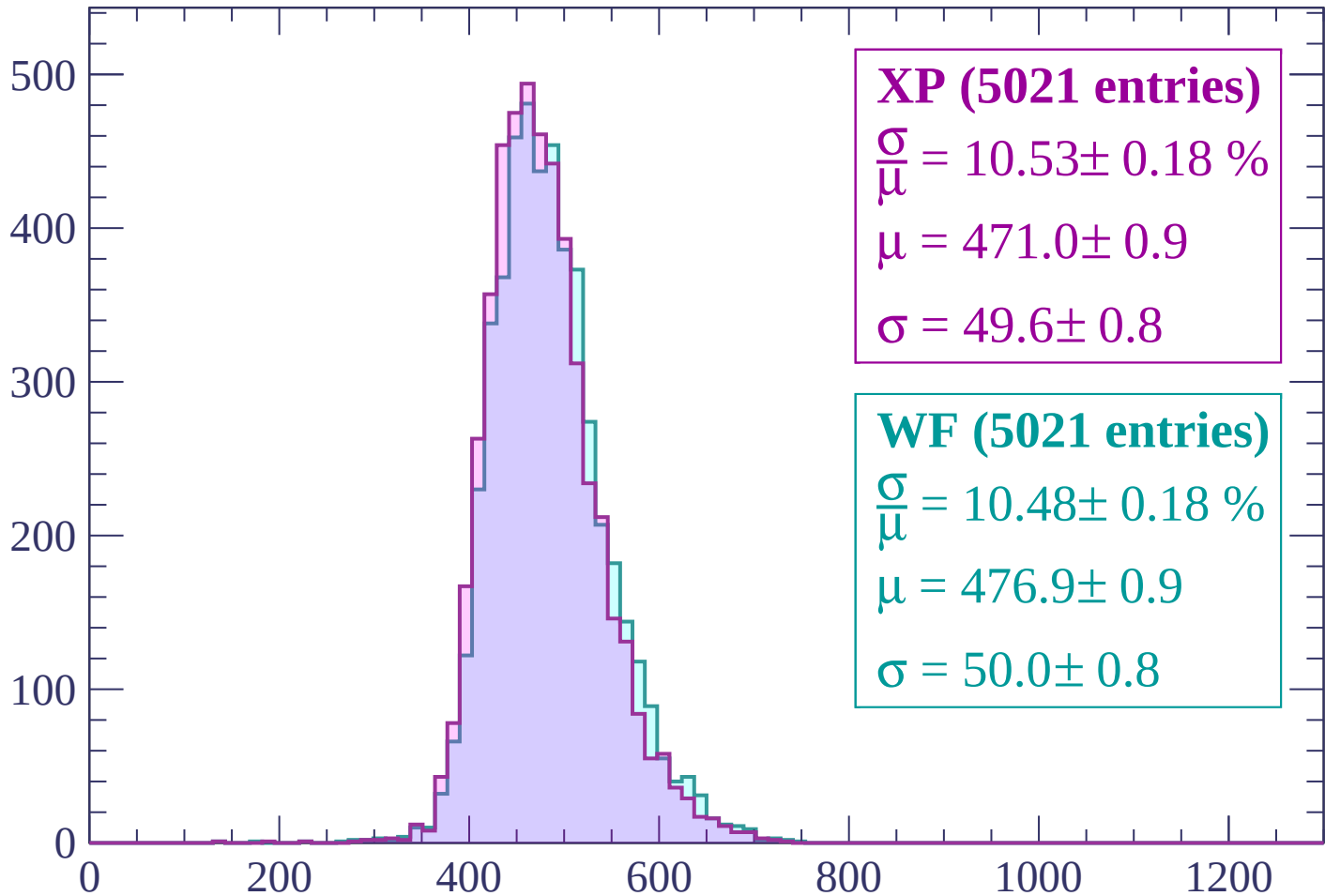
$$\mu = 479.7 \pm 1.0$$

$$\sigma = 54.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (5021 entries)

$\frac{\sigma}{\mu} = 10.53 \pm 0.18 \%$

$\mu = 471.0 \pm 0.9$

$\sigma = 49.6 \pm 0.8$

WF (5021 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.18 \%$

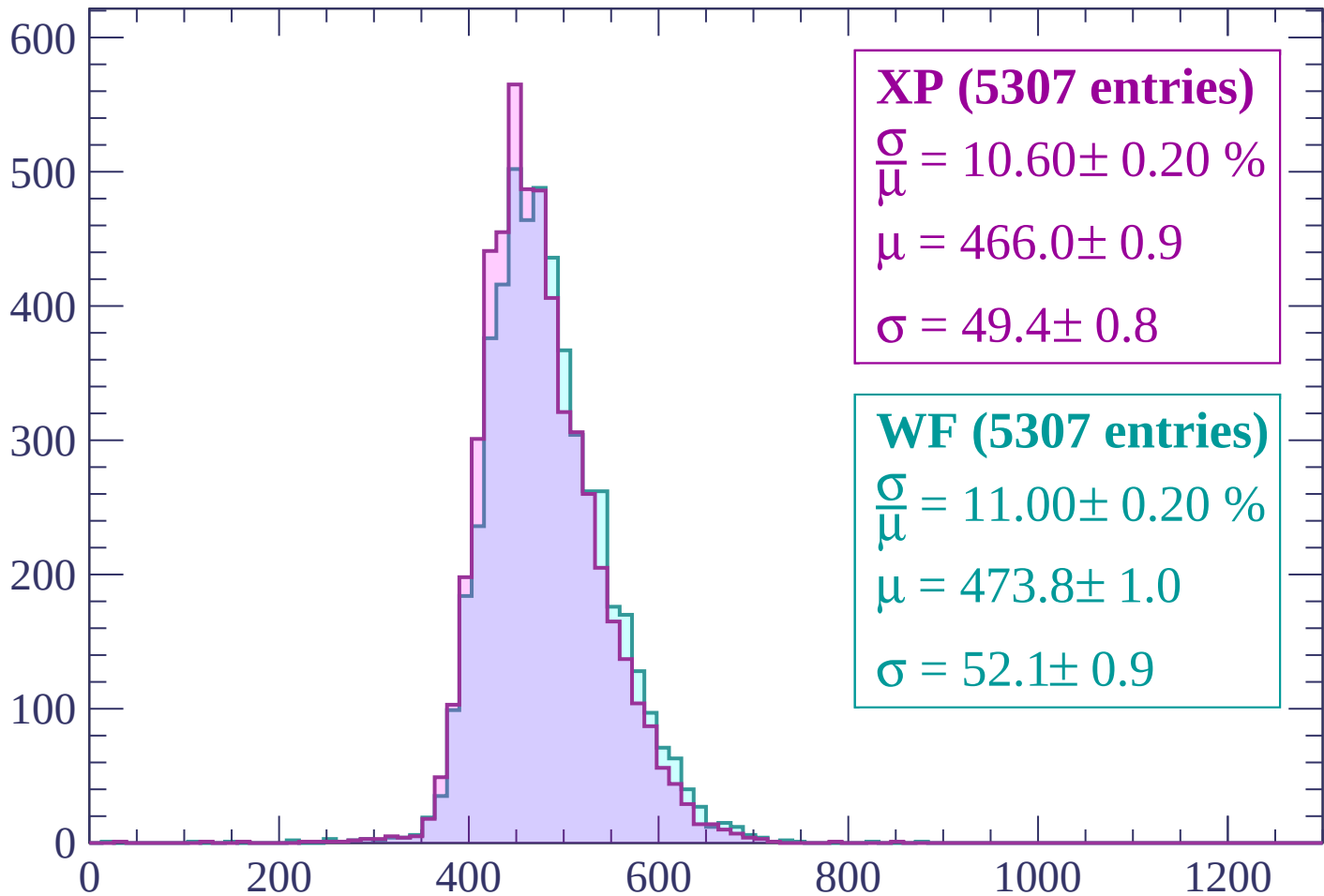
$\mu = 476.9 \pm 0.9$

$\sigma = 50.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (5307 entries)

$\frac{\sigma}{\mu} = 10.60 \pm 0.20 \%$

$\mu = 466.0 \pm 0.9$

$\sigma = 49.4 \pm 0.8$

WF (5307 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.20 \%$

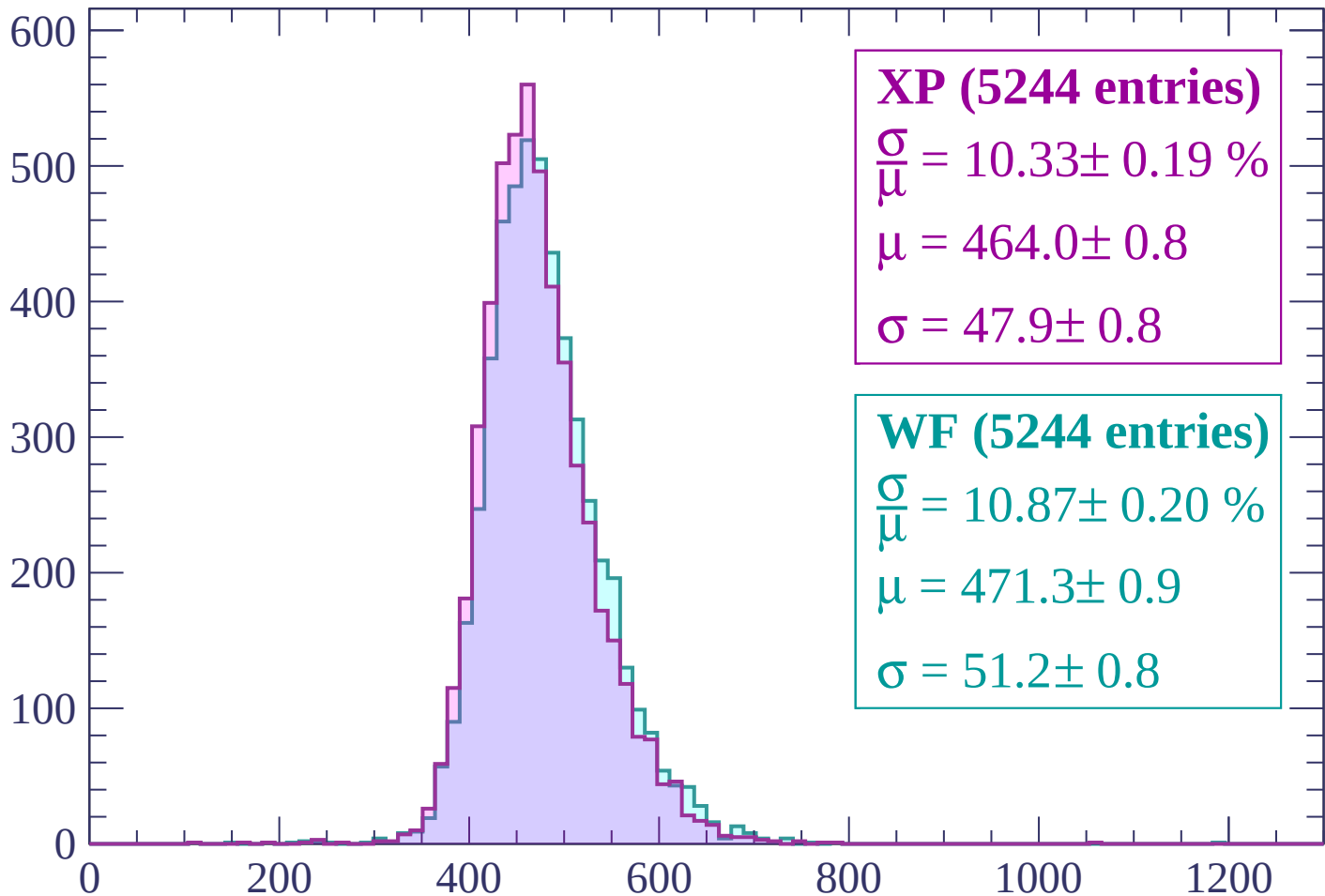
$\mu = 473.8 \pm 1.0$

$\sigma = 52.1 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (5244 entries)

$\frac{\sigma}{\mu} = 10.33 \pm 0.19 \%$

$\mu = 464.0 \pm 0.8$

$\sigma = 47.9 \pm 0.8$

WF (5244 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.20 \%$

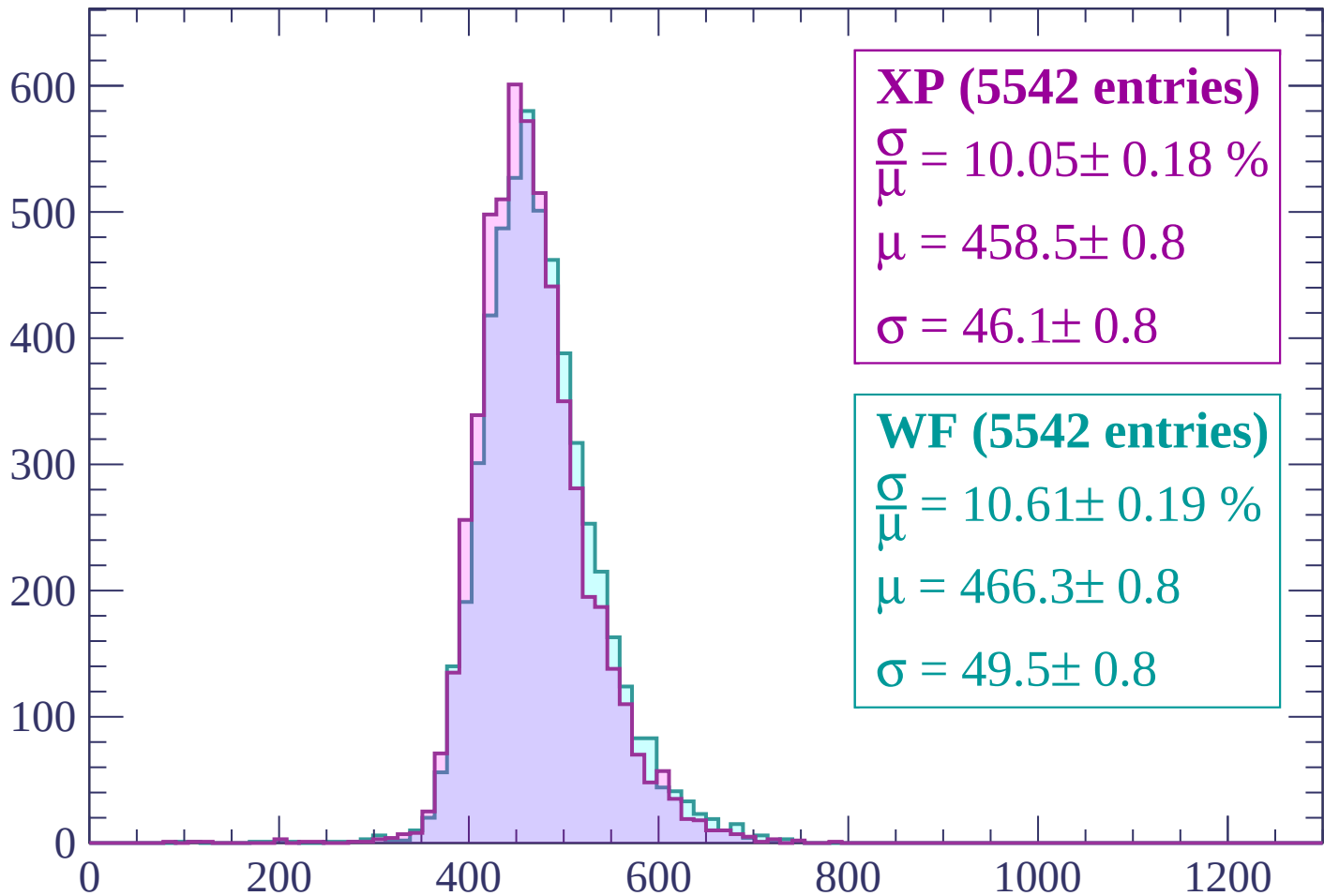
$\mu = 471.3 \pm 0.9$

$\sigma = 51.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (5542 entries)

$\frac{\sigma}{\mu} = 10.05 \pm 0.18 \%$

$\mu = 458.5 \pm 0.8$

$\sigma = 46.1 \pm 0.8$

WF (5542 entries)

$\frac{\sigma}{\mu} = 10.61 \pm 0.19 \%$

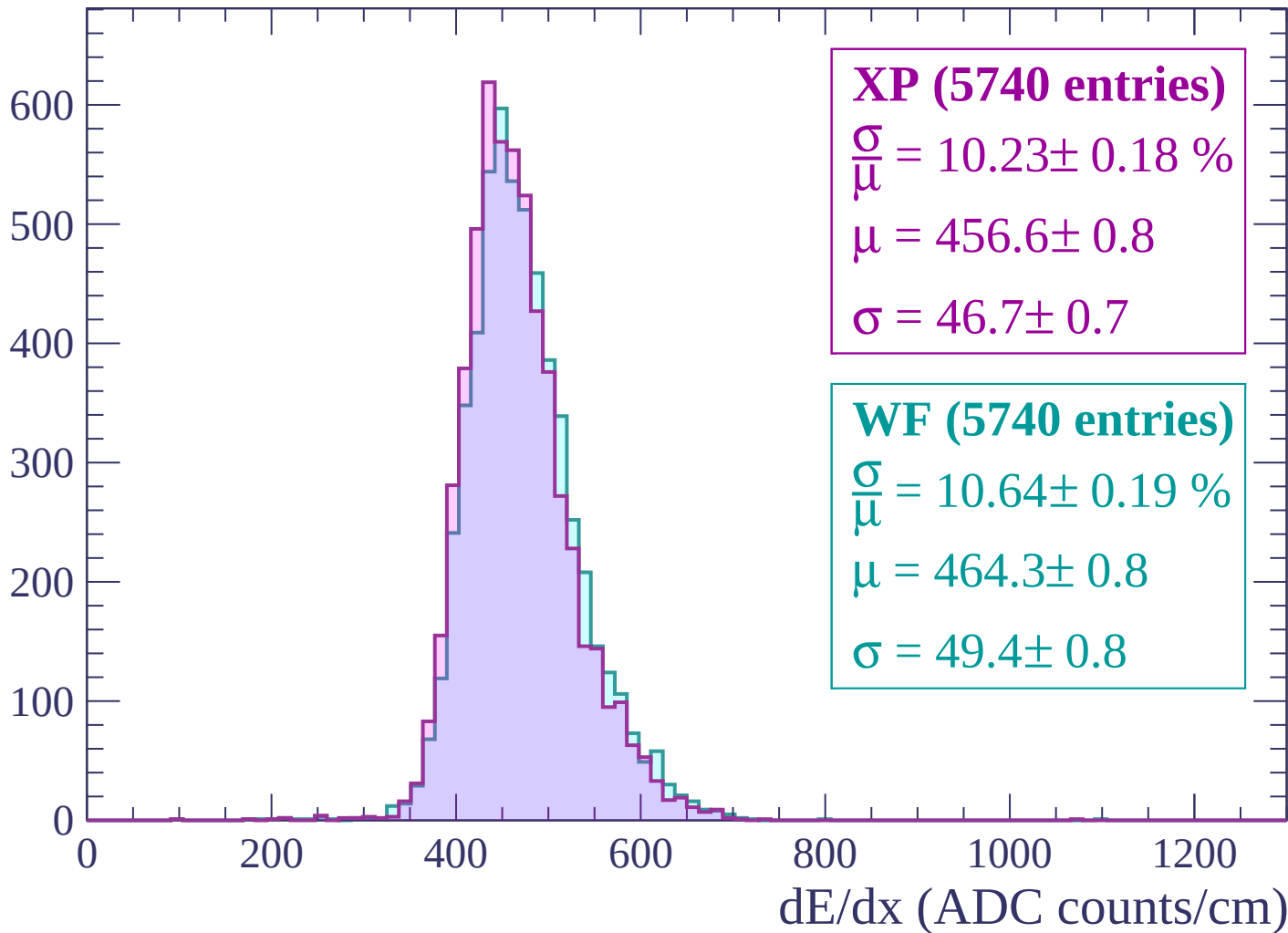
$\mu = 466.3 \pm 0.8$

$\sigma = 49.5 \pm 0.8$

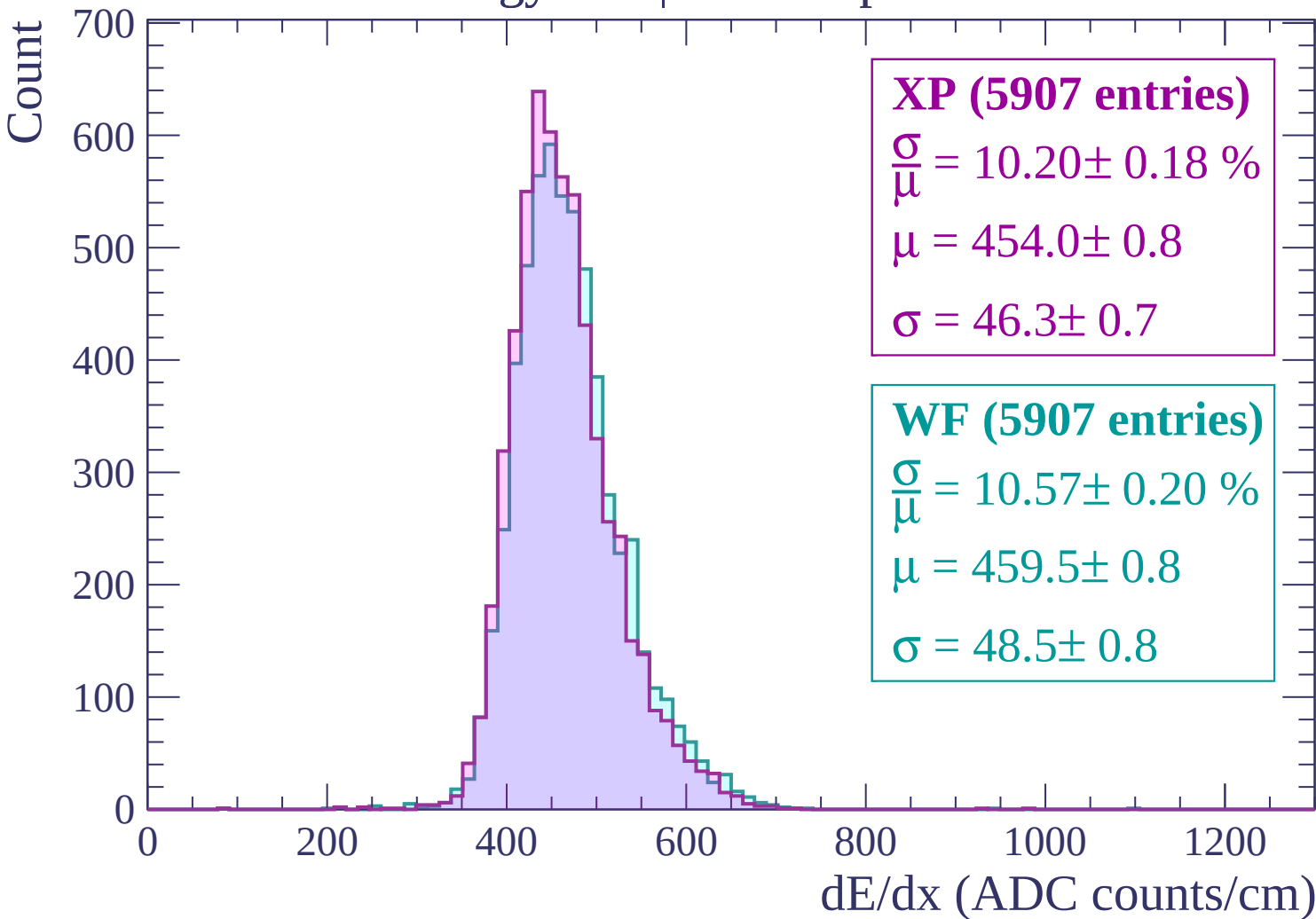
dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count

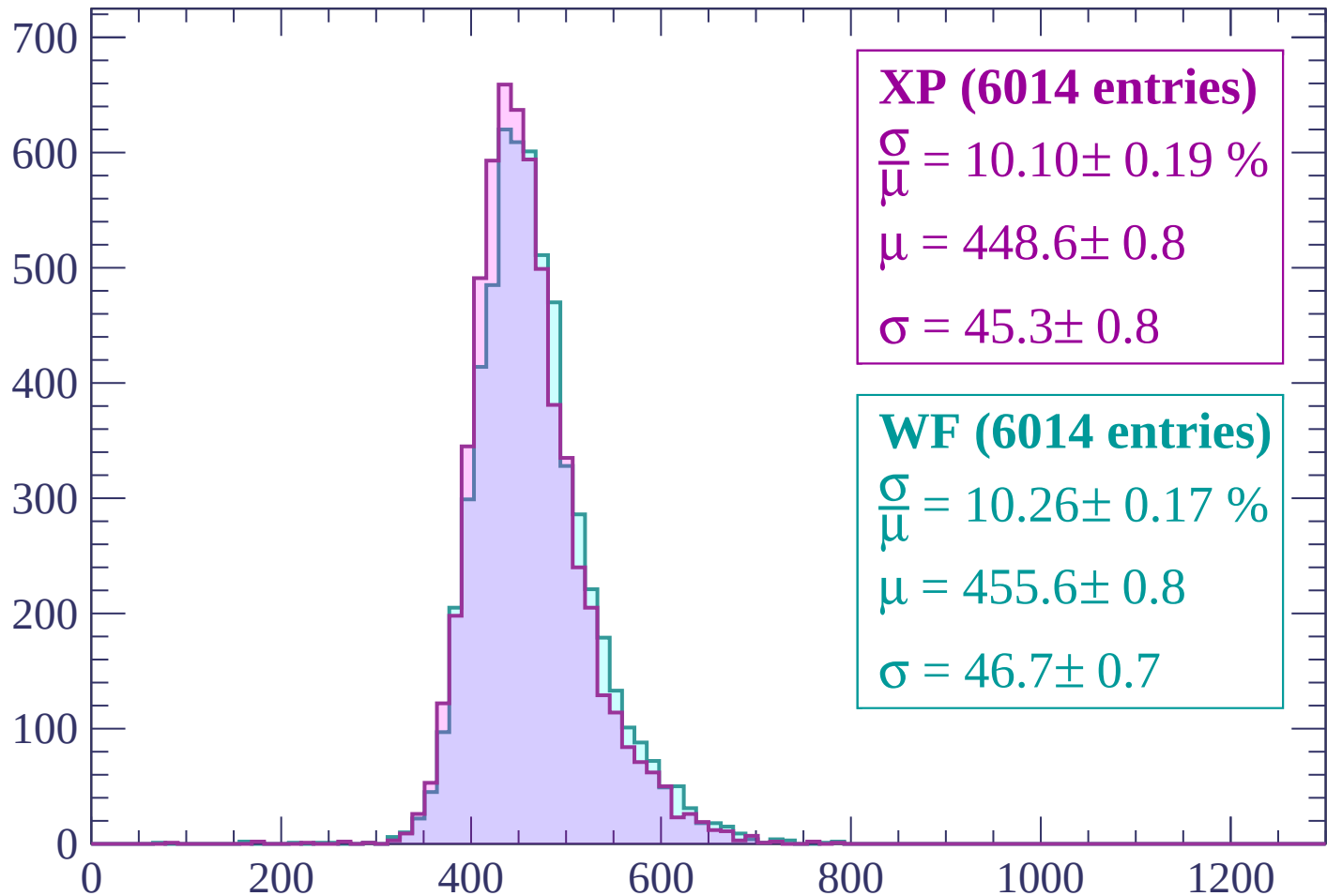


Energy loss | $-1260 < p < -1200$



Energy loss | $-1200 < p < -1140$

Count



XP (6014 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.19 \%$$

$$\mu = 448.6 \pm 0.8$$

$$\sigma = 45.3 \pm 0.8$$

WF (6014 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.17 \%$$

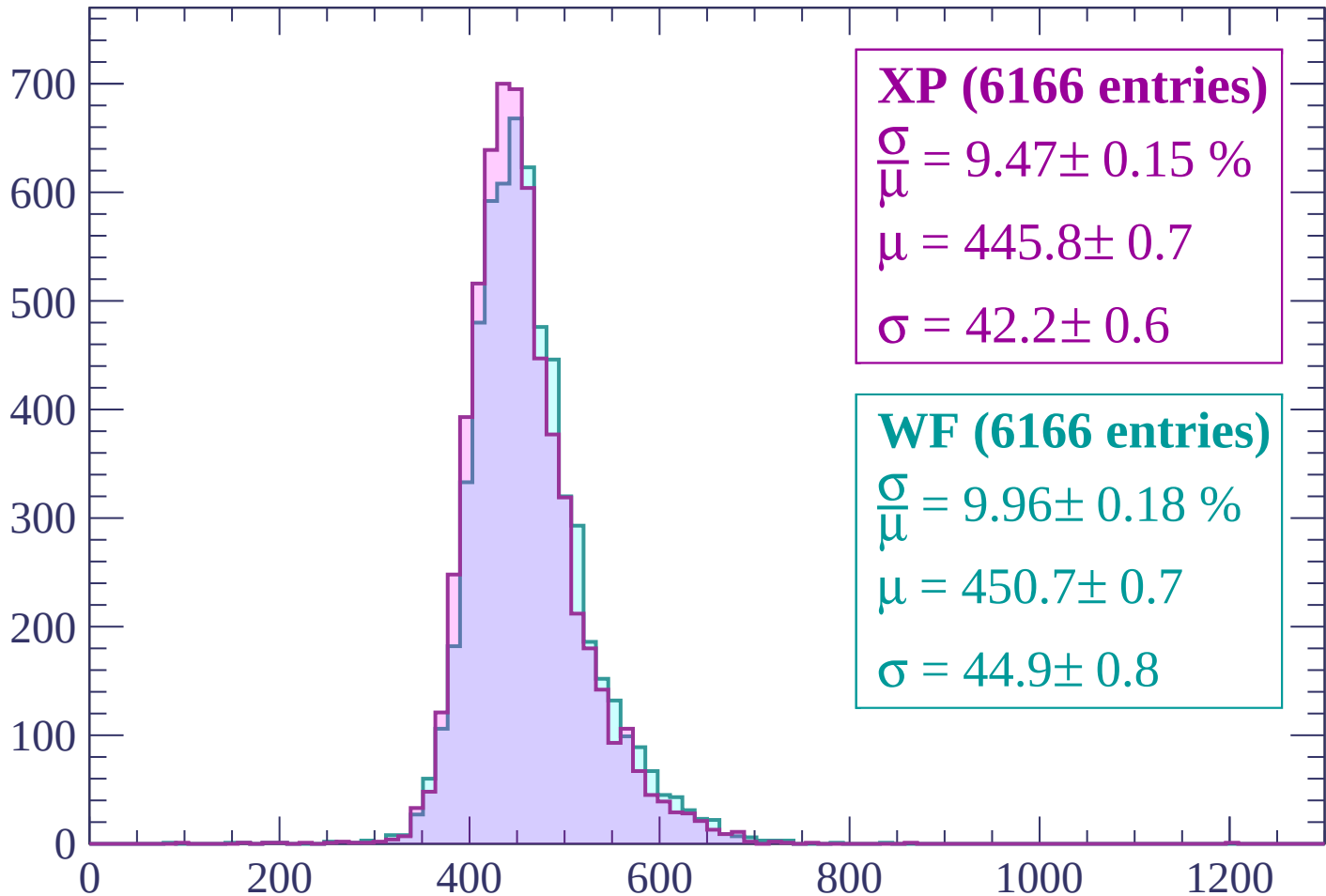
$$\mu = 455.6 \pm 0.8$$

$$\sigma = 46.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count



XP (6166 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.15 \%$$

$$\mu = 445.8 \pm 0.7$$

$$\sigma = 42.2 \pm 0.6$$

WF (6166 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.18 \%$$

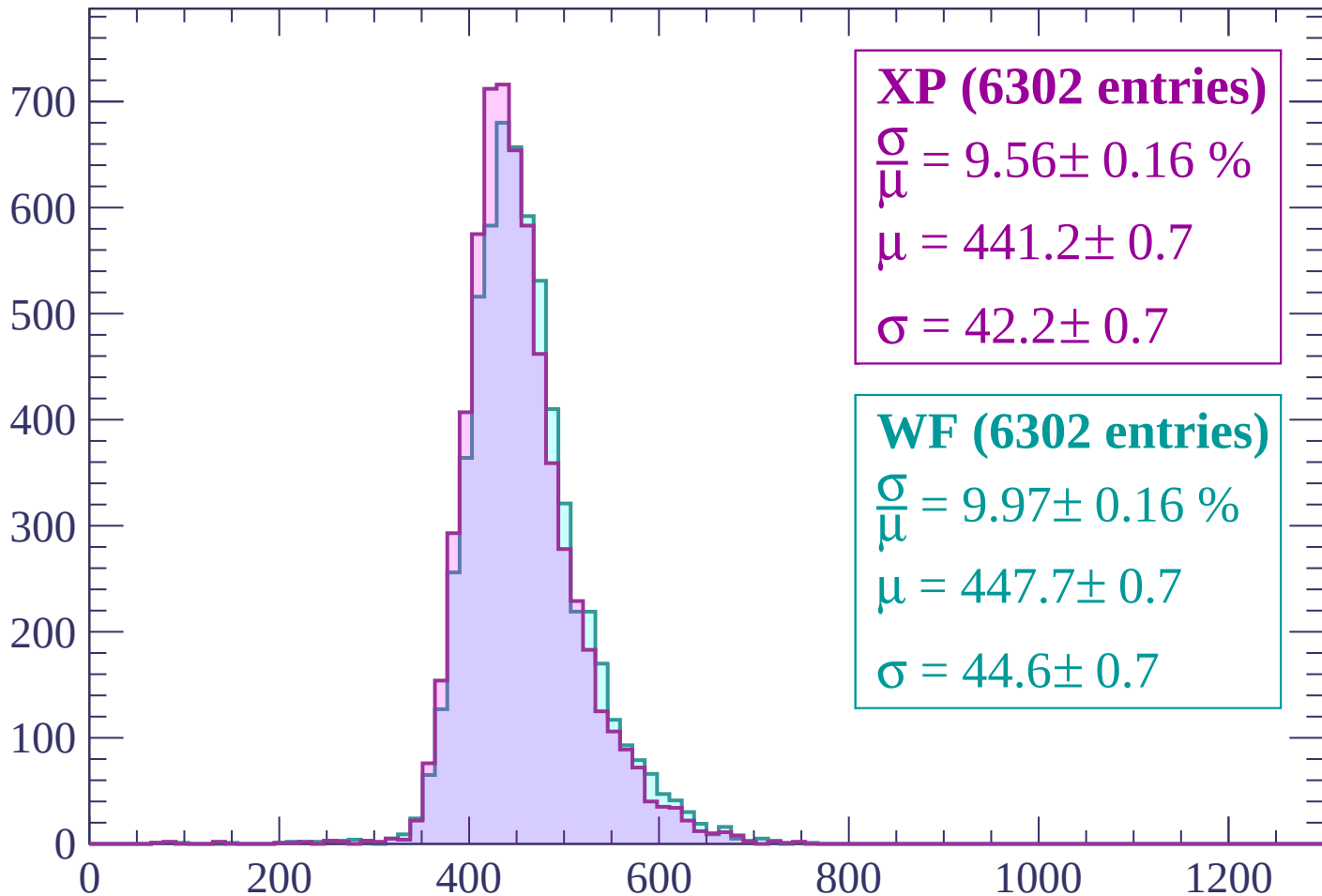
$$\mu = 450.7 \pm 0.7$$

$$\sigma = 44.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (6302 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.16 \%$$

$$\mu = 441.2 \pm 0.7$$

$$\sigma = 42.2 \pm 0.7$$

WF (6302 entries)

$$\frac{\sigma}{\mu} = 9.97 \pm 0.16 \%$$

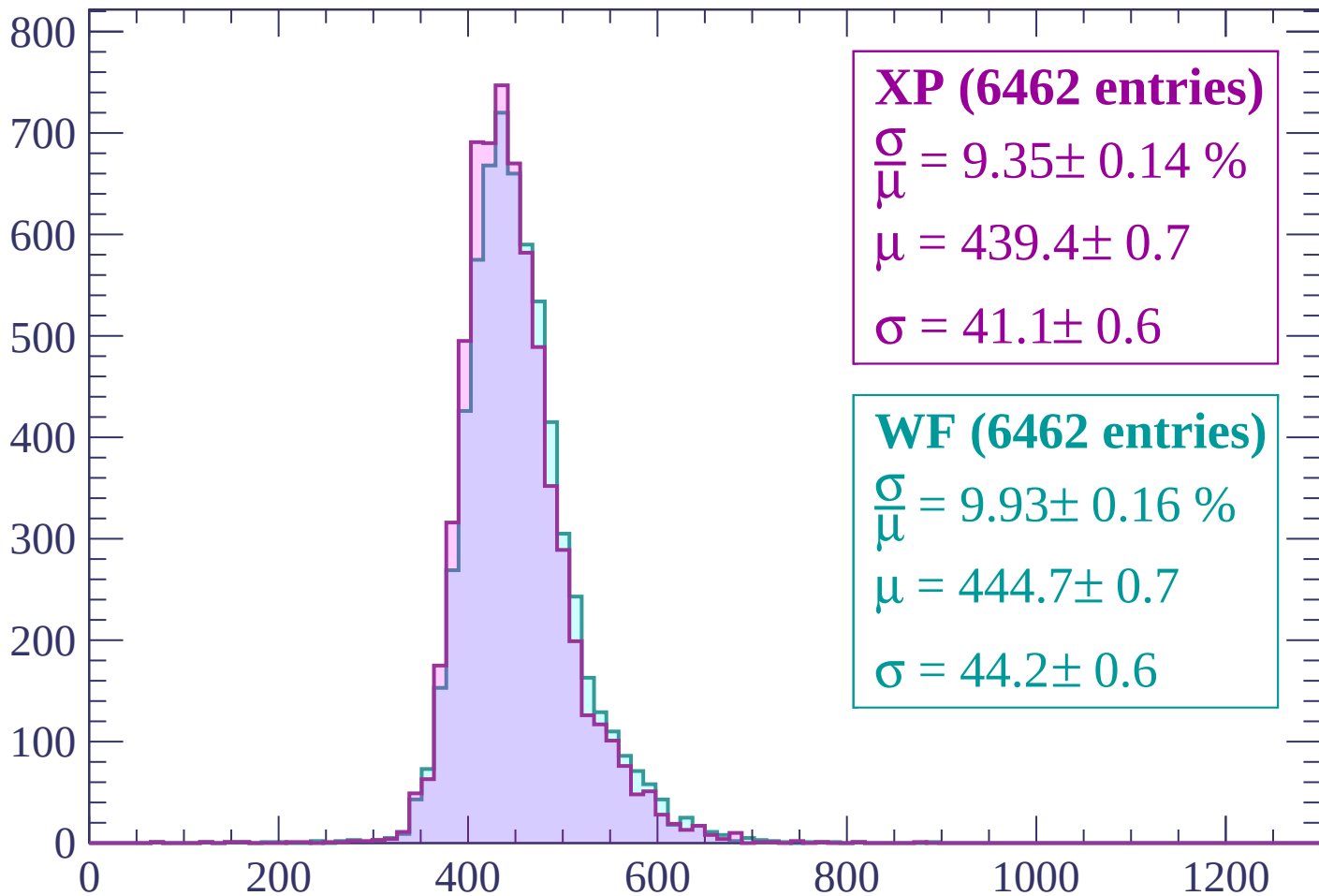
$$\mu = 447.7 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6462 entries)

$\frac{\sigma}{\mu} = 9.35 \pm 0.14 \%$

$\mu = 439.4 \pm 0.7$

$\sigma = 41.1 \pm 0.6$

WF (6462 entries)

$\frac{\sigma}{\mu} = 9.93 \pm 0.16 \%$

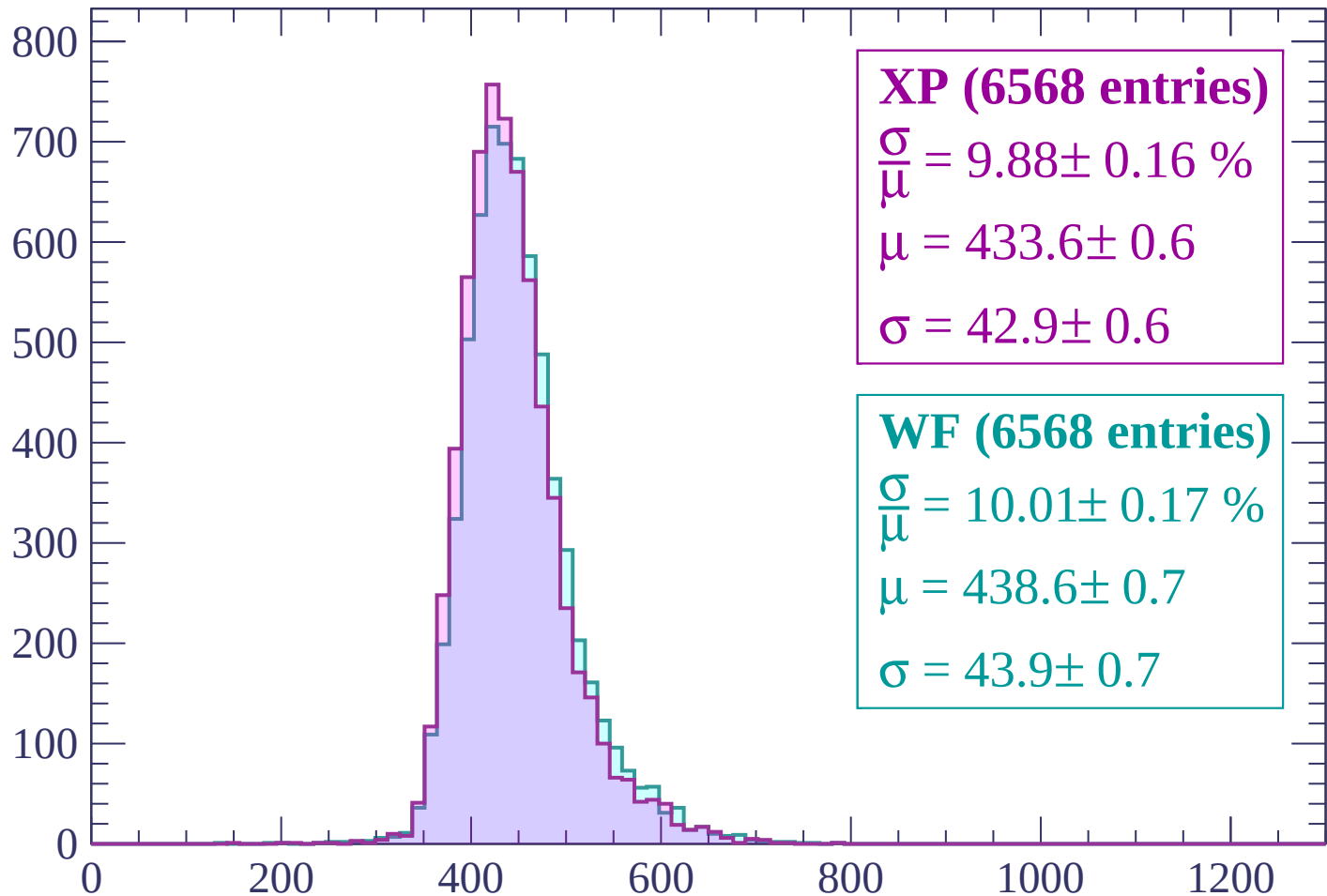
$\mu = 444.7 \pm 0.7$

$\sigma = 44.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (6568 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.16 \%$$

$$\mu = 433.6 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

WF (6568 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.17 \%$$

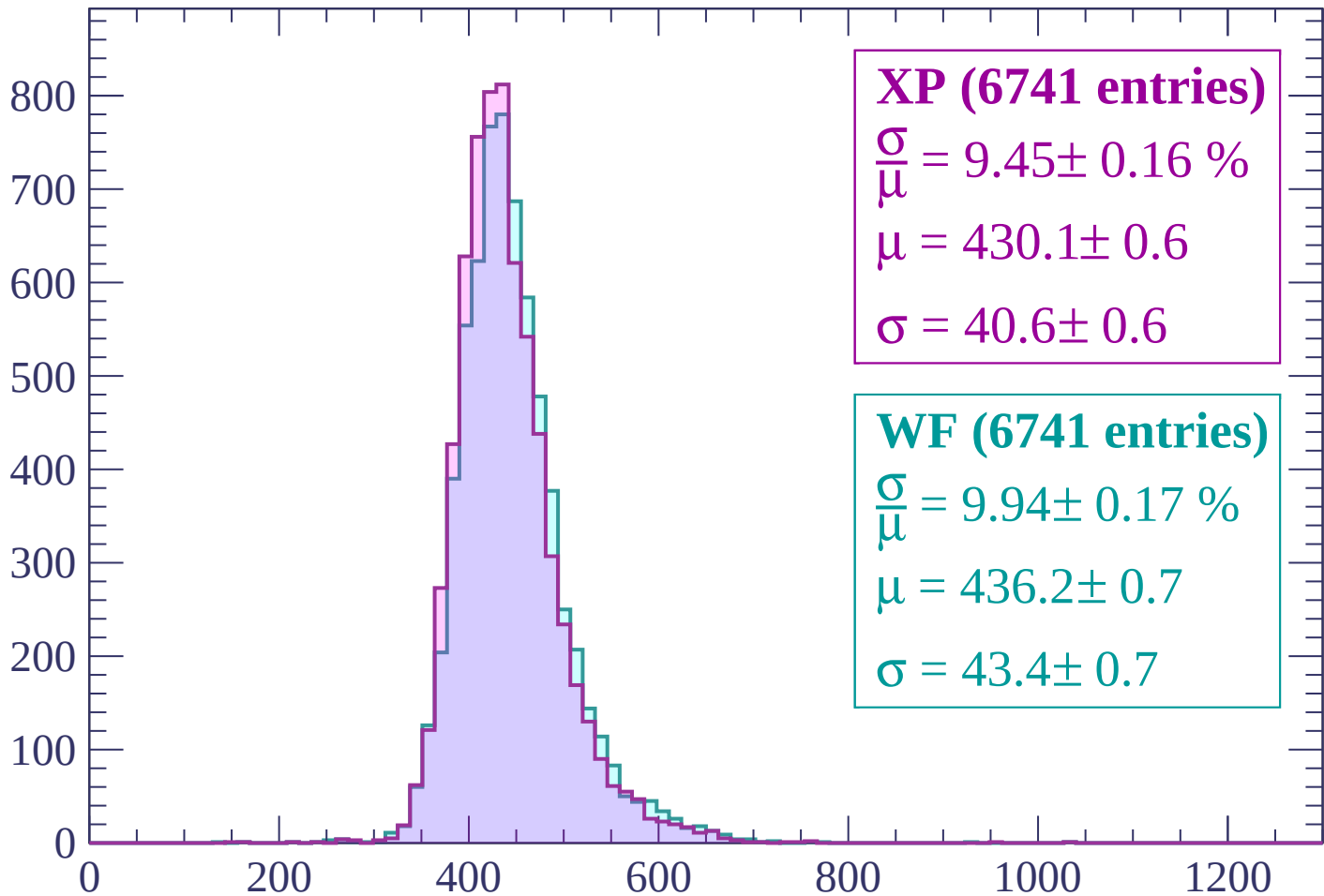
$$\mu = 438.6 \pm 0.7$$

$$\sigma = 43.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (6741 entries)

$\frac{\sigma}{\mu} = 9.45 \pm 0.16 \%$

$\mu = 430.1 \pm 0.6$

$\sigma = 40.6 \pm 0.6$

WF (6741 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$

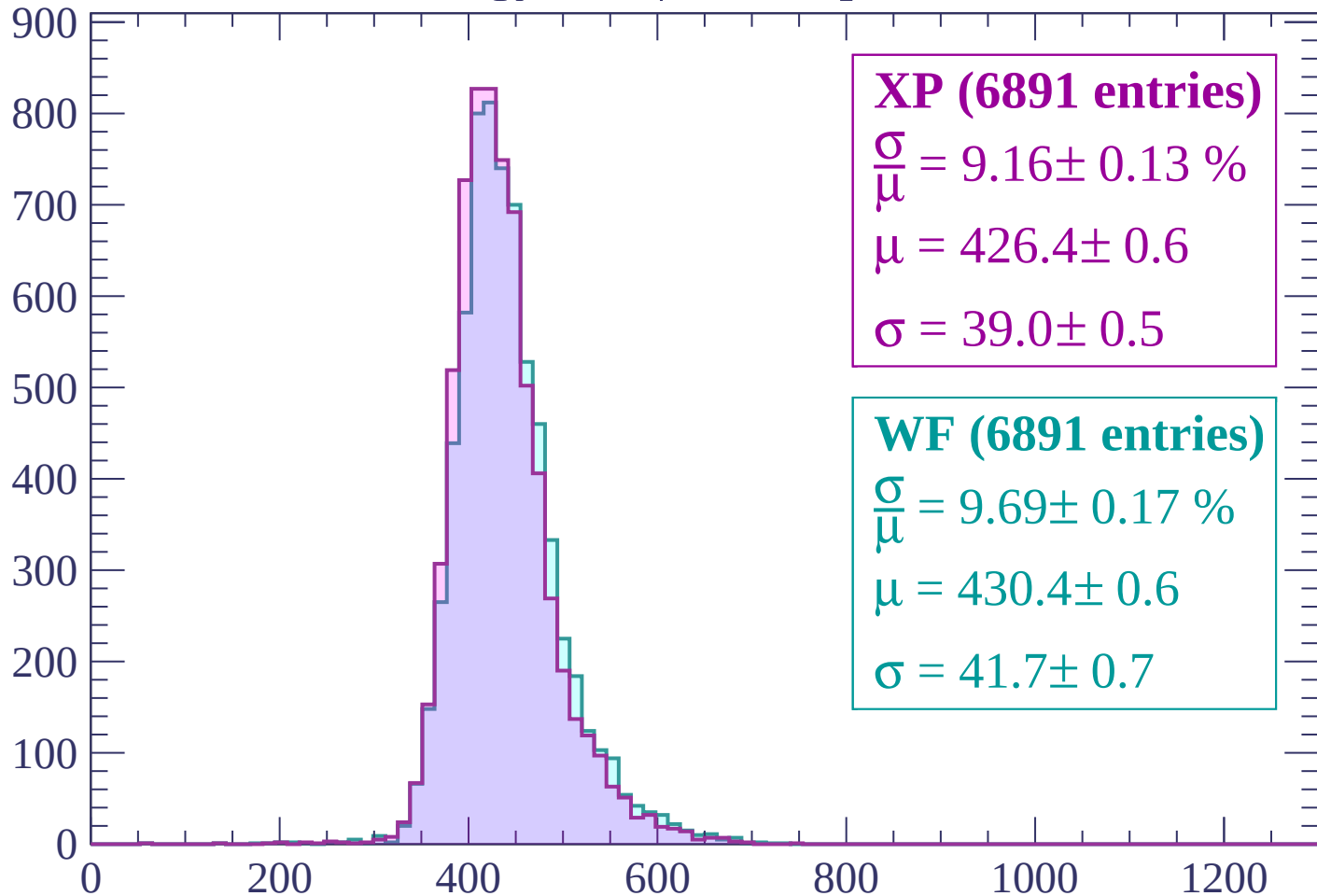
$\mu = 436.2 \pm 0.7$

$\sigma = 43.4 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (6891 entries)

$\frac{\sigma}{\mu} = 9.16 \pm 0.13 \%$

$\mu = 426.4 \pm 0.6$

$\sigma = 39.0 \pm 0.5$

WF (6891 entries)

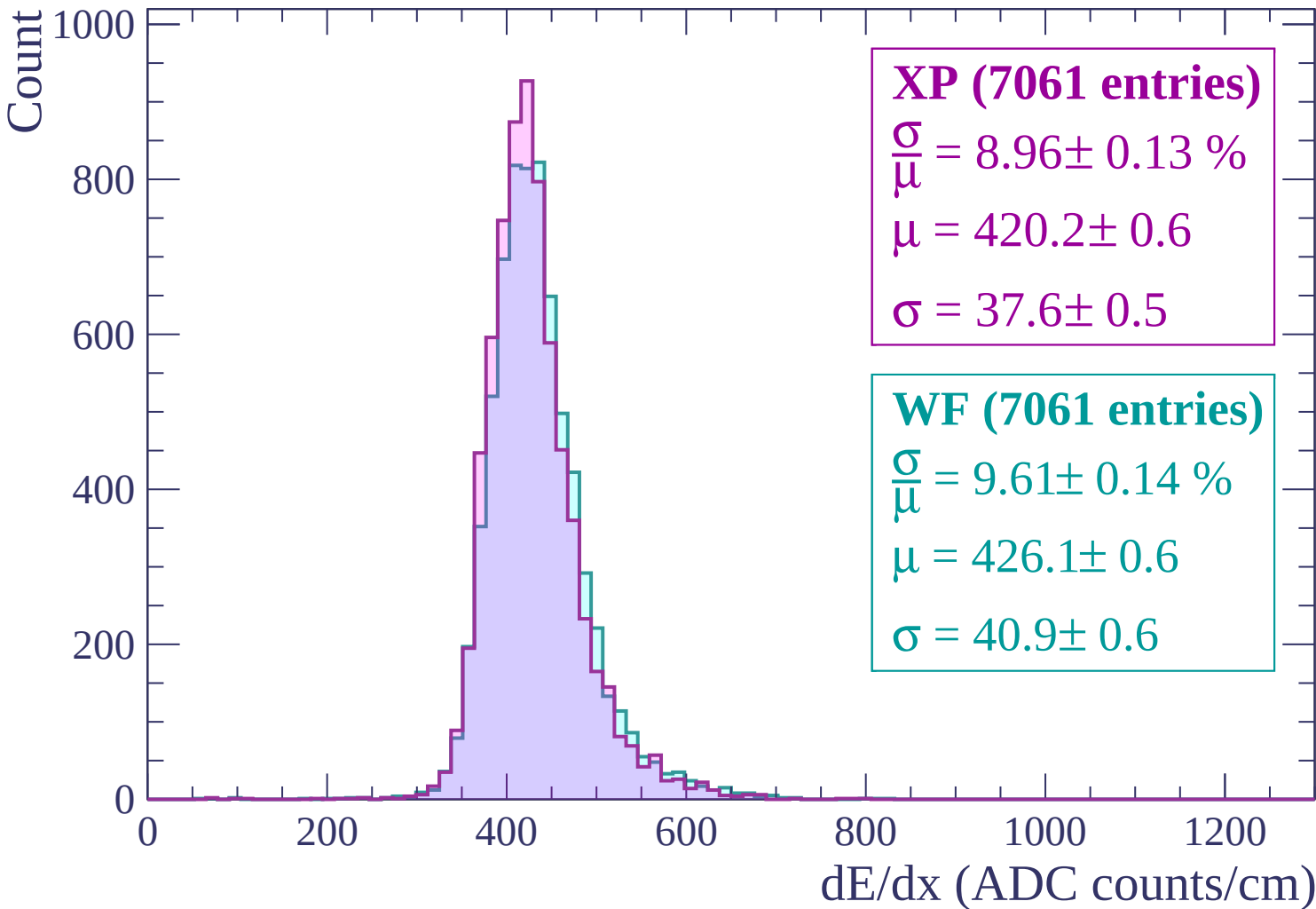
$\frac{\sigma}{\mu} = 9.69 \pm 0.17 \%$

$\mu = 430.4 \pm 0.6$

$\sigma = 41.7 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$



Energy loss | $-720 < p < -660$

Count

1000

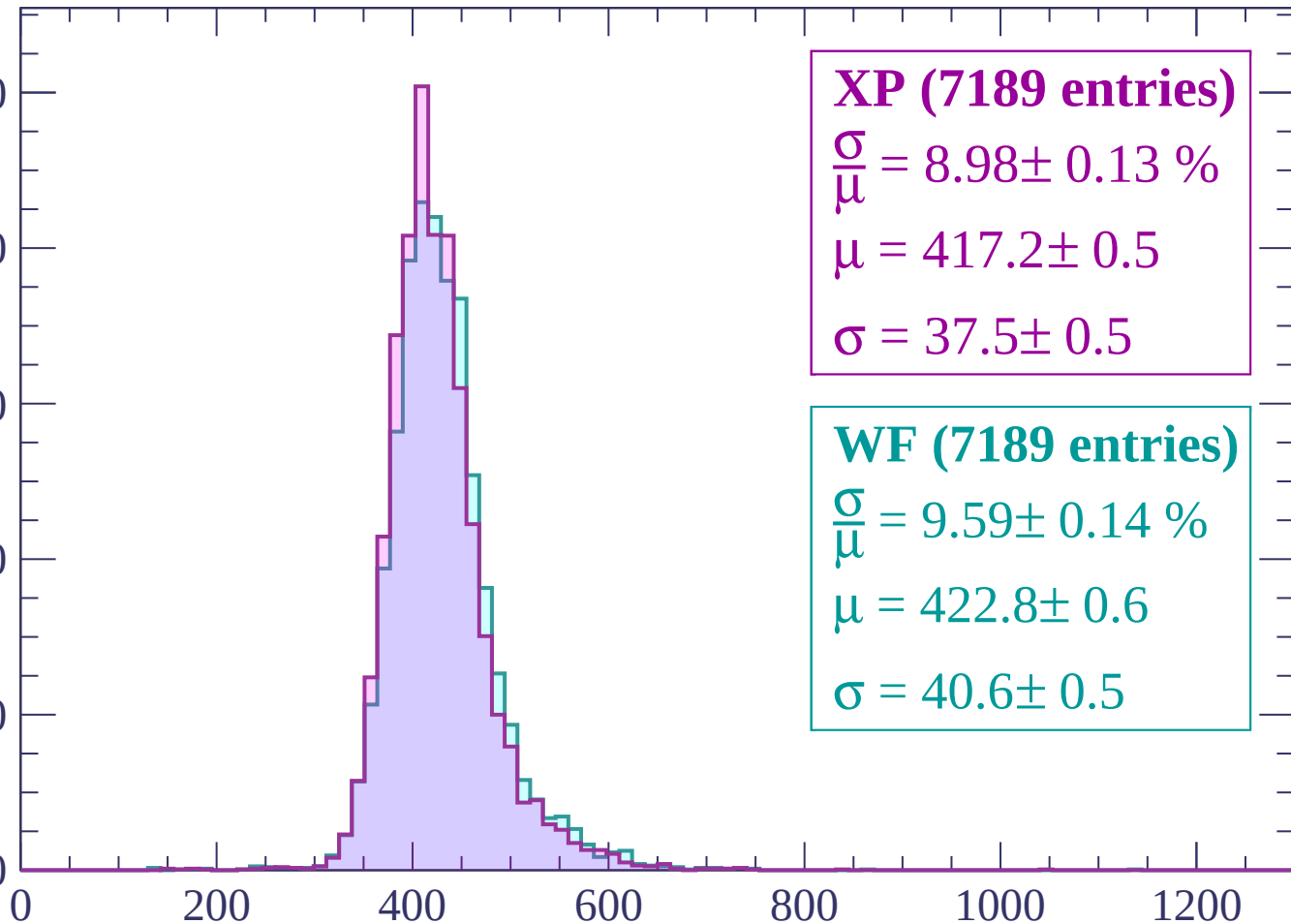
800

600

400

200

0



XP (7189 entries)

$\frac{\sigma}{\mu} = 8.98 \pm 0.13 \%$

$\mu = 417.2 \pm 0.5$

$\sigma = 37.5 \pm 0.5$

WF (7189 entries)

$\frac{\sigma}{\mu} = 9.59 \pm 0.14 \%$

$\mu = 422.8 \pm 0.6$

$\sigma = 40.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

1000

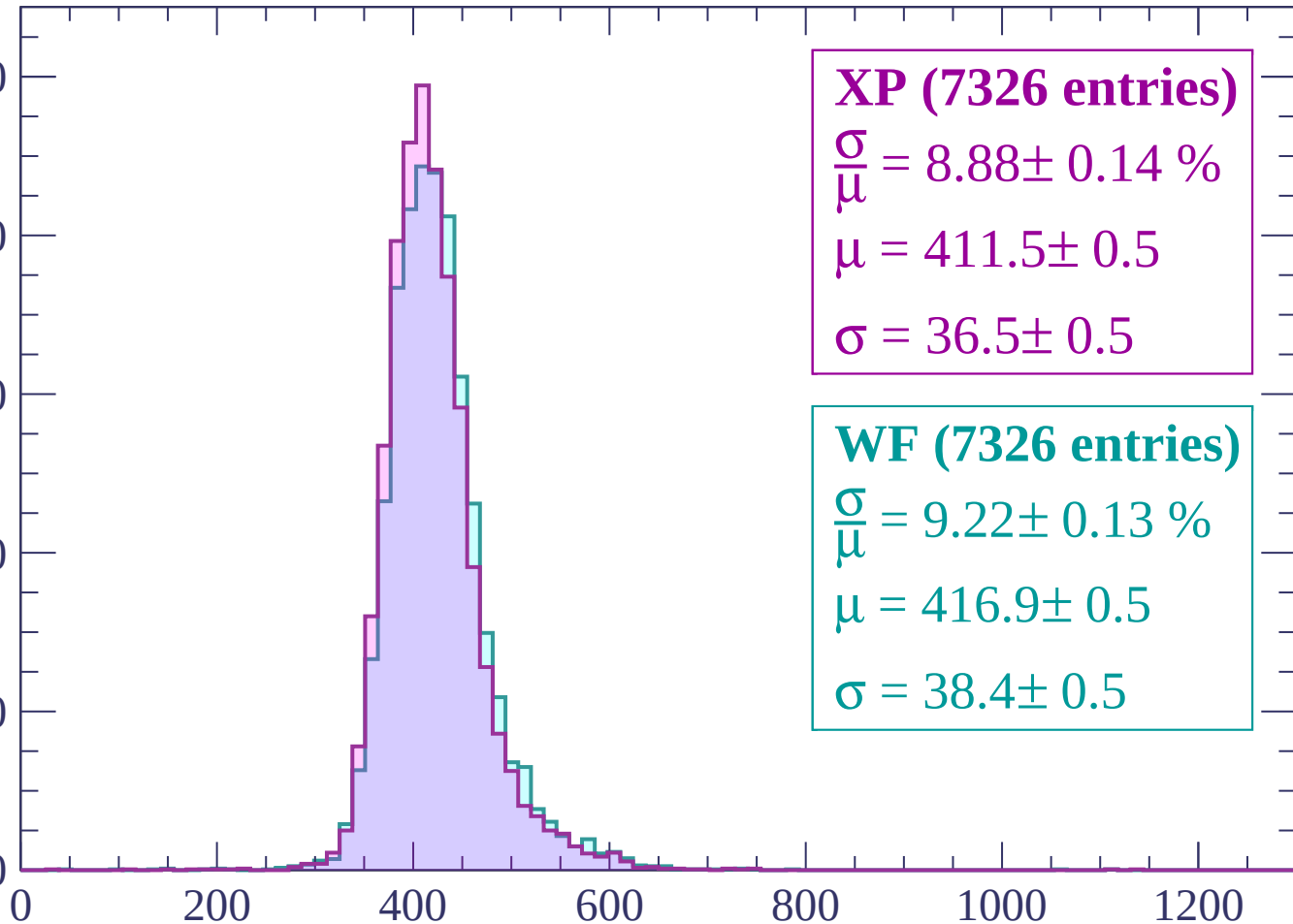
800

600

400

200

0



XP (7326 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.14 \%$

$\mu = 411.5 \pm 0.5$

$\sigma = 36.5 \pm 0.5$

WF (7326 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.13 \%$

$\mu = 416.9 \pm 0.5$

$\sigma = 38.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1000

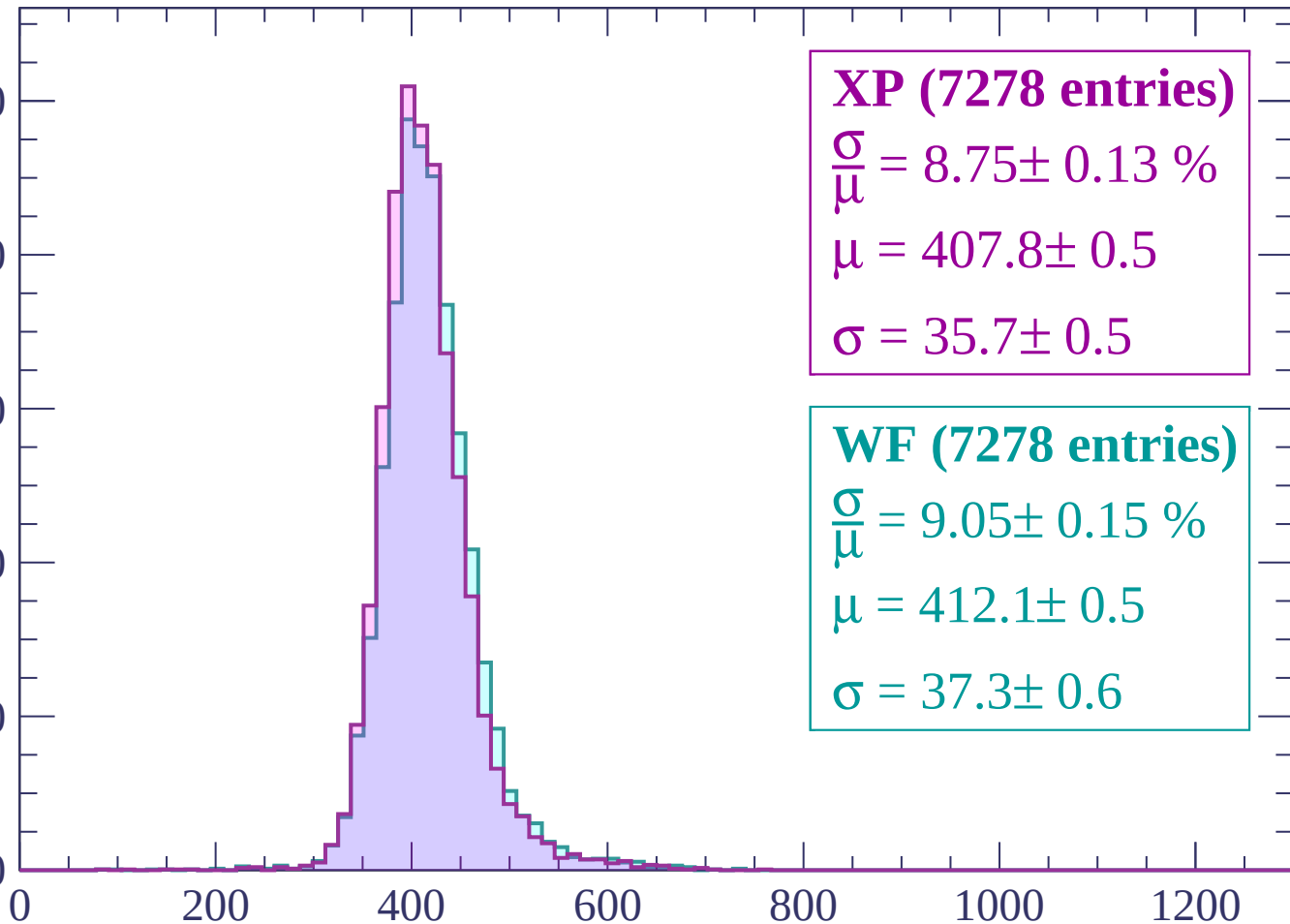
800

600

400

200

0



XP (7278 entries)

$\frac{\sigma}{\mu} = 8.75 \pm 0.13 \%$

$\mu = 407.8 \pm 0.5$

$\sigma = 35.7 \pm 0.5$

WF (7278 entries)

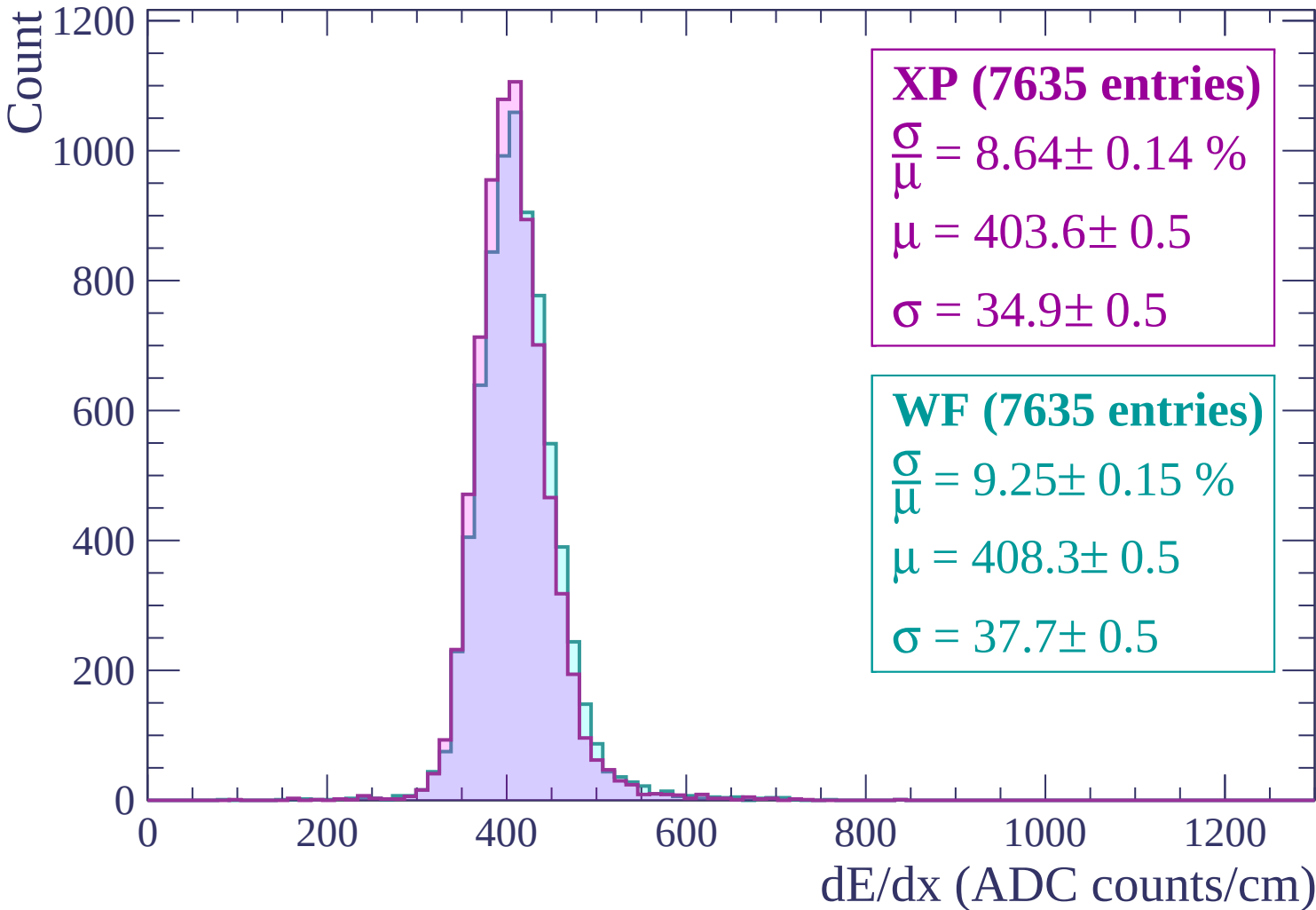
$\frac{\sigma}{\mu} = 9.05 \pm 0.15 \%$

$\mu = 412.1 \pm 0.5$

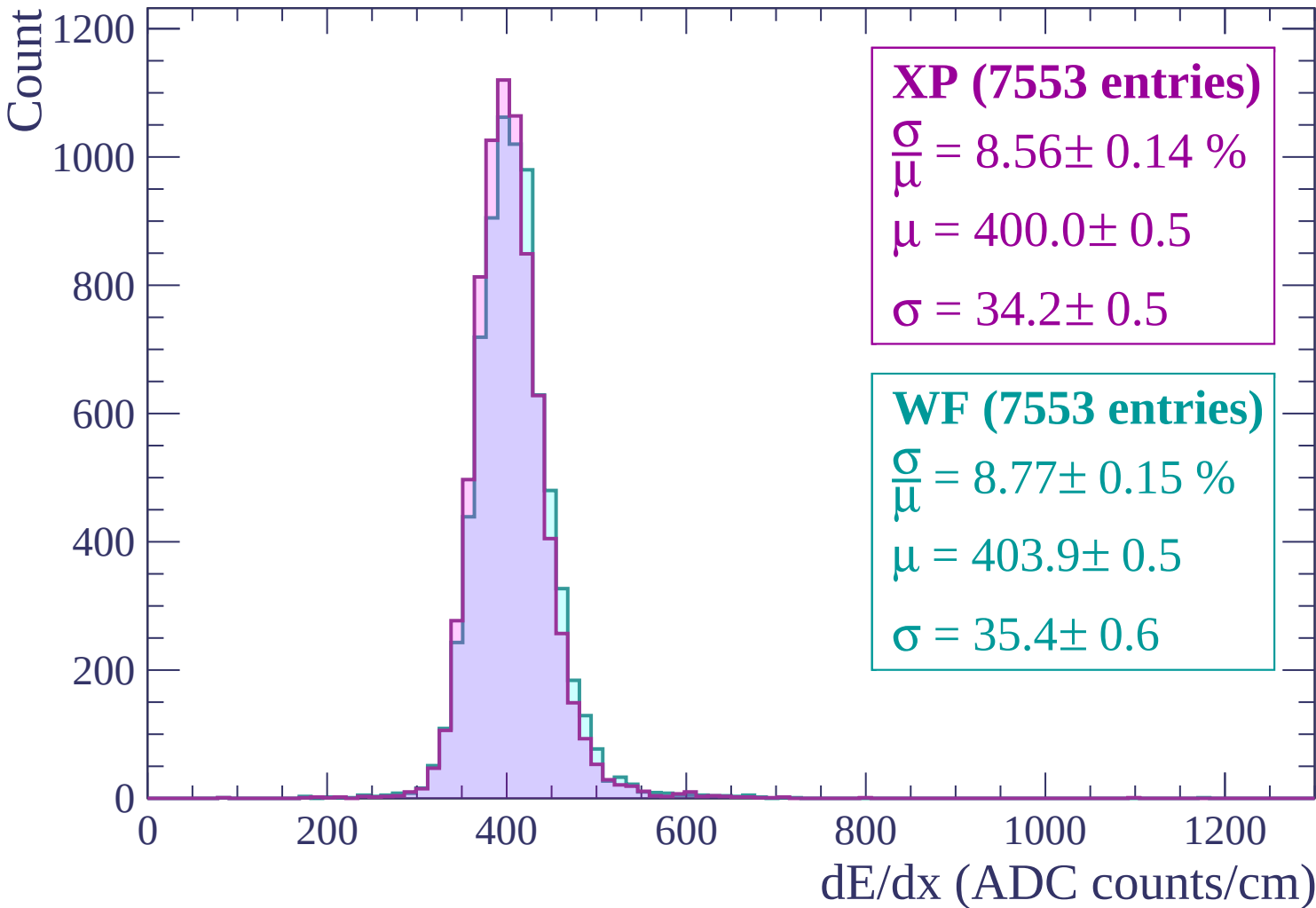
$\sigma = 37.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

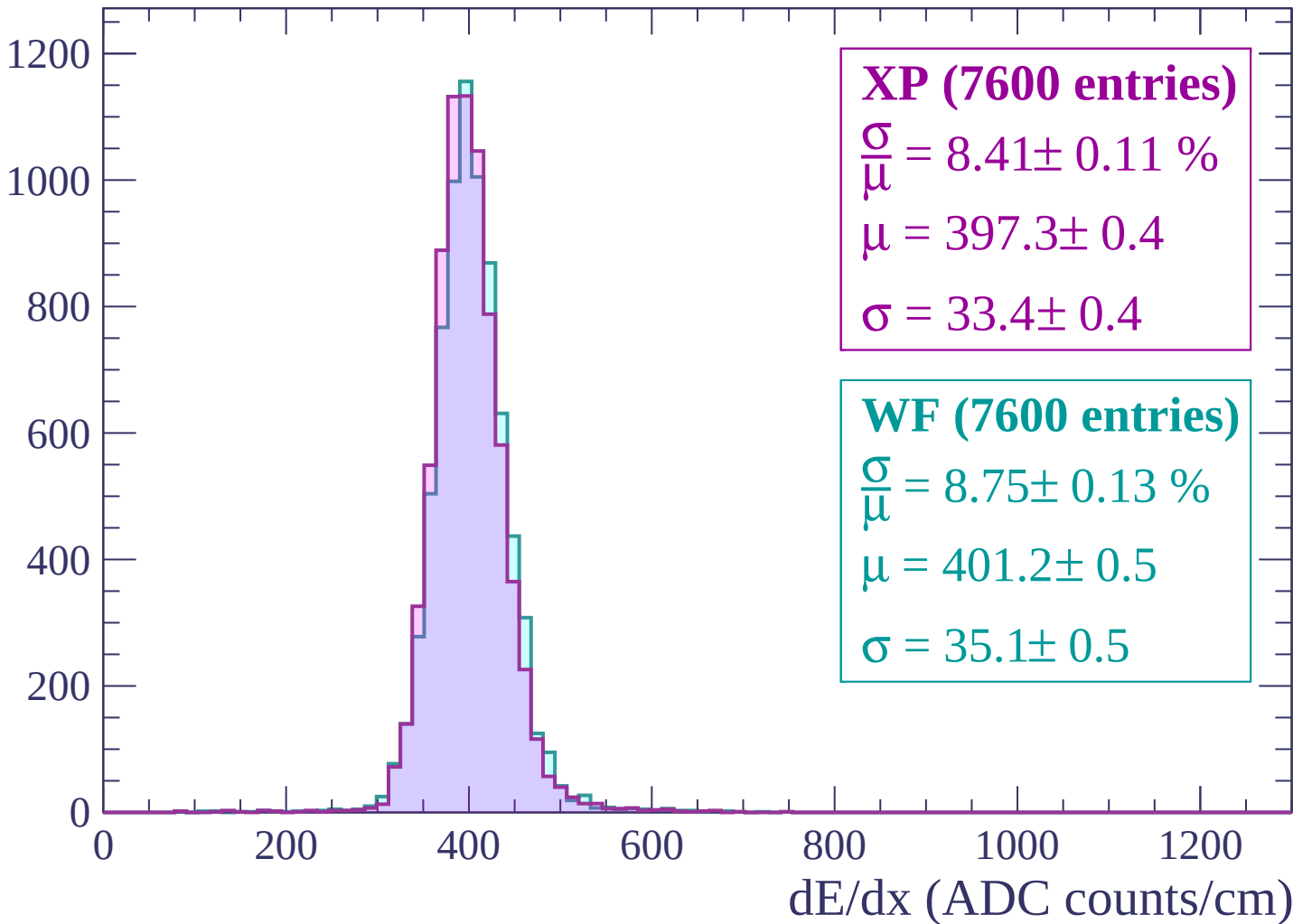


Energy loss | $-480 < p < -420$

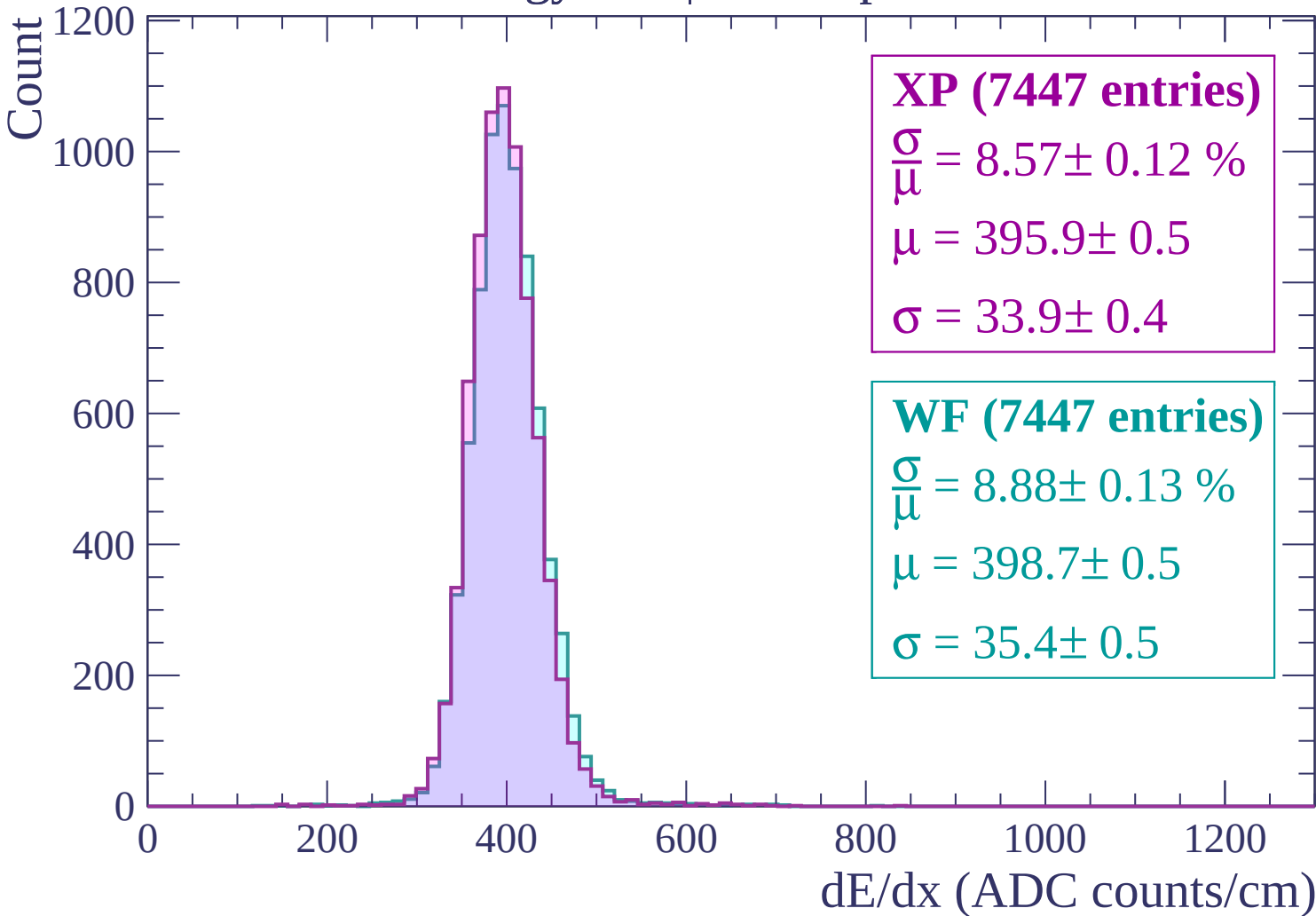


Energy loss | $-420 < p < -360$

Count



Energy loss | $-360 < p < -300$



Energy loss | $-300 < p < -240$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (6656 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.12 \%$$

$$\mu = 398.0 \pm 0.5$$

$$\sigma = 33.7 \pm 0.4$$

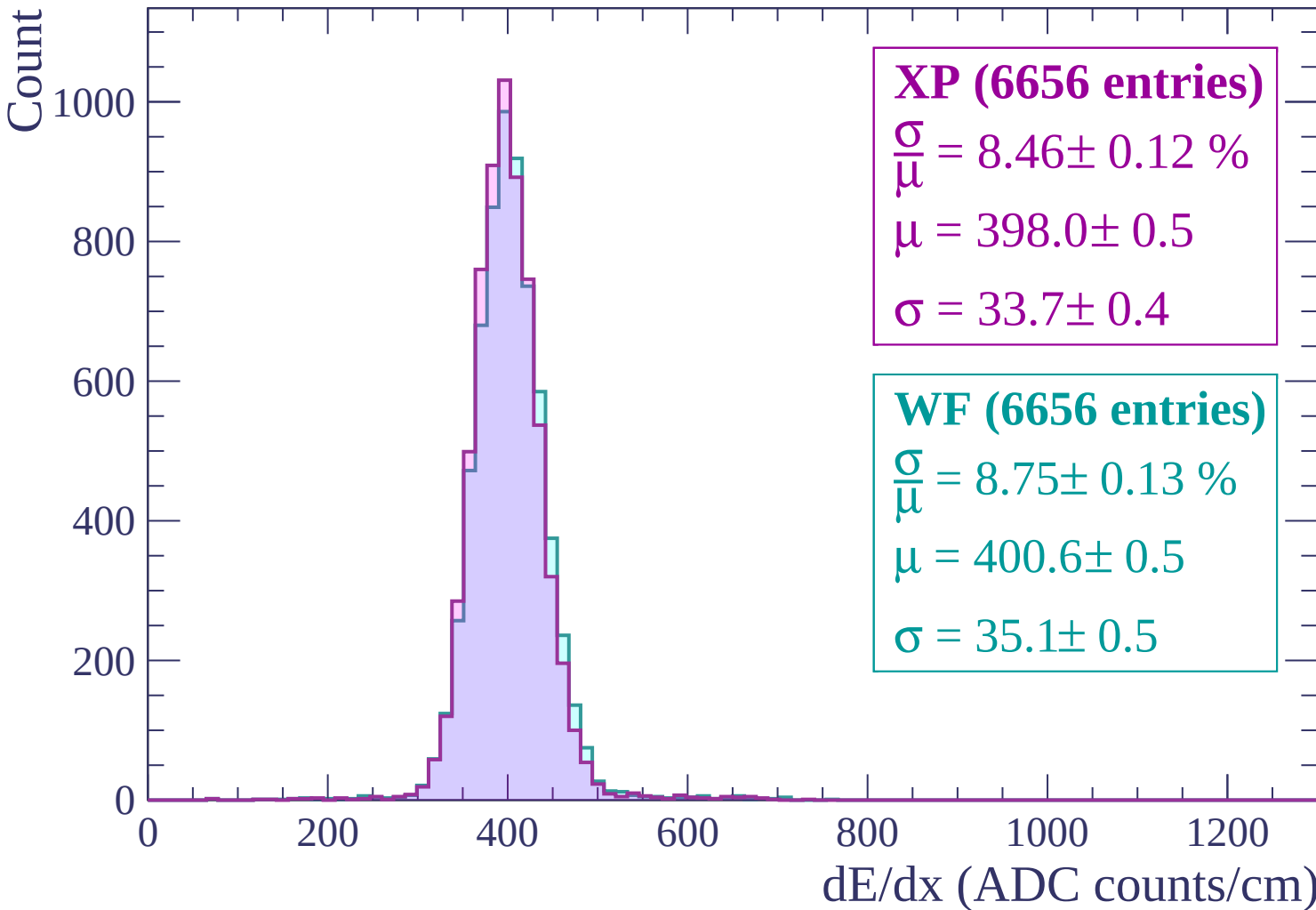
WF (6656 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.13 \%$$

$$\mu = 400.6 \pm 0.5$$

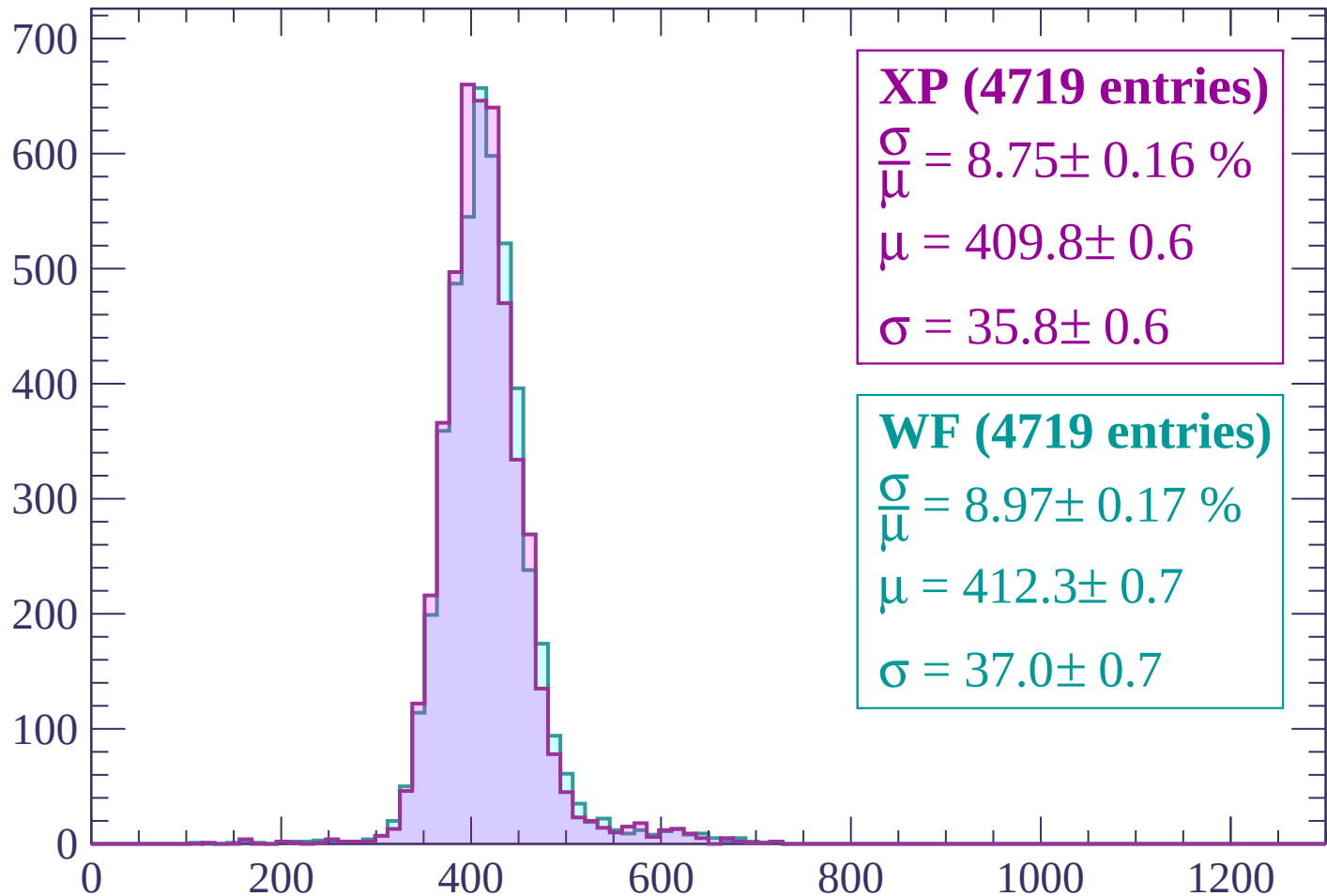
$$\sigma = 35.1 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $-240 < p < -180$

Count



XP (4719 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.16 \%$$

$$\mu = 409.8 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

WF (4719 entries)

$$\frac{\sigma}{\mu} = 8.97 \pm 0.17 \%$$

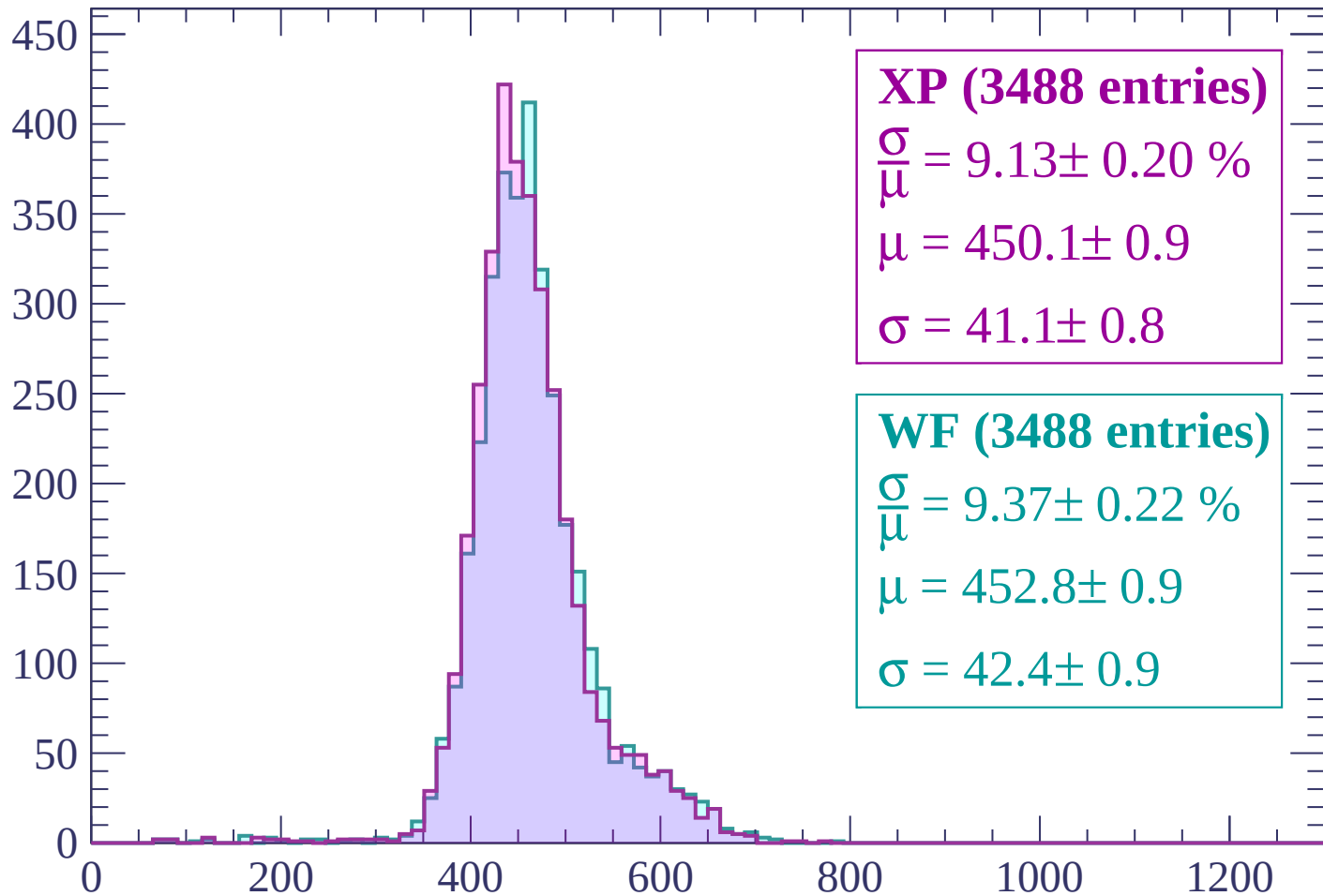
$$\mu = 412.3 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (3488 entries)

$$\frac{\sigma}{\mu} = 9.13 \pm 0.20 \%$$

$$\mu = 450.1 \pm 0.9$$

$$\sigma = 41.1 \pm 0.8$$

WF (3488 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.22 \%$$

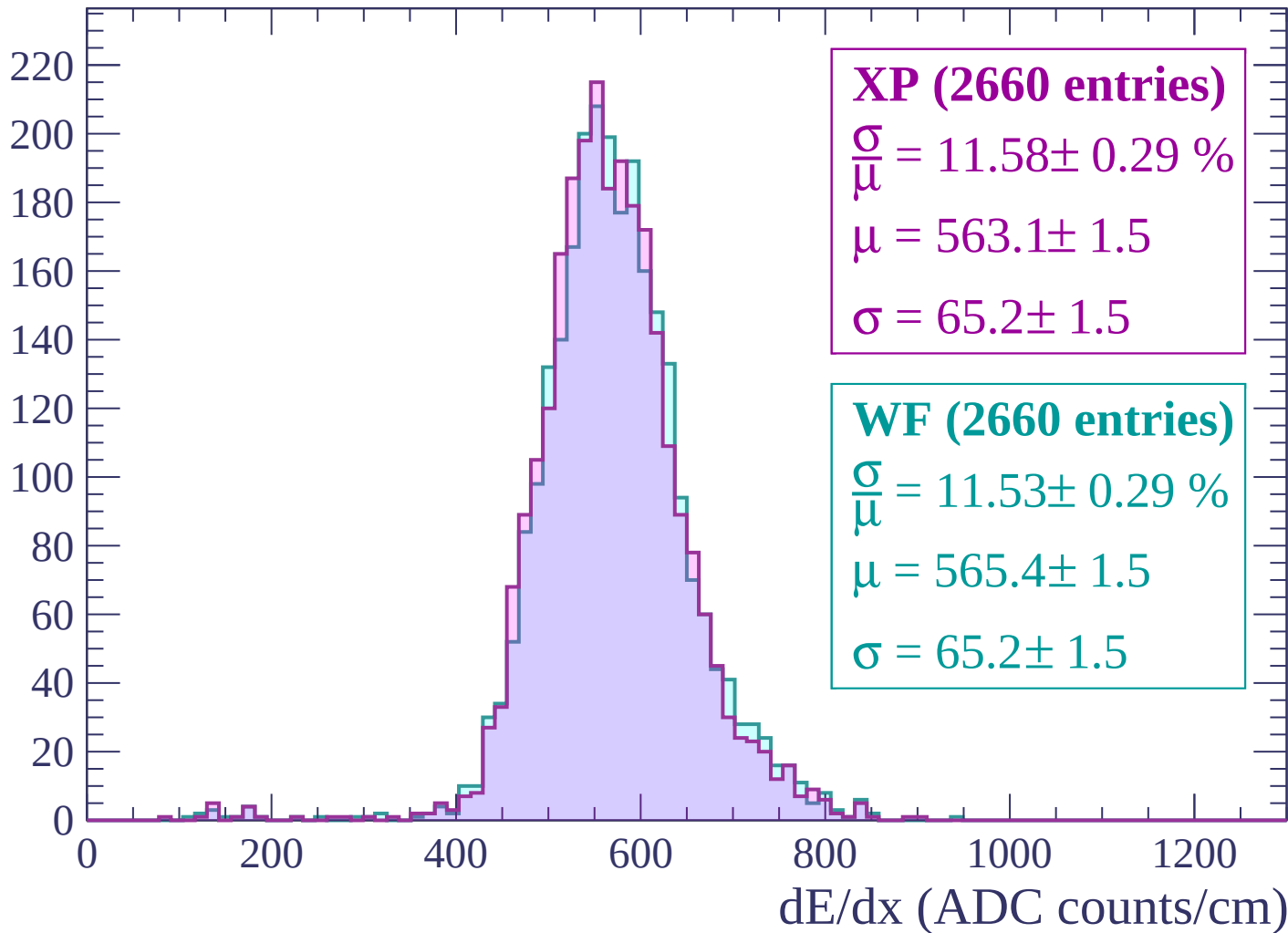
$$\mu = 452.8 \pm 0.9$$

$$\sigma = 42.4 \pm 0.9$$

dE/dx (ADC counts/cm)

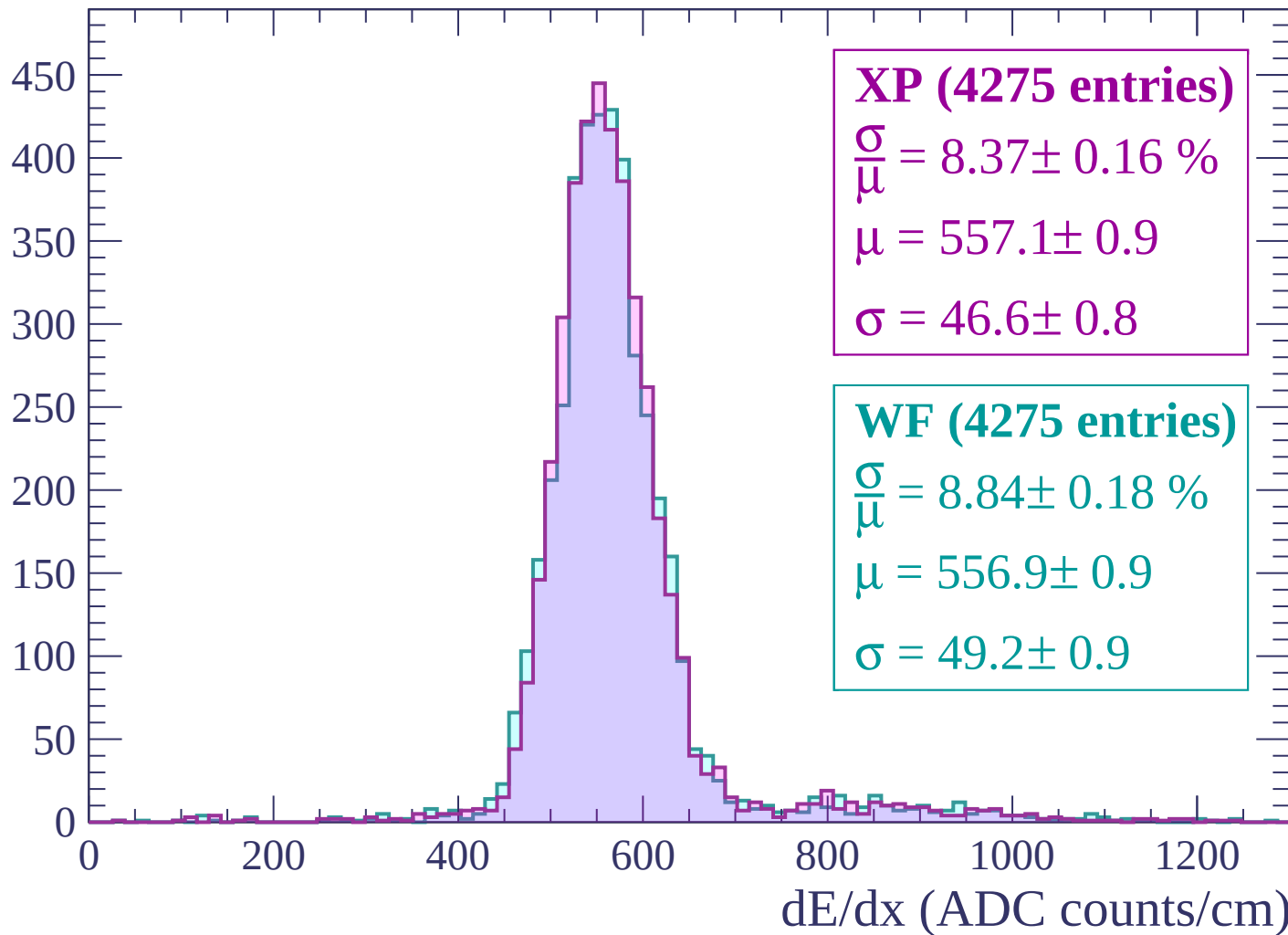
Energy loss | $-120 < p < -60$

Count



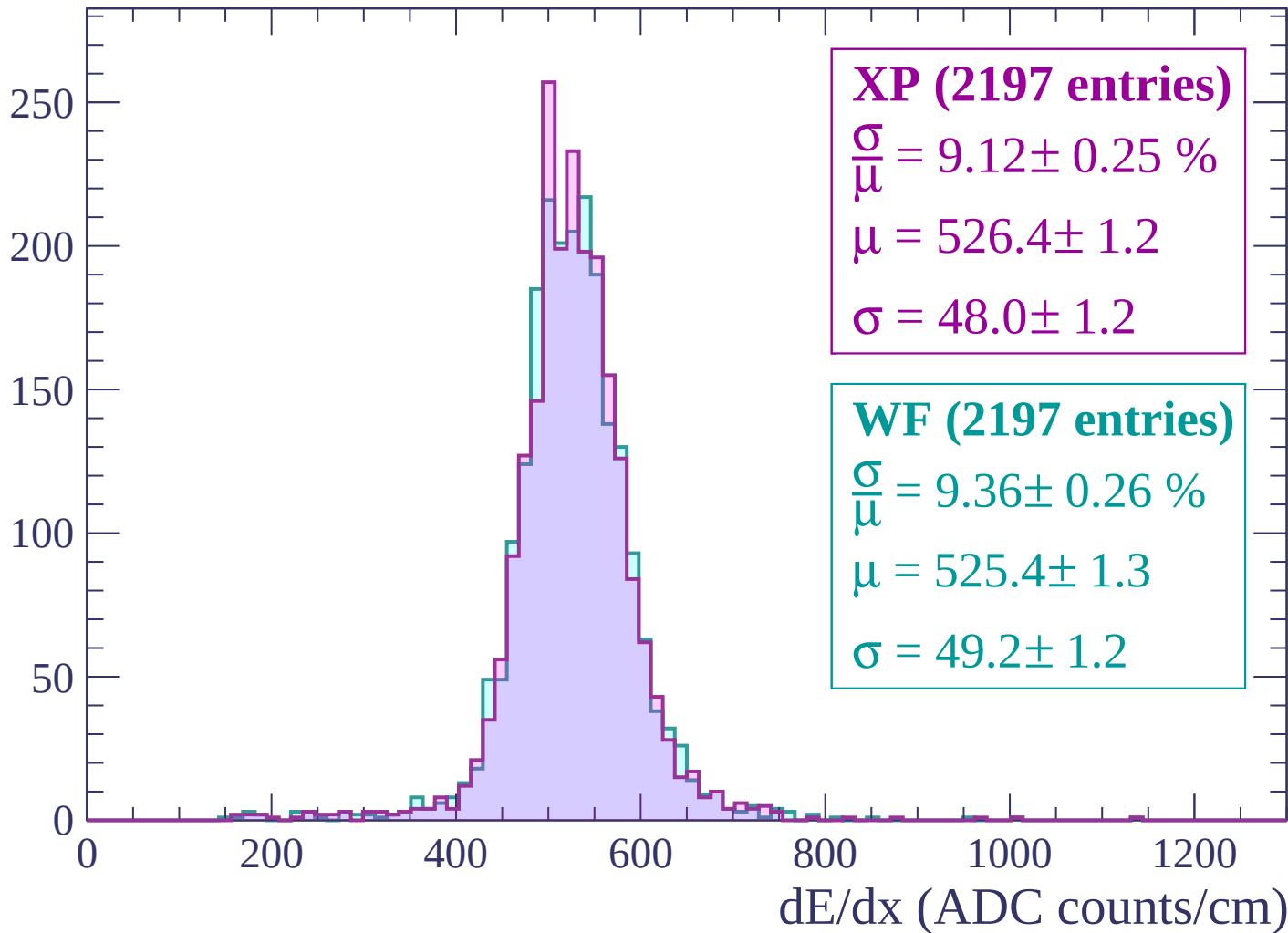
Energy loss | $-60 < p < 0$

Count



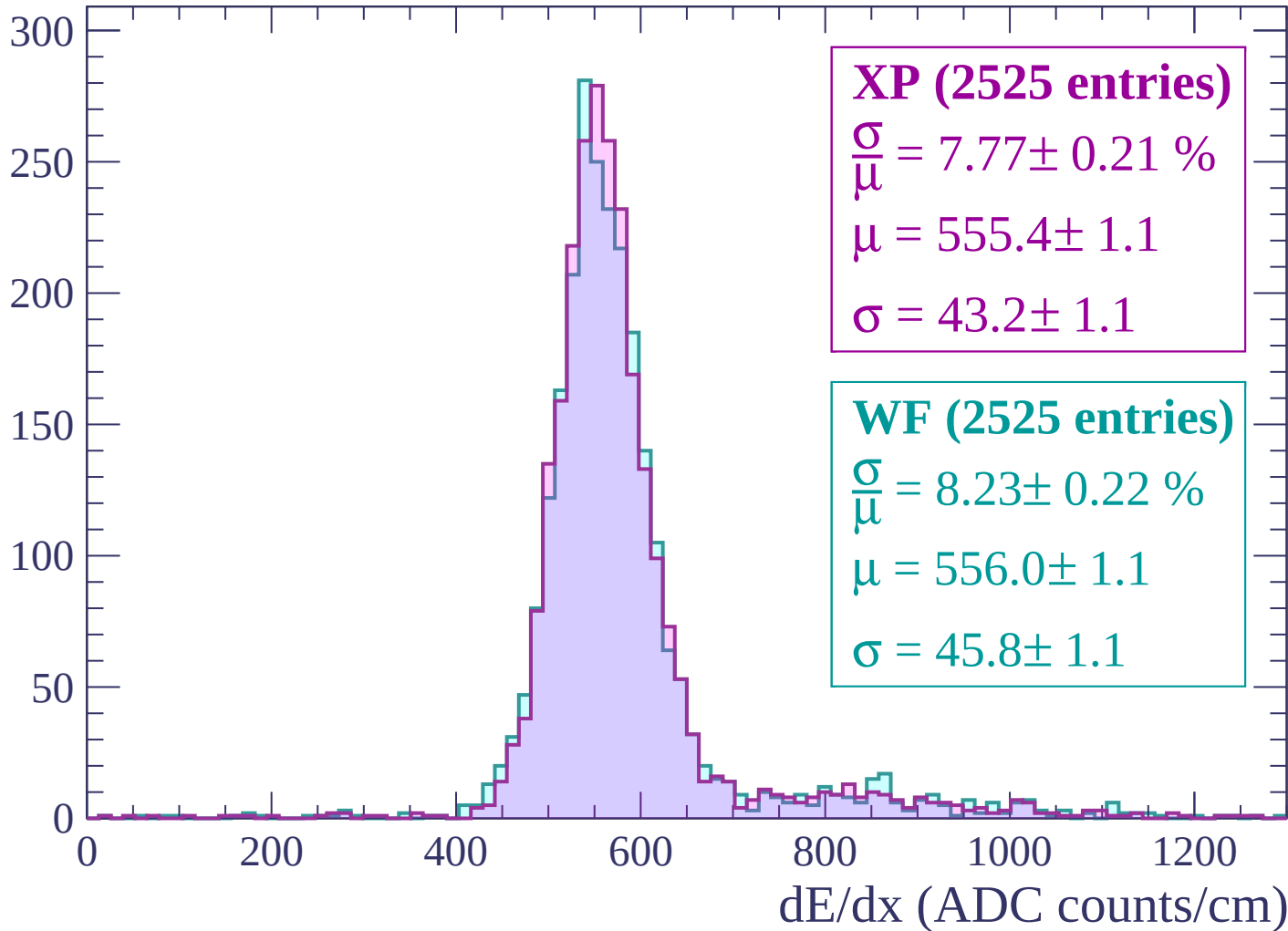
Energy loss | $0 < p < 60$

Count



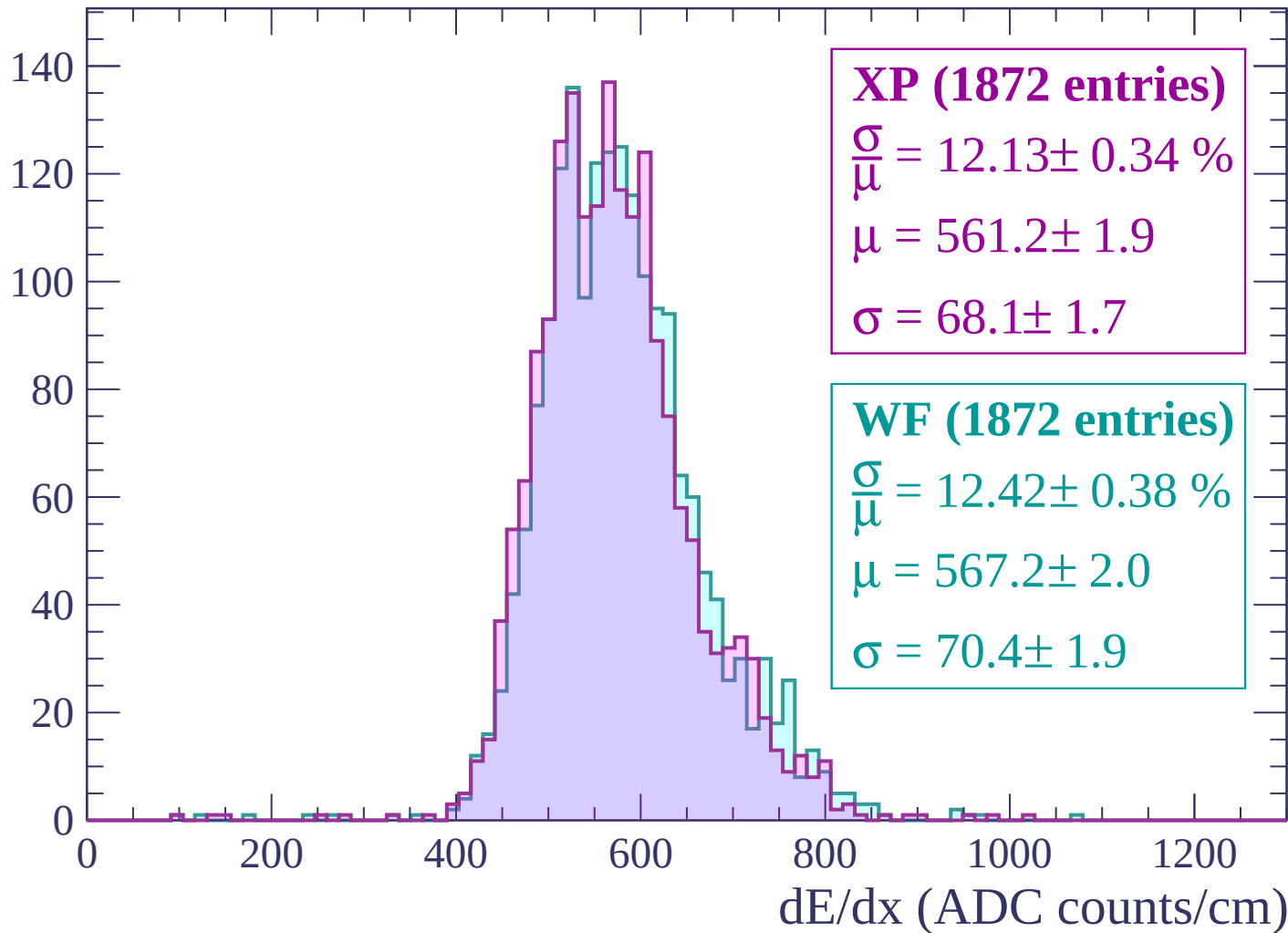
Energy loss | $60 < p < 120$

Count



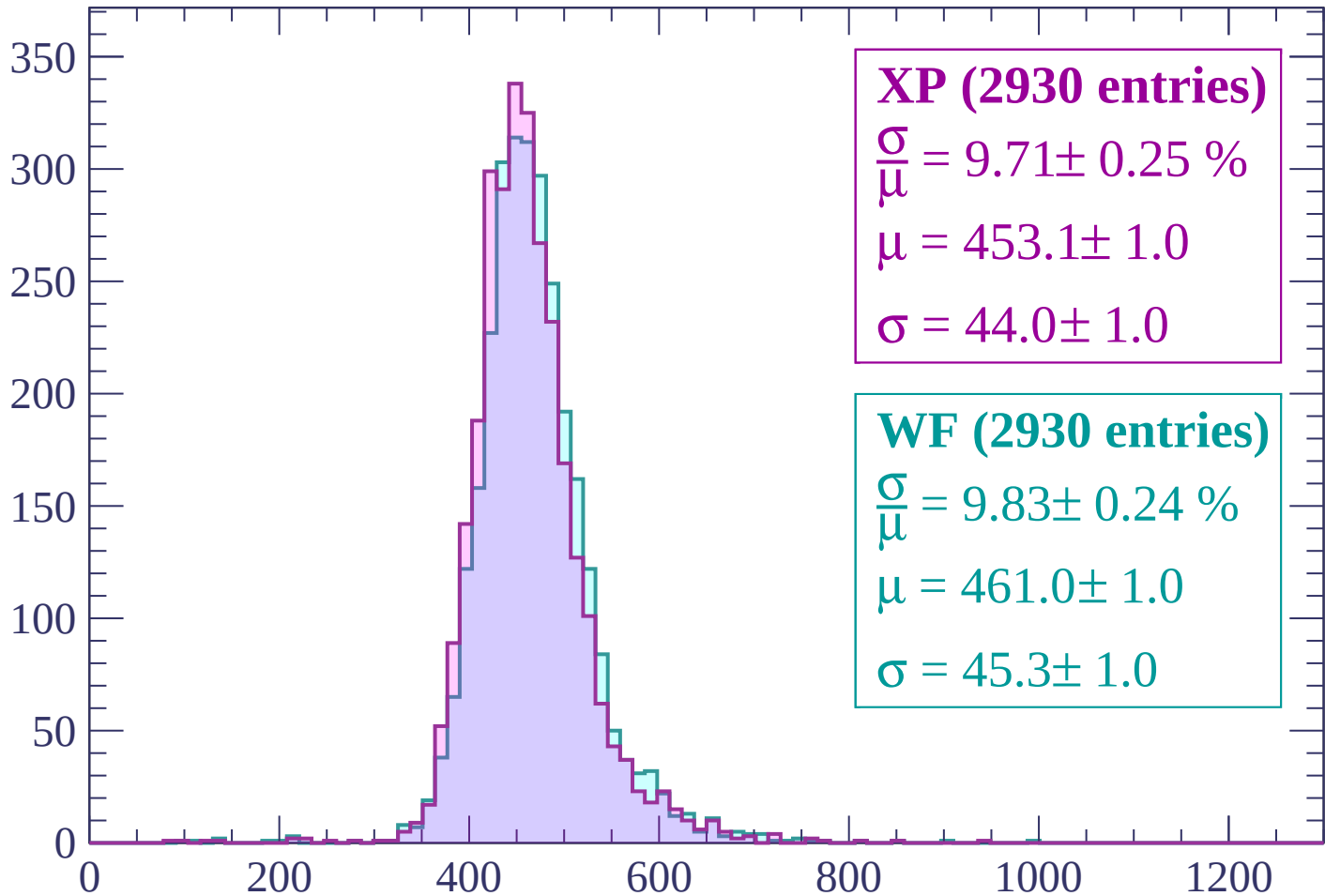
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (2930 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.25 \%$$

$$\mu = 453.1 \pm 1.0$$

$$\sigma = 44.0 \pm 1.0$$

WF (2930 entries)

$$\frac{\sigma}{\mu} = 9.83 \pm 0.24 \%$$

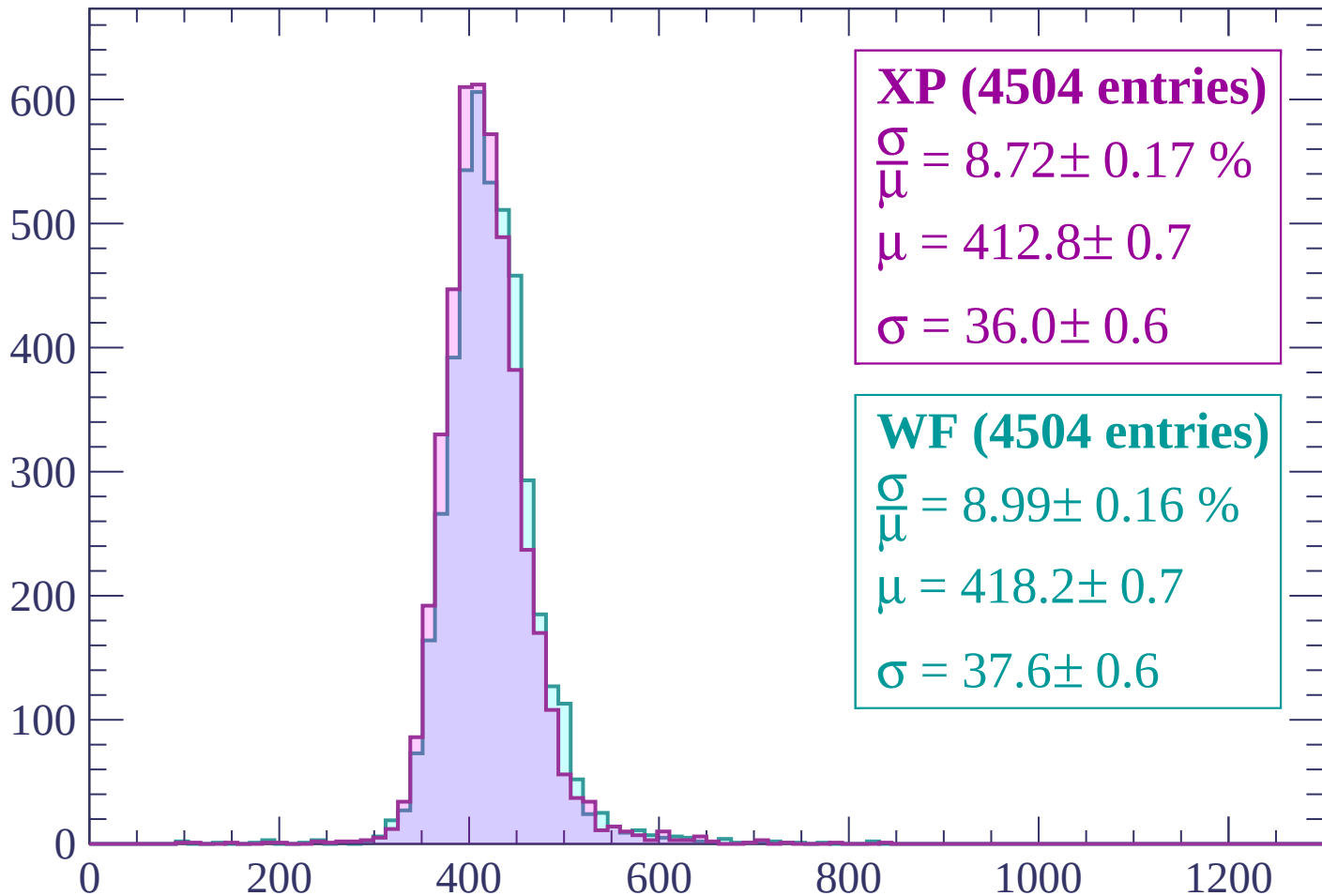
$$\mu = 461.0 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (4504 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.17 \%$$

$$\mu = 412.8 \pm 0.7$$

$$\sigma = 36.0 \pm 0.6$$

WF (4504 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

$$\mu = 418.2 \pm 0.7$$

$$\sigma = 37.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count

1000

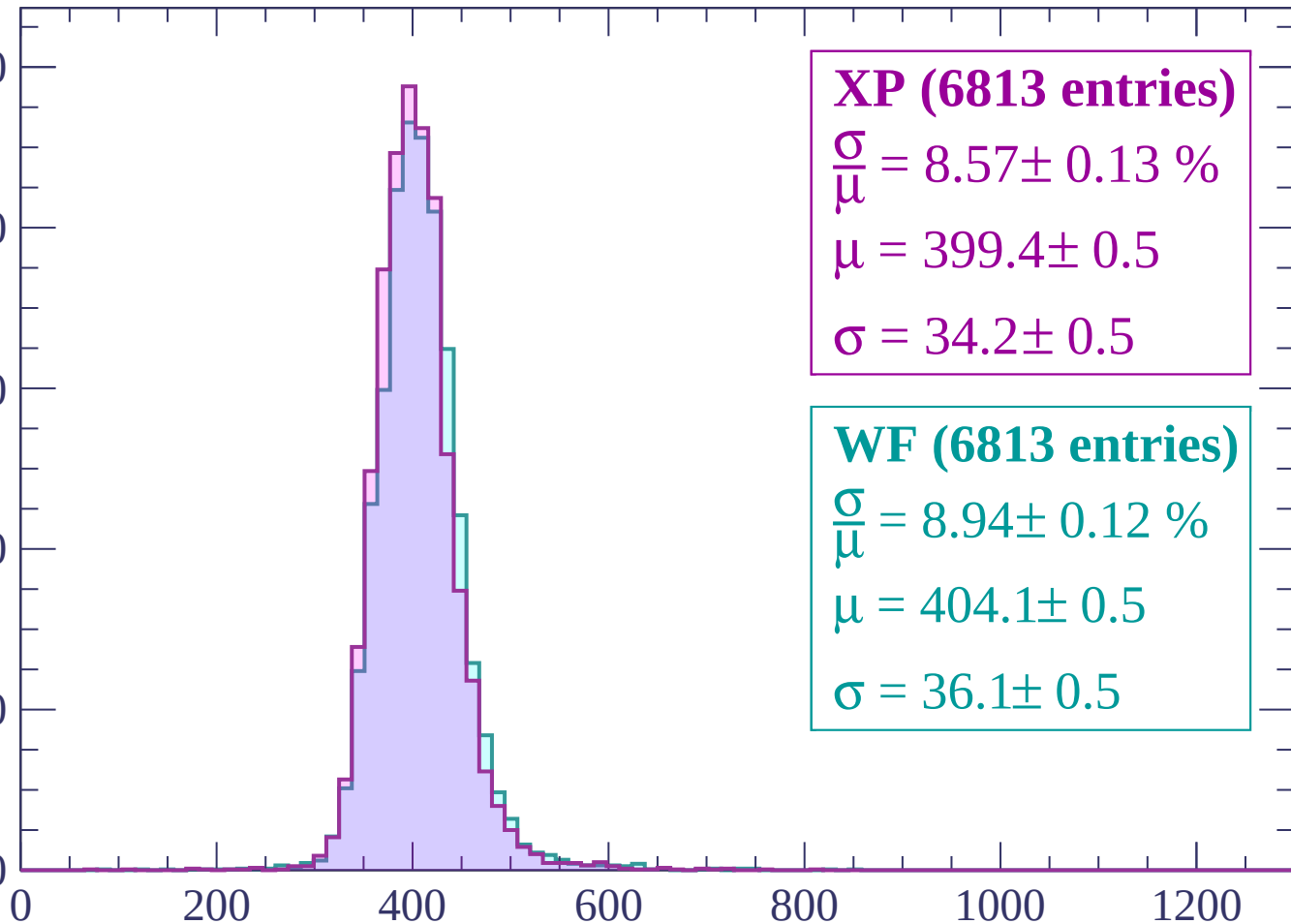
800

600

400

200

0



XP (6813 entries)

$\frac{\sigma}{\mu} = 8.57 \pm 0.13 \%$

$\mu = 399.4 \pm 0.5$

$\sigma = 34.2 \pm 0.5$

WF (6813 entries)

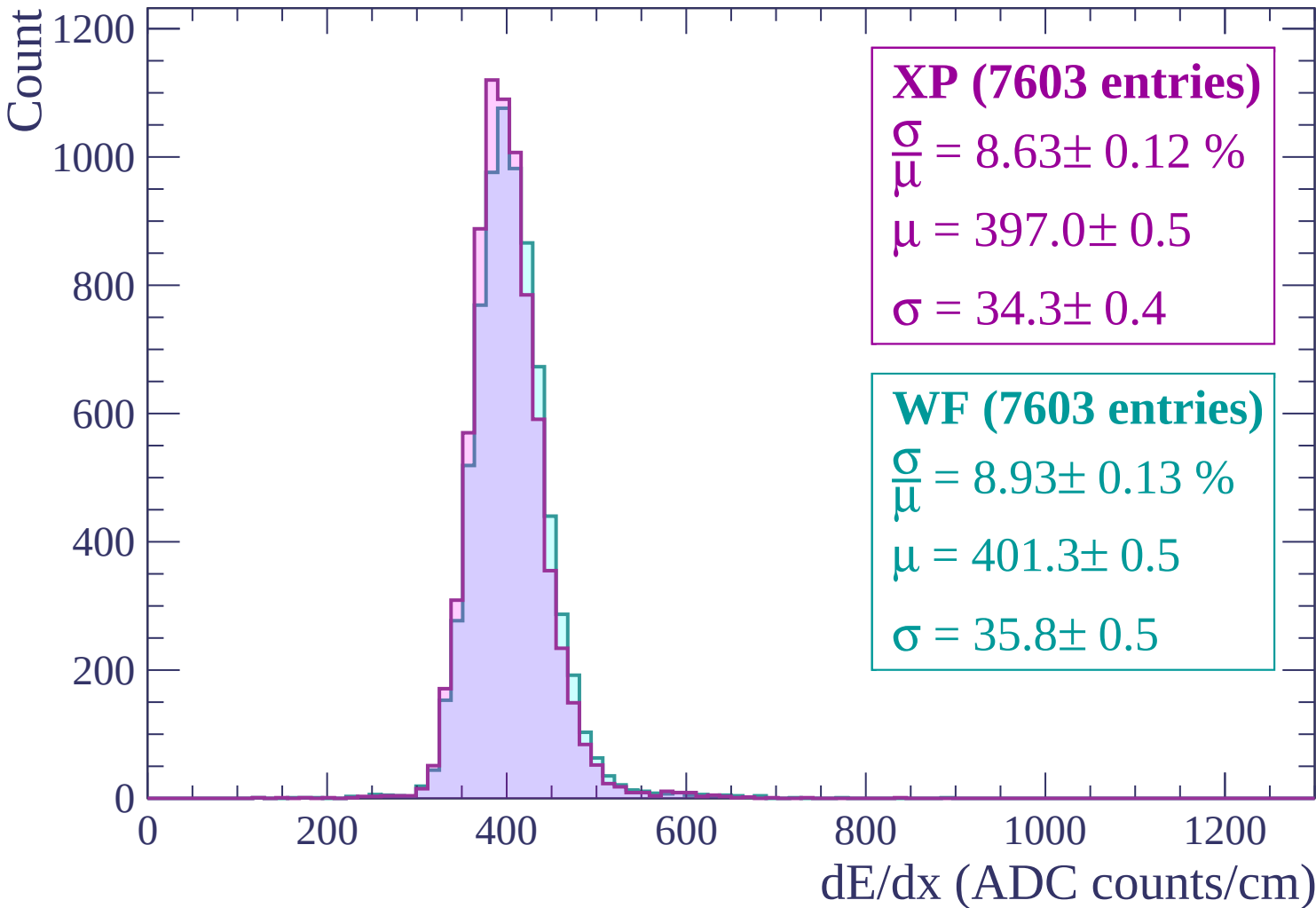
$\frac{\sigma}{\mu} = 8.94 \pm 0.12 \%$

$\mu = 404.1 \pm 0.5$

$\sigma = 36.1 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$



Energy loss | $420 < p < 480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8208 entries)

$\frac{\sigma}{\mu} = 8.78 \pm 0.12 \%$

$\mu = 398.5 \pm 0.5$

$\sigma = 35.0 \pm 0.4$

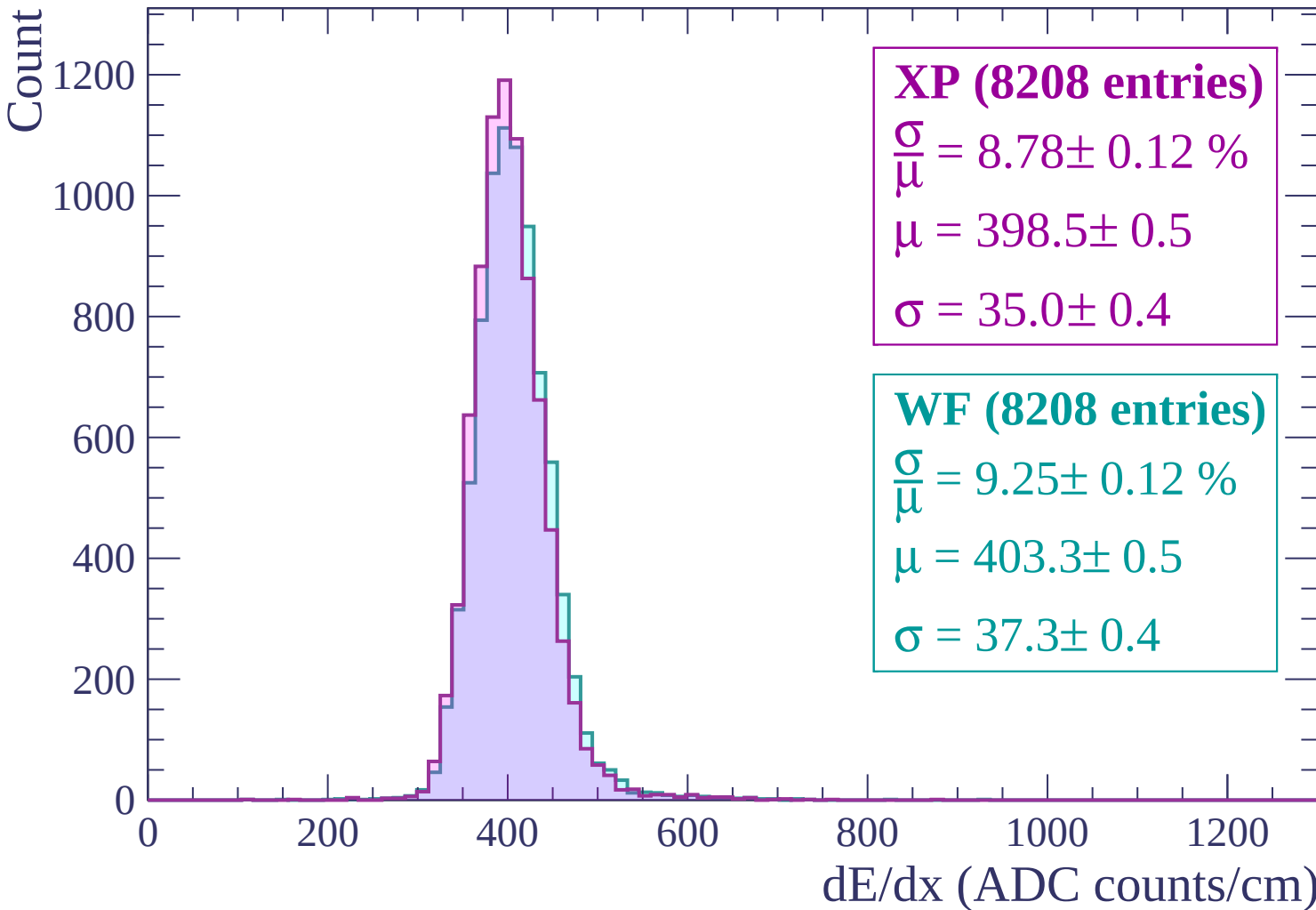
WF (8208 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.12 \%$

$\mu = 403.3 \pm 0.5$

$\sigma = 37.3 \pm 0.4$

dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$

Count

1200

1000

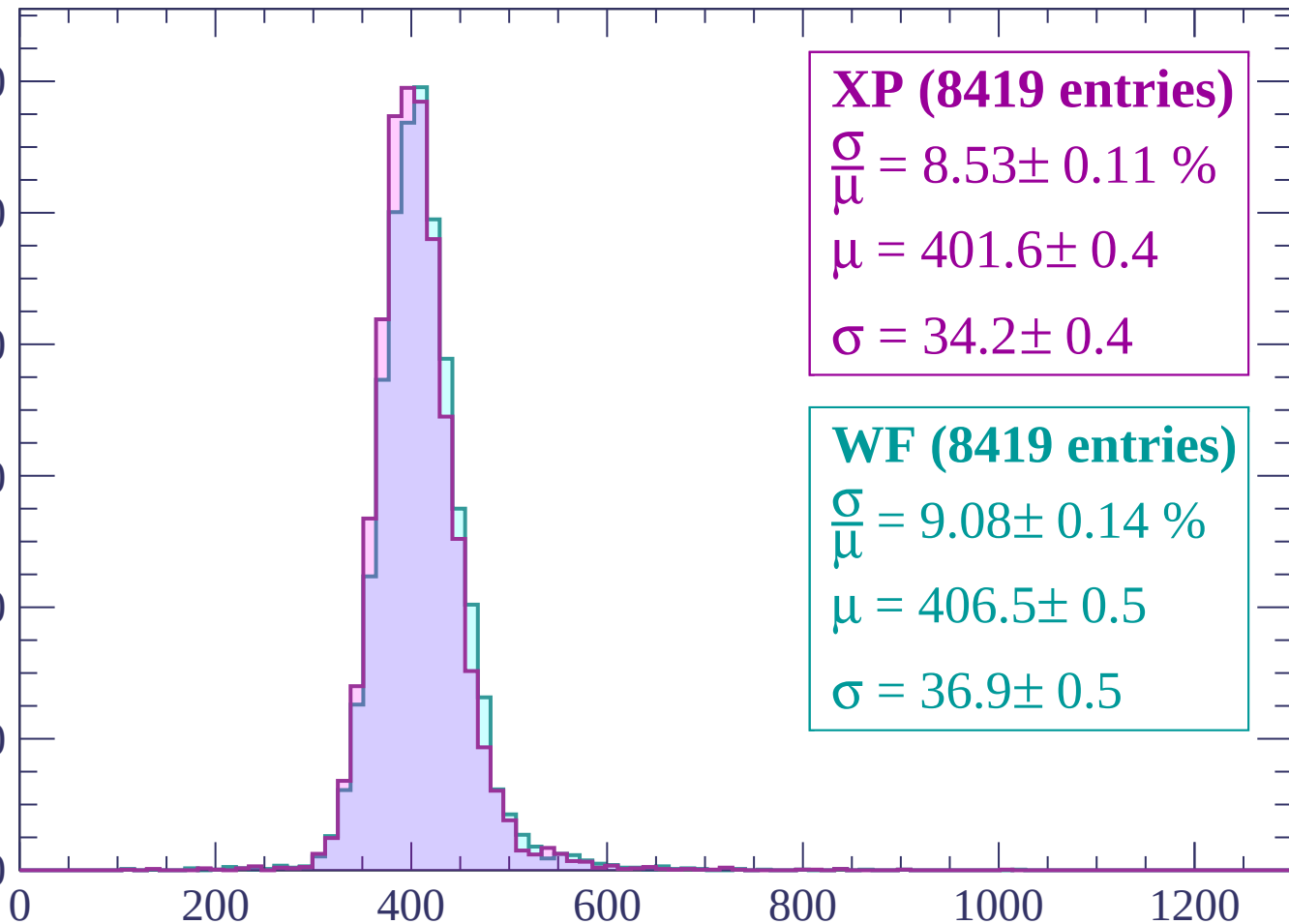
800

600

400

200

0



XP (8419 entries)

$\frac{\sigma}{\mu} = 8.53 \pm 0.11 \%$

$\mu = 401.6 \pm 0.4$

$\sigma = 34.2 \pm 0.4$

WF (8419 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.14 \%$

$\mu = 406.5 \pm 0.5$

$\sigma = 36.9 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count

1200
1000
800
600
400
200
0

XP (8512 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.13 \%$$

$$\mu = 404.3 \pm 0.5$$

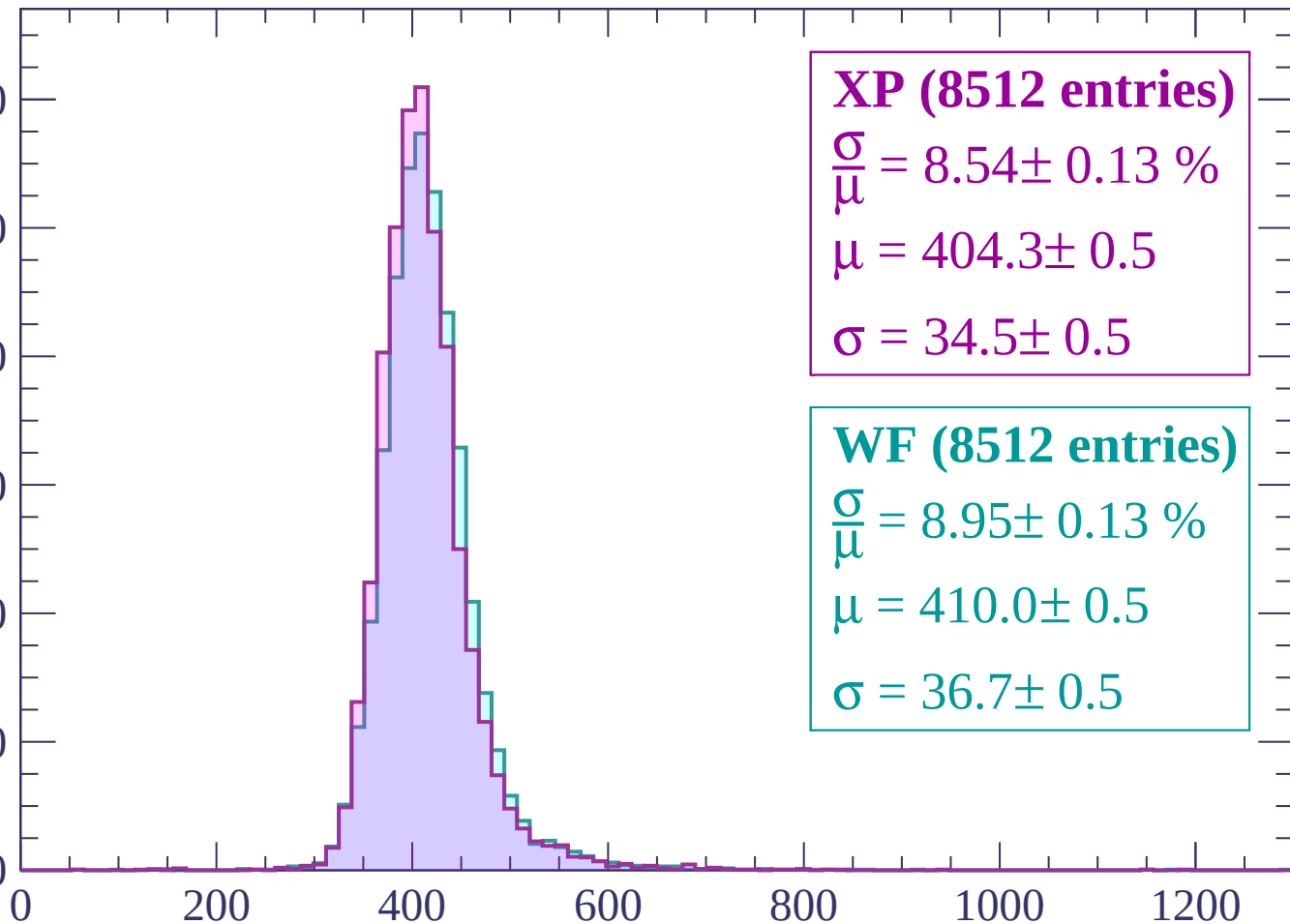
$$\sigma = 34.5 \pm 0.5$$

WF (8512 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$$

$$\mu = 410.0 \pm 0.5$$

$$\sigma = 36.7 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (8615 entries)

$\frac{\sigma}{\mu} = 8.70 \pm 0.12 \%$

$\mu = 409.6 \pm 0.5$

$\sigma = 35.7 \pm 0.5$

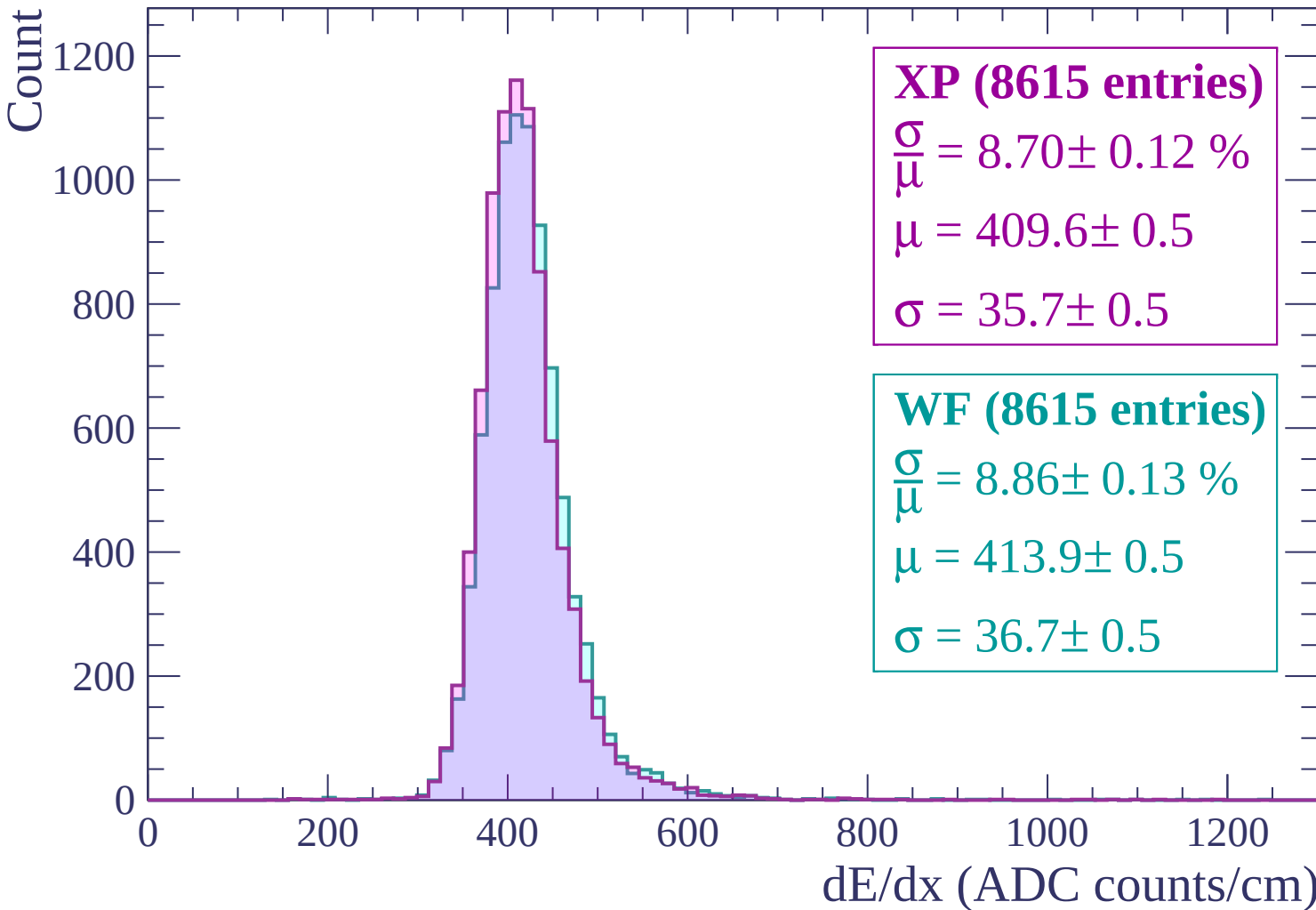
WF (8615 entries)

$\frac{\sigma}{\mu} = 8.86 \pm 0.13 \%$

$\mu = 413.9 \pm 0.5$

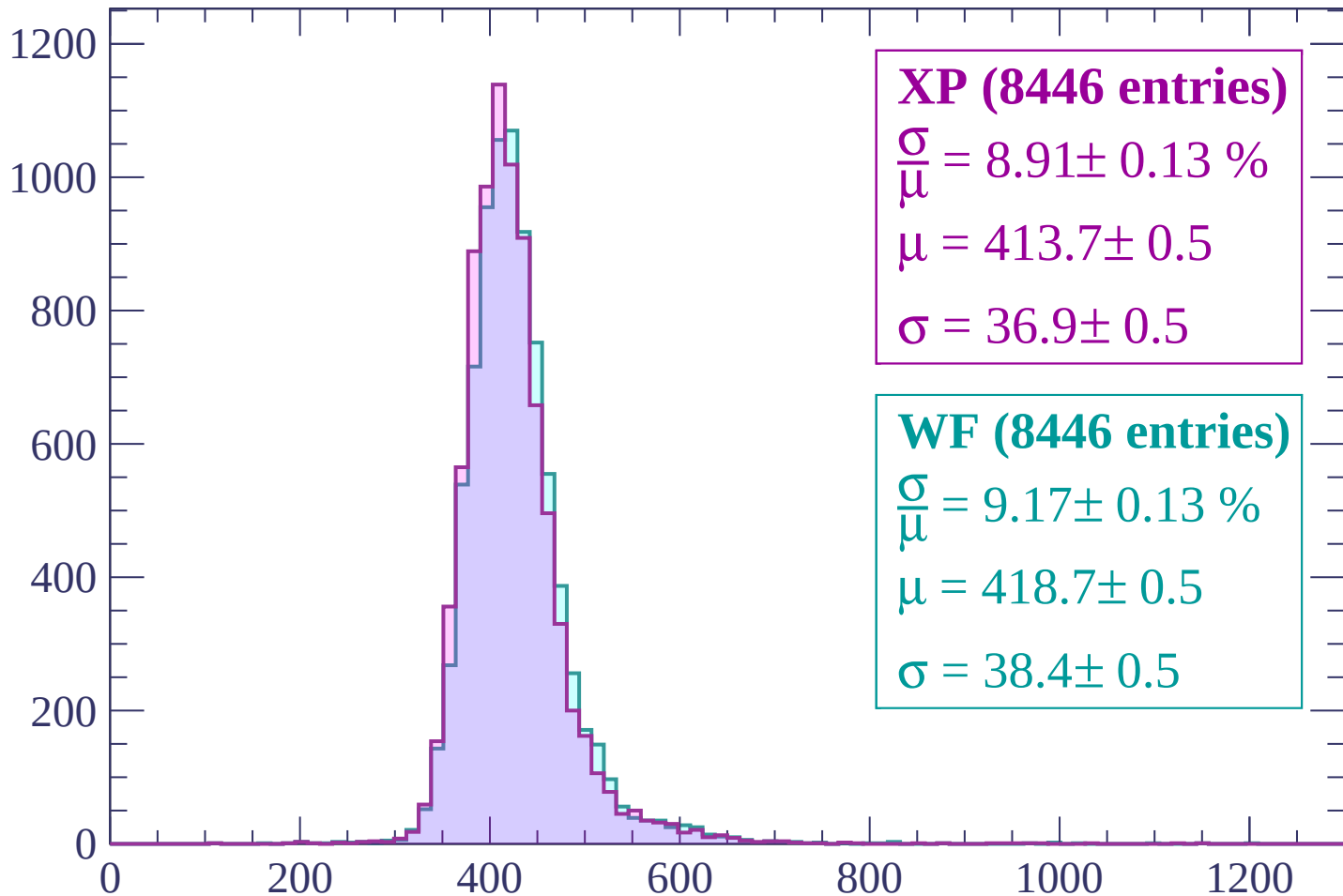
$\sigma = 36.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $660 < p < 720$

Count



XP (8446 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$

$\mu = 413.7 \pm 0.5$

$\sigma = 36.9 \pm 0.5$

WF (8446 entries)

$\frac{\sigma}{\mu} = 9.17 \pm 0.13 \%$

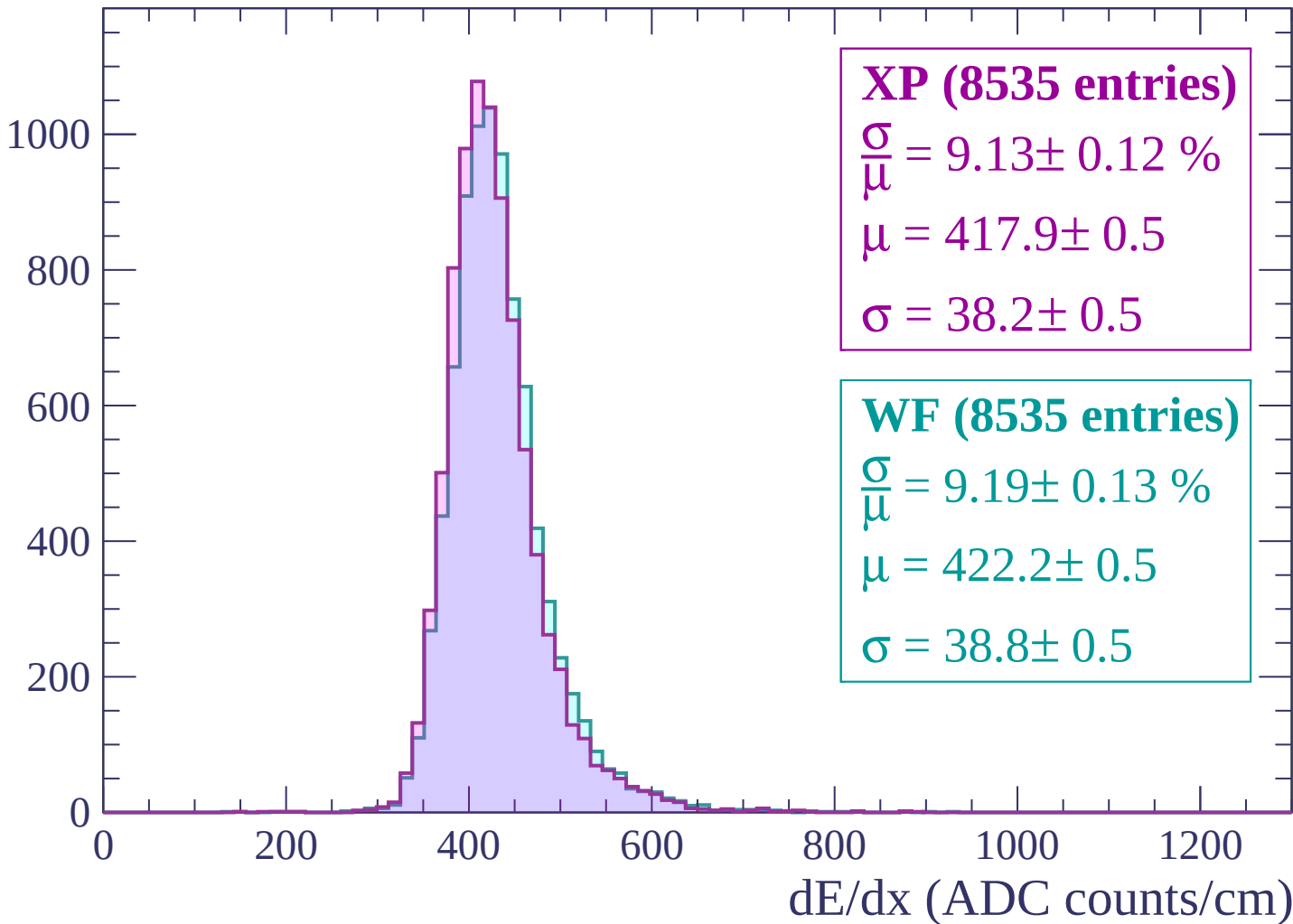
$\mu = 418.7 \pm 0.5$

$\sigma = 38.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



Energy loss | $780 < p < 840$

Count

1000

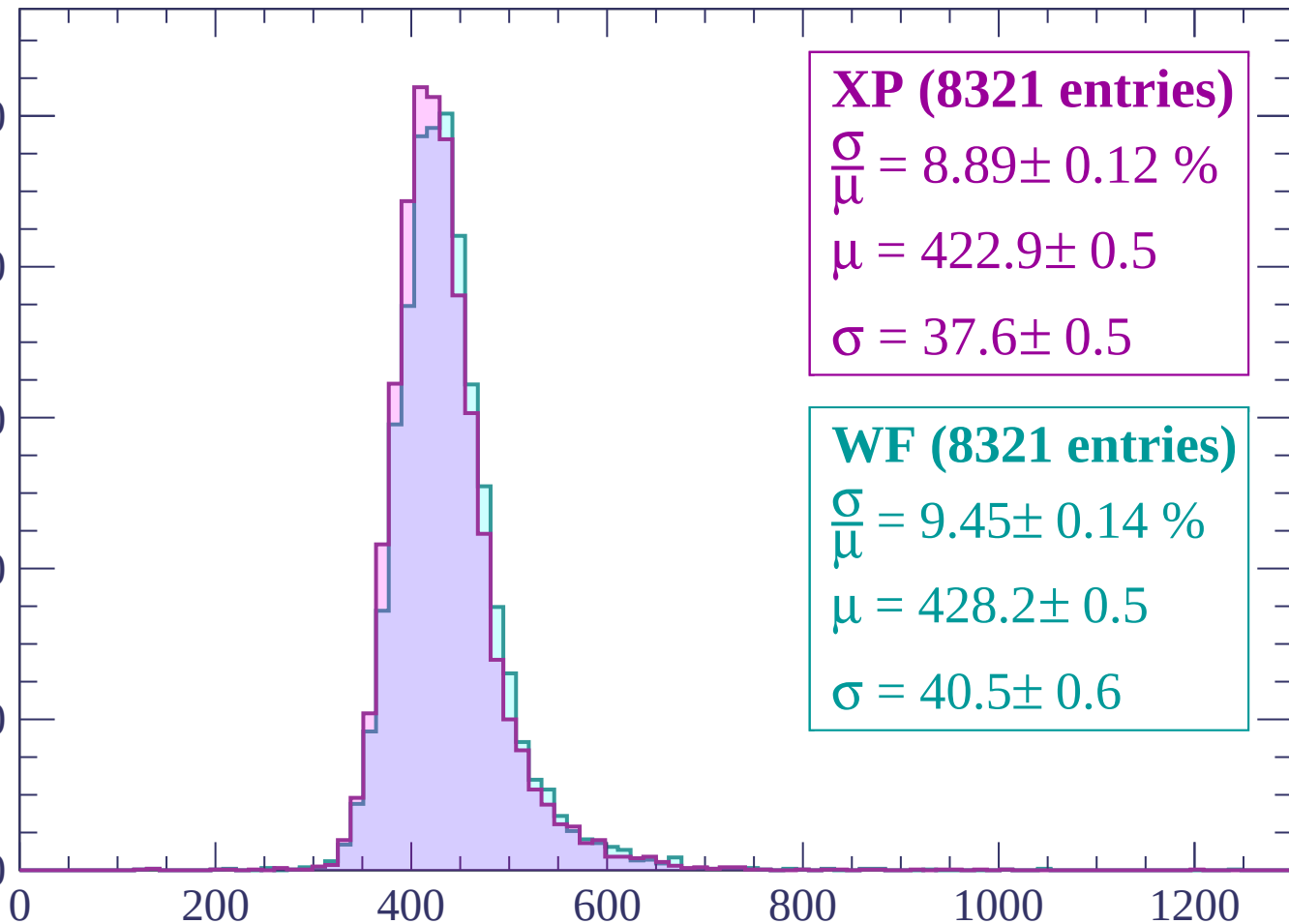
800

600

400

200

0



XP (8321 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.12 \%$

$\mu = 422.9 \pm 0.5$

$\sigma = 37.6 \pm 0.5$

WF (8321 entries)

$\frac{\sigma}{\mu} = 9.45 \pm 0.14 \%$

$\mu = 428.2 \pm 0.5$

$\sigma = 40.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count

1000

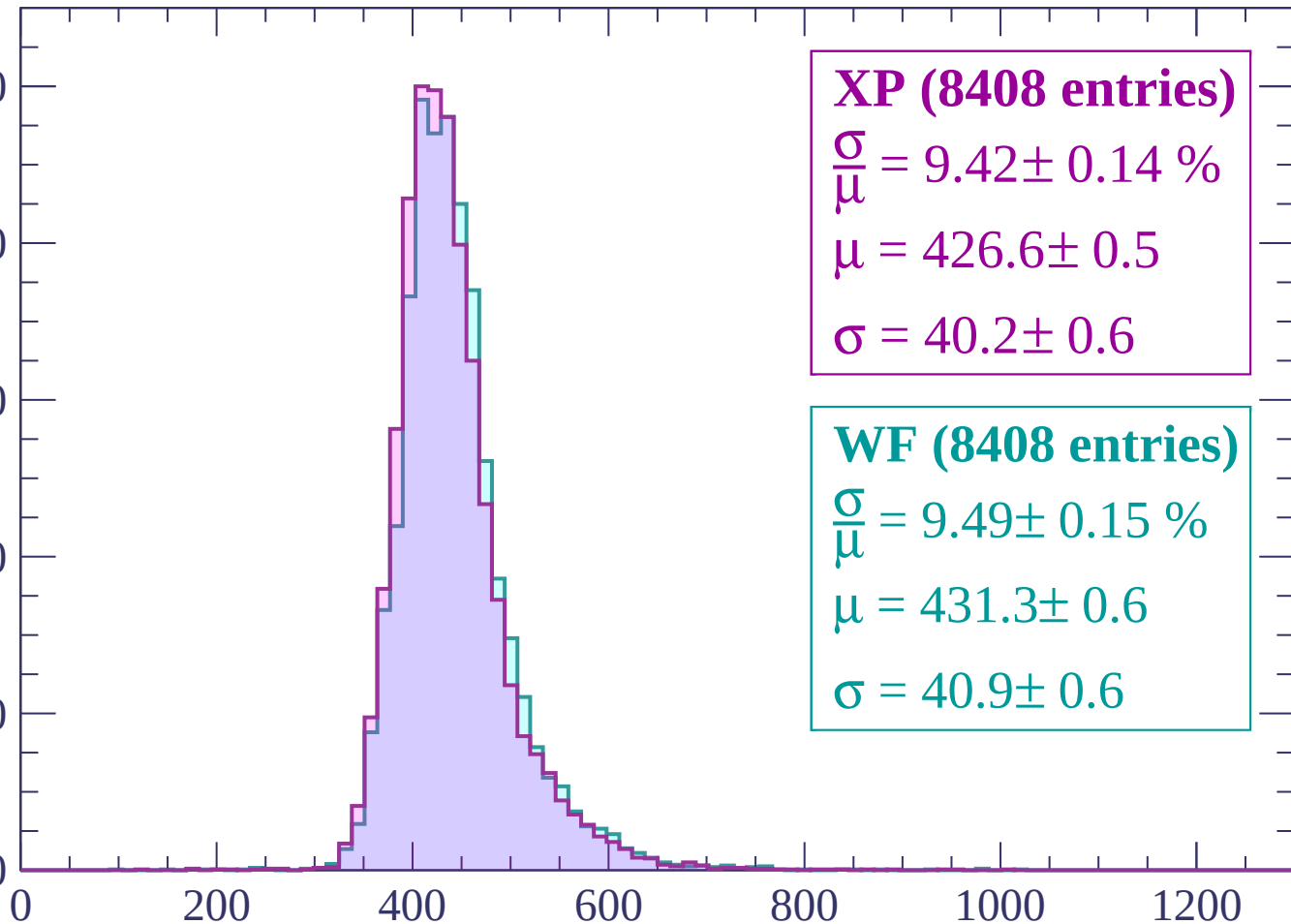
800

600

400

200

0



XP (8408 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.14 \%$

$\mu = 426.6 \pm 0.5$

$\sigma = 40.2 \pm 0.6$

WF (8408 entries)

$\frac{\sigma}{\mu} = 9.49 \pm 0.15 \%$

$\mu = 431.3 \pm 0.6$

$\sigma = 40.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count

1000

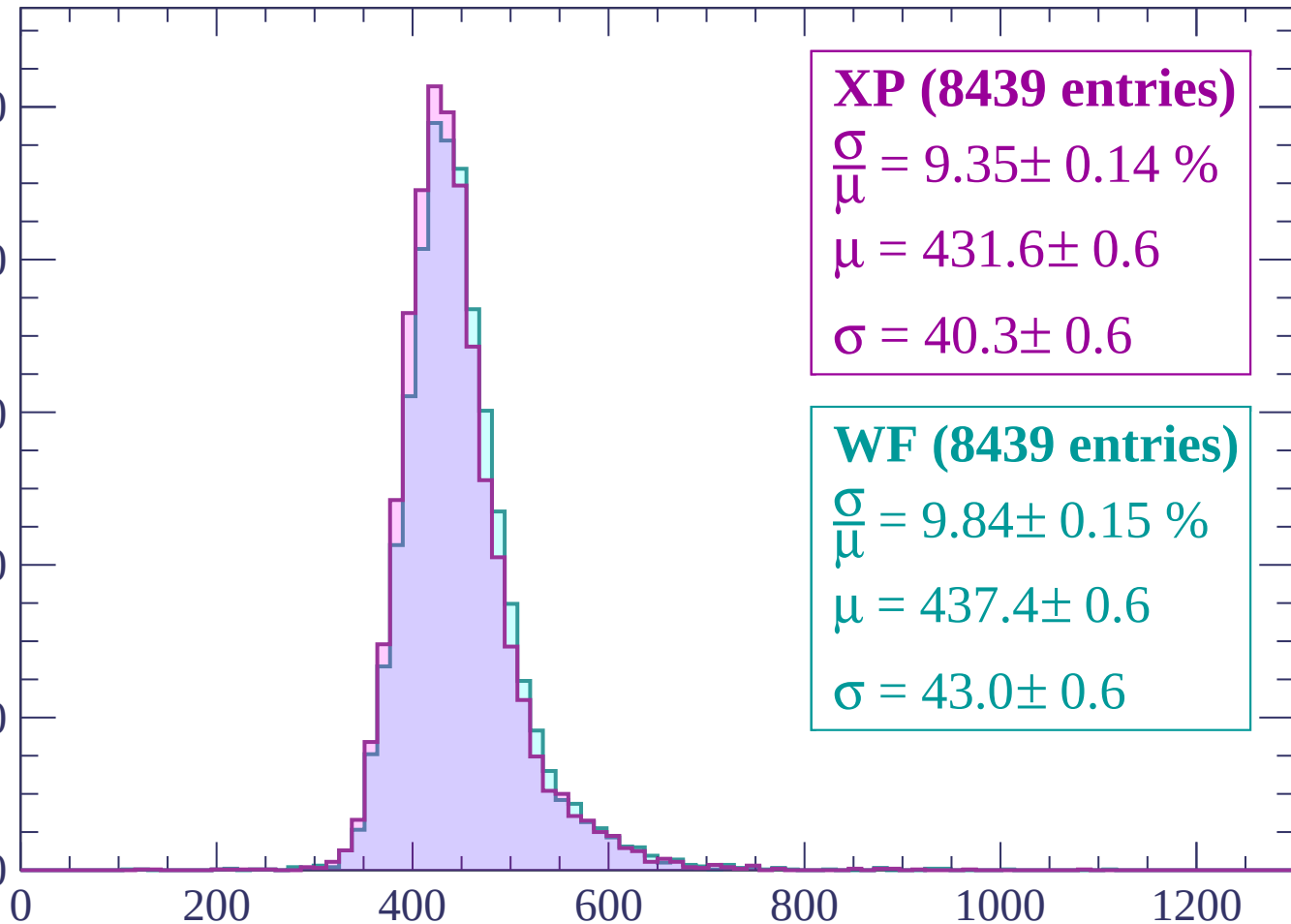
800

600

400

200

0



XP (8439 entries)

$\frac{\sigma}{\mu} = 9.35 \pm 0.14 \%$

$\mu = 431.6 \pm 0.6$

$\sigma = 40.3 \pm 0.6$

WF (8439 entries)

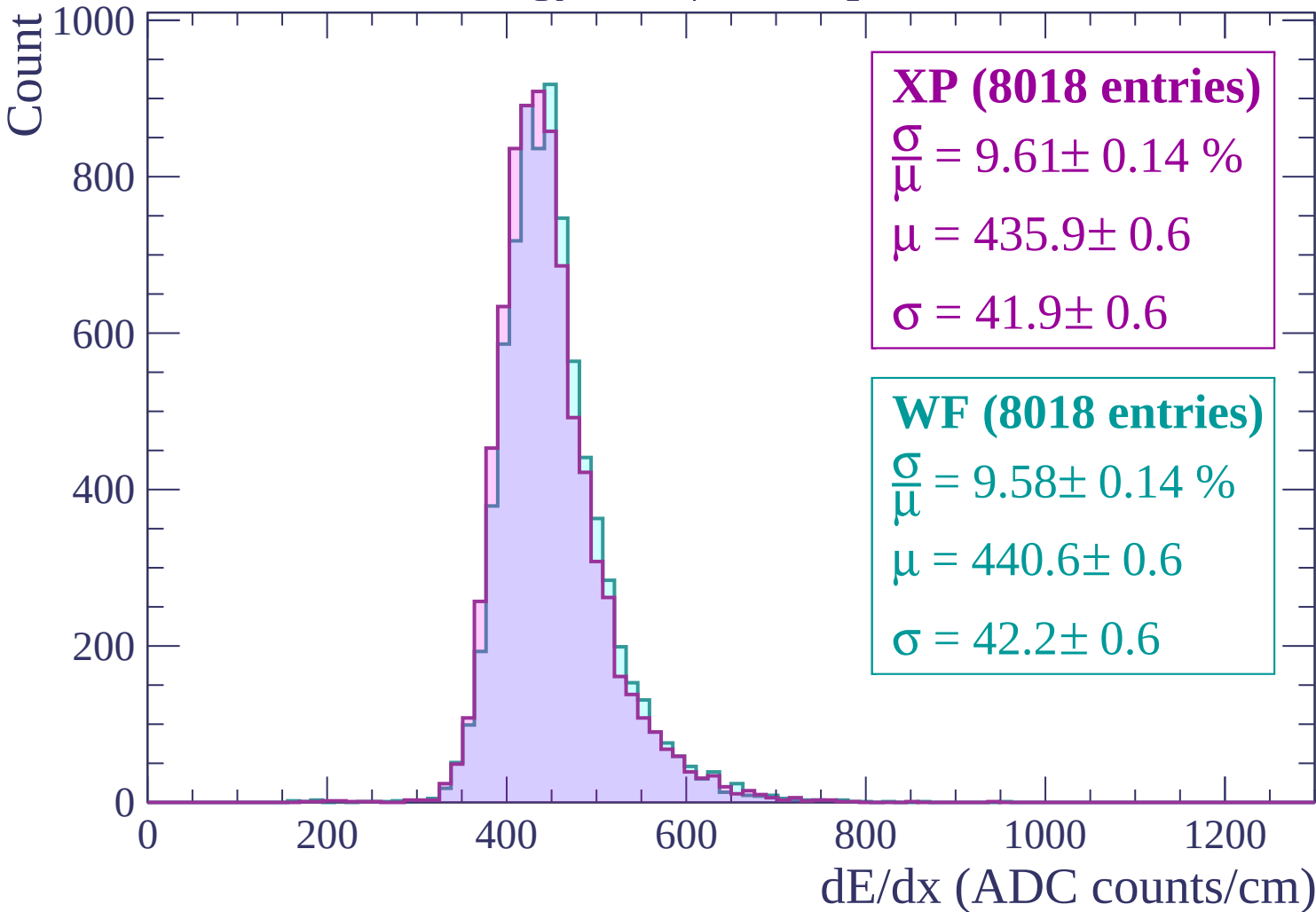
$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$

$\mu = 437.4 \pm 0.6$

$\sigma = 43.0 \pm 0.6$

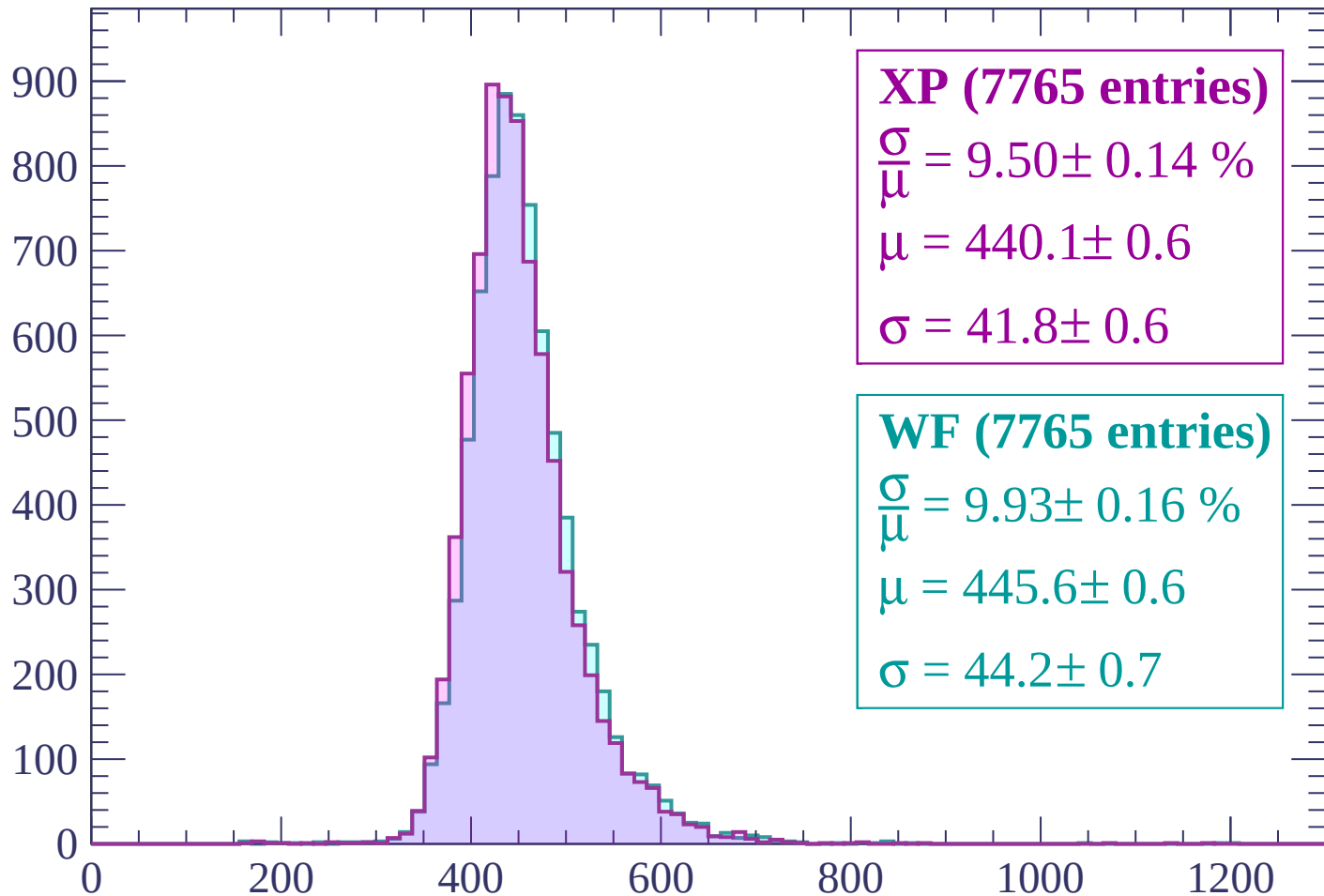
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$



Energy loss | $1020 < p < 1080$

Count



XP (7765 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.14 \%$$

$$\mu = 440.1 \pm 0.6$$

$$\sigma = 41.8 \pm 0.6$$

WF (7765 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.16 \%$$

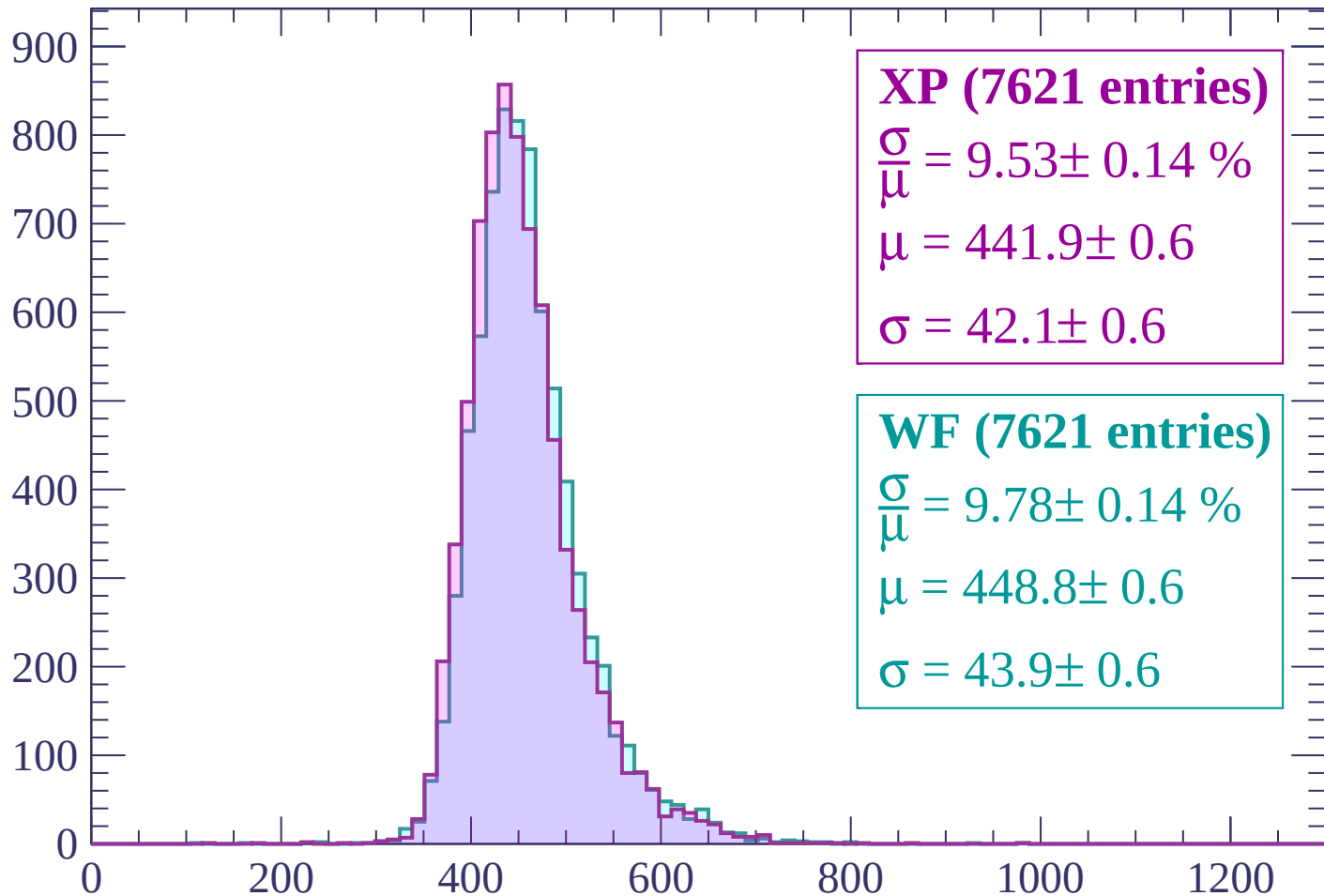
$$\mu = 445.6 \pm 0.6$$

$$\sigma = 44.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (7621 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.14 \%$$

$$\mu = 441.9 \pm 0.6$$

$$\sigma = 42.1 \pm 0.6$$

WF (7621 entries)

$$\frac{\sigma}{\mu} = 9.78 \pm 0.14 \%$$

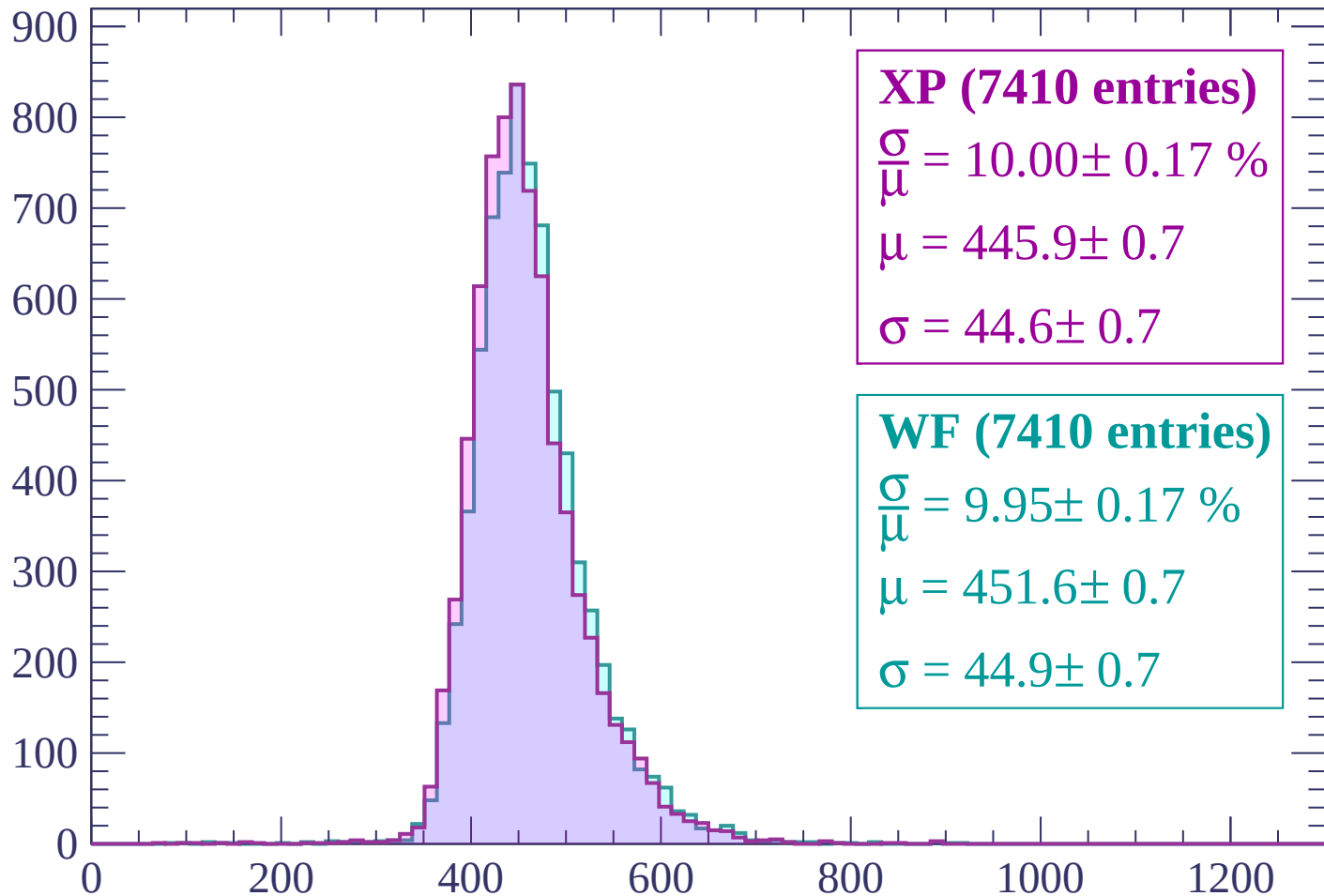
$$\mu = 448.8 \pm 0.6$$

$$\sigma = 43.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (7410 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.17 \%$$

$$\mu = 445.9 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

WF (7410 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.17 \%$$

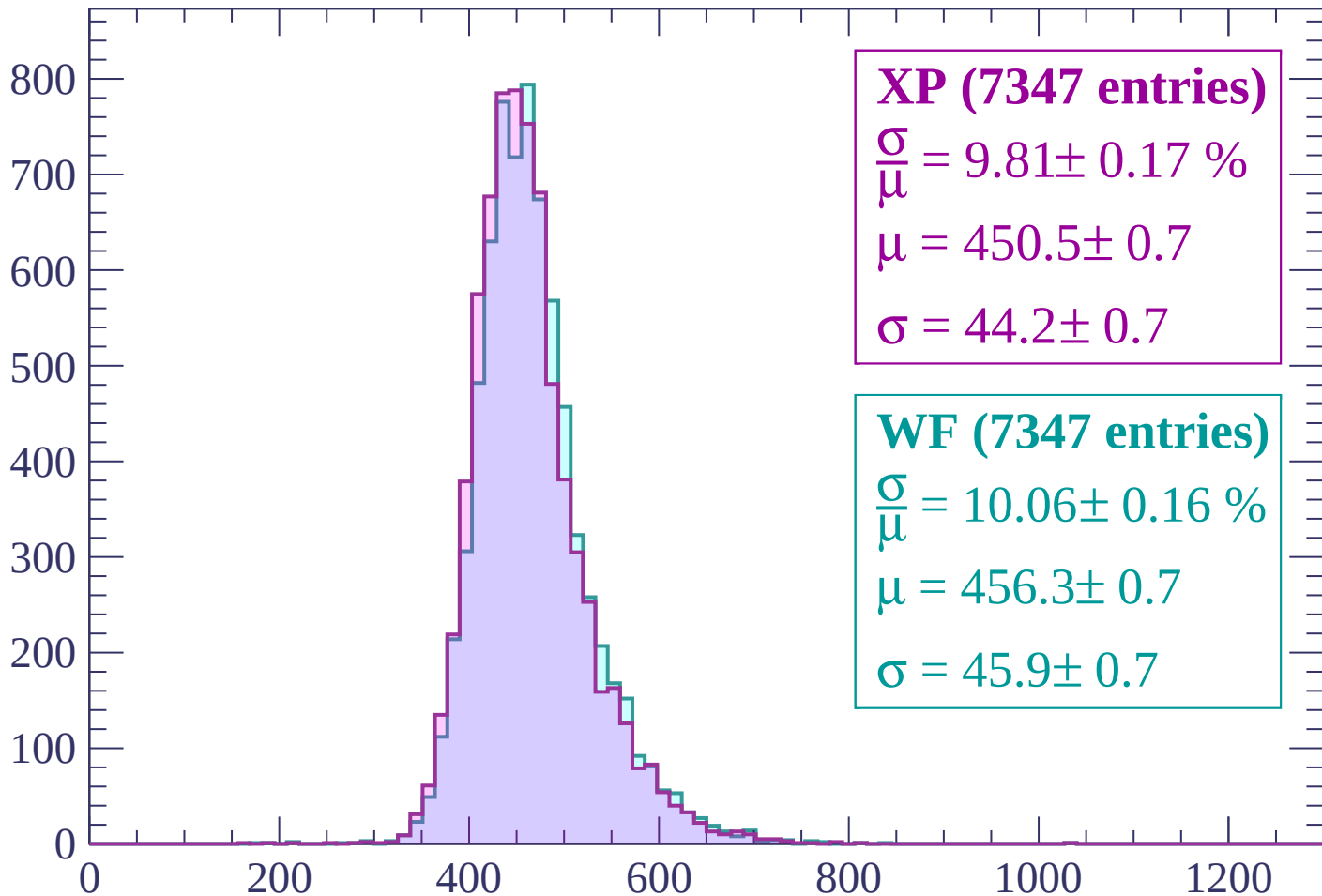
$$\mu = 451.6 \pm 0.7$$

$$\sigma = 44.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (7347 entries)

$$\frac{\sigma}{\mu} = 9.81 \pm 0.17 \%$$

$$\mu = 450.5 \pm 0.7$$

$$\sigma = 44.2 \pm 0.7$$

WF (7347 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.16 \%$$

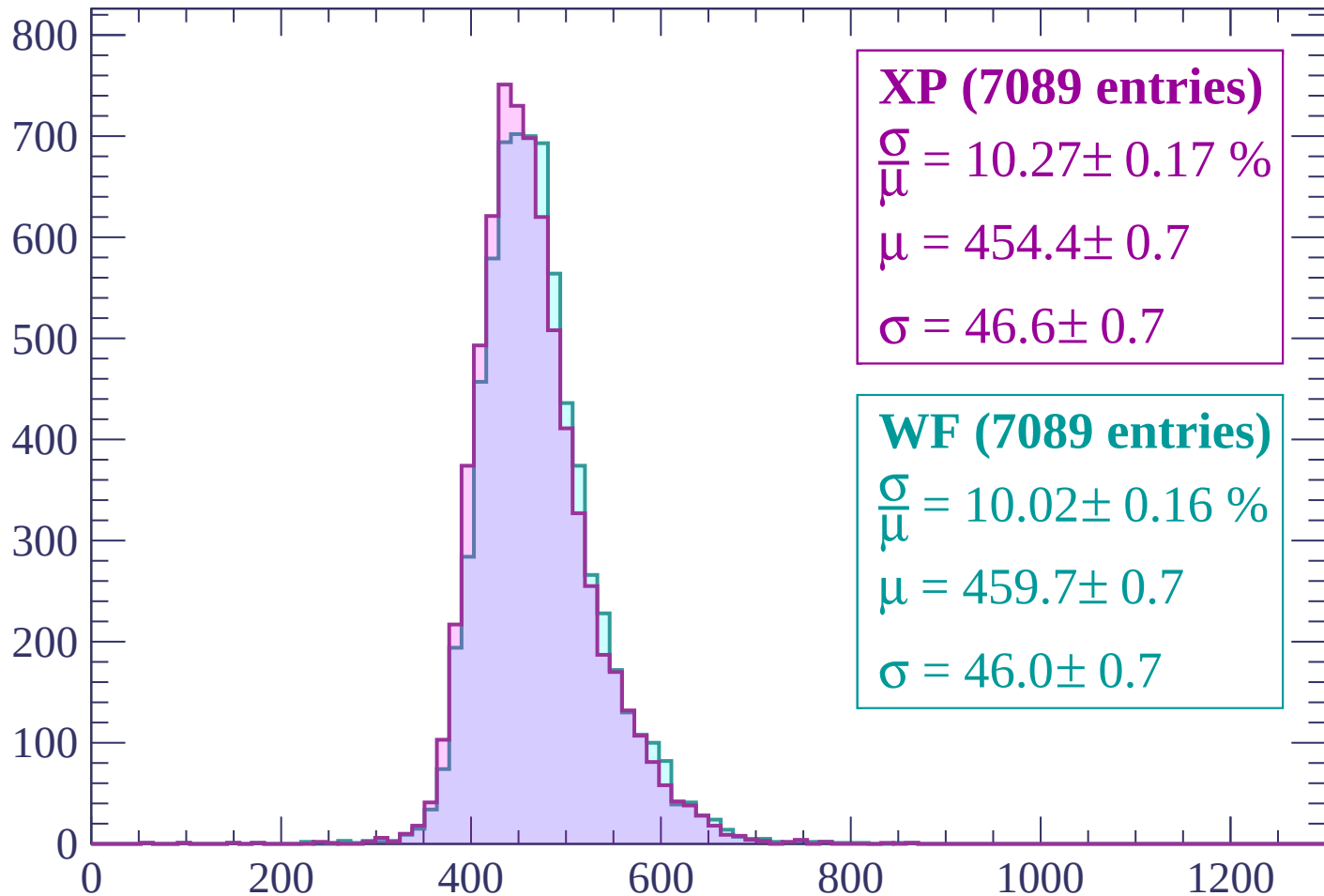
$$\mu = 456.3 \pm 0.7$$

$$\sigma = 45.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7089 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.17 \%$$

$$\mu = 454.4 \pm 0.7$$

$$\sigma = 46.6 \pm 0.7$$

WF (7089 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.16 \%$$

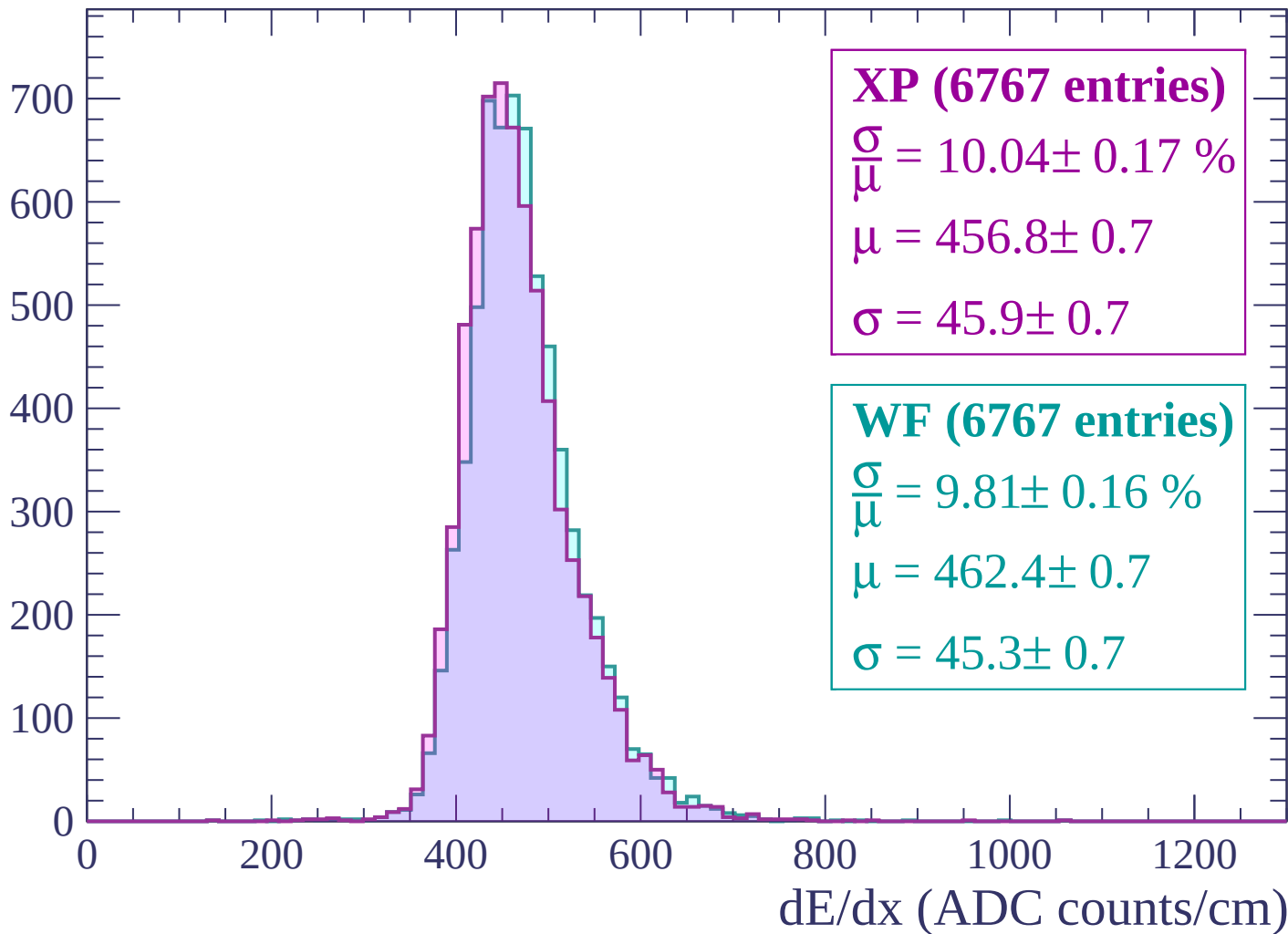
$$\mu = 459.7 \pm 0.7$$

$$\sigma = 46.0 \pm 0.7$$

dE/dx (ADC counts/cm)

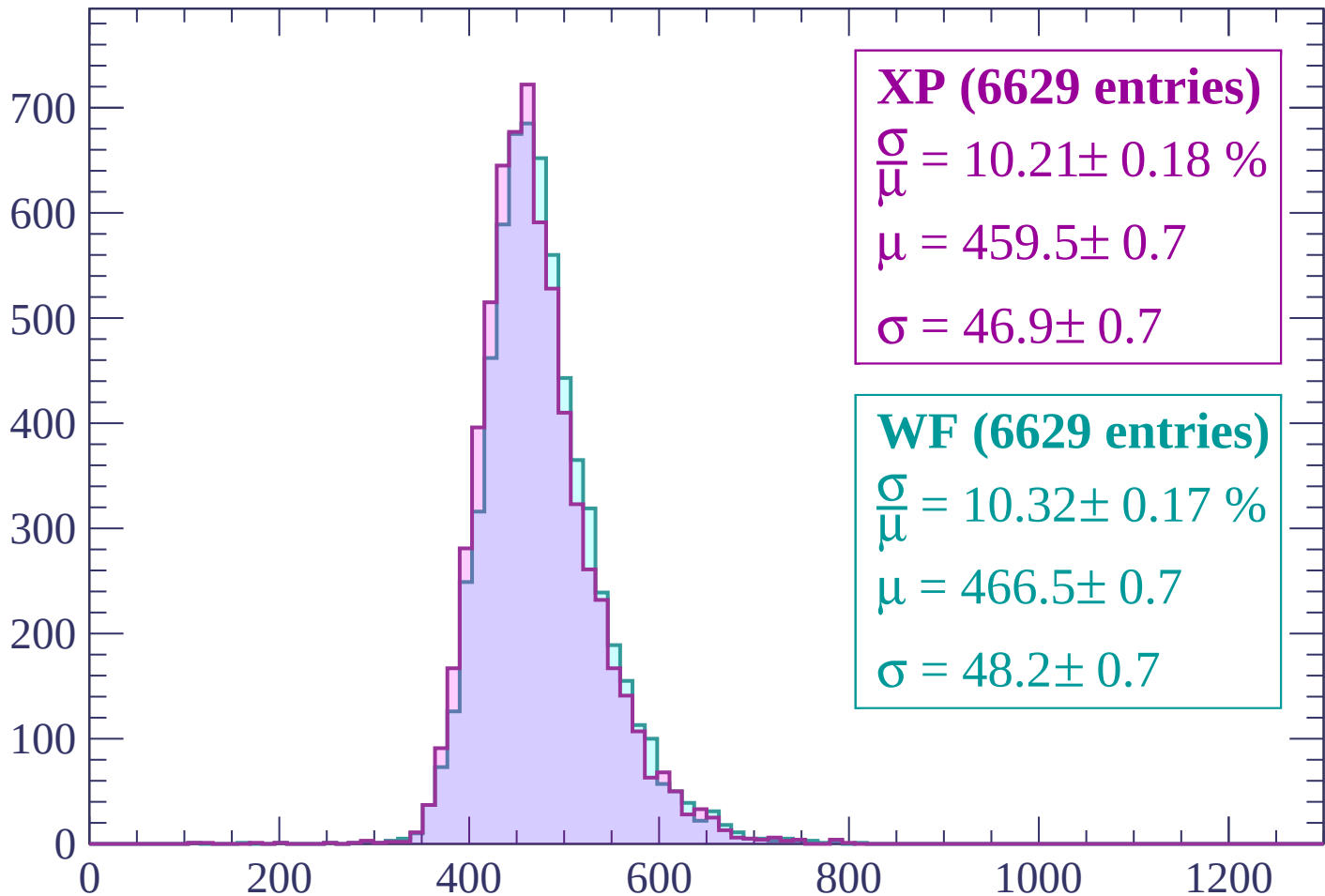
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (6629 entries)

$\frac{\sigma}{\mu} = 10.21 \pm 0.18 \%$

$\mu = 459.5 \pm 0.7$

$\sigma = 46.9 \pm 0.7$

WF (6629 entries)

$\frac{\sigma}{\mu} = 10.32 \pm 0.17 \%$

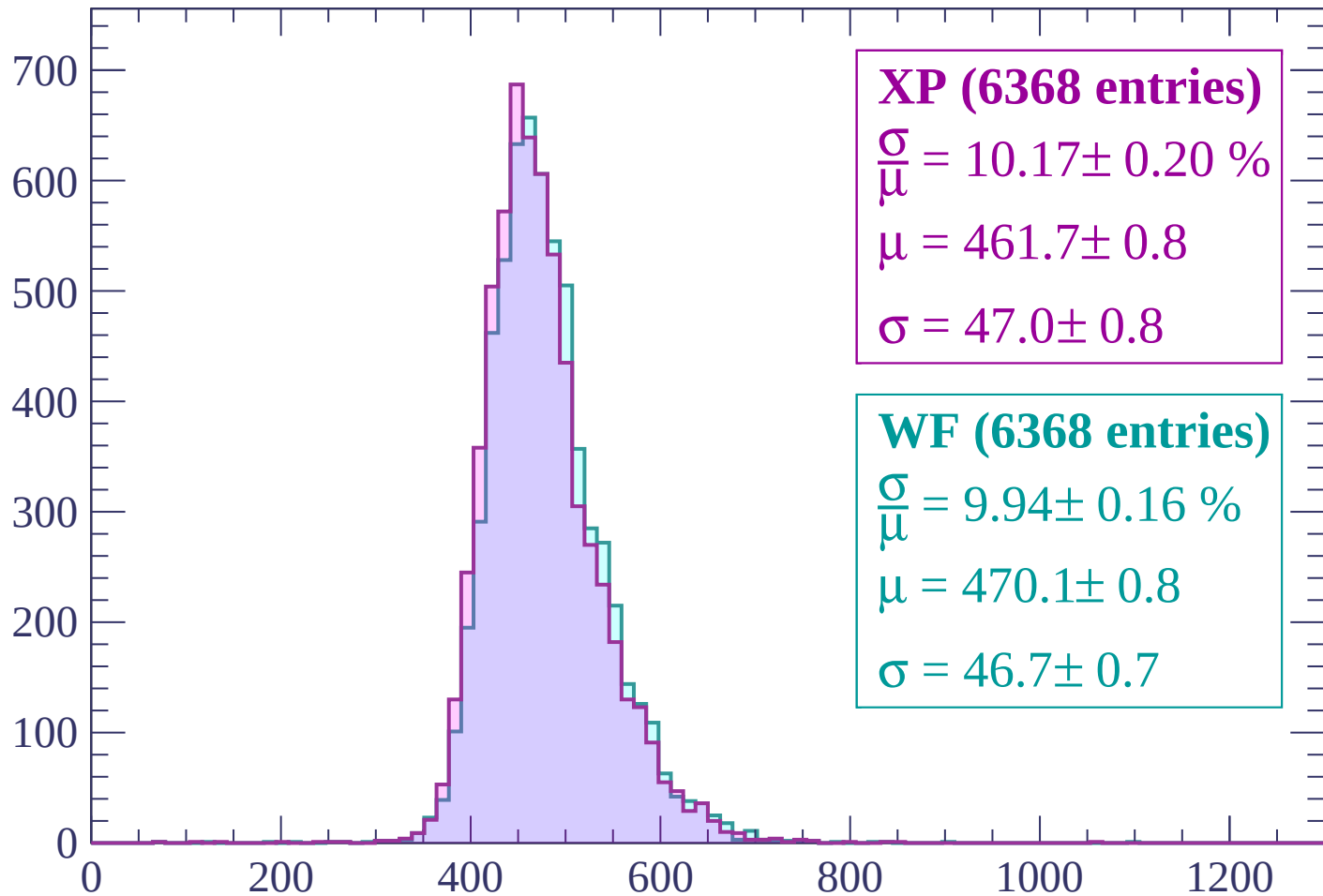
$\mu = 466.5 \pm 0.7$

$\sigma = 48.2 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (6368 entries)

$\frac{\sigma}{\mu} = 10.17 \pm 0.20 \%$

$\mu = 461.7 \pm 0.8$

$\sigma = 47.0 \pm 0.8$

WF (6368 entries)

$\frac{\sigma}{\mu} = 9.94 \pm 0.16 \%$

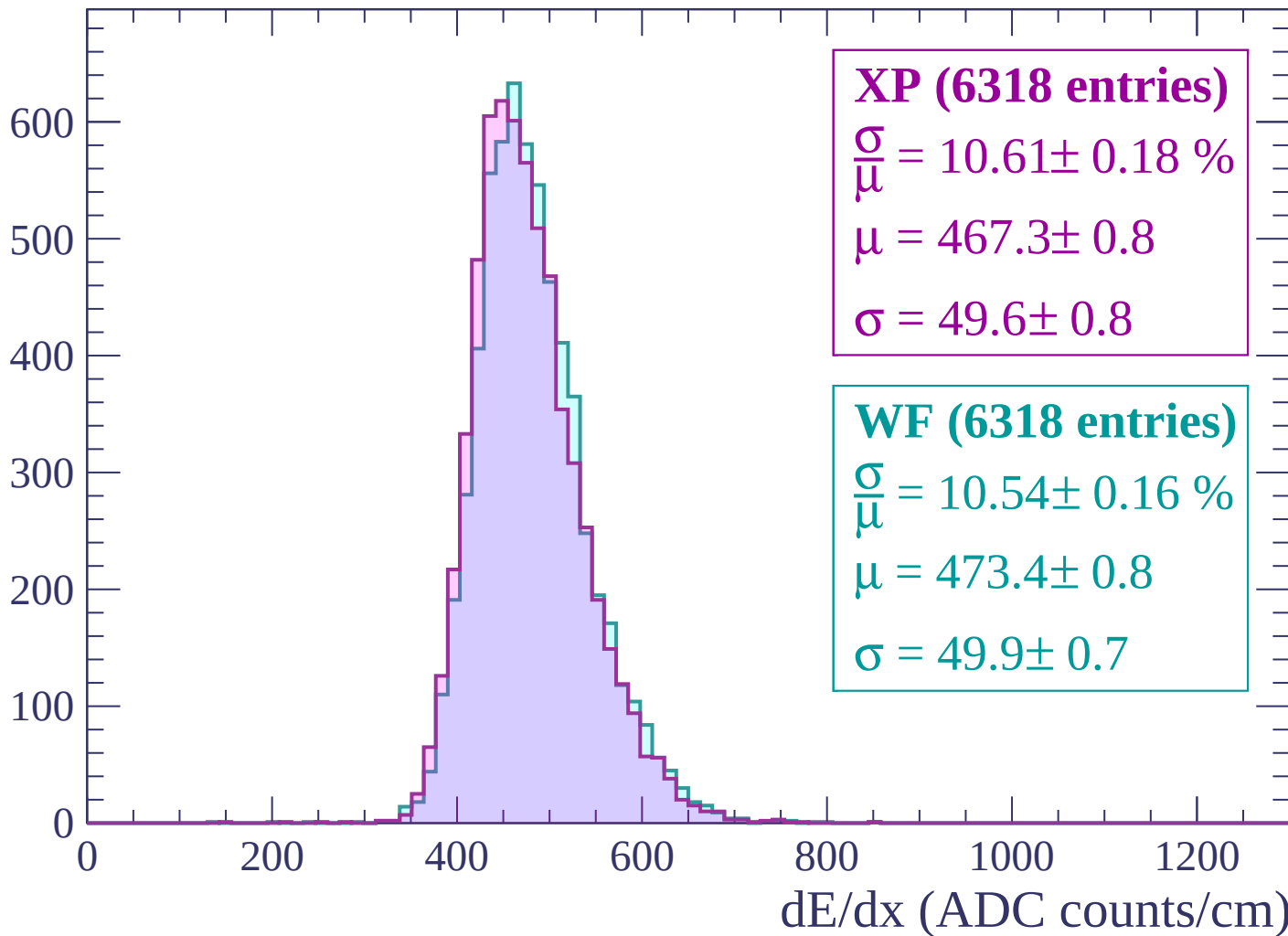
$\mu = 470.1 \pm 0.8$

$\sigma = 46.7 \pm 0.7$

dE/dx (ADC counts/cm)

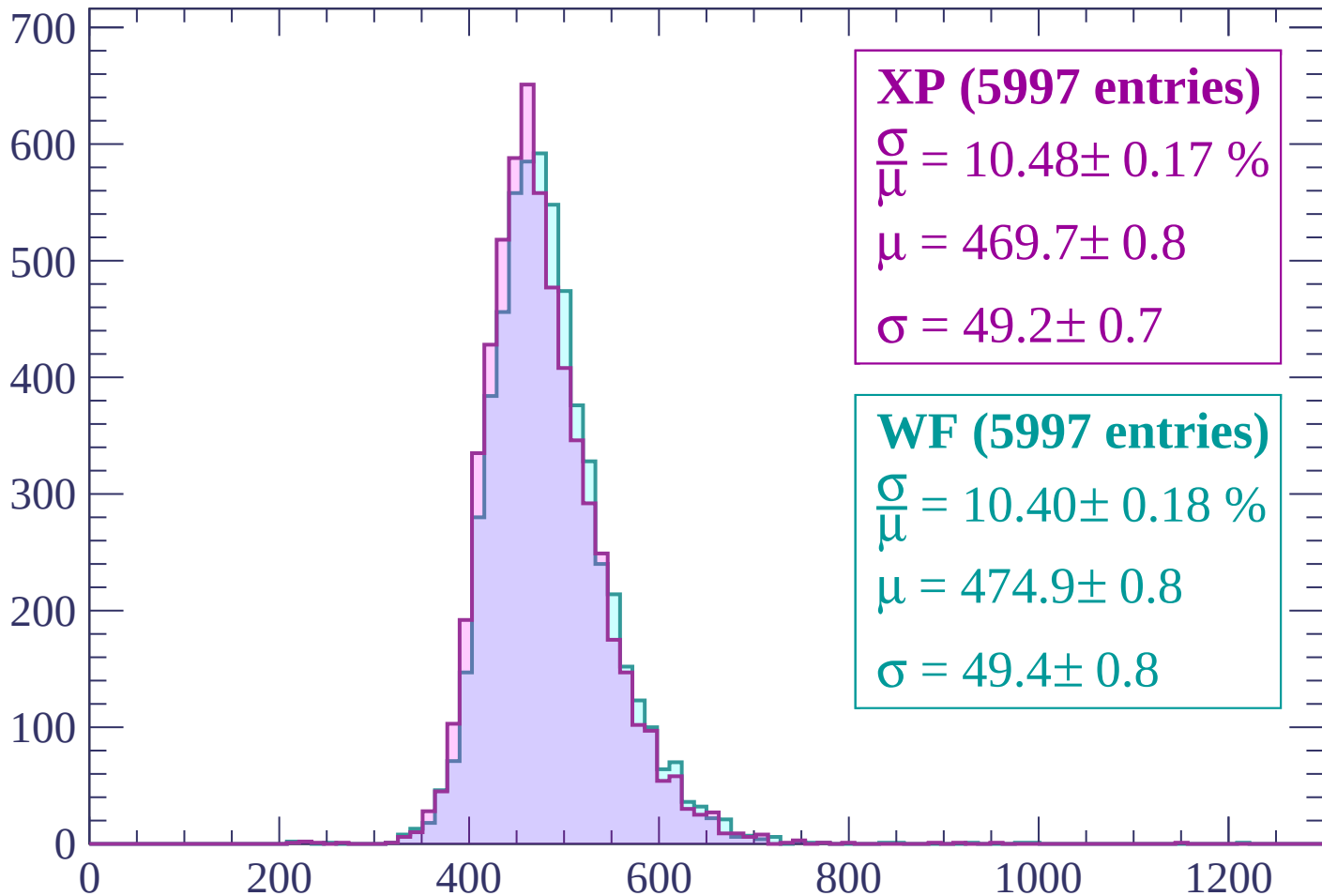
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (5997 entries)

$$\frac{\sigma}{\mu} = 10.48 \pm 0.17 \%$$

$$\mu = 469.7 \pm 0.8$$

$$\sigma = 49.2 \pm 0.7$$

WF (5997 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.18 \%$$

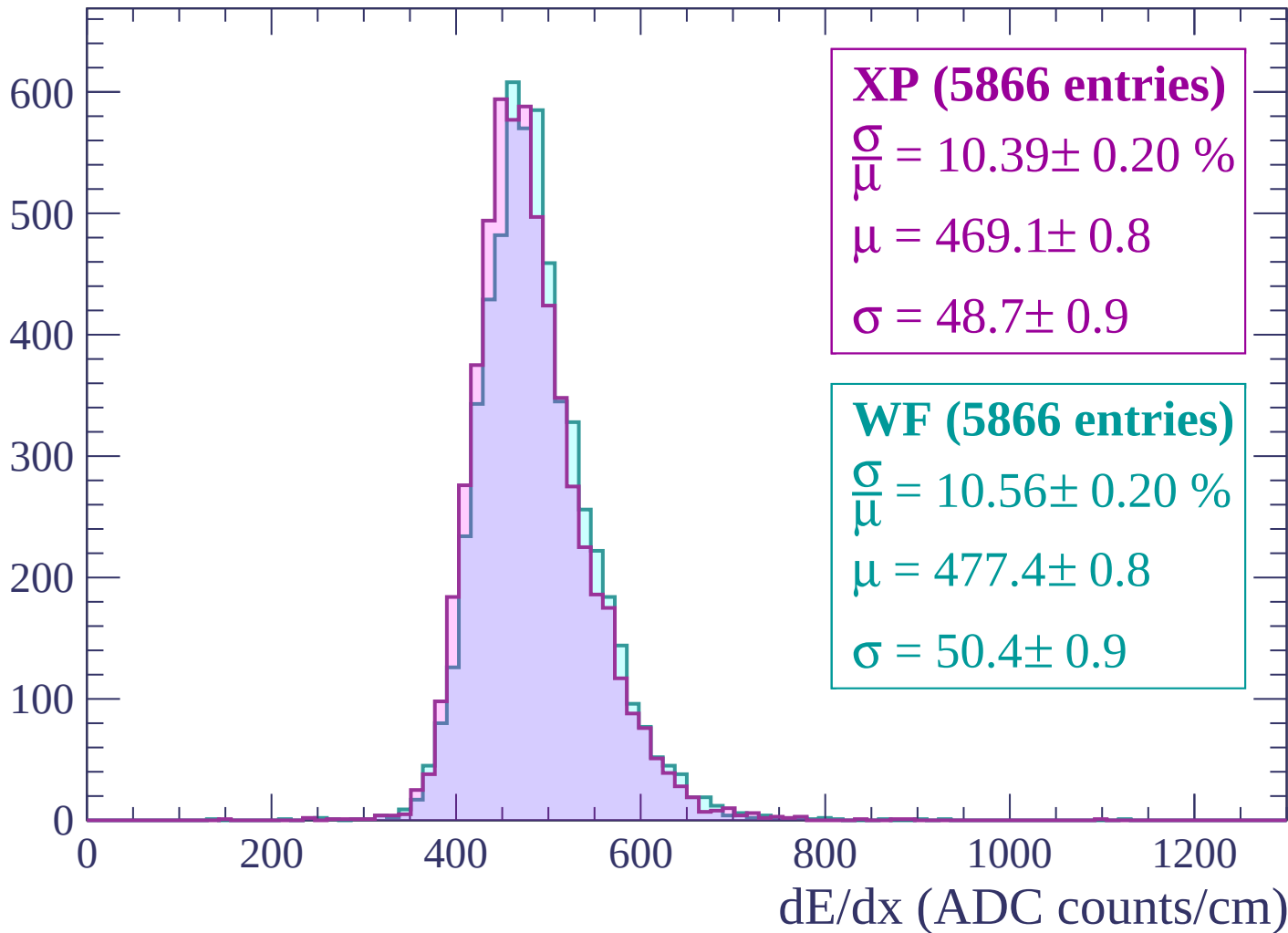
$$\mu = 474.9 \pm 0.8$$

$$\sigma = 49.4 \pm 0.8$$

dE/dx (ADC counts/cm)

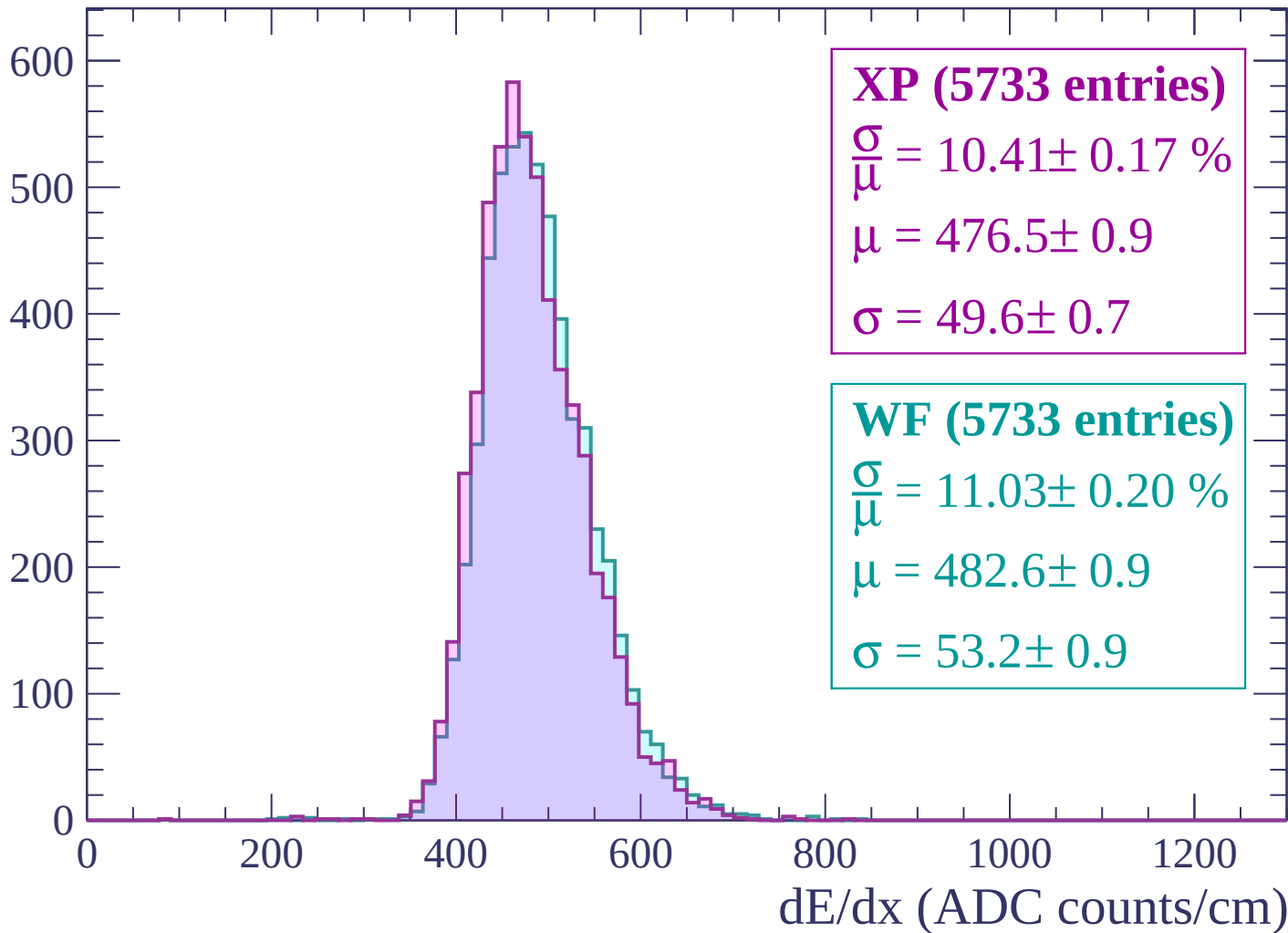
Energy loss | $1620 < p < 1680$

Count



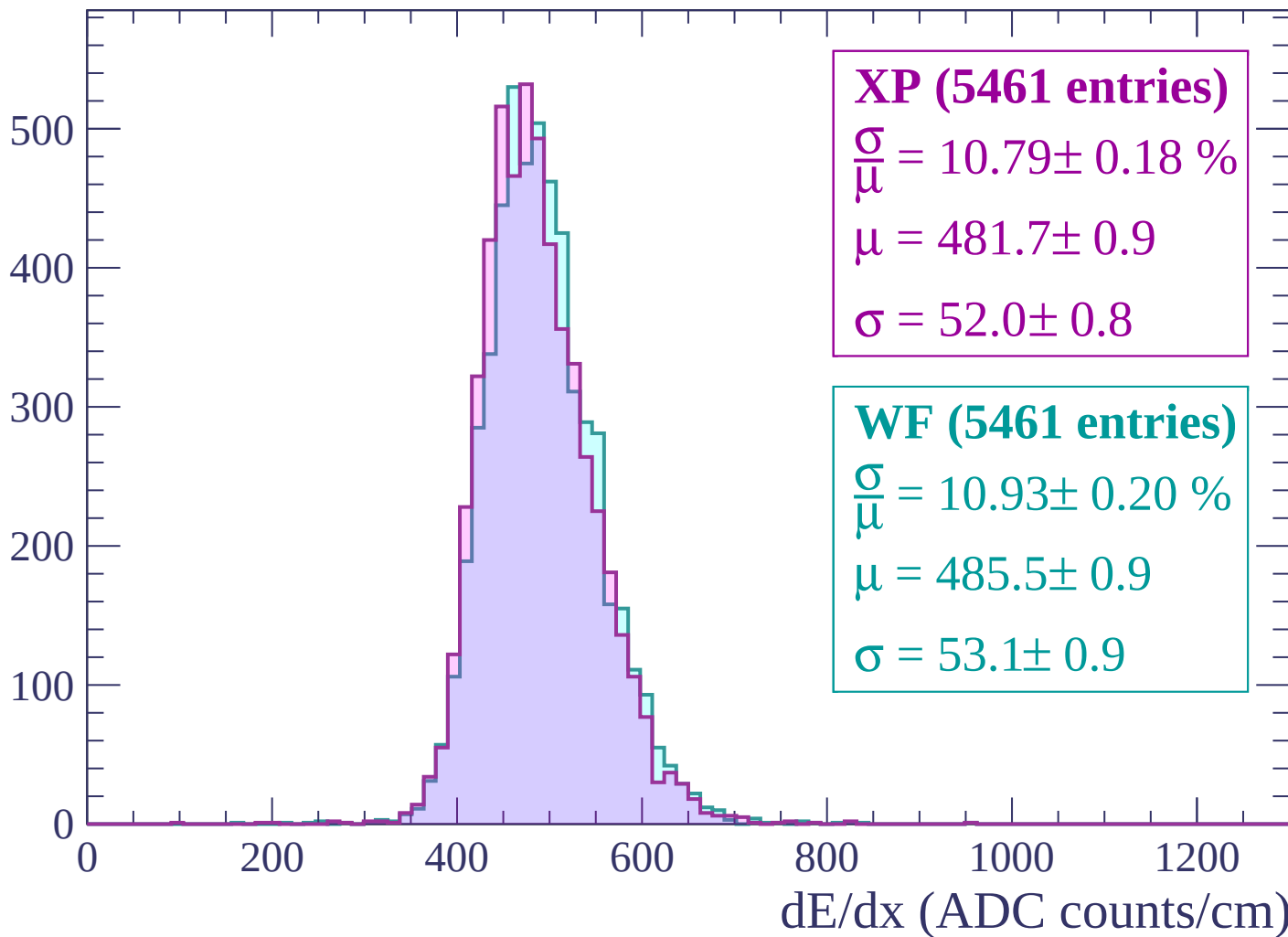
Energy loss | $1680 < p < 1740$

Count



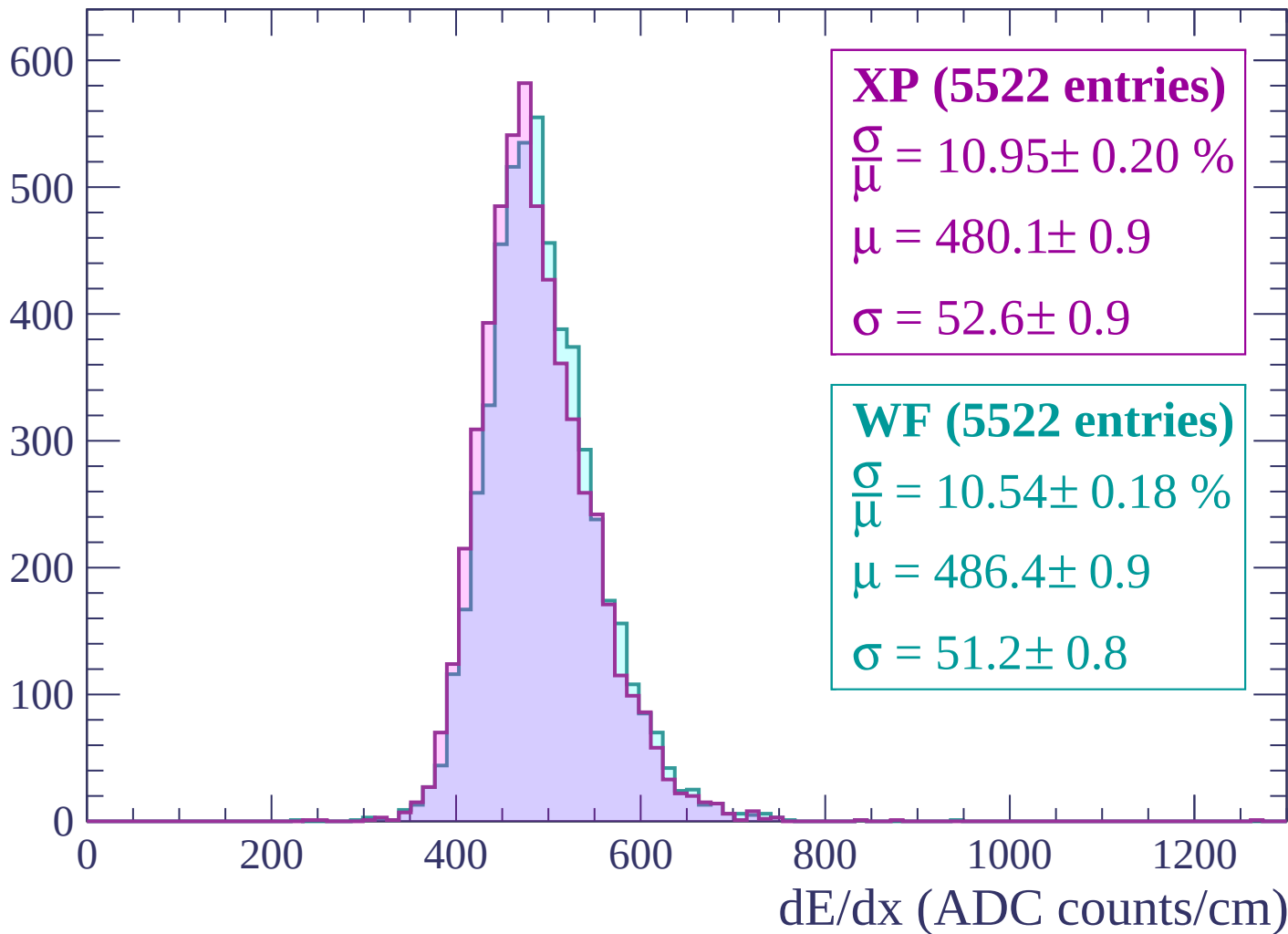
Energy loss | $1740 < p < 1800$

Count



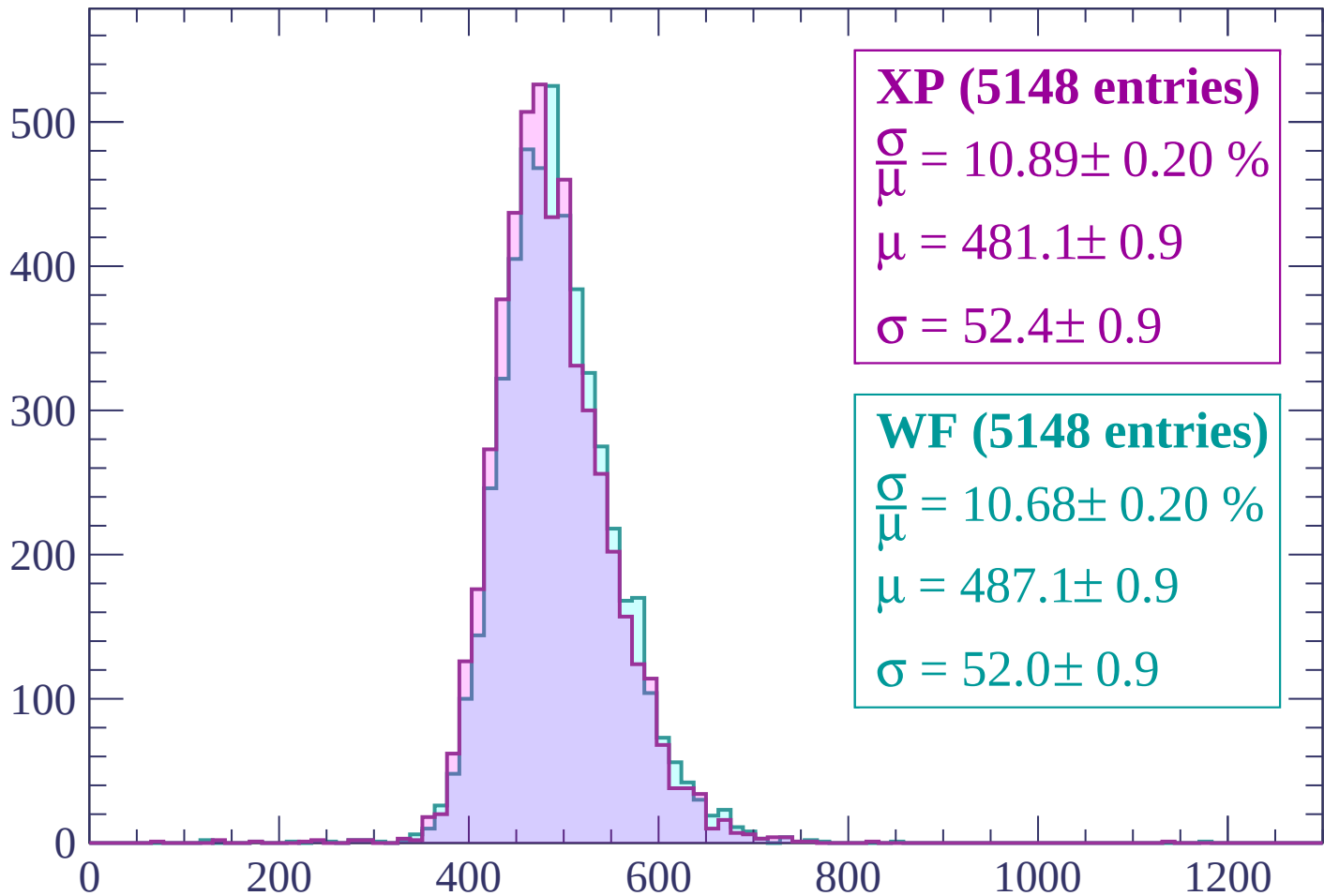
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (5148 entries)

$\frac{\sigma}{\mu} = 10.89 \pm 0.20 \%$

$\mu = 481.1 \pm 0.9$

$\sigma = 52.4 \pm 0.9$

WF (5148 entries)

$\frac{\sigma}{\mu} = 10.68 \pm 0.20 \%$

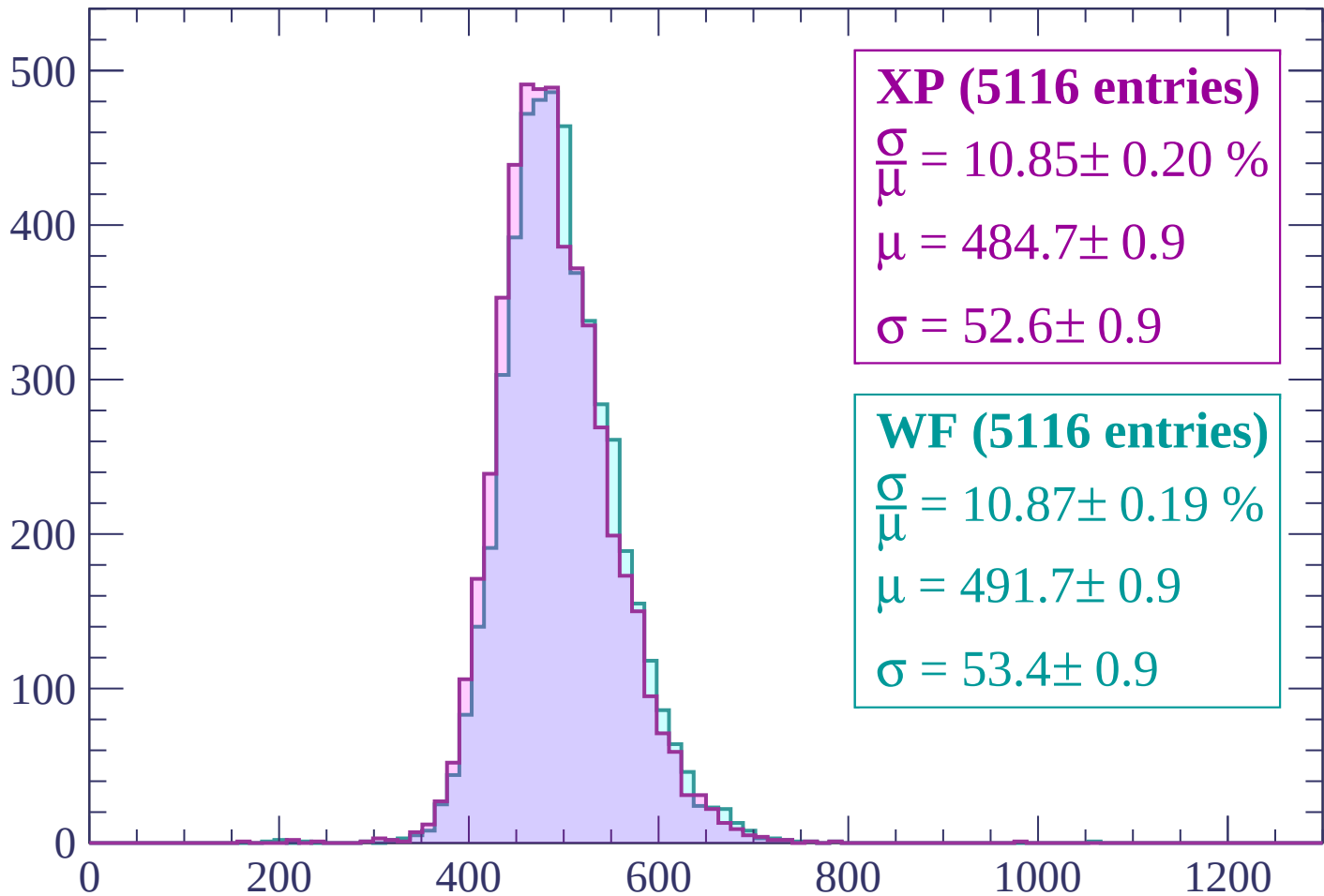
$\mu = 487.1 \pm 0.9$

$\sigma = 52.0 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1980$

Count



XP (5116 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.20 \%$$

$$\mu = 484.7 \pm 0.9$$

$$\sigma = 52.6 \pm 0.9$$

WF (5116 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.19 \%$$

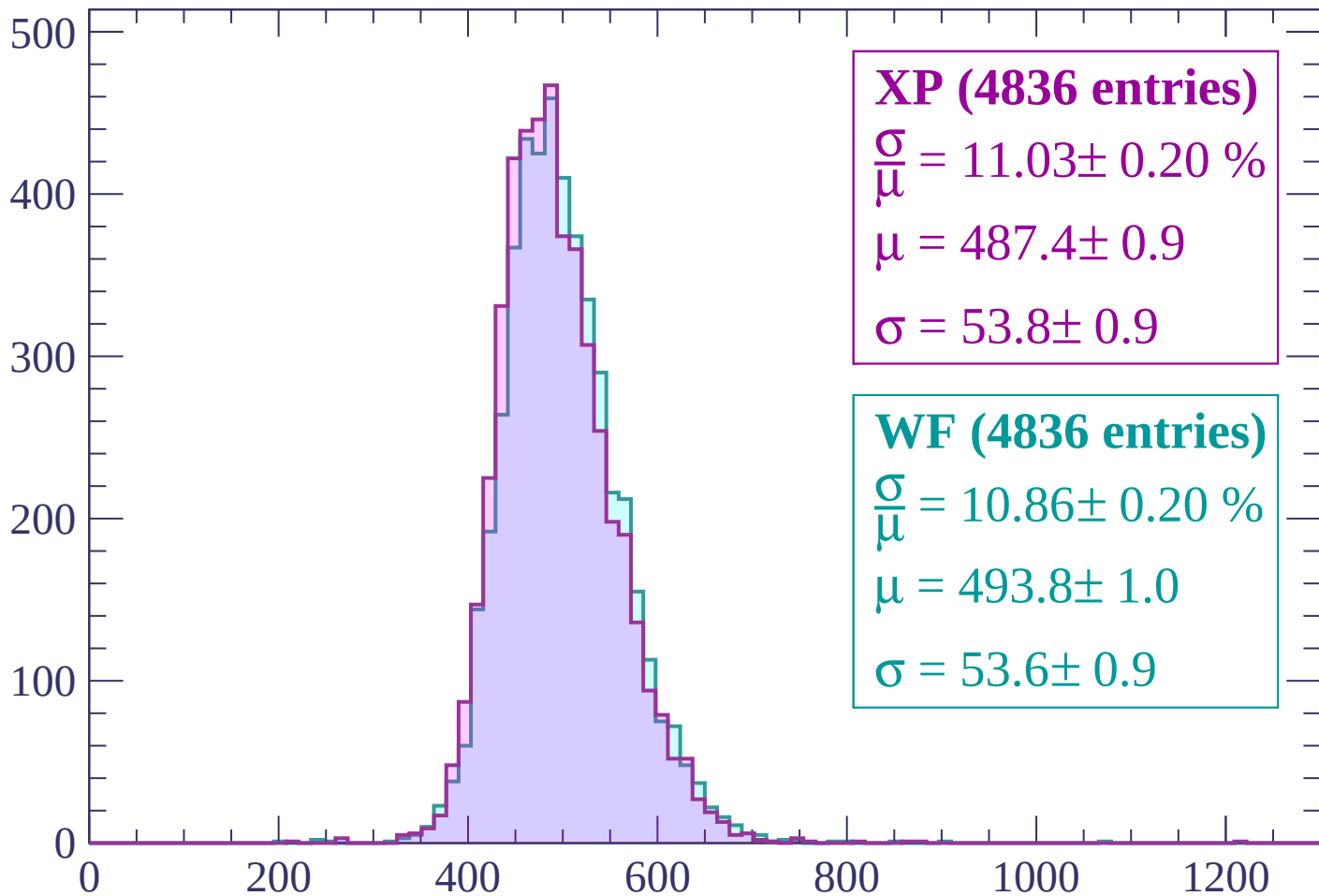
$$\mu = 491.7 \pm 0.9$$

$$\sigma = 53.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count



XP (4836 entries)

$\frac{\sigma}{\mu} = 11.03 \pm 0.20 \%$

$\mu = 487.4 \pm 0.9$

$\sigma = 53.8 \pm 0.9$

WF (4836 entries)

$\frac{\sigma}{\mu} = 10.86 \pm 0.20 \%$

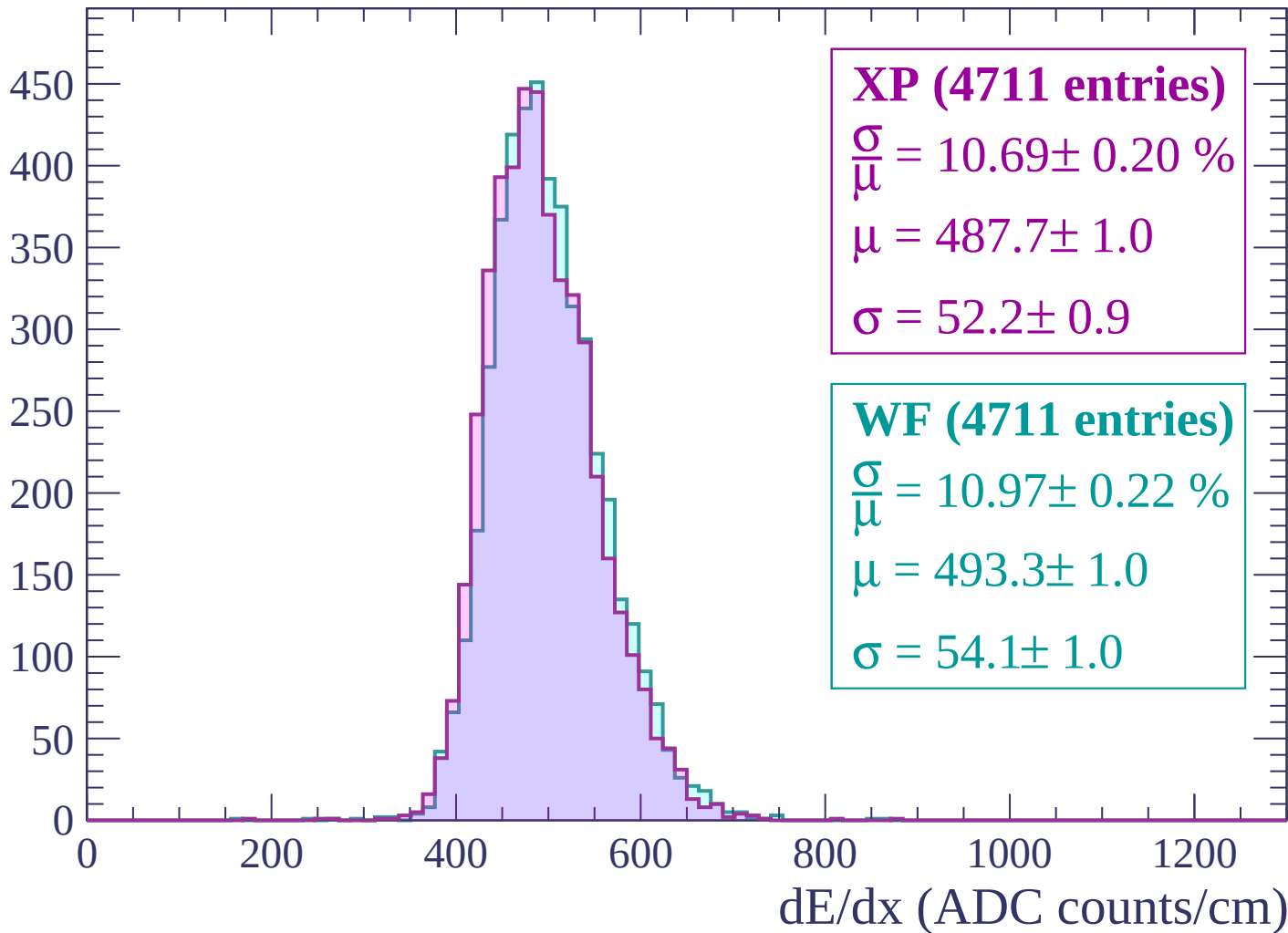
$\mu = 493.8 \pm 1.0$

$\sigma = 53.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

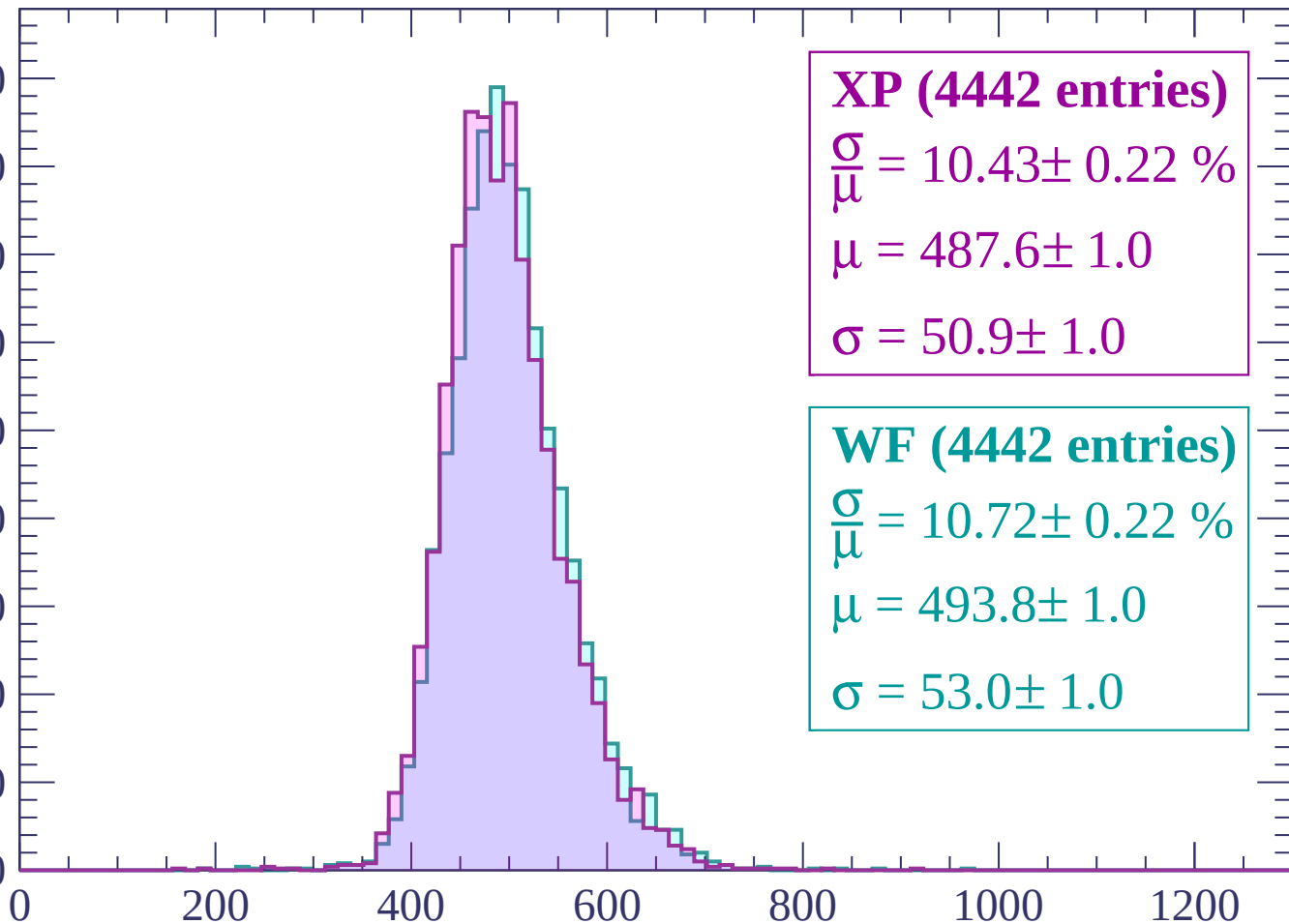
Count



Energy loss | $2100 < p < 2160$

Count

450
400
350
300
250
200
150
100
50
0



XP (4442 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.22 \%$

$\mu = 487.6 \pm 1.0$

$\sigma = 50.9 \pm 1.0$

WF (4442 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$

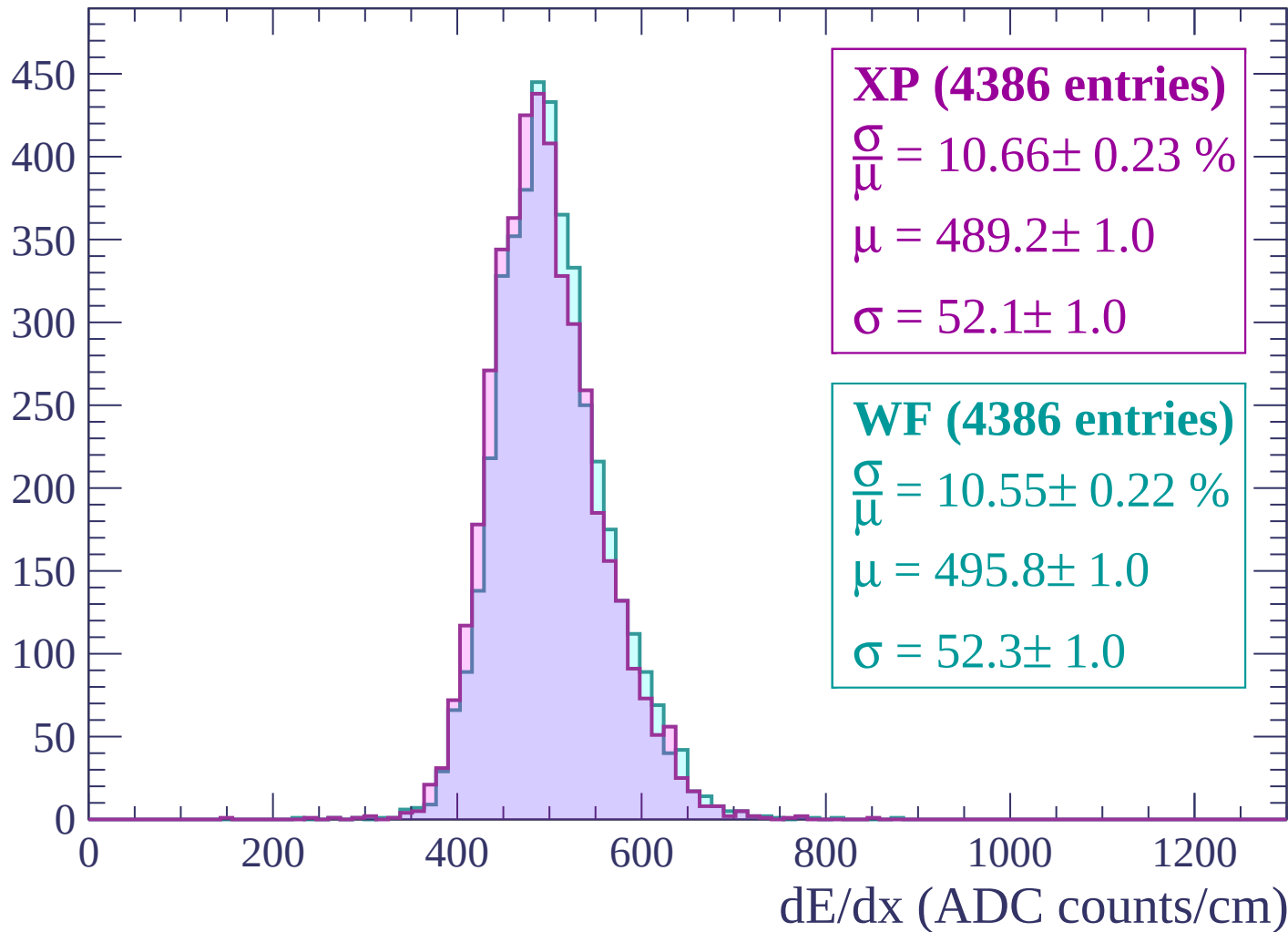
$\mu = 493.8 \pm 1.0$

$\sigma = 53.0 \pm 1.0$

dE/dx (ADC counts/cm)

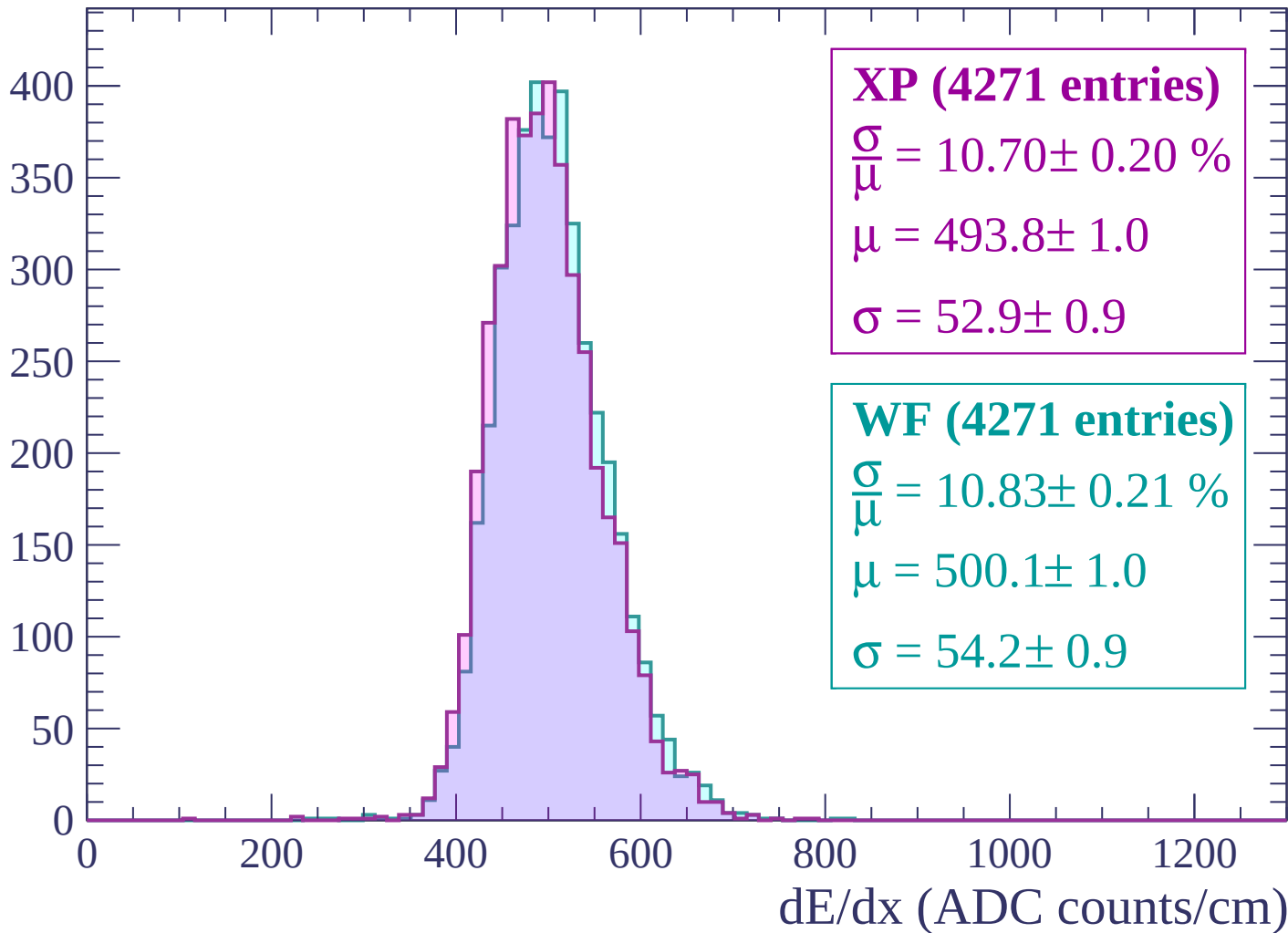
Energy loss | $2160 < p < 2220$

Count



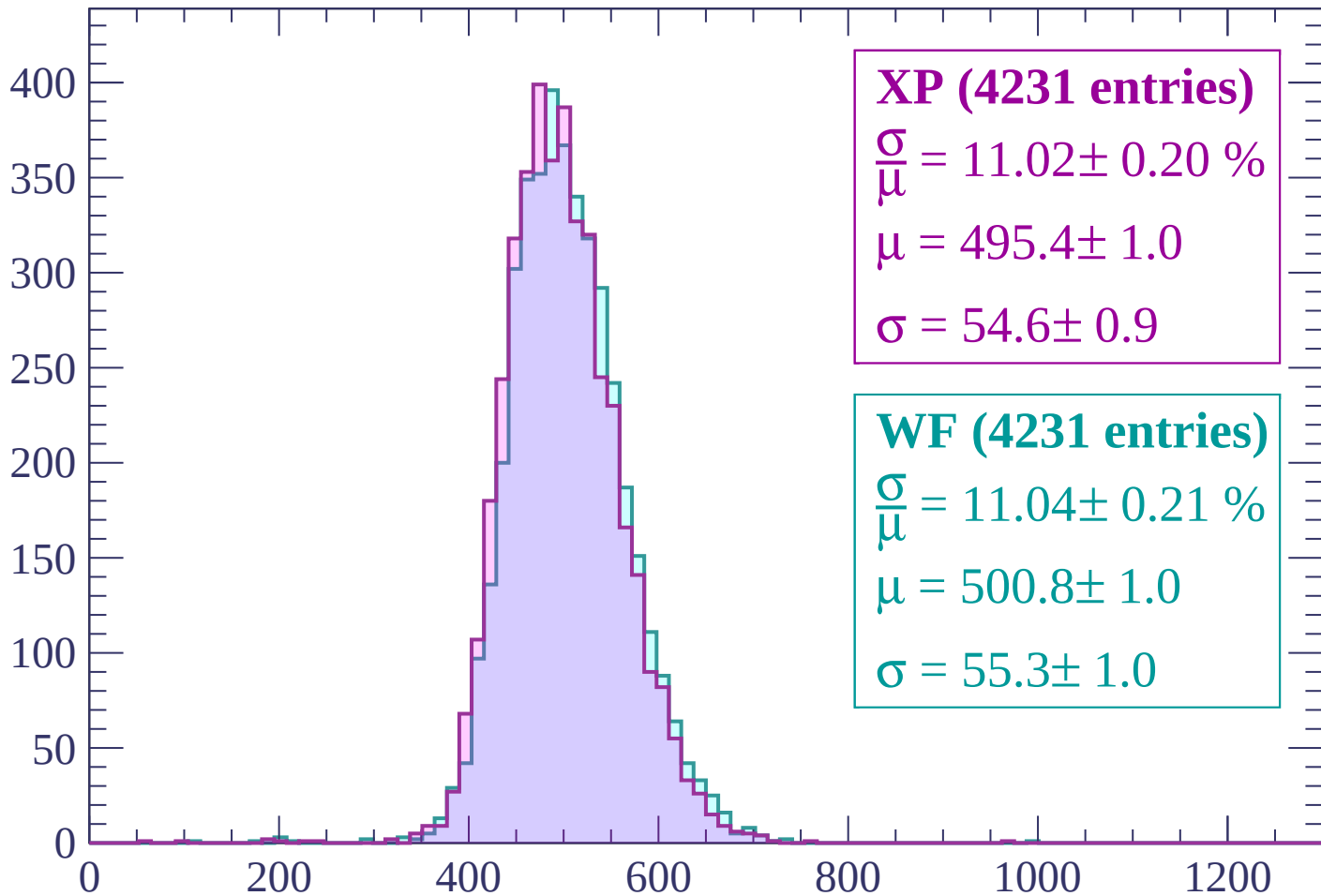
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



XP (4231 entries)

$\frac{\sigma}{\mu} = 11.02 \pm 0.20 \%$

$\mu = 495.4 \pm 1.0$

$\sigma = 54.6 \pm 0.9$

WF (4231 entries)

$\frac{\sigma}{\mu} = 11.04 \pm 0.21 \%$

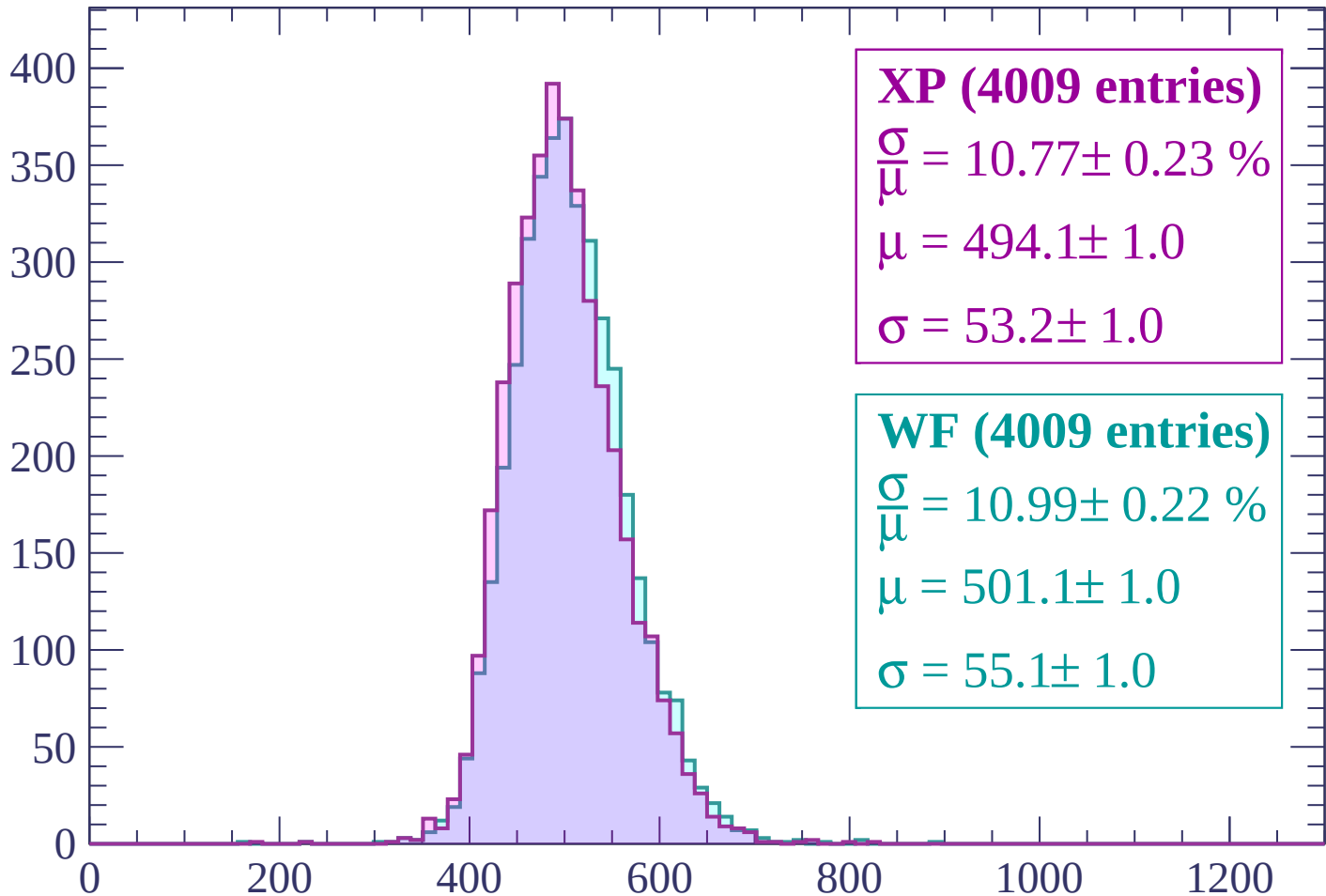
$\mu = 500.8 \pm 1.0$

$\sigma = 55.3 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP (4009 entries)

$\frac{\sigma}{\mu} = 10.77 \pm 0.23 \%$

$\mu = 494.1 \pm 1.0$

$\sigma = 53.2 \pm 1.0$

WF (4009 entries)

$\frac{\sigma}{\mu} = 10.99 \pm 0.22 \%$

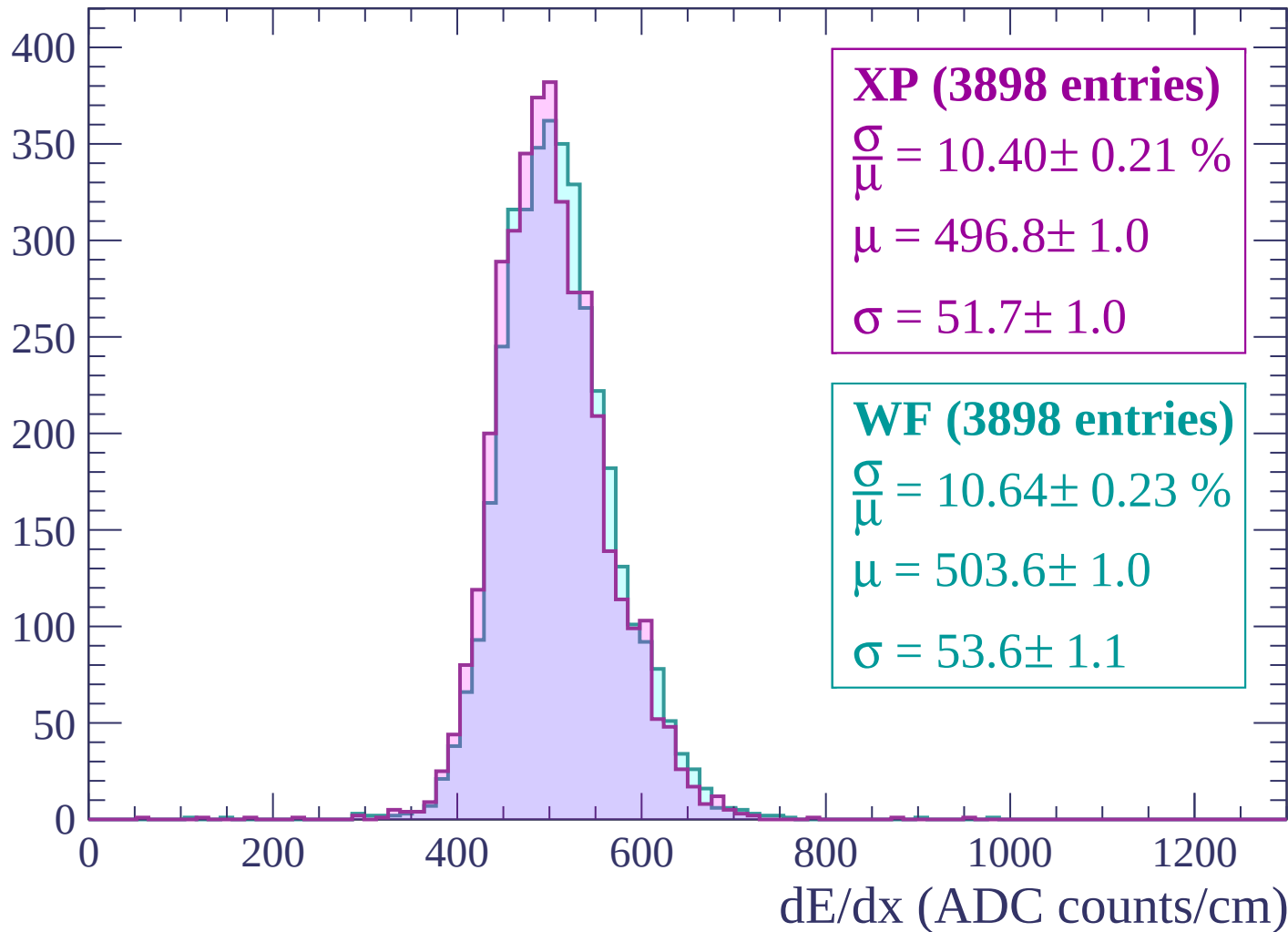
$\mu = 501.1 \pm 1.0$

$\sigma = 55.1 \pm 1.0$

dE/dx (ADC counts/cm)

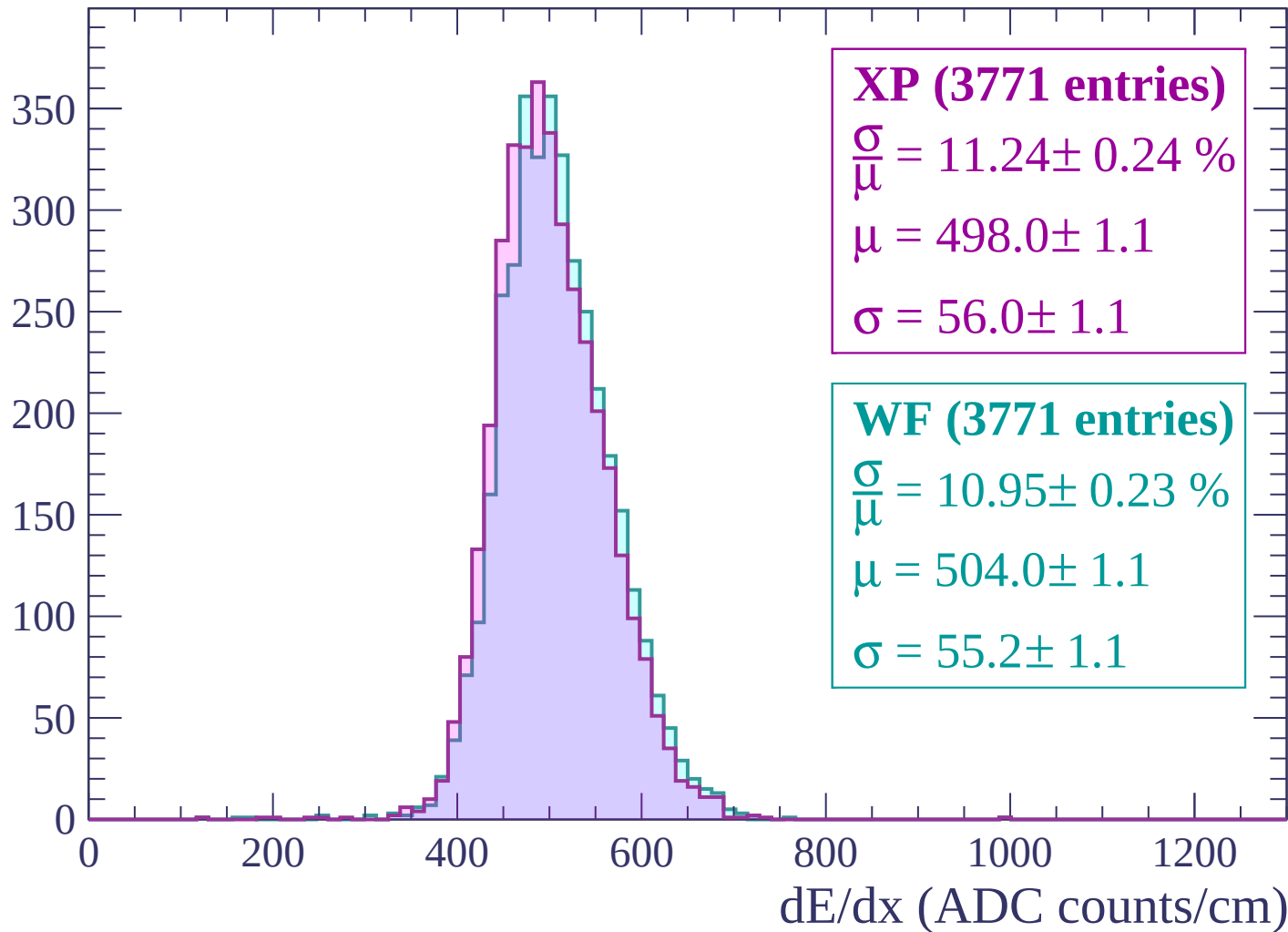
Energy loss | $2400 < p < 2460$

Count



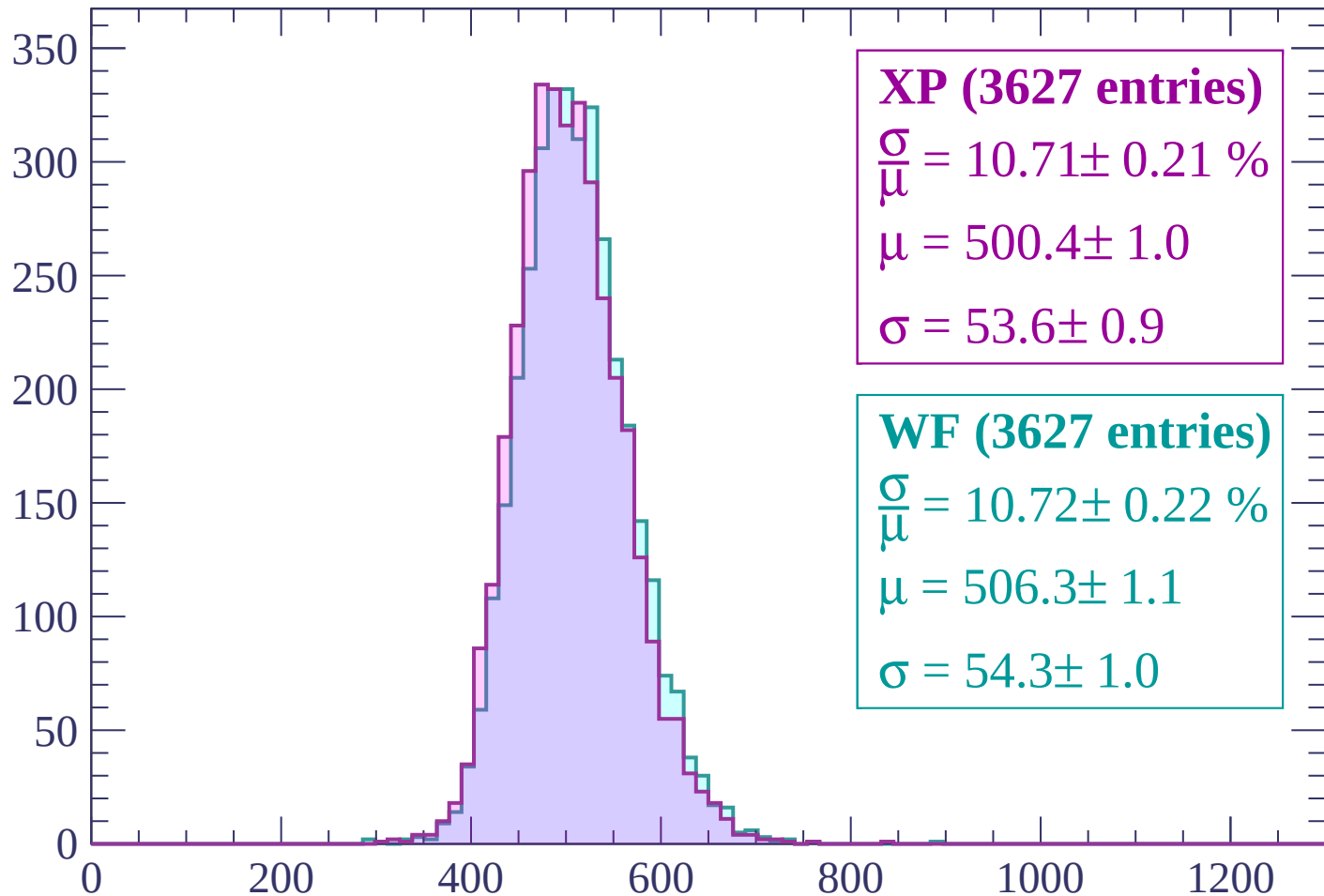
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (3627 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.21 \%$

$\mu = 500.4 \pm 1.0$

$\sigma = 53.6 \pm 0.9$

WF (3627 entries)

$\frac{\sigma}{\mu} = 10.72 \pm 0.22 \%$

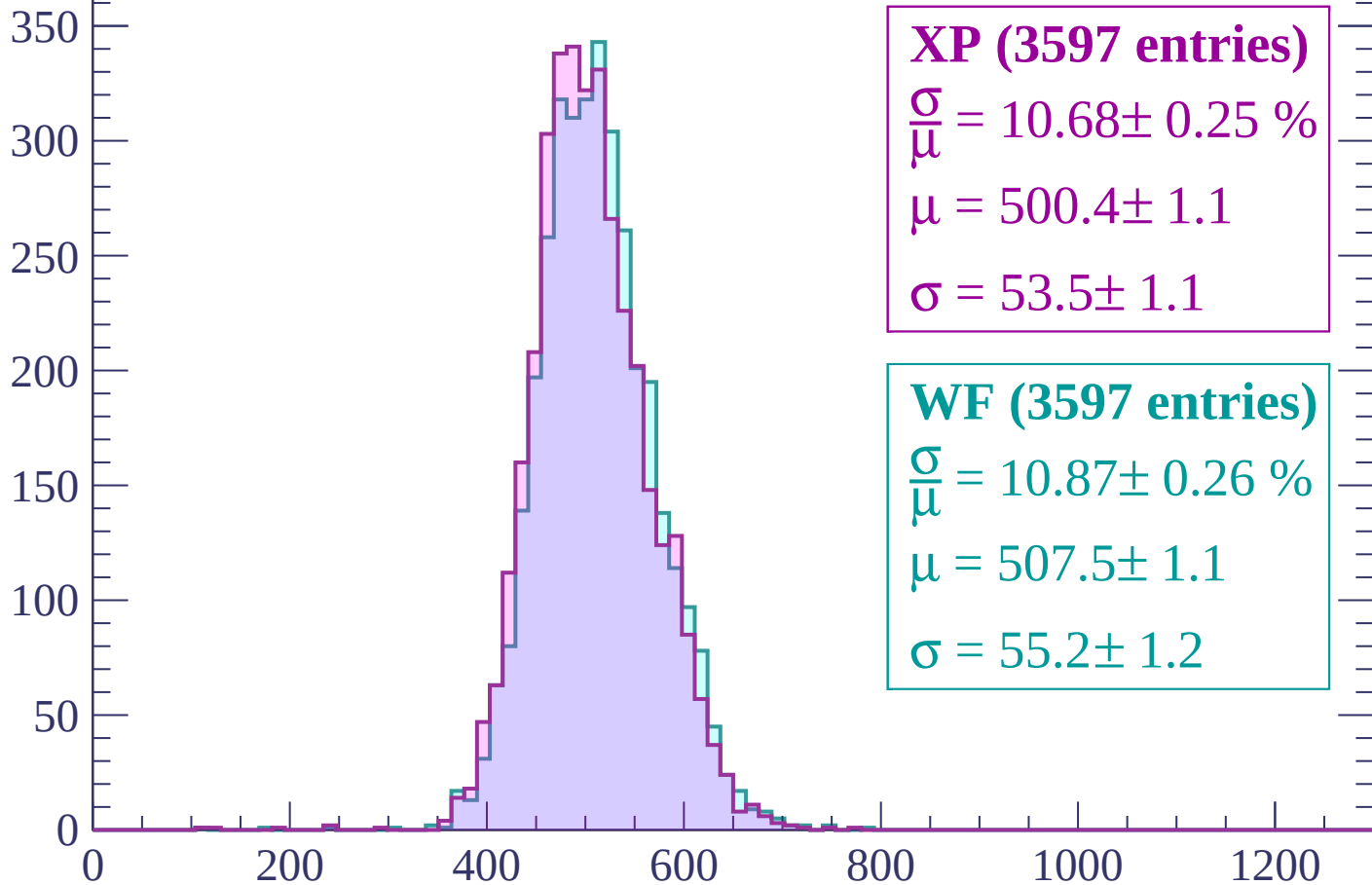
$\mu = 506.3 \pm 1.1$

$\sigma = 54.3 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (3597 entries)

$\frac{\sigma}{\mu} = 10.68 \pm 0.25 \%$

$\mu = 500.4 \pm 1.1$

$\sigma = 53.5 \pm 1.1$

WF (3597 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.26 \%$

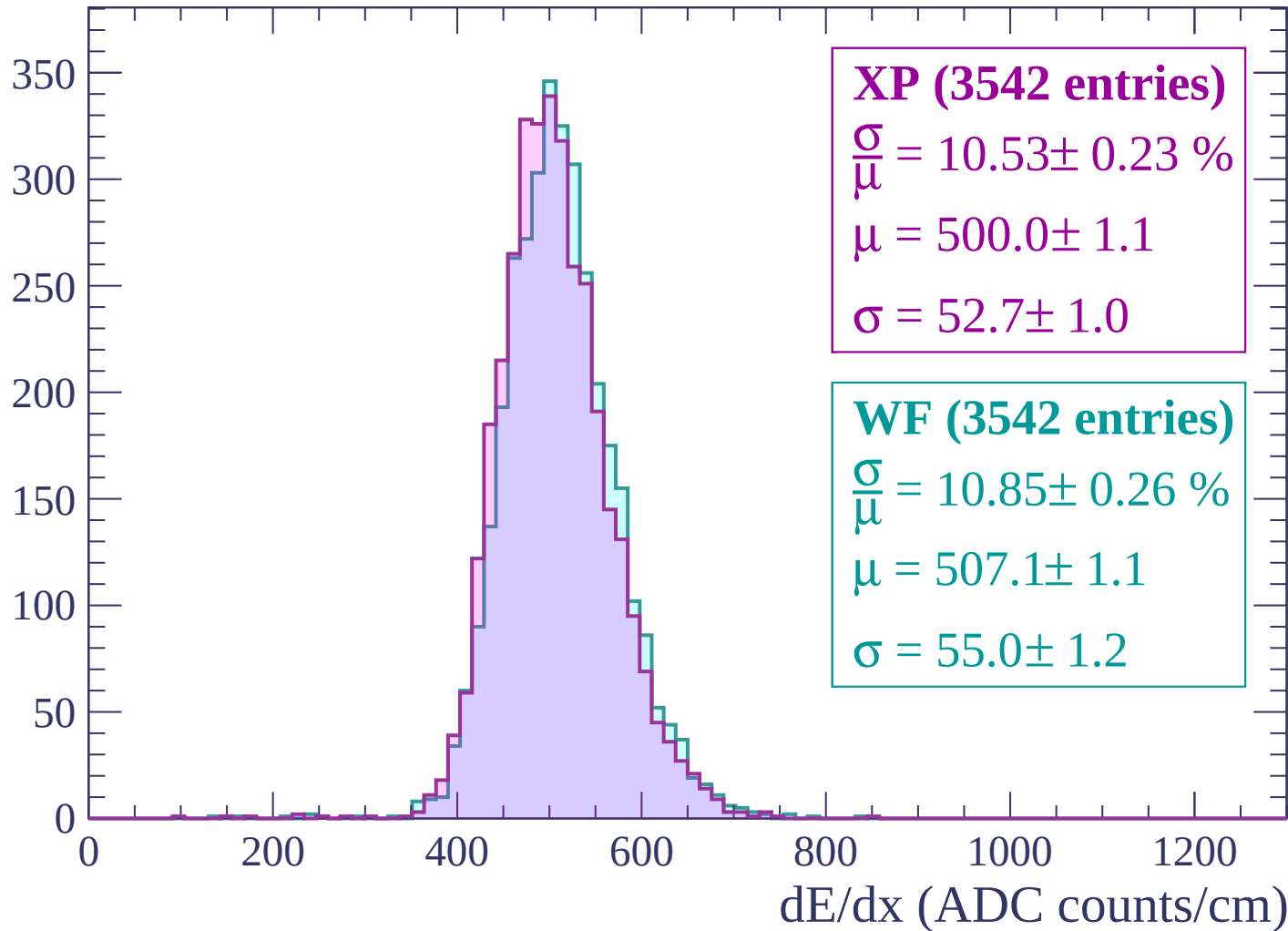
$\mu = 507.5 \pm 1.1$

$\sigma = 55.2 \pm 1.2$

dE/dx (ADC counts/cm)

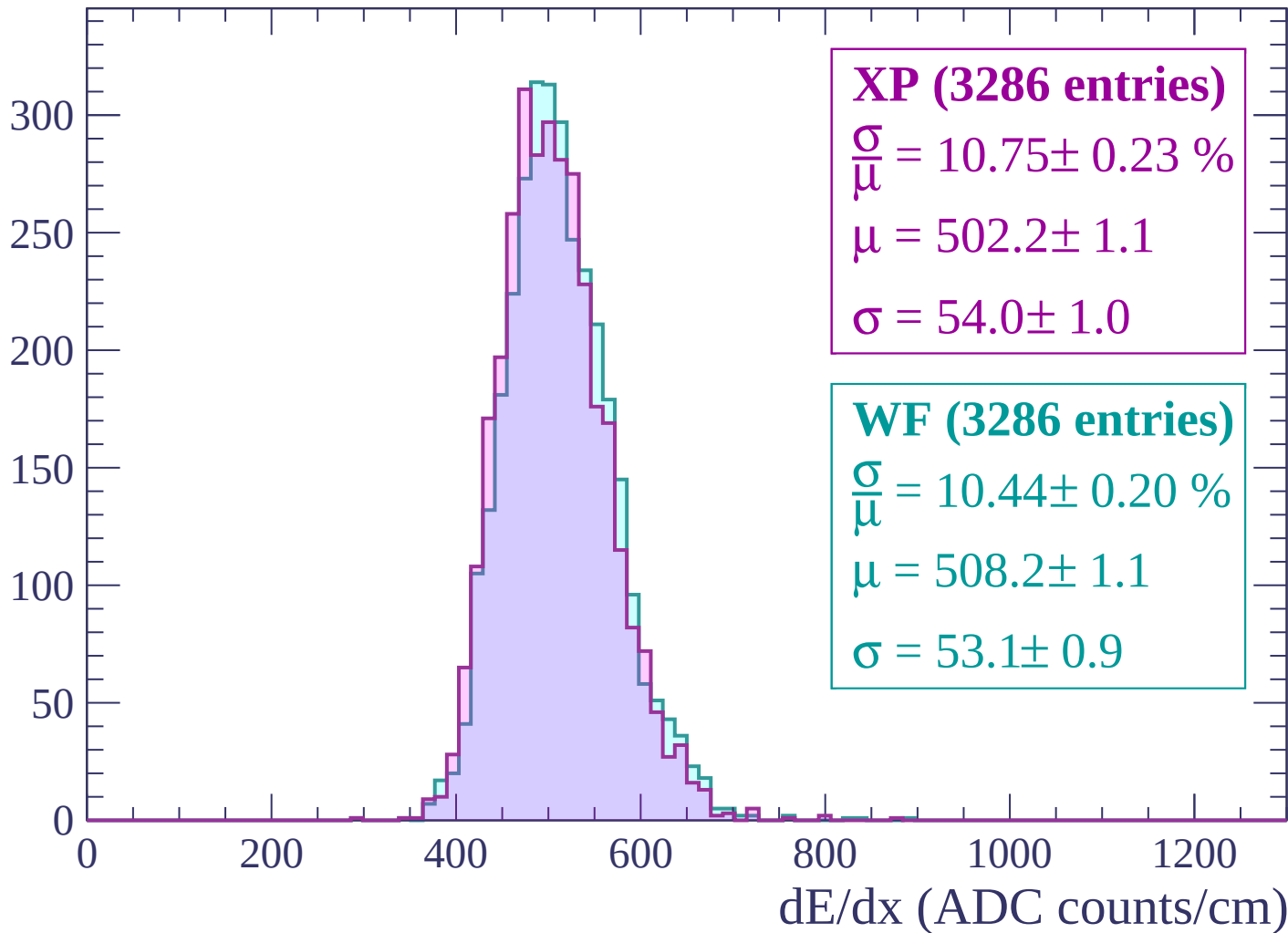
Energy loss | $2640 < p < 2700$

Count



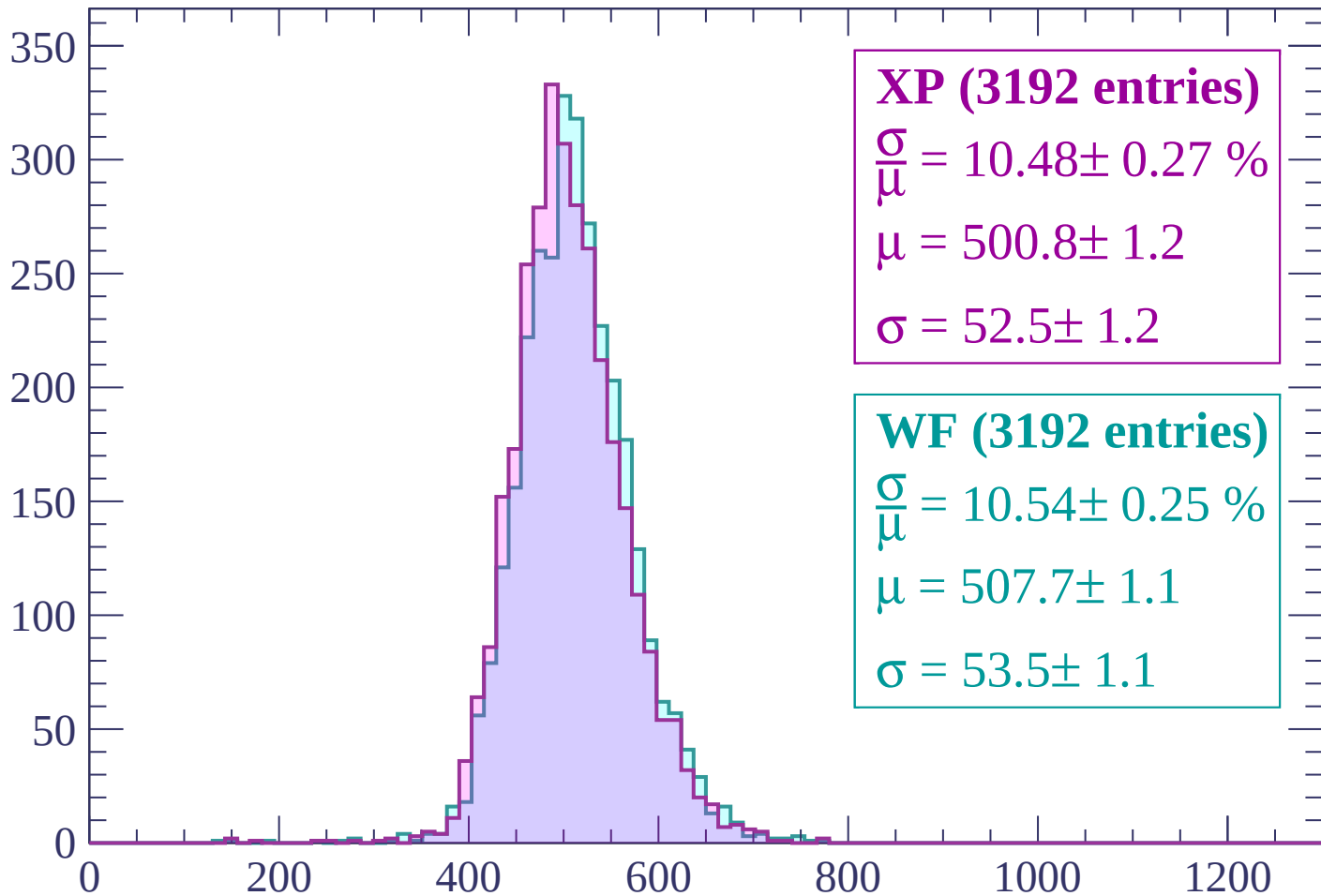
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (3192 entries)

$$\frac{\sigma}{\mu} = 10.48 \pm 0.27 \%$$

$$\mu = 500.8 \pm 1.2$$

$$\sigma = 52.5 \pm 1.2$$

WF (3192 entries)

$$\frac{\sigma}{\mu} = 10.54 \pm 0.25 \%$$

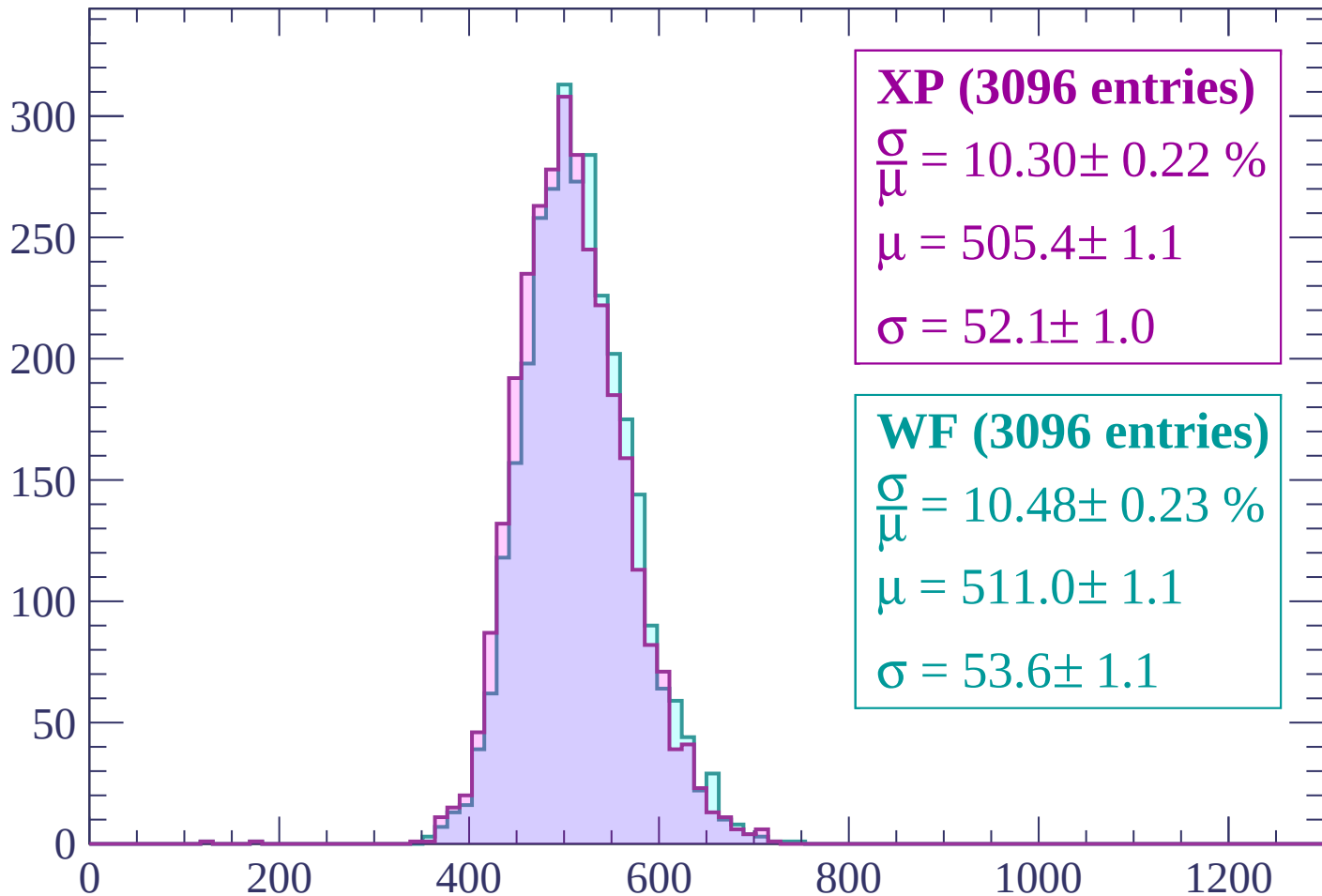
$$\mu = 507.7 \pm 1.1$$

$$\sigma = 53.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



XP (3096 entries)

$\frac{\sigma}{\mu} = 10.30 \pm 0.22 \%$

$\mu = 505.4 \pm 1.1$

$\sigma = 52.1 \pm 1.0$

WF (3096 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.23 \%$

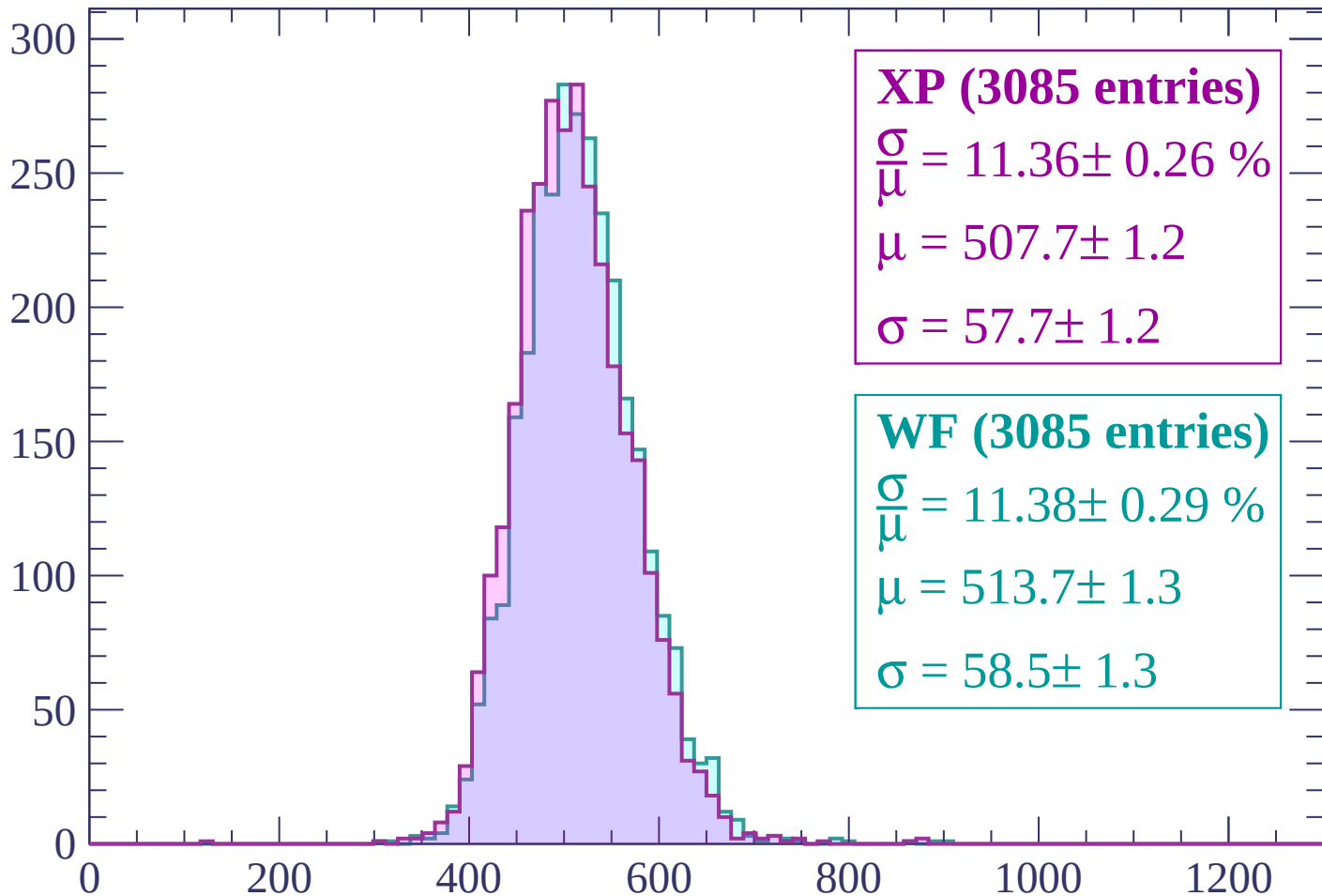
$\mu = 511.0 \pm 1.1$

$\sigma = 53.6 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (3085 entries)

$$\frac{\sigma}{\mu} = 11.36 \pm 0.26 \%$$

$$\mu = 507.7 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

WF (3085 entries)

$$\frac{\sigma}{\mu} = 11.38 \pm 0.29 \%$$

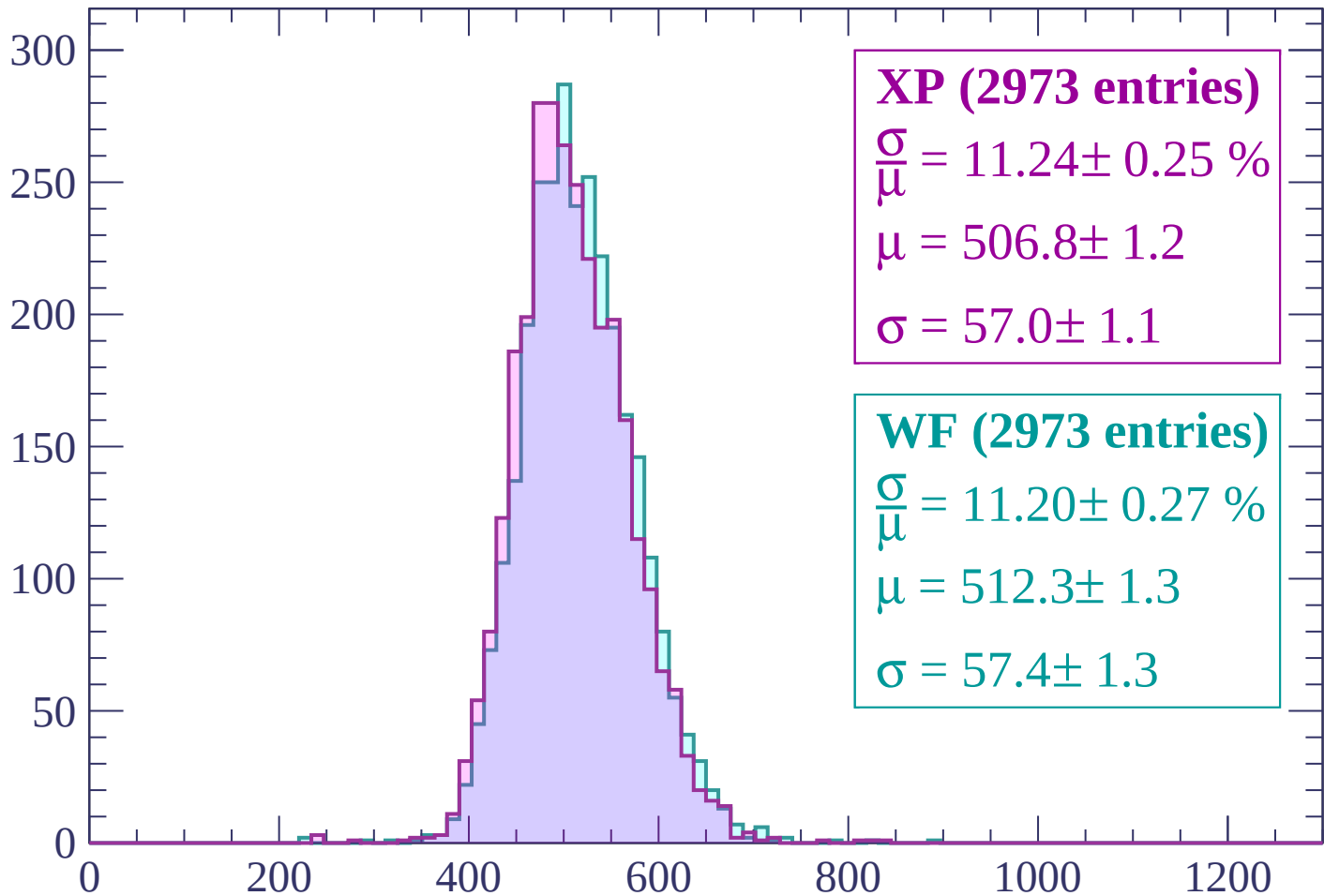
$$\mu = 513.7 \pm 1.3$$

$$\sigma = 58.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (2973 entries)

$$\frac{\sigma}{\mu} = 11.24 \pm 0.25 \%$$

$$\mu = 506.8 \pm 1.2$$

$$\sigma = 57.0 \pm 1.1$$

WF (2973 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.27 \%$$

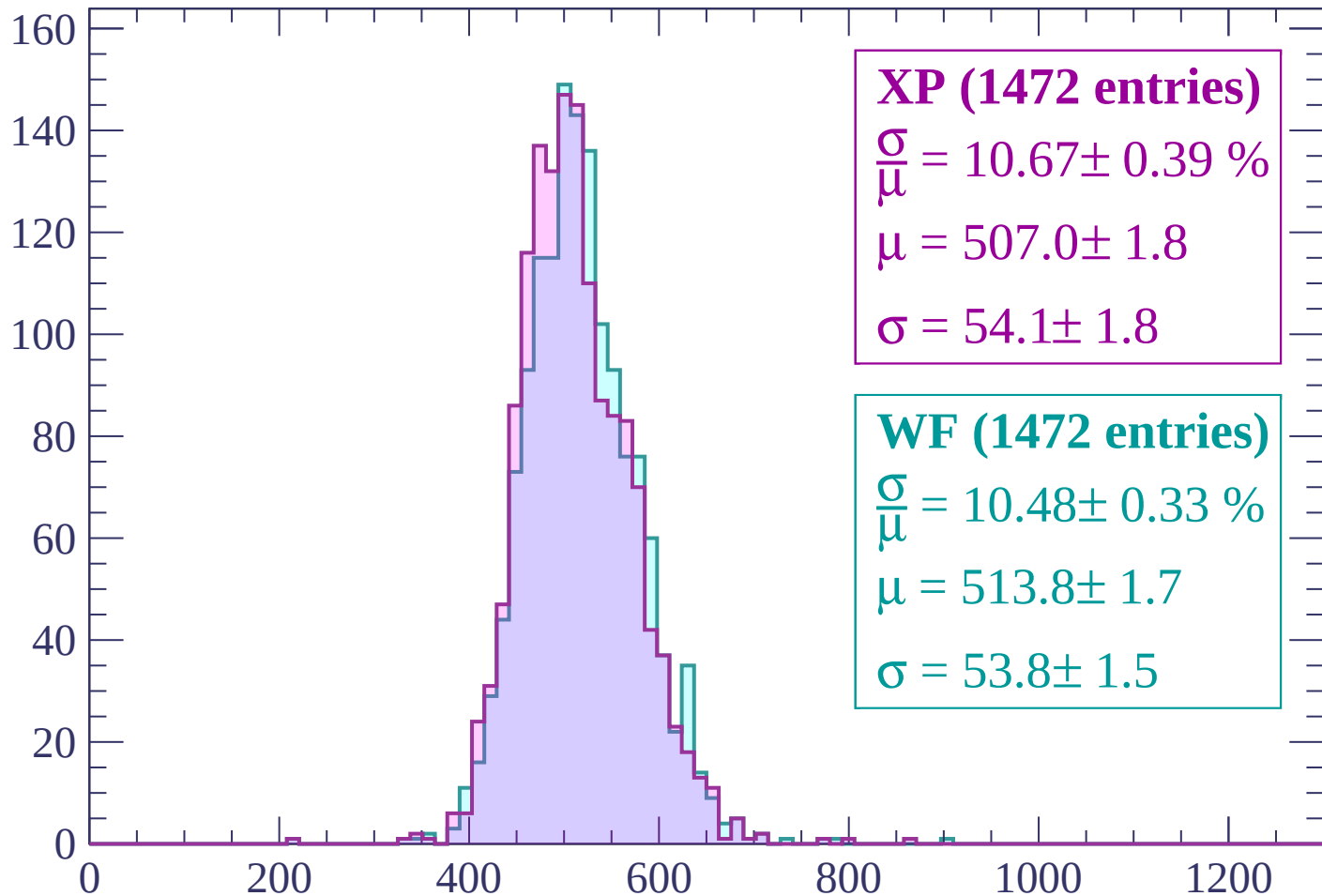
$$\mu = 512.3 \pm 1.3$$

$$\sigma = 57.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (1472 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.39 \%$

$\mu = 507.0 \pm 1.8$

$\sigma = 54.1 \pm 1.8$

WF (1472 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.33 \%$

$\mu = 513.8 \pm 1.7$

$\sigma = 53.8 \pm 1.5$

dE/dx (ADC counts/cm)

